Main Category	Category	Terms
Disease	OMIM_DISEASE	Breast cancer, somatic
Disease	OMIM_DISEASE	Colorectal cancer, somatic
Disease	OMIM_DISEASE	Ovarian cancer, somatic
Disease	OMIM_DISEASE	Proteus syndrome, somatic mosaic
Disease	OMIM_DISEASE	Cowden syndrome 6
Disease	UP_KW_DISEASE	Disease variant
Disease	UP_KW_DISEASE	Proto-oncogene
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Apoptosis
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Carbohydrate metabolism
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Glucose metabolism
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Glycogen biosynthesis
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Glycogen metabolism
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Neurogenesis
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Sugar transport
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Translation regulation
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transport
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Membrane
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Mitochondrion
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cell membrane
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Developmental protein
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Kinase

Main Category	Category	Terms
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Serine/threonine-protein kinase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Transferase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Developmental protein
Functional Annotations	UP_KW_PTM	Acetylation
Functional Annotations	UP_KW_PTM	Glycoprotein
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation
Functional Annotations	UP_KW_PTM	Disulfide bond
Functional Annotations	UP_KW_PTM	Isopeptide bond
Functional Annotations	UP_SEQ_FEATURE	ACT_SITE:Proton acceptor
Functional Annotations	UP_SEQ_FEATURE	CARBOHYD:O-linked (GlcNAc) serine; alternate
Functional Annotations	UP_SEQ_FEATURE	CARBOHYD:O-linked (GlcNAc) threonine
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Low complexity
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cter in ubiquitin)
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:AGC-kinase C-terminal
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:PH
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Protein kinase
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:E->K: Loss of membrane localization; when associated with Q-20.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->A: Impairs interaction with PtdIns(3,4,5)P3 and PtdIns(3,4)P2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->M: Abolished serine/ threonine-protein kinase activity.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->Q: Substantial reduction of phosphorylation at T-308 and S-473, loss of AKT activation, and loss of binding to PIP3

Main Category	Category	Terms
		as well as IGF1-induced membrane recruitment.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->Q: Substantial reduction of phosphorylation at T-308 and S-473, reduced AKT activation, and reduced binding to PIP3 as well as IGF1-induced membrane recruitment. Loss of membrane localization; when associated with K-17.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Slight increase of phosphorylation at T-308 and S-473.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Substantial reduction of ubiquitination, phosphorylation at T-308 and S-473, AKT activation as well as IGF1-induced membrane recruitment. Decrease in ubiquitination and phosphorylation at T-308 as well as impaired association with the membrane; when associated with K-17.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Substantial reduction of ubiquitination, phosphorylation at T-308 and S-473, AKT activation, loss of binding to PIP3 as well as IGF1-induced membrane recruitment.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:R->A: Impairs interaction with PtdIns(3,4,5)P3 and PtdIns(3,4)P2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:RCL->ACA: Abolished binding to cyclin-A, preventing phosphorylation by CDK2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:RDL->ADA: Abolished binding to cyclin-A, preventing phosphorylation by CDK2.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->D: 7-fold activation and 25-fold activation; when associated with D-308.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Reduces O-GlcNAc levels; Reduces O-GlcNAc levels even more; when associated with A-305.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Reduces O-GlcNAc levels; Reduces O-GlcNAc levels even more; when associated with A-312.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->D: 5-fold activation and 18-fold activation; when associated with D-473.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->Y: Abolishes phosphorylation at Thr-308.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->F: 55% inhibition of activation.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:Y->F: Significant loss of interaction with TNK2. Loss of membrane localization. Significant reduction in phosphorylation on Ser-473.

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	SITE:Cleavage; by caspase-3
Gene Ontology	GOTERM_BP_DIRECT	osteoblast differentiation
Gene Ontology	GOTERM_BP_DIRECT	maternal placenta development
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of endothelial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	cell migration involved in sprouting angiogenesis
Gene Ontology	GOTERM_BP_DIRECT	cytoplasmic translation
Gene Ontology	GOTERM_BP_DIRECT	complement receptor mediated signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	sphingosine-1-phosphate receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	glycogen metabolic process
Gene Ontology	GOTERM_BP_DIRECT	glycogen biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	regulation of glycogen biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	glucose metabolic process
Gene Ontology	GOTERM_BP_DIRECT	gluconeogenesis
Gene Ontology	GOTERM_BP_DIRECT	translational initiation
Gene Ontology	GOTERM_BP_DIRECT	regulation of translation
Gene Ontology	GOTERM_BP_DIRECT	protein phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	ubiquitin-dependent protein catabolic process
Gene Ontology	GOTERM_BP_DIRECT	protein import into nucleus
Gene Ontology	GOTERM_BP_DIRECT	NLS-bearing protein import into nucleus
Gene Ontology	GOTERM_BP_DIRECT	nitric oxide biosynthetic process

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	endoplasmic reticulum to Golgi vesicle-mediated transport
Gene Ontology	GOTERM_BP_DIRECT	Golgi to plasma membrane transport
Gene Ontology	GOTERM_BP_DIRECT	apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	activation-induced cell death of T cells
Gene Ontology	GOTERM_BP_DIRECT	inflammatory response
Gene Ontology	GOTERM_BP_DIRECT	response to oxidative stress
Gene Ontology	GOTERM_BP_DIRECT	signal transduction
Gene Ontology	GOTERM_BP_DIRECT	epidermal growth factor receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	G protein-coupled receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	germ cell development
Gene Ontology	GOTERM_BP_DIRECT	nervous system development
Gene Ontology	GOTERM_BP_DIRECT	cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	insulin receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	apoptotic mitochondrial changes
Gene Ontology	GOTERM_BP_DIRECT	cellular response to starvation
Gene Ontology	GOTERM_BP_DIRECT	response to heat
Gene Ontology	GOTERM_BP_DIRECT	response to hormone
Gene Ontology	GOTERM_BP_DIRECT	gene expression
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of autophagy
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of endothelial cell migration
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of gene expression

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of long-chain fatty acid import across plasma membrane
Gene Ontology	GOTERM_BP_DIRECT	fibroblast migration
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of fibroblast migration
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of sodium ion transport
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of glucose metabolic process
Gene Ontology	GOTERM_BP_DIRECT	regulation of neuron projection development
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of macroautophagy
Gene Ontology	GOTERM_BP_DIRECT	phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	cell migration
Gene Ontology	GOTERM_BP_DIRECT	protein ubiquitination
Gene Ontology	GOTERM_BP_DIRECT	peptidyl-serine phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	peptidyl-threonine phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	cytokine-mediated signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	mammalian oogenesis stage
Gene Ontology	GOTERM_BP_DIRECT	cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	protein catabolic process
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cell growth
Gene Ontology	GOTERM_BP_DIRECT	regulation of cell migration
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cell migration
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell migration

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	anaphase-promoting complex-dependent catabolic process
Gene Ontology	GOTERM_BP_DIRECT	SCF-dependent proteasomal ubiquitin-dependent protein catabolic process
Gene Ontology	GOTERM_BP_DIRECT	T cell costimulation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of protein ubiquitination
Gene Ontology	GOTERM_BP_DIRECT	regulation of myelination
Gene Ontology	GOTERM_BP_DIRECT	lipopolysaccharide-mediated signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	cellular response to nutrient levels
Gene Ontology	GOTERM_BP_DIRECT	TOR signaling
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of fatty acid beta-oxidation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of endodeoxyribonuclease activity
Gene Ontology	GOTERM_BP_DIRECT	response to food
Gene Ontology	GOTERM_BP_DIRECT	peripheral nervous system myelin maintenance
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of proteasomal ubiquitin-dependent protein catabolic process
Gene Ontology	GOTERM_BP_DIRECT	cellular response to insulin stimulus
Gene Ontology	GOTERM_BP_DIRECT	regulation of protein localization
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of peptidyl-serine phosphorylation
Gene Ontology	GOTERM_BP_DIRECT	cellular response to stress
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of toll-like receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of toll-like receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	toll-like receptor 4 signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	cellular response to amino acid starvation

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	response to fluid shear stress
Gene Ontology	GOTERM_BP_DIRECT	protein K29-linked ubiquitination
Gene Ontology	GOTERM_BP_DIRECT	intracellular signal transduction
Gene Ontology	GOTERM_BP_DIRECT	interleukin-18-mediated signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	cellular response to vascular endothelial growth factor stimulus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to decreased oxygen levels
Gene Ontology	GOTERM_BP_DIRECT	SREBP-SCAP complex retention in endoplasmic reticulum
Gene Ontology	GOTERM_BP_DIRECT	tRNA stabilization
Gene Ontology	GOTERM_BP_DIRECT	calcium import into the mitochondrion
Gene Ontology	GOTERM_BP_DIRECT	PERK-mediated unfolded protein response
Gene Ontology	GOTERM_BP_DIRECT	TORC1 signaling
Gene Ontology	GOTERM_BP_DIRECT	glucose homeostasis
Gene Ontology	GOTERM_BP_DIRECT	regulation of apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	Golgi to plasma membrane protein transport
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	proteasome-mediated ubiquitin-dependent protein catabolic process
Gene Ontology	GOTERM_BP_DIRECT	anoikis
Gene Ontology	GOTERM_BP_DIRECT	regulation of mRNA stability
Gene Ontology	GOTERM_BP_DIRECT	phosphatidylinositol 3-kinase/protein kinase B signal transduction
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of blood vessel endothelial cell migration
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of nitric oxide biosynthetic process

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cholesterol biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of fat cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cilium assembly
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of glycogen biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of Notch signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of proteolysis
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of translational initiation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of translational initiation
Gene Ontology	GOTERM_BP_DIRECT	nitric oxide metabolic process
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of D-glucose import
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of organ growth
Gene Ontology	GOTERM_BP_DIRECT	protein autophosphorylation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of lipid biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	insulin-like growth factor receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	behavioral response to pain
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of smooth muscle cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	striated muscle cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of protein metabolic process
Gene Ontology	GOTERM_BP_DIRECT	protein maturation
Gene Ontology	GOTERM_BP_DIRECT	excitatory postsynaptic potential

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	response to growth hormone
Gene Ontology	GOTERM_BP_DIRECT	regulation of vesicle-mediated transport
Gene Ontology	GOTERM_BP_DIRECT	mammary gland epithelial cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	labyrinthine layer blood vessel development
Gene Ontology	GOTERM_BP_DIRECT	protein localization to cilium
Gene Ontology	GOTERM_BP_DIRECT	response to UV-A
Gene Ontology	GOTERM_BP_DIRECT	response to growth factor
Gene Ontology	GOTERM_BP_DIRECT	protein localization to endoplasmic reticulum
Gene Ontology	GOTERM_BP_DIRECT	protein K11-linked ubiquitination
Gene Ontology	GOTERM_BP_DIRECT	cellular response to calcium ion
Gene Ontology	GOTERM_BP_DIRECT	cellular response to tumor necrosis factor
Gene Ontology	GOTERM_BP_DIRECT	cellular response to growth factor stimulus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to epidermal growth factor stimulus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to prostaglandin E stimulus
Gene Ontology	GOTERM_BP_DIRECT	tricarboxylic acid metabolic process
Gene Ontology	GOTERM_BP_DIRECT	establishment of protein localization to mitochondrion
Gene Ontology	GOTERM_BP_DIRECT	maintenance of protein location in mitochondrion
Gene Ontology	GOTERM_BP_DIRECT	cellular response to rapamycin
Gene Ontology	GOTERM_BP_DIRECT	protein K6-linked ubiquitination
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of release of cytochrome c from mitochondria
Gene Ontology	GOTERM_BP_DIRECT	cellular response to granulocyte macrophage colony-stimulating factor stimulus

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	execution phase of apoptosis
Gene Ontology	GOTERM_BP_DIRECT	vascular endothelial cell response to laminar fluid shear stress
Gene Ontology	GOTERM_BP_DIRECT	regulation of postsynapse organization
Gene Ontology	GOTERM_BP_DIRECT	tRNA (guanine-N7)-methylation
Gene Ontology	GOTERM_BP_DIRECT	regulation of tRNA methylation
Gene Ontology	GOTERM_BP_DIRECT	cellular response to oxidised low-density lipoprotein particle stimulus
Gene Ontology	GOTERM_BP_DIRECT	cGAS/STING signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of protein localization to lysosome
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cGAS/STING signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	beta-arrestin-dependent dopamine receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of G1/S transition of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of protein localization to nucleus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to peptide
Gene Ontology	GOTERM_BP_DIRECT	regulation of signal transduction by p53 class mediator
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cilium assembly
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of oxidative stress-induced intrinsic apoptotic signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of leukocyte cell-cell adhesion
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of protein localization to plasma membrane
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of protein maturation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of hydrogen peroxide-induced neuron intrinsic apoptotic signaling pathway
Gene	GOTERM_BP_DIRECT	negative regulation of PERK-mediated

Main Category	Category	Terms
Ontology		unfolded protein response
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of TORC1 signaling
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of TORC1 signaling
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of TORC2 signaling
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of protein localization to endoplasmic reticulum
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of anaphase-promoting complex-dependent catabolic process
Gene Ontology	GOTERM_BP_DIRECT	cellular response to nerve growth factor stimulus
Gene Ontology	GOTERM_BP_DIRECT	response to insulin-like growth factor stimulus
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of protein localization to cell surface
Gene Ontology	GOTERM_BP_DIRECT	regulation of type B pancreatic cell development
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of lymphocyte migration
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of extrinsic apoptotic signaling pathway in absence of ligand
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of intrinsic apoptotic signaling pathway
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	mitochondrion
Gene Ontology	GOTERM_CC_DIRECT	mitochondrial intermembrane space
Gene Ontology	GOTERM_CC_DIRECT	endoplasmic reticulum
Gene Ontology	GOTERM_CC_DIRECT	spindle
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene	GOTERM_CC_DIRECT	plasma membrane

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_CC_DIRECT	cell-cell junction
Gene Ontology	GOTERM_CC_DIRECT	cilium
Gene Ontology	GOTERM_CC_DIRECT	cell cortex
Gene Ontology	GOTERM_CC_DIRECT	microtubule cytoskeleton
Gene Ontology	GOTERM_CC_DIRECT	membrane
Gene Ontology	GOTERM_CC_DIRECT	lamellipodium
Gene Ontology	GOTERM_CC_DIRECT	vesicle
Gene Ontology	GOTERM_CC_DIRECT	protein-containing complex
Gene Ontology	GOTERM_CC_DIRECT	ciliary basal body
Gene Ontology	GOTERM_CC_DIRECT	postsynapse
Gene Ontology	GOTERM_CC_DIRECT	glutamatergic synapse
Gene Ontology	GOTERM_MF_DIRECT	nucleotide binding
Gene Ontology	GOTERM_MF_DIRECT	protein kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein serine/threonine kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein serine/threonine/tyrosine kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	calmodulin binding
Gene Ontology	GOTERM_MF_DIRECT	ATP binding
Gene Ontology	GOTERM_MF_DIRECT	phosphatidylinositol-3,4,5-trisphosphate binding
Gene Ontology	GOTERM_MF_DIRECT	kinase activity
Gene	GOTERM_MF_DIRECT	transferase activity

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_MF_DIRECT	enzyme binding
Gene Ontology	GOTERM_MF_DIRECT	kinase binding
Gene Ontology	GOTERM_MF_DIRECT	protein kinase binding
Gene Ontology	GOTERM_MF_DIRECT	nitric-oxide synthase regulator activity
Gene Ontology	GOTERM_MF_DIRECT	protein serine/threonine kinase inhibitor activity
Gene Ontology	GOTERM_MF_DIRECT	identical protein binding
Gene Ontology	GOTERM_MF_DIRECT	protein homodimerization activity
Gene Ontology	GOTERM_MF_DIRECT	phosphatidylinositol-3,4-bisphosphate binding
Gene Ontology	GOTERM_MF_DIRECT	14-3-3 protein binding
Gene Ontology	GOTERM_MF_DIRECT	potassium channel activator activity
Gene Ontology	GOTERM_MF_DIRECT	protein serine kinase activity
Gene Ontology	GOTERM_MF_DIRECT	TORC2 complex binding
General Annotations	CHROMOSOME	14
General Annotations	CYTOGENETIC_LOCATION	14q32.33
General Annotations	ENTREZ_GENE_SUMMARY	This gene encodes one of the three members of the human AKT serine-threonine protein kinase family which are often referred to as protein kinase B alpha, beta, and gamma. These highly similar AKT proteins all have an N-terminal pleckstrin homology domain, a serine/threonine-specific kinase domain and a C-terminal regulatory domain. These proteins are phosphorylated by phosphoinositide 3-kinase (PI3K). AKT/PI3K forms a key component of many signalling pathways that involve the binding of membrane-bound ligands such as receptor tyrosine kinases, G-protein coupled receptors, and integrin-linked kinase. These AKT proteins therefore regulate a wide variety of cellular functions including cell proliferation, survival,

Main Category	Category	Terms
		metabolism, and angiogenesis in both normal and malignant cells. AKT proteins are recruited to the cell membrane by phosphatidylinositol 3,4,5-trisphosphate (PIP3) after phosphorylation of phosphatidylinositol 4,5-bisphosphate (PIP2) by PI3K. Subsequent phosphorylation of both threonine residue 308 and serine residue 473 is required for full activation of the AKT1 protein encoded by this gene. Phosphorylation of additional residues also occurs, for example, in response to insulin growth factor-1 and epidermal growth factor. Protein phosphatases act as negative regulators of AKT proteins by dephosphorylating AKT or PIP3. The PI3K/AKT signalling pathway is crucial for tumor cell survival. Survival factors can suppress apoptosis in a transcription-independent manner by activating AKT1 which then phosphorylates and inactivates components of the apoptotic machinery. AKT proteins also participate in the mammalian target of rapamycin (mTOR) signalling pathway which controls the assembly of the eukaryotic translation initiation factor 4F (eIF4E) complex and this pathway, in addition to responding to extracellular signals from growth factors and cytokines, is disregulated in many cancers. Mutations in this gene are associated with multiple types of cancer and excessive tissue growth including Proteus syndrome and Cowden syndrome 6, and breast, colorectal, and ovarian cancers. Multiple alternatively spliced transcript variants have been found for this gene. [provided by RefSeq, Jul 2020]
General Annotations	UP_COMMENT_ACTIVITY_REGULATION	Three specific sites, one in the kinase domain (Thr-308) and the two other ones in the C-terminal regulatory region (Ser-473 and Tyr-474), need to be phosphorylated for its full activation (PubMed:15262962, PubMed:20481595, PubMed:21392984, PubMed:9512493, PubMed:9736715). Inhibited by pyrrolopyrimidine inhibitors like aniline triazole and spiroindoline (PubMed:18456494, PubMed:20810279). {ECO:0000269|PubMed:15262962, ECO:0000269|PubMed:18456494, ECO:0000269|PubMed:20481595, ECO:0000269|PubMed:20810279, ECO:0000269|PubMed:21392984, ECO:0000269|PubMed:9512493, ECO:0000269|PubMed:9736715, ECO:0000305|PubMed:20810279}.
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative splicing; Named isoforms=2; Name=1; IsoId=P31749-1; Sequence=Displayed; Name=2; IsoId=P31749-2; Sequence=VSP_056180;

Main Category	Category	Terms
General Annotations	UP_COMMENT_BIOPHYSICOCHEMICAL_PROPERTIES	Kinetic parameters: KM=52.8 uM for ATP (for purified and in vitro activated AKT1) {ECO:0000269|PubMed:16540465}; KM=0.5 uM for peptide substrate (for purified and in vitro activated AKT1) {ECO:0000269|PubMed:16540465}; KM=143.3 uM for ATP (for recombinant myristoylated AKT1 expressed and immunoprecipitated from Rat-1 cells) {ECO:0000269|PubMed:16540465}; KM=2.9 uM for peptide substrate (for recombinant myristoylated AKT1 expressed and immunoprecipitated from Rat-1 cells) {ECO:0000269|PubMed:16540465};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-seryl-[protein] + ATP = O-phospho-L-seryl-[protein] + ADP + H(+); Xref=Rhea:RHEA:17989, Rhea:RHEA-COMP:9863, Rhea:RHEA-COMP:11604, ChEBI:CHEBI:15378, ChEBI:CHEBI:29999, ChEBI:CHEBI:30616, ChEBI:CHEBI:83421, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000256|ARBA:ARBA00048679};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-seryl-[protein] + ATP = O-phospho-L-seryl-[protein] + ADP + H(+); Xref=Rhea:RHEA:17989, Rhea:RHEA-COMP:9863, Rhea:RHEA-COMP:11604, ChEBI:CHEBI:15378, ChEBI:CHEBI:29999, ChEBI:CHEBI:30616, ChEBI:CHEBI:83421, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000269|PubMed:12172553, ECO:0000269|PubMed:15861136, ECO:0000269|PubMed:16139227, ECO:0000269|PubMed:1718748, ECO:0000269|PubMed:1851997, ECO:0000269|PubMed:26440888, ECO:0000269|PubMed:29343641, ECO:0000269|PubMed:31548394, ECO:0000269|PubMed:32322062, ECO:0000269|PubMed:33594058};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-threonyl-[protein] + ATP = O-phospho-L-threonyl-[protein] + ADP + H(+); Xref=Rhea:RHEA:46608, Rhea:RHEA-COMP:11060, Rhea:RHEA-COMP:11605, ChEBI:CHEBI:15378, ChEBI:CHEBI:30013, ChEBI:CHEBI:30616, ChEBI:CHEBI:61977, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000256|ARBA:ARBA00047899};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-threonyl-[protein] + ATP = O-phospho-L-threonyl-[protein] + ADP + H(+); Xref=Rhea:RHEA:46608, Rhea:RHEA-COMP:11060, Rhea:RHEA-COMP:11605, ChEBI:CHEBI:15378, ChEBI:CHEBI:30013, ChEBI:CHEBI:30616, ChEBI:CHEBI:61977,

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		ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000269| PubMed:12172553, ECO:0000269| PubMed:16139227, ECO:0000269| PubMed:1718748, ECO:0000269| PubMed:1851997};
General Annotations	UP_COMMENT_DISEASE	Breast cancer (BC) [MIM:114480]: A common malignancy originating from breast epithelial tissue. Breast neoplasms can be distinguished by their histologic pattern. Invasive ductal carcinoma is by far the most common type. Breast cancer is etiologically and genetically heterogeneous. Important genetic factors have been indicated by familial occurrence and bilateral involvement. Mutations at more than one locus can be involved in different families or even in the same case. {ECO:0000269| PubMed:17611497}. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Colorectal cancer (CRC) [MIM:114500]: A complex disease characterized by malignant lesions arising from the inner wall of the large intestine (the colon) and the rectum. Genetic alterations are often associated with progression from premalignant lesion (adenoma) to invasive adenocarcinoma. Risk factors for cancer of the colon and rectum include colon polyps, long-standing ulcerative colitis, and genetic family history. Note=The gene represented in this entry may be involved in disease pathogenesis.
General Annotations	UP_COMMENT_DISEASE	Cowden syndrome 6 (CWS6) [MIM:615109]: A form of Cowden syndrome, a hamartomatous polyposis syndrome with age-related penetrance. Cowden syndrome is characterized by hamartomatous lesions affecting derivatives of ectodermal, mesodermal and endodermal layers, macrocephaly, facial trichilemmomas (benign tumors of the hair follicle infundibulum), acral keratoses, papillomatous papules, and elevated risk for development of several types of malignancy, particularly breast carcinoma in women and thyroid carcinoma in both men and women. Colon cancer and renal cell carcinoma have also been reported. Hamartomas can be found in virtually every organ, but most commonly in the skin, gastrointestinal tract, breast and thyroid. {ECO:0000269| PubMed:12176338, ECO:0000269| PubMed:23246288}. Note=The disease is caused by variants affecting the gene represented in this entry.
General	UP_COMMENT_DISEASE	Note=Genetic variations in AKT1 may play a

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Annotations		role in susceptibility to ovarian cancer.
General Annotations	UP_COMMENT_DISEASE	Proteus syndrome (PROTEUSS) [MIM:176920]: A highly variable, severe disorder of asymmetric and disproportionate overgrowth of body parts, connective tissue nevi, epidermal nevi, dysregulated adipose tissue, and vascular malformations. Many features of Proteus syndrome overlap with other overgrowth syndromes. {ECO:0000269|PubMed:18954143, ECO:0000269|PubMed:21793738}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DOMAIN	Binding of the PH domain to phosphatidylinositol 3,4,5- trisphosphate (PI(3,4,5)P3) following phosphatidylinositol 3-kinase alpha (PIK3CA) activity results in its targeting to the plasma membrane (PubMed:12176338, PubMed:12964941). PI(3,4,5)P3 is also required for phosphorylation at Thr-308 and subsequent activation (By similarity). The PH domain mediates interaction with TNK2 and Tyr-176 is also essential for this interaction (PubMed:20333297). {ECO:0000250|UniProtKB:P47196, ECO:0000269|PubMed:12176338, ECO:0000269|PubMed:12964941, ECO:0000269|PubMed:20333297}.
General Annotations	UP_COMMENT_DOMAIN	The AGC-kinase C-terminal mediates interaction with THEM4. {ECO:0000269|PubMed:11598301}.
General Annotations	UP_COMMENT_FUNCTION	AKT1 is one of 3 closely related serine/threonine-protein kinases (AKT1, AKT2 and AKT3) called the AKT kinase, and which regulate many processes including metabolism, proliferation, cell survival, growth and angiogenesis (PubMed:11882383, PubMed:15526160, PubMed:15861136, PubMed:21432781, PubMed:21620960, PubMed:31204173). This is mediated through serine and/or threonine phosphorylation of a range of downstream substrates (PubMed:11882383, PubMed:15526160, PubMed:21432781, PubMed:21620960, PubMed:29343641, PubMed:31204173). Over 100 substrate candidates have been reported so far, but for most of them, no isoform specificity has been reported (PubMed:11882383, PubMed:15526160, PubMed:21432781, PubMed:21620960). AKT is responsible of the regulation of glucose uptake by mediating insulin-induced translocation of the SLC2A4/GLUT4 glucose transporter to the cell surface (By similarity).

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		Phosphorylation of PTPN1 at 'Ser-50' negatively modulates its phosphatase activity preventing dephosphorylation of the insulin receptor and the attenuation of insulin signaling (By similarity). Phosphorylation of TBC1D4 triggers the binding of this effector to inhibitory 14-3-3 proteins, which is required for insulin-stimulated glucose transport (PubMed:11994271). AKT also regulates the storage of glucose in the form of glycogen by phosphorylating GSK3A at 'Ser-21' and GSK3B at 'Ser-9', resulting in inhibition of its kinase activity (By similarity). Phosphorylation of GSK3 isoforms by AKT is also thought to be one mechanism by which cell proliferation is driven (By similarity). AKT also regulates cell survival via the phosphorylation of MAP3K5 (apoptosis signal-related kinase) (PubMed:11154276). Phosphorylation of 'Ser-83' decreases MAP3K5 kinase activity stimulated by oxidative stress and thereby prevents apoptosis (PubMed:11154276). AKT mediates insulin-stimulated protein synthesis by phosphorylating TSC2 at 'Ser-939' and 'Thr-1462', thereby activating the mTORC1 signaling pathway, and leading to both phosphorylation of 4E-BP1 and in activation of RPS6KB1 (PubMed:12150915, PubMed:12172553). Also regulates the mTORC1 signaling pathway by catalyzing phosphorylation of CASTOR1 and DEPDC5 (PubMed:31548394, PubMed:33594058). AKT plays a role as key modulator of the AKT-mTOR signaling pathway controlling the tempo of the process of newborn neurons integration during adult neurogenesis, including correct neuron positioning, dendritic development and synapse formation (By similarity). Part of a positive feedback loop of mTORC2 signaling by mediating phosphorylation of MAPKAP1/SIN1, promoting mTORC2 activation (By similarity). AKT is involved in the phosphorylation of members of the FOXO factors (Forkhead family of transcription factors), leading to binding of 14-3-3 proteins and cytoplasmic localization (PubMed:10358075). In particular, FOXO1 is phosphorylated at 'Thr-24', 'Ser-256' and 'Ser-319' (PubMed:10358075). FOXO3 and FOXO4 are phosphorylated on equivalent sites (PubMed:10358075). AKT has an important role in the regulation of NF-kappa-B-dependent gene transcription and positively regulates the activity of CREB1 (cyclic AMP (cAMP)-response element binding protein) (PubMed:9829964). The phosphorylation of CREB1 induces the binding of accessory proteins that are

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		necessary for the transcription of pro-survival genes such as BCL2 and MCL1 (PubMed:9829964). AKT phosphorylates 'Ser-454' on ATP citrate lyase (ACLY), thereby potentially regulating ACLY activity and fatty acid synthesis (By similarity). Activates the 3B isoform of cyclic nucleotide phosphodiesterase (PDE3B) via phosphorylation of 'Ser-273', resulting in reduced cyclic AMP levels and inhibition of lipolysis (By similarity). Phosphorylates PIKFYVE on 'Ser-318', which results in increased PI(3)P-5 activity (By similarity). The Rho GTPase-activating protein DLC1 is another substrate and its phosphorylation is implicated in the regulation cell proliferation and cell growth (By similarity). Signals downstream of phosphatidylinositol 3-kinase (PI(3)K) to mediate the effects of various growth factors such as platelet-derived growth factor (PDGF), epidermal growth factor (EGF), insulin and insulin-like growth factor 1 (IGF1) (PubMed:12176338, PubMed:12964941). AKT mediates the antiapoptotic effects of IGF1 (By similarity). Essential for the SPATA13-mediated regulation of cell migration and adhesion assembly and disassembly (PubMed:19934221). May be involved in the regulation of the placental development (By similarity). Phosphorylates STK4/MST1 at 'Thr-120' and 'Thr-387' leading to inhibition of its: kinase activity, nuclear translocation, autophosphorylation and ability to phosphorylate FOXO3 (PubMed:17726016). Phosphorylates STK3/MST2 at 'Thr-117' and 'Thr-384' leading to inhibition of its: cleavage, kinase activity, autophosphorylation at Thr-180, binding to RASSF1 and nuclear translocation (PubMed:20086174). Phosphorylates SRPK2 and enhances its kinase activity towards SRSF2 and ACIN1 and promotes its nuclear translocation (PubMed:19592491). Phosphorylates RAF1 at 'Ser-259' and negatively regulates its activity (PubMed:10576742). Phosphorylation of BAD stimulates its pro-apoptotic activity (PubMed:10926925). Phosphorylates KAT6A at 'Thr-369' and this phosphorylation inhibits the interaction of KAT6A with PML and negatively regulates its acetylation activity towards p53/TP53 (PubMed:23431171). Phosphorylates palladin (PALLD), modulating cytoskeletal organization and cell motility (PubMed:20471940). Phosphorylates prohibitin (PHB), playing an important role in cell metabolism and proliferation (PubMed:18507042). Phosphorylates CDKN1A, for which phosphorylation at

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		'Thr-145' induces its release from CDK2 and cytoplasmic relocalization (PubMed:16982699). These recent findings indicate that the AKT1 isoform has a more specific role in cell motility and proliferation (PubMed:16139227). Phosphorylates CLK2 thereby controlling cell survival to ionizing radiation (PubMed:20682768). Phosphorylates PCK1 at 'Ser-90', reducing the binding affinity of PCK1 to oxaloacetate and changing PCK1 into an atypical protein kinase activity using GTP as donor (PubMed:32322062). Also acts as an activator of TMEM175 potassium channel activity in response to growth factors: forms the lysoK(GF) complex together with TMEM175 and acts by promoting TMEM175 channel activation, independently of its protein kinase activity (PubMed:32228865). Acts as a regulator of mitochondrial calcium uptake by mediating phosphorylation of MICU1 in the mitochondrial intermembrane space, impairing MICU1 maturation (PubMed:30504268). Acts as an inhibitor of tRNA methylation by mediating phosphorylation of the N-terminus of METTL1, thereby inhibiting METTL1 methyltransferase activity (PubMed:15861136). In response to LPAR1 receptor pathway activation, phosphorylates Rabin8/RAB3IP which alters its activity and phosphorylates WDR44 which induces WDR44 binding to Rab11, thereby switching Rab11 vesicular function from preciliary trafficking to endocytic recycling (PubMed:31204173). {ECO:0000250| UniProtKB:P31750, ECO:0000250| UniProtKB:P47196, ECO:0000269| PubMed:10358075, ECO:0000269| PubMed:10576742, ECO:0000269| PubMed:10926925, ECO:0000269| PubMed:11154276, ECO:0000269| PubMed:11994271, ECO:0000269| PubMed:12150915, ECO:0000269| PubMed:12172553, ECO:0000269| PubMed:12176338, ECO:0000269| PubMed:12964941, ECO:0000269| PubMed:15861136, ECO:0000269| PubMed:16139227, ECO:0000269| PubMed:16982699, ECO:0000269| PubMed:17726016, ECO:0000269| PubMed:18507042, ECO:0000269| PubMed:19592491, ECO:0000269| PubMed:19934221, ECO:0000269| PubMed:20086174, ECO:0000269| PubMed:20471940, ECO:0000269| PubMed:20682768, ECO:0000269| PubMed:23431171, ECO:0000269| PubMed:30504268, ECO:0000269| PubMed:31204173, ECO:0000269| PubMed:31548394, ECO:0000269|

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		PubMed:32228865, ECO:0000269| PubMed:32322062, ECO:0000269| PubMed:33594058, ECO:0000269| PubMed:9829964, ECO:0000303| PubMed:11882383, ECO:0000303| PubMed:15526160, ECO:0000303| PubMed:21432781, ECO:0000303| PubMed:21620960}.
General Annotations	UP_COMMENT_INTERACTION	A0A0S2Z3D6; Q5JR59-3: MTUS2; NbExp=3; IntAct=EBI-16430470, EBI-11522433; A0A0S2Z3D6; P56279: TCL1A; NbExp=3; IntAct=EBI-16430470, EBI-749995;
General Annotations	UP_COMMENT_INTERACTION	B0LPE5; P56279: TCL1A; NbExp=3; IntAct=EBI-10175397, EBI-749995;
General Annotations	UP_COMMENT_INTERACTION	P31749; Q6UY14-3: ADAMTSL4; NbExp=3; IntAct=EBI-296087, EBI-10173507; P31749; P31749: AKT1; NbExp=3; IntAct=EBI-296087, EBI-296087; P31749; P31751: AKT2; NbExp=3; IntAct=EBI-296087, EBI-296058; P31749; Q86V81: ALYREF; NbExp=5; IntAct=EBI-296087, EBI-347640; P31749; Q8IXJ9: ASXL1; NbExp=2; IntAct=EBI-296087, EBI-1646500; P31749; P54253: ATXN1; NbExp=6; IntAct=EBI-296087, EBI-930964; P31749; O95999: BCL10; NbExp=5; IntAct=EBI-296087, EBI-958922; P31749; Q9H0C5: BTBD1; NbExp=3; IntAct=EBI-296087, EBI-935503; P31749; Q9UQM7: CAMK2A; NbExp=3; IntAct=EBI-296087, EBI-1383687; P31749; P20248: CCNA2; NbExp=2; IntAct=EBI-296087, EBI-457097; P31749; Q16543: CDC37; NbExp=5; IntAct=EBI-296087, EBI-295634; P31749; Q8TAP6: CEP76; NbExp=3; IntAct=EBI-296087, EBI-742887; P31749; Q92793: CREBBP; NbExp=3; IntAct=EBI-296087, EBI-81215; P31749; Q15438: CYTH1; NbExp=3; IntAct=EBI-296087, EBI-997830; P31749; Q96D03: DDIT4L; NbExp=3; IntAct=EBI-296087, EBI-742054; P31749; P26358: DNMT1; NbExp=6; IntAct=EBI-296087, EBI-719459; P31749; Q1W6H9: FAM110C; NbExp=2; IntAct=EBI-296087, EBI-3942563; P31749; Q12778: FOXO1; NbExp=2; IntAct=EBI-296087, EBI-1108782; P31749; O43524: FOXO3; NbExp=3; IntAct=EBI-296087, EBI-1644164; P31749; P06241: FYN; NbExp=3; IntAct=EBI-296087, EBI-515315; P31749; P22466: GAL; NbExp=3; IntAct=EBI-296087, EBI-6624768; P31749; P49841: GSK3B; NbExp=4; IntAct=EBI-296087, EBI-373586; P31749; Q96LI6-3: HSFY2; NbExp=3;

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		IntAct=EBI-296087, EBI-25830912; P31749; P08238: HSP90AB1; NbExp=3; IntAct=EBI-296087, EBI-352572; P31749; P42858: HTT; NbExp=3; IntAct=EBI-296087, EBI-466029; P31749; Q5S007: LRRK2; NbExp=6; IntAct=EBI-296087, EBI-5323863; P31749; Q99683: MAP3K5; NbExp=2; IntAct=EBI-296087, EBI-476263; P31749; P10636: MAPT; NbExp=2; IntAct=EBI-296087, EBI-366182; P31749; Q00987: MDM2; NbExp=4; IntAct=EBI-296087, EBI-389668; P31749; Q14696: MESD; NbExp=3; IntAct=EBI-296087, EBI-6165891; P31749; P42345: MTOR; NbExp=4; IntAct=EBI-296087, EBI-359260; P31749; Q9BZQ8: NIBAN1; NbExp=2; IntAct=EBI-296087, EBI-6916466; P31749; P04150: NR3C1; NbExp=5; IntAct=EBI-296087, EBI-493507; P31749; O15530: PDPK1; NbExp=4; IntAct=EBI-296087, EBI-717097; P31749; Q96S96: PEBP4; NbExp=2; IntAct=EBI-296087, EBI-8563667; P31749; P19174: PLCG1; NbExp=9; IntAct=EBI-296087, EBI-79387; P31749; O60437: PPL; NbExp=2; IntAct=EBI-296087, EBI-368321; P31749; P62136: PPP1CA; NbExp=4; IntAct=EBI-296087, EBI-357253; P31749; P67775: PPP2CA; NbExp=4; IntAct=EBI-296087, EBI-712311; P31749; P30153: PPP2R1A; NbExp=2; IntAct=EBI-296087, EBI-302388; P31749; P17612: PRKACA; NbExp=3; IntAct=EBI-296087, EBI-476586; P31749; Q05513: PRKCZ; NbExp=2; IntAct=EBI-296087, EBI-295351; P31749; Q8WUY3: PRUNE2; NbExp=3; IntAct=EBI-296087, EBI-743880; P31749; P23443: RPS6KB1; NbExp=3; IntAct=EBI-296087, EBI-1775921; P31749; Q15047: SETDB1; NbExp=9; IntAct=EBI-296087, EBI-79691; P31749; Q96EB6: SIRT1; NbExp=5; IntAct=EBI-296087, EBI-1802965; P31749; Q13043: STK4; NbExp=13; IntAct=EBI-296087, EBI-367376; P31749; Q9UNE7-1: STUB1; NbExp=5; IntAct=EBI-296087, EBI-15687717; P31749; P56279: TCL1A; NbExp=8; IntAct=EBI-296087, EBI-749995; P31749; Q9NYB0: TERF2IP; NbExp=2; IntAct=EBI-296087, EBI-750109; P31749; Q5T1C6: THEM4; NbExp=4; IntAct=EBI-296087, EBI-7684443; P31749; Q92547: TOPBP1; NbExp=2; IntAct=EBI-296087, EBI-308302; P31749; P53804: TTC3; NbExp=4; IntAct=EBI-296087, EBI-2681313; P31749; P10599: TXN; NbExp=3; IntAct=EBI-296087, EBI-594644; P31749; P08670: VIM;

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		NbExp=29; IntAct=EBI-296087, EBI-353844; P31749; Q6GPH4: XAF1; NbExp=4; IntAct=EBI-296087, EBI-2815120; P31749; P17028: ZNF24; NbExp=3; IntAct=EBI-296087, EBI-707773; P31749; P29067: Arrb2; Xeno; NbExp=2; IntAct=EBI-296087, EBI-1636616; P31749; Q83730: m005R; Xeno; NbExp=3; IntAct=EBI-296087, EBI-6859930; P31749; P03165: X; Xeno; NbExp=3; IntAct=EBI-296087, EBI-7683985; P31749-1; Q15118-1: PDK1; NbExp=2; IntAct=EBI-12562306, EBI-12562315;
General Annotations	UP_COMMENT_PTM	Acetylated on Lys-14 and Lys-20 by the histone acetyltransferases EP300 and KAT2B. Acetylation results in reduced phosphorylation and inhibition of activity. Deacetylated at Lys-14 and Lys-20 by SIRT1. SIRT1-mediated deacetylation relieves the inhibition. {ECO:0000269|PubMed:21775285}.
General Annotations	UP_COMMENT_PTM	Cleavage by caspase-3/CASP3 (By similarity). Cleaved at the caspase-3 consensus site Asp-462 during apoptosis, resulting in down- regulation of the AKT signaling pathway and decreased cell survival (PubMed:23152800). {ECO:0000250|UniProtKB:P31750, ECO:0000269|PubMed:23152800}.
General Annotations	UP_COMMENT_PTM	O-GlcNAcylation at Thr-305 and Thr-312 inhibits activating phosphorylation at Thr-308 via disrupting the interaction between AKT1 and PDPK1. O-GlcNAcylation at Ser-473 also probably interferes with phosphorylation at this site. {ECO:0000269|PubMed:14761976, ECO:0000269|PubMed:15047712, ECO:0000269|PubMed:15718470, ECO:0000269|PubMed:16266983, ECO:0000269|PubMed:17013611, ECO:0000269|PubMed:18456494, ECO:0000269|PubMed:20333297, ECO:0000269|PubMed:20481595, ECO:0000269|PubMed:20978158, ECO:0000269|PubMed:21464307, ECO:0000269|PubMed:23799035, ECO:0000269|PubMed:8978681, ECO:0000269|PubMed:9512493, ECO:0000269|PubMed:9736715}.
General Annotations	UP_COMMENT_PTM	Phosphorylation on Thr-308, Ser-473 and Tyr-474 is required for full activity (PubMed:12149249, PubMed:15047712, PubMed:15262962, PubMed:16266983, PubMed:18456494, PubMed:20481595, PubMed:20978158, PubMed:8978681, PubMed:9512493, PubMed:9736715). Phosphorylation of the activation loop at

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		Thr-308 by PDPK1/PDK1 is a prerequisite for full activation (PubMed:9512493). Phosphorylation by mTORC2 in response to growth factors plays a key role in AKT1 activation: mTORC2 phosphorylates different sites depending on the context, such as Thr- 450, Ser-473, Ser-477 or Thr-479, thereby facilitating subsequent phosphorylation of the activation loop by PDPK1/PDK1 (PubMed:15718470, PubMed:24670654). Phosphorylation at Ser-473 by mTORC2 promotes ubiquitination and degradation by the proteasome (By similarity). Also phosphorylated at Ser-477 and Thr-479 by CDK2, facilitating subsequent phosphorylation of the activation loop by PDPK1/PDK1 (PubMed:24670654). Activated TNK2 phosphorylates it on Tyr-176 resulting in its binding to the anionic plasma membrane phospholipid PA (PubMed:20333297). This phosphorylated form localizes to the cell membrane, where it is targeted by PDPK1 and PDPK2 for further phosphorylations on Thr-308 and Ser-473 leading to its activation (PubMed:20333297). Phosphorylated at Thr-308 and Ser-473 by IKBKE and TBK1 (PubMed:21464307). Ser-473 phosphorylation is enhanced by interaction with AGAP2 isoform 2 (PIKE- A) (PubMed:14761976). Ser-473 phosphorylation is enhanced in focal cortical dysplasias with Taylor-type balloon cells (PubMed:17013611). Ser-473 phosphorylation is enhanced by signaling through activated FLT3 (PubMed:16266983). Ser-473 is dephosphorylated by PHLPP (PubMed:28147277). Dephosphorylated at Thr-308 and Ser-473 by PP2A phosphatase (PubMed:21329884). The phosphorylated form of PPP2R5B is required for bridging AKT1 with PP2A phosphatase (PubMed:21329884). Ser-473 is dephosphorylated by CPPED1, leading to termination of signaling (PubMed:23799035). AIM2 acts as an inhibitor of AKT1 by inhibiting phosphorylation Ser-473: AIM2 acts both by inhibiting the activity of PRKDC/DNA-PK kinase and promoting dephosphorylation by PP2A phosphatase (By similarity). {ECO:0000250|UniProtKB:P31750, ECO:0000269|PubMed:12149249, ECO:0000269|PubMed:14761976, ECO:0000269|PubMed:15047712, ECO:0000269|PubMed:15262962, ECO:0000269|PubMed:15718470, ECO:0000269|PubMed:16266983, ECO:0000269|PubMed:17013611,

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		ECO:0000269|PubMed:18456494, ECO:0000269|PubMed:20333297, ECO:0000269|PubMed:20481595, ECO:0000269|PubMed:20978158, ECO:0000269|PubMed:21329884, ECO:0000269|PubMed:21464307, ECO:0000269|PubMed:23799035, ECO:0000269|PubMed:24670654, ECO:0000269|PubMed:28147277, ECO:0000269|PubMed:8978681, ECO:0000269|PubMed:9512493, ECO:0000269|PubMed:9736715}.
General Annotations	UP_COMMENT_PTM	Ubiquitinated; undergoes both 'Lys-48'- and 'Lys-63'-linked polyubiquitination. TRAF6-induced 'Lys-63'-linked AKT1 ubiquitination is critical for phosphorylation and activation (PubMed:19713527). When ubiquitinated, it translocates to the plasma membrane, where it becomes phosphorylated (PubMed:20059950). When fully phosphorylated and translocated into the nucleus, undergoes 'Lys-48'-polyubiquitination catalyzed by TTC3, leading to its degradation by the proteasome (PubMed:20059950). Also ubiquitinated by TRIM13 leading to its proteasomal degradation (PubMed:21333377). Phosphorylated, undergoes 'Lys-48'-linked polyubiquitination preferentially at Lys-284 catalyzed by MUL1, leading to its proteasomal degradation (PubMed:22410793). Ubiquitinated via 'Lys-48'-linked polyubiquitination by ZNRF1, leading to its degradation by the proteasome (By similarity). {ECO:0000250|UniProtKB:P31750, ECO:0000269|PubMed:19713527, ECO:0000269|PubMed:20059950, ECO:0000269|PubMed:21333377, ECO:0000269|PubMed:22410793}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the protein kinase superfamily. AGC Ser/Thr protein kinase family. RAC subfamily. {ECO:0000256|ARBA:ARBA00006935}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the protein kinase superfamily. AGC Ser/Thr protein kinase family. RAC subfamily. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cytoplasm {ECO:0000250|UniProtKB:P31750}. Nucleus {ECO:0000269|PubMed:20333297}. Cell membrane {ECO:0000269|PubMed:20333297}. Mitochondrion intermembrane space {ECO:0000250|UniProtKB:P31750}. Note=Nucleus after activation by integrin-linked protein kinase 1 (ILK1). Nuclear translocation is enhanced by interaction with TCL1A. Phosphorylation on Tyr-176 by

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		TNK2 results in its localization to the cell membrane where it is targeted for further phosphorylations on Thr-308 and Ser-473 leading to its activation and the activated form translocates to the nucleus. Colocalizes with WDFY2 in intracellular vesicles (PubMed:16792529). Also localizes to mitochondrial intermembrane space in response to rapamycin treatment (By similarity). {ECO:0000250| UniProtKB:P31750, ECO:0000269| PubMed:16792529}.
General Annotations	UP_COMMENT_SUBUNIT	Interacts with BTBD10 (By similarity). Interacts with KCTD20 (By similarity). Interacts (via the C-terminus) with CCDC88A (via its C-terminus). Interacts with GRB10; the interaction leads to GRB10 phosphorylation thus promoting YWHAE-binding (By similarity). Interacts with AGAP2 (isoform 2/PIKE-A); the interaction occurs in the presence of guanine nucleotides. Interacts with AKTIP. Interacts (via PH domain) with MTCP1, TCL1A and TCL1B. Interacts with CDKN1B; the interaction phosphorylates CDKN1B promoting 14-3-3 binding and cell-cycle progression. Interacts with MAP3K5 and TRAF6. Interacts with BAD, PPP2R5B, STK3 and STK4. Interacts (via PH domain) with SIRT1. Interacts with SRPK2 in a phosphorylation-dependent manner. Interacts with RAF1. Interacts with TRIM13; the interaction ubiquitinates AKT1 leading to its proteasomal degradation. Interacts with TNK2 and CLK2. Interacts (via the C-terminus) with THEM4 (via its C-terminus). Interacts with and phosphorylated by PDPK1. Interacts with PA2G4 (By similarity). Interacts with KIF14; the interaction is detected in the plasma membrane upon INS stimulation and promotes AKT1 phosphorylation (PubMed:24784001). Interacts with FAM83B; activates the PI3K/AKT signaling cascade (PubMed:23676467). Interacts with WDFY2 (via WD repeats 1-3) (PubMed:16792529). Forms a complex with WDFY2 and FOXO1 (By similarity). Interacts with FAM168A (PubMed:23251525). Interacts with SYAP1 (via phosphorylated form and BSD domain); this interaction is enhanced in a mTORC2-mediated manner in response to epidermal growth factor (EGF) stimulation and activates AKT1 (PubMed:23300339). Interacts with PKHM3 (By similarity). Interacts with FKBP5/ FKBP51; promoting interaction between Akt/AKT1 and PHLPP1, thereby enhancing dephosphorylation and subsequent activation of Akt/AKT1 (PubMed:28147277). Interacts with TMEM175; leading to

Main Category	Category	Terms
		formation of the lysoK(GF) complex (PubMed:32228865). Acts as a negative regulator of the cGAS-STING pathway by mediating phosphorylation of CGAS during mitosis, leading to its inhibition (PubMed:26440888). {ECO:0000250|UniProtKB:P31750, ECO:0000250|UniProtKB:P47196, ECO:0000269|PubMed:10576742, ECO:0000269|PubMed:10716693, ECO:0000269|PubMed:10926925, ECO:0000269|PubMed:10983986, ECO:0000269|PubMed:11154276, ECO:0000269|PubMed:11598301, ECO:0000269|PubMed:11839817, ECO:0000269|PubMed:12042314, ECO:0000269|PubMed:12176338, ECO:0000269|PubMed:12244301, ECO:0000269|PubMed:12964941, ECO:0000269|PubMed:14749367, ECO:0000269|PubMed:14761976, ECO:0000269|PubMed:15118108, ECO:0000269|PubMed:16139227, ECO:0000269|PubMed:16417524, ECO:0000269|PubMed:16792529, ECO:0000269|PubMed:17726016, ECO:0000269|PubMed:17932490, ECO:0000269|PubMed:19592491, ECO:0000269|PubMed:19713527, ECO:0000269|PubMed:20086174, ECO:0000269|PubMed:20333297, ECO:0000269|PubMed:20682768, ECO:0000269|PubMed:21329884, ECO:0000269|PubMed:21333377, ECO:0000269|PubMed:21775285, ECO:0000269|PubMed:22629392, ECO:0000269|PubMed:23251525, ECO:0000269|PubMed:23300339, ECO:0000269|PubMed:23676467, ECO:0000269|PubMed:24784001, ECO:0000269|PubMed:26440888, ECO:0000269|PubMed:28147277, ECO:0000269|PubMed:32228865}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Expressed in prostate cancer and levels increase from the normal to the malignant state (at protein level). Expressed in all human cell types so far analyzed. The Tyr-176 phosphorylated form shows a significant increase in expression in breast cancers during the progressive stages i.e. normal to hyperplasia (ADH), ductal carcinoma in situ (DCIS), invasive ductal carcinoma (IDC) and lymph node metastatic (LNMM) stages. {ECO:0000269|PubMed:1718748, ECO:0000269|PubMed:17932490, ECO:0000269|PubMed:20333297}.
Identifiers	ENTREZ_GENE_ID	207

Main Category	Category	Terms
Identifiers	ENSEMBL_GENE_ID	ENSG00000142208
Identifiers	UNIPROT_ID	X2CV47_HUMAN
Identifiers	UNIPROT_ID	X2CVF3_HUMAN
Identifiers	UNIPROT_ID	G3V3X1_HUMAN
Identifiers	UNIPROT_ID	G3V2I6_HUMAN
Identifiers	UNIPROT_ID	B3KVH4_HUMAN
Identifiers	UNIPROT_ID	B0LPE5_HUMAN
Identifiers	UNIPROT_ID	AKT1_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BHK3_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BHJ3_HUMAN
Identifiers	UNIPROT_ID	A0A804HJM6_HUMAN
Identifiers	UNIPROT_ID	A0A142IKA9_HUMAN
Identifiers	UNIPROT_ID	A0A0S2Z3D6_HUMAN
Identifiers	UNIPROT_ID	A0A087WY56_HUMAN
Interactions	DRUGBANK	ATP
Interactions	DRUGBANK	Arsenic trioxide
Interactions	DRUGBANK	Genistein
Interactions	DRUGBANK	Inositol 1,3,4,5-Tetrakisphosphate
Interactions	DRUGBANK	Resveratrol
Interactions	DRUGBANK	Archexin
Interactions	DRUGBANK	Enzastaurin
Interactions	DRUGBANK	Perifosine
Interactions	DRUGBANK	N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine
Interactions	DRUGBANK	5-(5-chloro-7H-pyrrolo[2,3-d]pyrimidin-4-yl)-4,5,6,7-tetrahydro-1H-imidazo[4,5-c]pyridine
Interactions	PUBCHEM	Gsk 690693
Interactions	PUBCHEM	Mk-2206
Interactions	PUBCHEM	Capivasertib
Interactions	PUBCHEM	Afuresertib
Interactions	PUBCHEM	Uprosertib
Interactions	UP_KW_LIGAND	ATP-binding

Main Category	Category	Terms
Interactions	UP_KW_LIGAND	Nucleotide-binding
Literature	GENERIF_SUMMARY	Binding of CTMP to PKBalpha reduces its activity by inhibiting phosphorylation on serine 473 and threonine 308.
Literature	GENERIF_SUMMARY	regulated in platelets by collage receptor glycoprotein VI
Literature	GENERIF_SUMMARY	Absence of mutations in the pleckstrin homology (PH) domain of protein kinase B (PKB/Akt) in malignant melanoma.
Literature	GENERIF_SUMMARY	Immunohistochemical localization of phosphorylated AKT/PKB in multiple myeloma cells.
Literature	GENERIF_SUMMARY	Akt enhances Mdm2-mediated ubiquitination and degradation of p53.
Literature	GENERIF_SUMMARY	Identification of 14-3-3zeta as a protein kinase B/Akt substrate.
Literature	GENERIF_SUMMARY	Phosphorylation of HDM2 by Akt, and protein binding
Literature	GENERIF_SUMMARY	IGF-I protects the cells from apoptosis by blocking the activation of caspases, which may be responsible for the loss of FAK and Akt.
Literature	GENERIF_SUMMARY	Akt promotes cell-cycle progression through the mechanisms of phosphorylation-dependent 14-3-3 binding to p27(Kip1) and cytoplasmic localization.
Literature	GENERIF_SUMMARY	AKT activation delays radiation-induced apoptosis, allowing the DNA repair mechanism more time to remove cyclobutane thymine dimers
Literature	GENERIF_SUMMARY	Different cellular localization, translocation, and insulin-induced phosphorylation of PKBalpha in HepG2 cells and hepatocytes
Literature	GENERIF_SUMMARY	This study shows that activation of Akt by pervanadate or serum is associated with tyrosine phosphorylation of Akt.
Literature	GENERIF_SUMMARY	3' phosphoinositide lipid-dependent translocation of PKB to the plasma membrane promotes serine 473 phosphorylation, which is, in turn, necessary for PDK1-mediated phosphorylation of threonine 308 and, consequentially, full PKB activation.
Literature	GENERIF_SUMMARY	determination of high-resolution structure of the pleckstrin homology domain of bound to phosphatidylinositol (3,4,5)-trisphosphate
Literature	GENERIF_SUMMARY	connective tissue growth factor induced

Main Category	Category	Terms
		fibronectin production, cell migration, and cytoskeletal rearrangement are associated with recruitment of Src and phosphorylation of p42/44 MAPK and protein kinase B
Literature	GENERIF_SUMMARY	These data indicate that Akt may contribute to tumor-cell proliferation by phosphorylation and cytosolic retention of p27(kip1), thus relieving CDK2 from p27-induced inhibition.
Literature	GENERIF_SUMMARY	Data show that activation of protein kinase B (PKB)/Akt, contributes to resistance to antiproliferative signals and breast cancer progression in part by impairing the nuclear import and action of p27.
Literature	GENERIF_SUMMARY	Data show that cytoplasmic relocalization of p27(kip1), secondary to Akt-mediated phosphorylation, inactivates the growth inhibitory properties of p27(kip1) and sustains the proliferation of breast cancer cells.
Literature	GENERIF_SUMMARY	PECAM-1 involvement through Akt/PKB activation in starvation-induced transendothelial migration of CD34+CD14+ circulating precursors
Literature	GENERIF_SUMMARY	data demonstrate that Rho/ROCK pathway negatively regulates eNOS phosphorylation through inhibition of protein kinase B (PKB), whereas it downregulates eNOS expression independent of PKB
Literature	GENERIF_SUMMARY	chemotherapeutic drugs exhibited their cytotoxic effects in part by down-regulating Akt signaling following TRADD expression
Literature	GENERIF_SUMMARY	We conclude that normal HERG function in HEK293 cells requires basal activity of PKB. Our data represent the first evidence that PKB phosphorylation regulates K(+) channels.
Literature	GENERIF_SUMMARY	Decreased phosphorylation of protein kinase B and erk1/erk2 in neutrophils from patients with myelodysplastic syndrome
Literature	GENERIF_SUMMARY	This protein protects HL60 leukemia cells from TRAIL-induced apoptosis through a mechanism involving NF-kappaB activation and cFLIP(L) up-regulation.
Literature	GENERIF_SUMMARY	Increased phosphorylation of this protein was observed in A431 clonal variants.
Literature	GENERIF_SUMMARY	The protein kinase Akt induces epithelial mesenchymal transition and promotes enhanced motility and invasiveness of squamous cell carcinoma lines.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	activation of Notch1 signaling mediates p53 function in HPV16 E6 and E7 cell transformation via phosphatidylinositol(PI3K)-PKB/Akt pathway
Literature	GENERIF_SUMMARY	Akt is activated by adrenomedullin
Literature	GENERIF_SUMMARY	AKT1 is regulated by JIP1
Literature	GENERIF_SUMMARY	results suggest that TRB3 promotes glucose output from liver under fasting conditions by binding to and interfering with Akt phosphorylation in response to residual insulin signaling
Literature	GENERIF_SUMMARY	Akt regulates basic helix-loop-helix transcription factor-coactivator complex formation and activity during neuronal differentiation
Literature	GENERIF_SUMMARY	phosphorylation by Akt regulates the antiapoptotic function of PED/PEA-15 at least in part by controlling the stability of PED/PEA-15
Literature	GENERIF_SUMMARY	There are clear indications of a cross-talk between PKB and important signaling molecules downstream of the T cell receptor that modulate the thresholds of thymocyte selection and T cell activation.
Literature	GENERIF_SUMMARY	Akt has a pivotal role in the regulation of endometrial cancer cell survival through the up-regulation of a specific inhibitor of apoptosis protein
Literature	GENERIF_SUMMARY	PI3K/Akt is essential for protecting human keratinocytes against UV-induced apoptosis, whereas NF-kappaB pathway provides little, if any, protective role
Literature	GENERIF_SUMMARY	The activity of p38 MAP Kinase is regulated by AKT1 in neoplasms.
Literature	GENERIF_SUMMARY	AKT is upregulated in prostate cancer and that expression is correlated with tumor progression
Literature	GENERIF_SUMMARY	protein kinase B/AKT mediates regulation of survivin levels by integrins
Literature	GENERIF_SUMMARY	AKT and MCAM are reciprocally regulated.
Literature	GENERIF_SUMMARY	Data show that inhibition of protein kinase B (PKB) involves atypical protein kinase C zeta, which physically interacts with PKB in unstimulated cells.
Literature	GENERIF_SUMMARY	dysregulation of the phosphoinositol-3-kinase/Akt pathway causes Lhermitte-Duclos disease

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	AKT1 activation disrupts breast acinar architecture and enhances proliferation in an mTOR-dependent manner.
Literature	GENERIF_SUMMARY	HER2/neu predominantly uses Akt to suppress RARE binding activity, which may be one mechanism by which HER2/neu induces ATRA resistance in breast cancer cells
Literature	GENERIF_SUMMARY	Gi, but not Gq or G12/13, signaling pathways are required for activation of Akt in platelets
Literature	GENERIF_SUMMARY	AKT has a role in CXCL16 signaling that induces cell-cell adhesion and aortic smooth muscle cell proliferation
Literature	GENERIF_SUMMARY	There is a down regulation of the serine/threonine kinase, Akt/PKB, concurrent with elevated endogenous GSK3beta kinase activity in familial Alzheimer's disease beta-amyloid precursor protein expressing cells.
Literature	GENERIF_SUMMARY	we propose that an increased expression level of the HMGA2 protein is closely associated with the malignant phenotype in the pancreatic exocrine system.
Literature	GENERIF_SUMMARY	demonstration of a specific role for S1P3 in S1P-induced Akt activation
Literature	GENERIF_SUMMARY	results suggest that matrix metalloproteinase-1 can stimulate dephosphorylation of Akt protein and neuronal death through a non-proteolytic mechanism that involves changes in integrin alpha2beta1 signaling
Literature	GENERIF_SUMMARY	Akt negatively regulates the cell aging of endothelial cells via a p53/p21-dependent pathway.
Literature	GENERIF_SUMMARY	Data show that respiratory syncytial virus stimulates sphingosine kinase activity in lung epithelial cells, leading to activation of antiapoptotic sphingosine 1-phosphate and subsequently activation of Akt and extracellular signal-related kinase.
Literature	GENERIF_SUMMARY	Ft1 protein interacts directly with PKB, enhancing the phosphorylation of both of its regulatory sites by promoting its interaction with the upstream kinase PDK1
Literature	GENERIF_SUMMARY	role for Ins(1,3,4,5,6)P5 as a specific inhibitor of the PI 3-K/Akt signalling pathway, that may sensitize cancer cells to the action of commonly used anticancer drugs
Literature	GENERIF_SUMMARY	important role for nuclear activation of Akt 1 in thyroid cancer progression

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Thus a physiological concentration of insulin stimulated Akt-1 & Akt-2 phosphorylation in human skeletal muscle in the absence of hyperglycemia, but Akt-2 expression is impaired in muscle of obese patients with atypical diabetes with severe hyperglycemia.
Literature	GENERIF_SUMMARY	These results suggest that short-term exposure to TNF-alpha augments insulin effects through protein kinase B-alpha and glycogen synthase kinase-3 beta, whereas long-term exposure causes insulin resistance in HepG2 cells.
Literature	GENERIF_SUMMARY	Expression of constitutively active PKB alpha abrogates dexamethasone stimulation of hPDK4 promoter activity.
Literature	GENERIF_SUMMARY	ET-1 enhances the ability of lung fibroblasts to contract extracellular matrix through the Akt/phosphoinositide 3-kinase (PI3-kinase) pathway to induce de novo protein synthesis of proteins that contribute to a contractile phenotype.
Literature	GENERIF_SUMMARY	Results report that a novel AMPK catalytic subunit family member, ARK5, plays a key role in tumor malignancy downstream of Akt.
Literature	GENERIF_SUMMARY	The positive regulatory role of constitutively active PKB on TCR responsiveness, subsequent T cell division, and effector function is linked to a negative regulatory mechanism on the nuclear accumulation/ shuttling of NFAT and NF-kappa B proteins.
Literature	GENERIF_SUMMARY	activated Akt and Akt2 have roles in progression of pancreatic ductal adenocarcinoma
Literature	GENERIF_SUMMARY	estradiol may exert part of its proliferative and antiapoptotic effects by a nongenomic manner through the Akt signaling pathway.
Literature	GENERIF_SUMMARY	Overexpressed recombinant human AKT participates in the settings of endothelial dysfunction in SHR rats by impaired membrane localization suggesting that AKT is involved in endothelium dysfunction in hypertension.
Literature	GENERIF_SUMMARY	Data suggest that the expansion of AKT-driven prostate epithelial cells requires mTOR-dependent survival signaling and activation of hypoxia inducible factor (HIF)-1 alpha.
Literature	GENERIF_SUMMARY	T cells from hAKT1 transgenic mice had an elevated production of IFN-gamma at baseline that was maintained during sepsis,

Main Category	Category	Terms
		while IL-4 had little change.
Literature	GENERIF_SUMMARY	High cell density blocks EGF-dependent cell cycle progression by inhibiting EGF signaling at the level of EGF-dependent Akt activation rather than at the level of EGFR activation.
Literature	GENERIF_SUMMARY	Akt and P70S6K phosphorylation and Gadd45 levels are modulated by GPx-1 in tumor cells
Literature	GENERIF_SUMMARY	Akt1 pathway is persistently activated in head and neck squamous cell carcinoma
Literature	GENERIF_SUMMARY	Akt is required and sufficient to mediate Ang1-induced EC survival in response to growth factor depletion. Blocking Akt function abolishes angiopoietin 1 (Ang1), mediated EC survival, and activating Akt rescues a Tie2 blockade-induced EC apoptosis
Literature	GENERIF_SUMMARY	Genetic distributions of 5 single nucleotide polymorphisms studied did not deviate from Hardy-Weinberg equilibrium in patient and control groups.
Literature	GENERIF_SUMMARY	The mechanism of insulin receptor substrate-2, protein kinase B, and anti-apoptotic PKB substrate control of beta-cell growth and survival, and whether these may be targeted therapeutically to delay the onset of type 2 diabetes are reviewed.
Literature	GENERIF_SUMMARY	in primary cultures, Akt selectively phosphorylates tau at S214 rather than T212 so tau S214 may participate in Akt-mediated anti-apoptotic signaling.
Literature	GENERIF_SUMMARY	These results establish a role for FOXN1 in initiation of terminal differentiation and implicate Akt in subsequent events.
Literature	GENERIF_SUMMARY	In Goto-Kakizaki rat soleus muscle, chronic administration of antioxidant alpha -lipoic-acid partly ameliorated the diabetes-related deficit in glucose metabolism including the enzymes Akt/PKB and PI-3 kinase.
Literature	GENERIF_SUMMARY	cAMP-mediated apoptosis was largely independent of protein kinase A (PKA) and exchange protein directly activated by cAMP (EPAC), but associated with inhibition of the phosphatidylinositol 3-kinase (PI3K)/AKT pathway
Literature	GENERIF_SUMMARY	Sp1 has a role in Akt-mediated induction of VEGF expression through an HIF-1-independent mechanism
Literature	GENERIF_SUMMARY	TNF-alpha, IL-10, and IL-6 production in Kupffer cells following LPS or lipoteichoic

Main Category	Category	Terms
		acid stimulation may create a basis for understanding how the balance between pro- and antiinflammatory cytokines is regulated in the liver following bacterial infections
Literature	GENERIF_SUMMARY	The rapid activation of PI3K-Akt/PKB-mTOR-p70(S6K) cascade by T3 provides a new molecular mechanism for thyroid hormone action
Literature	GENERIF_SUMMARY	demonstrated that TRAIL mediates the recruitment of PI-3K/Akt and NF-B/IB pathways in leukaemic cells, namely Jurkat T cells
Literature	GENERIF_SUMMARY	Akt kinase activity can be inhibited by a peptide spanning the betaA strand of the proto-oncogene TCL1
Literature	GENERIF_SUMMARY	Data provide evidence that Akt plays a central role in the anti-apoptotic effect of estrogen in vascular endothelial cells.
Literature	GENERIF_SUMMARY	evidence of a functional interplay among integrin receptors, PKCepsilon, and protein kinase B (PKB/Akt) in recurrent CWR-R1 prostate cancer cells
Literature	GENERIF_SUMMARY	Phosphorylated form of Akt-1 is most likely involved in the progression of prostate cancer and is an excellent biomarker for biochemical recurrence.
Literature	GENERIF_SUMMARY	The AKT1 is a susceptibility gene for Japanese schizophrenia.
Literature	GENERIF_SUMMARY	R-Ras plays a key role in cell migration by locally regulating the switch from Rac to Rho activity after membrane protrusion and adhesion
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	shows that breast tumor kinase interacts with protein kinase B/Akt, a serine/threonine kinase involved in cell growth and survival
Literature	GENERIF_SUMMARY	protein kinase B/Akt phosphorylation is stimulated by mechanical stretch in epidermal cells via angiotensin II type 1 receptor and epidermal growth factor receptor
Literature	GENERIF_SUMMARY	Resistin induces HASMC proliferation through both ERK 1/2 and Akt signaling pathways.
Literature	GENERIF_SUMMARY	PKB/Akt activation in response to insulin or ionizing radiation is mediated through ATM
Literature	GENERIF_SUMMARY	AKT1 regulates the cell division cycle.

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Literature	GENERIF_SUMMARY	there is a negative feedback loop in apoptosis involving E2F and AKT
Literature	GENERIF_SUMMARY	These data suggest that amiloride sensitizes both tumor cells to TRAIL-induced apoptosis by promoting Akt dephosphorylation and caspase-8 activation via the intracellular acidification.
Literature	GENERIF_SUMMARY	GSK3 and PKB/Akt have roles in the integrin-mediated regulation of PTHrP, IL-6 and IL-8 in pancreatic cancer
Literature	GENERIF_SUMMARY	Gab1 tyrosine phosphorylation is stimulated by flow shear stress to mediate protein kinase B and endothelial nitric-oxide synthase activation in endothelial cells
Literature	GENERIF_SUMMARY	Akt kinase plays important survival function in T lymphocytes, which involves the regulation of FasL expression and consequent apoptotic signaling.
Literature	GENERIF_SUMMARY	activation of the PI3K/Akt pathway identified as an anti-inflammatory signal that may contribute to the establishment of Salmonella typhimurium in the intestine
Literature	GENERIF_SUMMARY	Pim-2 and Akt-1 are critical components of overlapping but independent pathways, either of which is sufficient to promote the growth and survival of nontransformed hematopoietic cells
Literature	GENERIF_SUMMARY	AKT and IL-15 may be important in extranodal NK/T-cell lymphoma tumor survival
Literature	GENERIF_SUMMARY	results show show that target of rapamycin (TOR) kinase and its associated protein rictor are necessary for AKT/PKB Ser473 phosphorylation and that a reduction in rictor or mammalian TOR (mTOR) expression inhibited an Akt/PKB effector
Literature	GENERIF_SUMMARY	there is a phosphorylation-regulated complex formation of Grb10 with 14-3-3 and Akt
Literature	GENERIF_SUMMARY	S6 kinase 1 is essential for the control of muscle cytoplasmic volume by Akt and mTOR
Literature	GENERIF_SUMMARY	S1P inhibits male germ cell apoptosis independently of its receptors, possibly by inhibiting the transcription factor NF-kappaB and Akt phosphorylation
Literature	GENERIF_SUMMARY	persistent activation of PI3K results in Akt-dependent sequestration of FoxO1 outside the nucleus of T cells interacting with APCs; this compartmentalization process can

Main Category	Category	Terms
		affect T cell growth after Ag recognition
Literature	GENERIF_SUMMARY	EGF and hypoxia induce CXCR4 in non-small cell lung cancer, a process regulated by the PI3-kinase/PTEN/AKT/mTOR signaling pathway and activation of HIF-1alpha
Literature	GENERIF_SUMMARY	activated Akt is positively correlated with the protein expression of the transcription/ translation factor Y-box binding protein-1 (YB-1) in primary breast cancer; activated Akt binds to and phosphorylates the YB-1 cold shock domain at Ser102
Literature	GENERIF_SUMMARY	arfaptin 2 phosphorylation at Ser260 by Akt inhibits PolyQ-huntingtin-induced toxicity by rescuing proteasome impairment
Literature	GENERIF_SUMMARY	C1 domain-containing phosphatase and TENsin homologue (C1-TEN) appears to be a novel intracellular phosphatase that negatively regulates the Akt/Protein kinase B signaling cascade
Literature	GENERIF_SUMMARY	Akt activation, but not ERK1/2 activation, is required for SDF-1alpha-induced migration of epitheloid carcinoma cells
Literature	GENERIF_SUMMARY	transcriptional induction of HO-1 gene expression by AEBSF is mediated via activation of a PKB, p38 MAPK signaling pathway
Literature	GENERIF_SUMMARY	insulin induces GATA2 phosphorylation on serine 401 in a PI-3K/Akt-dependent manner, impairing GATA2 translocation to the nucleus and its DNA binding activity
Literature	GENERIF_SUMMARY	In postmortem extracts from human brain, Akt is down-regulated by a caspase-3 cleavage, establishing a progressive alteration of Akt during pathological progression of Huntington's disease.
Literature	GENERIF_SUMMARY	Differential involvement of protein kinase pathways in regulating apoptosis induction by different forms of selenium in prostate tumor cells.
Literature	GENERIF_SUMMARY	However, muscle glycogen availability appears to contribute to regulation of the Akt pathway, which may influence cellular growth and adaptation in response to resistance exercise in a low-glycogen state.
Literature	GENERIF_SUMMARY	Tumor sections frmo colorectal cancer patients showed elevated expression levels of AKT1, correlating with enhanced cytoplasmic/nuclear expression of beta-catenin.
Literature	GENERIF_SUMMARY	Thus, Akt1 contains a functional NES and

Main Category	Category	Terms
		mutation of the NES results in nuclear-predominant Akt1 activation that is sufficient to induce migration.
Literature	GENERIF_SUMMARY	matrix-derived mechanical forces sensed by beta1 integrin are capable of modulating ILK activity which regulates fibroblast viability via an Akt-dependent mechanism
Literature	GENERIF_SUMMARY	JIP1 and JIP3, have a cross-talk that leads to the regulation of the ASK1-SEK1-JNK signal during glucose deprivation; cross-talk between JIP3 and JIP1 is mediated through SEK1-JNK2 and Akt1.
Literature	GENERIF_SUMMARY	PKB mediates c-erbB2-induced epithelial beta1 integrin conformational inactivation through Rho-independent F-actin rerrangements.
Literature	GENERIF_SUMMARY	Cyr61 expression provides cytoprotection in hyperoxia-induced pulmonary epithelial cell death and that this effect was in part mediated via the Akt signaling pathway
Literature	GENERIF_SUMMARY	in breast cancer patients, Akt activation is associated with tumour proliferation and poor prognosis, particularly in the subset of patients with ErbB2-overexpressing tumours
Literature	GENERIF_SUMMARY	Erythropoietin stimulated Akt phosphorylation in a dose-dependent manner (1-100 U/ml).
Literature	GENERIF_SUMMARY	RET/PTC is able to phosphorylate the Y315 residue of PKB, an event that results in maximal activation of PKB for RET/PTC-induced thyroid tumorigenesis.
Literature	GENERIF_SUMMARY	Altogether, our data demonstrate that Akt1 participates in a negative regulatory feedback loop by interacting with the JIP1 scaffold protein.
Literature	GENERIF_SUMMARY	suggests that AKT plays a role in the activation of prosurvival pathways in HTLV-1-transformed cells, possibly through NF-kappaB activation and inhibition of p53 transcription activity
Literature	GENERIF_SUMMARY	activation of Akt/PKB survival pathway is caused by PDGF
Literature	GENERIF_SUMMARY	activation of SREBP-1 by Akt leads to the induction of key enzymes of the cholesterol and fatty acid biosynthesis pathways, and thus membrane lipid biosynthesis
Literature	GENERIF_SUMMARY	Study demonstrated that phosphorylated-Akt-negative and PTEN-positive expression was a predictor of survival for patients with advanced endometrial carcinoma.

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Literature	GENERIF_SUMMARY	PKC412 suppresses Akt kinase activation and induces apoptosis in myeloma cell lines
Literature	GENERIF_SUMMARY	The replication of association of variants in the AKT1 gene in a family sample with similar ethnical background as in the original study adds further evidence for involvement of AKT1 in development of schizophrenic disorders.
Literature	GENERIF_SUMMARY	activation of mTOR by Akt-mediated cellular energy and inhibition of AMPK is the predominant pathway by which Akt activates mTOR in vivo
Literature	GENERIF_SUMMARY	demonstrates that attenuation of the PI3K-Akt pathway in tumors lacking TSC2 contributes to their benign nature
Literature	GENERIF_SUMMARY	overexpression of NBS1 is an oncogenic event that contributes to transformation through the activation of PI3-kinase/Akt
Literature	GENERIF_SUMMARY	in platelets glucose transport through GLUT3 is regulated by changes in surface expression and affinity modulation, which are both under control of PKB
Literature	GENERIF_SUMMARY	findings demonstrate that Akt/PKB activation was significantly elevated in cases of primary breast carcinoma with HER2/neu overexpression
Literature	GENERIF_SUMMARY	Semaphorin 4D/plexin-B1 induces endothelial cell migration through the activation of PYK2, Src, and the phosphatidylinositol 3-kinase-Akt pathway
Literature	GENERIF_SUMMARY	Introduction of a pH1 vector producing shRNA (short hairpin RNA) that targets p110beta abolished PIA [N6-(2-phenylisopropyl)adenosine]-induced Akt activation
Literature	GENERIF_SUMMARY	T-cad overexpression in vascular endothelial cells protects against stress-induced apoptosis through activation of the PI3K/Akt/mTOR survival signal pathway and concomitant suppression of the p38 MAPK proapoptotic pathway
Literature	GENERIF_SUMMARY	exogenous CXCL12 induced Akt1 phosphorylation is indispensable for proMMP-9 secretion, migration, and invasion of prostate cancer cells.
Literature	GENERIF_SUMMARY	PARP inhibition-induced Akt activation is dominantly responsible for the cytoprotection in oxidative stress
Literature	GENERIF_SUMMARY	AKT1-targeting ribozymes effective in downregulation of the isozyme and in

Main Category	Category	Terms
		sensitizing cells to anticancer chemotherapeutic agents
Literature	GENERIF_SUMMARY	PI3K-Akt signaling has a role in pulmonary metastatic nodule formation of osteosarcoma
Literature	GENERIF_SUMMARY	NF-kappaB1, and not c-Rel, is the critical signaling molecule downstream of the PI3K-PTEN-PKB signaling axis that regulates lymphocyte homeostasis.
Literature	GENERIF_SUMMARY	results presented in this paper provide the first direct evidence that PI3K-mediated Akt activation in adenovirus-infected corneal cells may contribute to viral pathogenesis by the prolongation of cell viability
Literature	GENERIF_SUMMARY	Akt phosphorylation of acinus on serine 422 and 573 results in its resistance to caspase cleavage in the nucleus and the inhibition of acinus-dependent chromatin condensation.
Literature	GENERIF_SUMMARY	Caveolin-1 is more expressed in cancer tissues than normal colon and related with Akt-1, not with Epidermal Growth Factor Receptor expression in colorectal cancer tissues, which suggests that signaling for caveolin-1 affects Akt-1 activation.
Literature	GENERIF_SUMMARY	Akt was rapidly phosphorylated by glimepiride and antisense oligonucleotides directed to Akt completely inhibited glimepiride-induced NO production. These data demonstrate that glimepiride induces NO production in HCAECs by activating PI3-kinase and Akt.
Literature	GENERIF_SUMMARY	Novel role for Akt phosphorylation in regulating the late stages of exocytosis and suggest which is achieved byphosphorylation of cysteine string protein on serine 10.
Literature	GENERIF_SUMMARY	There is constitutive activation of Akt in the majority of primary human lymphomas and hematopoietic cell lines which support its proposed key role in lymphoma cell survival.
Literature	GENERIF_SUMMARY	CKIepsilon-induced down-regulation of PI3K/Akt signaling through PTEN leads to amplified sensitivity to apoptosis.
Literature	GENERIF_SUMMARY	XIAP expression in colorectal cancer is regulated by hepatocyte growth factor/C-met pathway via Akt signaling
Literature	GENERIF_SUMMARY	Ser(87) of Bim(EL) is an important regulatory site that is targeted by Akt to attenuate the pro-apoptotic function of Bim(EL), thereby promoting cell survival

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Ang II increases albumin endocytosis through an AT(2) receptor mediated by activation of PKB in the plasma membrane, which depends on the basal activity of the phosphatidyl-inositol 3-kinase.
Literature	GENERIF_SUMMARY	In this study, we provide evidence for isoform-specific positive and negative roles for Akt1 and -2 in regulating growth factor-stimulated phenotypes in breast epithelial cells
Literature	GENERIF_SUMMARY	Akt activation may be involved in cell survival and extracellular matrix remodelling in human endometrium and decidua.
Literature	GENERIF_SUMMARY	Protein kinase B signaling pathways mediate CD36 expression in response to oxidized low density lipoprotein (oxLDL).
Literature	GENERIF_SUMMARY	Results suggest that the antiapoptotic effects of erythropoietin in neuronal cells require the combinatorial activation of multiple signaling pathways, including STAT5, AKT, and potentially MAP kinase.
Literature	GENERIF_SUMMARY	Data report that cepharanthine induces apoptosis in HuH-7 cells through activation of JNK1/2 and the downregulation of Akt.
Literature	GENERIF_SUMMARY	Activation of Rac by heregulin beta1 is mediated not only by ErbB3 and ErbB2 but also by transactivation of EGFR, and it is independent of ErbB4.
Literature	GENERIF_SUMMARY	Both activation of JNK and inhibition of Akt play a role in translocation of Nur77 from the nucleus to the cytoplasm.
Literature	GENERIF_SUMMARY	Data suggest that endothelial activation of Akt suppresses lesion formation via increased NO production, preservation of functional endothelial layer, and suppression of inflammatory and proliferative changes in the vascular wall.
Literature	GENERIF_SUMMARY	data show that inhibition of PI3K/protein kinase B (PKB) increased endostatin-induced apoptosis, and that endostatin-induced cell death is physiologically linked to PKB-mediated cell survival through caspase-8.
Literature	GENERIF_SUMMARY	hAR is a direct target of LEF-1/TCF transcriptional regulation in PCa cells; expression of the hAR protein is suppressed by a degradation pathway regulated by cross-talk of Wnt, Akt, and PP2A
Literature	GENERIF_SUMMARY	acute UV exposure downregulates TSP1 expression via PI3K-Akt activation in human keratinocytes

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	CARMA1 complex is required for induction of NF-kappaB by Akt, Akt plays a modulatory role in NF-kappaB induction
Literature	GENERIF_SUMMARY	14-3-3sigma stabilizes p27 Kip1 by inhibiting the activity of PKB/Akt
Literature	GENERIF_SUMMARY	reactive oxygen species generated by Nox4, at least in part, transmit cell survival signals through the AKT-ASK1 pathway in pancreatic cancer cells and their depletion leads to apoptosis.
Literature	GENERIF_SUMMARY	Overexpression of TSC2 rescues the migration phenotype of myr-Akt1-expressing tumor cells, and high levels of TSC2 in breast cancer patients correlate with increased metastasis and reduced survival.
Literature	GENERIF_SUMMARY	The activation and regulation of Akt and NF-kappa B in prostatic cancer cell line 1-LN are reported.
Literature	GENERIF_SUMMARY	Akt regulates its kinase activity through autophosphorylation
Literature	GENERIF_SUMMARY	VCP is an essential target of Akt signaling
Literature	GENERIF_SUMMARY	These results indicate that tryptase can activate PI3K-PKB pathway and enhance IL-8 expression.
Literature	GENERIF_SUMMARY	BAF155 and potentially INI1 are substrates for Akt phosphorylation
Literature	GENERIF_SUMMARY	The data support the significance of regulated Rac signaling in mediating Schwann cell-axon interaction and suggest that controlling Rac activity as a possible therapy for schwannomas.
Literature	GENERIF_SUMMARY	Findings did not confirm the association of AKT1 SNP2/3/4 TCG haplotype with the risk of schizophrenia, but showed the evidence of association with a different haplotype, AKT1 five-SNP AGCAG haplotype, with the risk of schizophrenia in Iranian population.
Literature	GENERIF_SUMMARY	angiostatins K1-3, K1-4 and K1-4.5 mediate anti-angiogenesis in a process involvingp53, FasL, AKT and mRNA deregulation
Literature	GENERIF_SUMMARY	It was found that an active process mediates AKT nuclear translocation as shown by fusing AKT with GFP3 protein.
Literature	GENERIF_SUMMARY	PI3K/AKT, MAPK/ERK and NFkappabeta signalling have roles in the maintenance of human embryonic stem cell pluripotency and viability
Literature	GENERIF_SUMMARY	These results suggest a critical role for the

Main Category	Category	Terms
		CS amplitude and the balance between Rac and Rho in mechanochemical regulation of lung EC barrier.
Literature	GENERIF_SUMMARY	C5b-9 regulation of the cell cycle activation in aortic endothelial cells through Akt pathway is dependent on inactivation of FOXO1
Literature	GENERIF_SUMMARY	Overexpression of dominant-active Akt1 restored HIF-1alpha expression
Literature	GENERIF_SUMMARY	The O-GlcNAc-modification on Akt1 may play a role in Akt1 nuclear localization.
Literature	GENERIF_SUMMARY	LRRC4 plays a major role in suppressing U251 cell proliferation by regulating the extracellular signal-regulated kinase (ERK)/Akt/NF-kappaBp65, STAT3, and JNK2/c-Jun pathways.
Literature	GENERIF_SUMMARY	results demonstrate that PI3-K/Akt pathway is activated constitutively in Burkitt's lymphoma cell line, P3HR-1; it promotes cell growth and the lytic cycle cascade of Epstein-Barr virus downstream of ZEBRA
Literature	GENERIF_SUMMARY	Data show that inorganic phosphate controls cell growth by activating ERK1/2 cascades and by facilitating the translocation of Mnk1 from cytosol into nucleus through an Akt-mediated MEK pathway.
Literature	GENERIF_SUMMARY	PYK2 mediates anti-apoptotic AKT signaling in response to benzo[a]pyrene diol epoxide in mammary epithelial cells
Literature	GENERIF_SUMMARY	AKT1 play an important role in tumor growth by inducing angiogenesis and by increasing HIF-1alpha and VEGF expression. This work provides a better understanding of the molecular mechanism of human cancer induced by the activation of PI3K signaling.
Literature	GENERIF_SUMMARY	PKCepsilon activates Akt via DNA-PK to mediate its antiapoptotic function
Literature	GENERIF_SUMMARY	Acidic pH exposure protects HEMEC through induction of Hsps and activation of MAPK and PI3 kinase pathway.
Literature	GENERIF_SUMMARY	The functional response of mesangial cells to treatment with CTGF was associated with the phosphorylation of Akt/protein kinase B (PKB) and resultant phosphorylation of a number of Akt/PKB substrates, including FKHR and p27(Kip-1).
Literature	GENERIF_SUMMARY	Akt activation was detected at very early, prenecrotic stages of disease pathogenesis, and maximal activation was observed

Main Category	Category	Terms
		during peak stages of muscle hypertrophy.
Literature	GENERIF_SUMMARY	The Akt expression pathway is marker of outcome in glioblastoma.
Literature	GENERIF_SUMMARY	Akt-and CREB-mediated prostate cancer cell proliferation is inhibited by Nexrutine, a Phellodendron amurense extract
Literature	GENERIF_SUMMARY	Results describe the roles of MEK1/ERK and AKT/PKB pathways on the effects of granulocyte macrophage colony-stimulating factor (GM-CSF) and M-CSF on tumor progression of lung cancer.
Literature	GENERIF_SUMMARY	Akt overactivation prevents the nuclear translocation of ERK1/2 and the AngII-induced proliferation through interaction with and stabilization of endogenous PEA-15.
Literature	GENERIF_SUMMARY	In vitro, LY294002 can inhibit protein expression of PKB in SACC-83 cells and SACC-LM cells.
Literature	GENERIF_SUMMARY	Inhibition of AKT1 is sufficient to affect cell migration, invasion, and proliferation.
Literature	GENERIF_SUMMARY	HO-1 and Akt exert codependent cytoprotective effects against oxidative stress-induced apoptosis in human aortic smooth muscle cells.
Literature	GENERIF_SUMMARY	SHIP2 substitutes for PTEN in the acute regulation of PKB in PC3 cells but not other prostate cell lines, where PTEN may share this role with further PIP3-degrading mechanisms.
Literature	GENERIF_SUMMARY	Hsp90 inhibition transiently activates Src kinase and promotes Src-dependent p85 PI3K and Akt and Erk activation
Literature	GENERIF_SUMMARY	The phosphoinositide 3-kinase (PI3K)/Akt pathway is commonly activated in cancer; therefore, we investigated its role in hypoxia-inducible factor-1alpha (HIF-1alpha) regulation.
Literature	GENERIF_SUMMARY	Akt regulates nuclear/cytoplasmic localization of tuberin.
Literature	GENERIF_SUMMARY	P-Akt expression also provides a potential mechanistic link between these pediatric tumors and adult malignancies.
Literature	GENERIF_SUMMARY	Leptin stimulates proliferation and inhibits apoptosis in colon cancer cells. This effect involves JAK2, PI3 kinase and JNK and activation of the oncogenic transcription factors signal transducer and activator of transcription

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	We demonstrate, for the first time in human skeletal muscle, that the regulation of Akt and its downstream signalling pathways GSK-3beta, mTOR and Foxo1 are associated with both the skeletal muscle hypertrophy and atrophy processes.
Literature	GENERIF_SUMMARY	histone deacetylase inhibitors not only inhibit deacetylase activity but also stimulate active NF-kappaB transcription and cell survival through signaling pathways involving Akt and increased p300 acetyltransferase activity
Literature	GENERIF_SUMMARY	AKT-induced BAD phosphorylation and its subsequent cytoplasmic sequestration by 14-3-3zeta is a major mechanism responsible for the postponement of UVB-induced apoptosis in human keratinocytes.
Literature	GENERIF_SUMMARY	Akt in thyroid tumorigenesis.
Literature	GENERIF_SUMMARY	Cross-talk between AKT and estrogen receptor beta-mediated transcription pathways may represent an important aspect that may influence breast cancer response to endocrine therapy.
Literature	GENERIF_SUMMARY	Results reveal that the SIN1-rictor-mTOR function in Akt-Ser473 phosphorylation is required for TORC2 function in cell survival but is dispensable for TORC1 function.
Literature	GENERIF_SUMMARY	Data show that UVB irradiation increases PTEN/Akt phosphorylation in dermal fibroblasts, and inhibition of PTEN and activation of Akt by phosphorylation are involved in UVB-induced MMP-1 and -3 secretions through upregulation of AP-1 activity.
Literature	GENERIF_SUMMARY	The cAMP-stimulated inverse ROS response in granulocytes from type 1 and type 2 diabetic patients may be due to a change in signaling pathways from cAMP/protein kinase A to PKB.
Literature	GENERIF_SUMMARY	These data suggest that IGF-1R and PKCdelta are required to stimulate PKB phosphorylation in response to BMOV in HepG2 cells.
Literature	GENERIF_SUMMARY	The data suggest that in response to hydrogen peroxide, two pathways are activated in Jurkat T lymphocytes that converge to result in the phosphorylation of Akt on S473 and T308.
Literature	GENERIF_SUMMARY	This study fails to support reduced signaling of the AKT-GSK3beta molecular cascade in schizophrenia.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Akt activity decreases dramatically during the course of neutrophil death.
Literature	GENERIF_SUMMARY	analysis of the LLKIL motif in CXCR2, which is required for full IL-8 ligand-induced activation of Erk, Akt, and chemotaxis in HL60 cells
Literature	GENERIF_SUMMARY	We showed Akt activation to significantly correlate with HER2 overexpression or LOH at the PTEN gene locus while inversely correlating with the PR expression.
Literature	GENERIF_SUMMARY	vascular endothelial growth factor expression is induced through the glucocorticoid receptor-related phosphatidylinositol 3-kinase/Akt and beta-catenin/T-cell factor-dependent pathway in human endothelial cells
Literature	GENERIF_SUMMARY	Akt1 and Akt2 have opposing roles in Rac/Pak signaling and cell migration
Literature	GENERIF_SUMMARY	impaired cell cycle and death sensitivity, are explained by a specific defective activation of Akt that impairs the expression of Survivin
Literature	GENERIF_SUMMARY	AKT prevents VEGF during internl ribosome entry site activity in myeloma cells during mTOR inhibition.
Literature	GENERIF_SUMMARY	the primary signaling pathway for LPC-dependent dendrite formation in human melanocytes involves the activation of PKCzeta and PKCzeta phosphorylation is Rac dependent
Literature	GENERIF_SUMMARY	using akt1, the evidence is supported that akt signalling plays a significant role in the chemoresistant phenotype in glioma
Literature	GENERIF_SUMMARY	In organotypic culture, AKT mediates changes related to cell shape and size with an expansion of the differentiated compartment.
Literature	GENERIF_SUMMARY	Mice transfected with human AKT1 modified with myristoylation show increases in the susceptibility of these mice to the induction of mammary tumors of epithelial origin by the carcinogen DMBA.
Literature	GENERIF_SUMMARY	S6K1 regulates GSK3 under conditions of mTOR-dependent feedback inhibition of Akt
Literature	GENERIF_SUMMARY	chemoattractant-stimulated superoxide production can be amplified by a positive feedback loop in which p67(phox) targets Vav1-mediated Rac activation
Literature	GENERIF_SUMMARY	Luteolin inhibits IGF-1R/AKT signaling, these

Main Category	Category	Terms
		results provideing a new insight into the mechanisms that luteolin inhibits testicular cancer cells.
Literature	GENERIF_SUMMARY	phosphorylation is a mechanism for regulation of insulin receptor substrate-1/2, Akt, and ERK1/2
Literature	GENERIF_SUMMARY	These data identify actin as a new functional target of Akt signaling.
Literature	GENERIF_SUMMARY	Akt activation is not associated with differences in survival or smoking status in bronchioalveolar carcinoma.
Literature	GENERIF_SUMMARY	The current data suggest that, following exposure to zinc, the sequential activation of Akt and GSK-3beta plays an important role directing hippocampal neural precursor cell death.
Literature	GENERIF_SUMMARY	Novel c-Akt-regulated genes contributing to Scatter factor-mediated antineoplastic drug resistance.
Literature	GENERIF_SUMMARY	TGF-beta1 co-ordinately and independently activates the FAK and AKT protein kinase pathways to confer an anoikis-resistant phenotype to myofibroblasts.
Literature	GENERIF_SUMMARY	inhibition of ITD/Flt3 activity did not prevent the phosphorylation of ERK, STAT5 or Akt in some primary AML cells. In parallel, in these cells, Flt3 and ERK or Akt cooperate to regulate cell survival
Literature	GENERIF_SUMMARY	These data reveals for the first time that PDK1 and PKB may differentially activate NF-kappaB, and that TPCK may subserve a useful anti-inflammatory function by inhibiting IKKbeta.
Literature	GENERIF_SUMMARY	These results indicate that increased hexosamine biosynthetic pathway flux in human astroglial cells results in a rapid, short-term phosphorylation of Akt that is likely a result of increased endoplasmic reticulum stress.
Literature	GENERIF_SUMMARY	Results show that coculture with ECs enhances VSMC adhesion and spreading by up-regulating beta(1)-integrin expression and activating the PI3K/Akt pathway. Interaction between ECs and VSMCs serves a role in vascular homeostasis and remodeling.
Literature	GENERIF_SUMMARY	Akt1 is a critical regulator of dendritic cell (DC) lifespan, and its manipulation in DCs can improve the clinical efficacy of DC-based tumor vaccines

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	A novel function of the AKT pathway is demonstrated in prostate cancer progression; constitutively active AKT promotes neuroendocrine differentiation and dominant negative AKT inhibits it.
Literature	GENERIF_SUMMARY	Myxoma virus (MV) M-T5 host range protein is functionally interchangeable with the host PIKE-A protein and that the activation of host Akt by either M-T5 or PIKE-A is critical for the permissiveness of cancer cells for MV.
Literature	GENERIF_SUMMARY	VEGF-A, STAT5 and AKT are downregulated in acute myeloid leukemia blasts of patients treated with SU5416
Literature	GENERIF_SUMMARY	Treatment with oleandrin facilitated nuclear translocation of FKHR in human, but not murine cells by dephosphorylating Akt.
Literature	GENERIF_SUMMARY	hypoxia-mediated suppression of Akt may induce caspase-mediated IRS-1 cleavage
Literature	GENERIF_SUMMARY	In sum, using a quantitative approach to study Akt activation identified ligand-dependent limits for the use of T308 or S473 as proxies for kinase activity.
Literature	GENERIF_SUMMARY	PI3K/Akt activation may lead to the development of chemoresistance in AML blasts through a mechanism involving a p53-dependent suppression of MRP1 expression.
Literature	GENERIF_SUMMARY	These data indicate that the aberrant expression of PTEN contributes to the activation of the PI3kinase/Akt pathway and its transcription factor mediators in glioma.
Literature	GENERIF_SUMMARY	protein kinase B, LKB1, and AMP-activated protein kinase have roles in activation of lipoprotein lipase by glucose-dependent insulinotropic polypeptide in adipocytes
Literature	GENERIF_SUMMARY	Akt2 was unable to induce CREB phosphorylation at Ser-133 in vivo and CREB target gene expression.
Literature	GENERIF_SUMMARY	Akt/Bad pathway generates a progressive resistance to apoptosis, at a time HTLV-I genes expression is silenced.
Literature	GENERIF_SUMMARY	These results indicate that resveratrol seems to exert its growth-inhibitory/ apoptotic effect on the breast cancer cell line MCF7 via the Akt-caspase-9 pathway.
Literature	GENERIF_SUMMARY	These results demonstrate that hypoxic condition-and high cell density-induced expression of Redd1 is mediated by coactivation of Sp1 and HIF-1alpha

Main Category	Category	Terms
		downstream of the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	siRNA constructs of AKT1 may effectively down-regulate the expression of AKT1 and reverse the resistant phenotype of gastric cancer cells
Literature	GENERIF_SUMMARY	results delineate a novel force-activated inside-out Src/PI3K/FAK/Akt pathway by which cancer cells regulate their own adhesion.
Literature	GENERIF_SUMMARY	The data suggest that aberrant activation of PI3K/Akt pathway plays an extensive role in thyroid tumorigenesis, particularly in FTC and ATC, and promotes progression of BTA to FTC and to ATC as the genetic alterations of this pathway accumulate.
Literature	GENERIF_SUMMARY	Akt thus serves as a molecular switch that increases angiogenesis and the generation of superoxide, fostering more aggressive tumor behavior.
Literature	GENERIF_SUMMARY	This study demonstrates a cachexia-associated loss of Akt-dependent signalling in human skeletal muscle with decreased activity of regulators of protein synthesis and a disinhibition of protein degradation.
Literature	GENERIF_SUMMARY	Par6alpha-mediated inhibition of insulin-dependent glycogen synthesis in C2C12 cells depends on the direct interaction of Par6alpha with aPKC and on aPKC-mediated T34 phosphorylation of Akt1.
Literature	GENERIF_SUMMARY	We conclude, therefore, that estrogen opens BK(Ca) channels in HCASMC by stimulating nNOS via a transduction sequence involving PI3-kinase and Akt.
Literature	GENERIF_SUMMARY	advances in the understanding of TCL1-Akt functional interaction in order to clarify the biological action of the proto-oncogene TCL1 family {review]
Literature	GENERIF_SUMMARY	Elevated and highly expressed levels in both laarge B-cell lymphoma and Hodgkin lymphoma.
Literature	GENERIF_SUMMARY	study provides at best weak support for the hypothesis that AKT1 is a susceptibility gene for schizophrenia
Literature	GENERIF_SUMMARY	Mechanism to selectively terminate Akt-signaling pathways through the differential inactivation of specific Akt isoforms by specific PHLPP isoforms.
Literature	GENERIF_SUMMARY	It was demonstrated that activation of phosphatidylinositol 3-kinase (PI3K) and

Main Category	Category	Terms
		Akt1 facilitated the nuclear translocation of Akt1, but the phosphorylation at threonine 308 and serine 473 was not prerequisite.
Literature	GENERIF_SUMMARY	These results indicate that the phosphatidylinositol 3-kinase/Akt/FoxO1 pathway participates in Ang II suppression of hSR-BI/CLA-1 expression and suggests that the endothelial receptor for hSR-BI/CLA-1 is downregulated by the renin-angiotensin system.
Literature	GENERIF_SUMMARY	receptor and nonreceptor PTKs have roles in modulating H2O2-induced ERK1/2 and PKB signaling [review]
Literature	GENERIF_SUMMARY	These findings implicate Akt in upstream events leading to BRCA1 nuclear localization and function.
Literature	GENERIF_SUMMARY	Fak/Src signaling to the PI3-K/Akt-1 and MEK/Erk pathways undergoes a differentiation state-specific uncoupling in enterocytes.
Literature	GENERIF_SUMMARY	KGF triggers negative feedback between ERK1/2 and AKT pathways to regulate cell proliferation/differention.
Literature	GENERIF_SUMMARY	These results indicated that the activation of CXCR4 and its signaling pathways (MEK1/2 and Akt) are essential for CXCL12-induced cholangiocarcinoma cell invasion.
Literature	GENERIF_SUMMARY	NF-kappa-B inhibition enhances CASP3 degradation of Akt1 and apoptosis in response to campothecin.
Literature	GENERIF_SUMMARY	PI3K/ILK/Akt/NF-kappaB axis is a promising target for therapeutic intervention in renal cell carcinoma.
Literature	GENERIF_SUMMARY	Lipid raft cholesterol regulates apoptotic cell death in prostate cancer cells through EGFR-mediated Akt and ERK pathways.
Literature	GENERIF_SUMMARY	IGF-1 rescue of AR toxicity is diminished by alanine substitutions at the Akt consensus site
Literature	GENERIF_SUMMARY	PKB/Akt inhibits ceramide-induced apoptosis in neuroblastoma cells by blocking AIF translocation.
Literature	GENERIF_SUMMARY	Proinvasive activity of BMP7 through SMAD4/src-independent and ERK/Rac/JNK-dependent signaling pathways in colon cancer cells is reported.
Literature	GENERIF_SUMMARY	vascular endothelial growth factor by Akt and mammalian target of rapamycin inhibitors in cell lines derived from

Main Category	Category	Terms
		childhood solid tumors
Literature	GENERIF_SUMMARY	Results show a a significant correlation between the EGFR L858R mutation and the expression of p-Akt and suggest that the activation of Akt is dependent on EGFR mutation pattern.
Literature	GENERIF_SUMMARY	In normal human skin, phospho-Akt appeared in the basal epidermal layer, but in psoriasis, phospho-Akt expression was low in the epidermis.
Literature	GENERIF_SUMMARY	EGFR/Akt pathway mitigated G2/M arrest in human HaCaT keratinocytes and normal human keratinocytes treated with low doses of UVA irradiation.
Literature	GENERIF_SUMMARY	Data show that Chlamydia trachomatis activates Rac and promotes its interaction with WAVE2 and Abi-1 to activate the Arp2/3 complex resulting in the induction of actin cytoskeletal rearrangements that are required for invasion.
Literature	GENERIF_SUMMARY	Activation and phosphorylation of the extracellular signal-regulated kinase 1/2 and AKT pathways are involved in cervical adenocarcinoma metastases to pelvic lymph nodes.
Literature	GENERIF_SUMMARY	Rho inactivation allows the activity of Rac to become dominant leading to stimulation of the phosphoinositide 3-kinase/Akt/IkappaB kinase/nuclear factor-kappaB prosurvival pathway.
Literature	GENERIF_SUMMARY	The AKT signaling pathway was directly involved in the effects induced by siRNA targeting LMP1.
Literature	GENERIF_SUMMARY	Incubation in the presence of Remnant lipoproteins (TG-rich lipoproteins)for 48 h induced the proliferation of prostate cancer PC-3 cells more significantly than prostate cancer LNCaP cells and human prostate stromal cells
Literature	GENERIF_SUMMARY	there is a link between Akt activation, repair of DNA damage, and radioresistance in glioblastoma
Literature	GENERIF_SUMMARY	Effects of endurance exercise training on insulin signaling and AKT in human skeletal muscle
Literature	GENERIF_SUMMARY	IGF-1 regulates ether-a-go-go channel activity in breast neoplasms, via an Akt pathway to promote cell proliferation.
Literature	GENERIF_SUMMARY	Tiam1-deficient lymphomas showed increased Rac activity, suggesting that the

Main Category	Category	Terms
		lack of Tiam1 is compensated by alternative Rac-activating mechanisms that lead to increased progression of PI3-kinase-induced T-lymphomas.
Literature	GENERIF_SUMMARY	3,3'-diindolylmethane -induced cell growth inhibition and apoptosis induction are partly mediated through the regulation of Akt/FOXO3a/GSK-3beta/beta-catenin/AR signaling
Literature	GENERIF_SUMMARY	Reduced activation of NF-kappaB via impaired PI3K/Akt activation under increased TNF-alpha levels could result in increased apoptosis of vitiliginous keratinocytes.
Literature	GENERIF_SUMMARY	Akt1 is cleaved in vitro at the caspase-3 consensus site DQDD(456; Resistance to cleavage at site DAKE(398) (within the kinase domain) in response to phosphorylation suggests a possible mechanism by which the anti-apoptotic role of Akt1 is regulated.
Literature	GENERIF_SUMMARY	PKB/Akt may regulate specific cellular responses to growth factors such as insulin under adverse conditions such as hypoxia
Literature	GENERIF_SUMMARY	Gastrointestinal stromal tumor secondary KIT mutations can be associated with KIT hyperactivation and imatinib resistance, in which AKT PI3-kinase is a crucial survival pathway.
Literature	GENERIF_SUMMARY	Phosphorylated Akt activity is associated with invasive ductal carcinoma of the breast.
Literature	GENERIF_SUMMARY	Results suggest that aberrant activation of PI3K-Akt pathway may contribute to increased cell invasiveness and facilitate prostate cancer progression.
Literature	GENERIF_SUMMARY	mTOR regulates the phosphorylation and activation of Akt in endothelial cells and a major effect of mTOR inhibition in endothelial cells is to suppress Akt-inducible pro-survival signals
Literature	GENERIF_SUMMARY	P-Rex1 links mTOR signaling to Rac activation and cell migration
Literature	GENERIF_SUMMARY	LA's ability to inhibit apoptosis and proliferation of ECs could beneficially affect endothelial dysfunction, which precedes manifestation of late diabetic vascular complications.
Literature	GENERIF_SUMMARY	peripherin is a novel substrate for Akt in vivo and its phosphorylation may play a role in motor nerve regeneration

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Akt/PKB interacts with the histone H3 methyltransferase SETDB1 and coordinates to silence gene expression.
Literature	GENERIF_SUMMARY	analysis of calcium/calmodulin binding to the pleckstrin homology domain of AKT1
Literature	GENERIF_SUMMARY	IGF-I and insulin promoted the growth of human acute myeloid leukemia blasts in vitro and activated the phosphoinositide 3-kinase (PI3K)/Akt and the extracellular signal-regulated kinase (Erk) pathways
Literature	GENERIF_SUMMARY	Chromogranin A can increase prostatic cancer cell survival through AKT up-regulatoin.
Literature	GENERIF_SUMMARY	Induction of apoptosis through the inhibition of Akt signaling pathway is one of the anti-proliferative mechanisms of DHEA in certain tumors.
Literature	GENERIF_SUMMARY	CD44-induced cell migration is dependent on its complex formation with Lyn and its consequent regulation of AKT phosphorylation in colon cancer cells.
Literature	GENERIF_SUMMARY	The inhibition of Akt and S6K1 phosphorylation by PRR5 knock down correlates with reduction in the expression level of platelet-derived growth factor receptor beta (PDGFRbeta).
Literature	GENERIF_SUMMARY	These results demonstrate that ""pure"" isoprenoids and genistein differentially impact cap-dependent translation in tumor cell lines."
Literature	GENERIF_SUMMARY	loss of PTEN expression, together with increased Akt phosphorylation, contributes to progression and recurrence of prostate cancer
Literature	GENERIF_SUMMARY	activation of the PI3K-Akt pathway during liver oncogenesis may be at least partially responsible for the elimination of HBV replication from tumor cells
Literature	GENERIF_SUMMARY	E17K mutation activates AKT1 by means of pathological localization to the plasma membrane, stimulates downstream signalling, transforms cells and induces leukaemia in mice; this mechanism indicates a direct role of AKT1 in human cancer
Literature	GENERIF_SUMMARY	In the absence of EGF, p38MAPK-activated AKT is necessary for HER-2 overexpressing human breast cancer cells to survive and to form colonies in soft agar.
Literature	GENERIF_SUMMARY	CTMP induces translocation of Akt to the

Main Category	Category	Terms
		membrane and thereby increases the level of Akt phosphorylation. As a result, CTMP enhances various cellular activities that are principally mediated by the PI3-kinase/Akt pathway.
Literature	GENERIF_SUMMARY	a subpopulation of Akt is cholesterol sensitive and the oncogenic effects conferred by myristoylation arise, in part, from the tendency of the membrane-targeted form of the protein to reside in cholesterol-rich membrane microdomains
Literature	GENERIF_SUMMARY	IKK alpha controls mTOR kinase activity in Akt-active, PTEN-null prostate cancer cells, with less involvement by IKK beta
Literature	GENERIF_SUMMARY	AKT amplification and the mTOR/p70S6K1 pathway play an important role in human lung cancer cells acquiring CDDP resistance
Literature	GENERIF_SUMMARY	Phosphoinositide 3-kinase-independent non-genomic signals transit from the androgen receptor to Akt1 in membrane raft microdomains
Literature	GENERIF_SUMMARY	BCR/ABL-Y177 plays an essential role in Ras and Akt activation and in human hematopoietic progenitor transformation in chronic myelogenous leukemia
Literature	GENERIF_SUMMARY	NGF stimulates endothelial cell invasion and cord formation by augmenting MMP-2 via the PI3K/Akt signaling pathway and AP-2 transcription factor, which may be responsible for triggering angiogenesis.
Literature	GENERIF_SUMMARY	PTEN dominantly inhibits Akt activation, the coexistence of high levels of the PTEN protein with enhanced Akt activation suggests the existence of novel mechanisms which attenuate PTEN
Literature	GENERIF_SUMMARY	prolonged Akt activation is an important signaling pathway for breast cancer cells expressing CXCR4 and is necessary for CXCL12-dependent cell migration
Literature	GENERIF_SUMMARY	Expression analysis of additional genes, AKT1, NOG and its antagonist BMP4, which interact downstream to FGFR1, demonstrated expression differences between primary rhabdomyosarcoma tumors and normal skeletal muscles
Literature	GENERIF_SUMMARY	Activation of AKT may play an important role in the development of diffuse large B cell lymphoma. It is closely related to the control of cell cycle and proliferation, but is not associated with apoptosis.
Literature	GENERIF_SUMMARY	activation of PRPK is mediated by another

Main Category	Category	Terms
		kinase, Akt/PKB, which phosphorylates PRPK at Ser250
Literature	GENERIF_SUMMARY	These results suggested that PKB enhanced DNMT1 stability and maintained DNA methylation and chromatin structure, which might contribute to cancer cell growth.
Literature	GENERIF_SUMMARY	Overexpression of Akt1 upregulates glycogen synthase activity and phosphorylation of mTOR in IRS-1 knockdown HepG2 cells.(
Literature	GENERIF_SUMMARY	Activation of AKT, possibly through the PI3K-AKT pathway, is an important component of ASCC tumorigenesis that contributes to MDM2 and TP53 accumulation in the nucleus.
Literature	GENERIF_SUMMARY	The role of P-AKT in multidrug resistance of gastric cancer cells and the possible underlying mechanisms are here investigated
Literature	GENERIF_SUMMARY	Akt blocks MstI-triggered FOXO3 nuclear translocation by phosphorylating MstI, promoting cell survival.
Literature	GENERIF_SUMMARY	down-regulation of Akt induced cytotoxicity of WM cells in the bone marrow microenvironment and inhibited migration and adhesion in vitro.
Literature	GENERIF_SUMMARY	Concluded that Wnt5a is a natural ligand of Fzd3 that triggers the PI3K/Akt signal and promotes adhesion of human dermal fibroblasts.
Literature	GENERIF_SUMMARY	Cutaneous human papillomaviruses down-regulate AKT1.
Literature	GENERIF_SUMMARY	Activation of PKB is associated with astrocytic tumor pathogenesis and progression
Literature	GENERIF_SUMMARY	Use of resveratrol or inhibition of Akt phosphorylation may represent an important therapeutic approach in combination with conventional therapies for the treatment of prostate cancer.
Literature	GENERIF_SUMMARY	We found significantly decreased RNA levels in prefrontal cortex of schizophrenic individuals, consistent with reduced AKT1 protein levels reported in schizophrenic brain.
Literature	GENERIF_SUMMARY	the AKT1 PH domain and hinge region as functional domains which jointly permit AKT1 translocation and phosphorylation in response to extracellular pressure

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Inhibition of PI3K signaling by PTEN inhibits tumor angiogenesis and growth. AKT is the downstream target of PI3K in controlling angiogenesis and tumor growth.
Literature	GENERIF_SUMMARY	IGF-1 activates the IGF receptor/IRS/PI3K/PKB pathway, and that PI3Kalpha is essential for the potentiatory effect of IGF-1 on platelet responses
Literature	GENERIF_SUMMARY	plexin-B1 promotes endothelial cell motility through RhoA and ROK by regulating the integrin-dependent signaling networks that result in the activation of PI3K and Akt
Literature	GENERIF_SUMMARY	Akt inhibition primes TRAIL-resistant cells to TRAIL-induced apoptosis.
Literature	GENERIF_SUMMARY	Platelet PAR-1/PAR-4 stimulation causes rapid Akt phosphorylation downstream of PLC, whereas with continuous stimulation, ADP and PI3K are required for maintaining Akt phosphorylation.
Literature	GENERIF_SUMMARY	important novel role for PKB in regulation of cell fate choices during hematopoietic lineage commitment
Literature	GENERIF_SUMMARY	Triple-negative breast cancers are characterized by increased phosphorylation of Akt kinase
Literature	GENERIF_SUMMARY	PKB[alpha] and/or PKB[gamma] and not PKB[beta] alone are involved in gemcitabine resistance mechanisms.
Literature	GENERIF_SUMMARY	Constitutive Akt activation is required for EGF to prevent neuroendocrine differentiation of prostate cancer cells.
Literature	GENERIF_SUMMARY	VEGF regulates angiopoietin-Tie2 signaling by inducing proteolytic cleavage and shedding of Tie2 via a novel PI3K/Akt-dependent pathway.
Literature	GENERIF_SUMMARY	Enhanced angiotensin converting enzyme 2 regulates the insulin/Akt signalling pathway by blockade of macrophage migration inhibitory factor expression.
Literature	GENERIF_SUMMARY	findings suggest that cigarette smoking is associated with the formation of LDL subfraction L5, which inhibits endothelial progenitor cell differentiation by impairing Akt phosphorylation via the LOX-1 receptor
Literature	GENERIF_SUMMARY	Akt1 and Akt2 mediate GPIb-IX signaling via the cGMP-dependent signaling pathway.
Literature	GENERIF_SUMMARY	HER-2/neu transcriptionally activates Jab1 expression via the AKT/beta-catenin pathway in breast cancer cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	AKT1 is a susceptibility gene for schizophrenia in the Chinese population and may play no major role in the therapeutic response to antipsychotics or in chlorpromazine-induced extrapyramidal syndrome.
Literature	GENERIF_SUMMARY	A novel point mutation (E17K) in the pleckstrin homology domain (PHD) of the AKT1 gene in human cancers was discussed.
Literature	GENERIF_SUMMARY	High levels of p-AKT expression occurred independently of the presence of PTEN or PIK3CA mutations in endometrial cancer.
Literature	GENERIF_SUMMARY	pleiotrophin potentiates cardiomyocyte cell death, at least partially, through inhibition of AKT signaling
Literature	GENERIF_SUMMARY	The activation of AKT and NF-kappaB, but not STAT3, significantly contributes to the progression of breast cancer.
Literature	GENERIF_SUMMARY	Akt and Mcl-1 are major components of a survival pathway that can be activated in CLL B cells by antigen stimulation.
Literature	GENERIF_SUMMARY	Mst1 and its caspase cleavage products are direct inhibitors of Akt
Literature	GENERIF_SUMMARY	p53 is an indispensable component of cellular signaling system which is regulated by caveolin-1 expression, involving Akt activation and increase in cyclin D1, thereby promoting proliferation of breast cancer cells.
Literature	GENERIF_SUMMARY	hypoxia could inhibit the SDF-1-dependent migration of HL-60 via blocking of Akt activation.
Literature	GENERIF_SUMMARY	modulation of phosphatidylinositol 3-kinase/Akt/GSK3beta signaling cascades can be beneficial for protecting or facilitating recovery from cellular LeTx intoxication in cells that depend on basal MEK1 activity for proliferation.
Literature	GENERIF_SUMMARY	Akt1 is a crucial regulator of osteoblasts and osteoclasts by promoting their differentiation and survival to maintain bone mass and turnover
Literature	GENERIF_SUMMARY	Parainfluenza virus 5 V protein interacts with Akt, a serine/threonine kinase, also known as protein kinase B.
Literature	GENERIF_SUMMARY	BCL2 functions as an activator of the AKT signaling pathway in pancreatic cancer.
Literature	GENERIF_SUMMARY	Impaired insulin signaling through Akt and

Main Category	Category	Terms
		AS160 in part explains insulin resistance at the molecular level in skeletal muscle in polycystic ovary syndrome.
Literature	GENERIF_SUMMARY	PI3-K/Akt pathways have a dual role in both survival and cell death processes depending on the stimulus.
Literature	GENERIF_SUMMARY	results indicate that shRNA targeting of tNOX inhibits the growth of cervical cancer cells, and reduces cell migration via a decrease in the membrane association of Rac.
Literature	GENERIF_SUMMARY	Our data suggest a crucial role of Ki-Ras
Literature	GENERIF_SUMMARY	Pim-1 is a crucial facet of cardioprotection downstream of Akt
Literature	GENERIF_SUMMARY	Levels of phospho-Akt are decreased in Caki cells overexpressing Par-4. inactivation of Akt is important in cellular pathways affected by Par-4.
Literature	GENERIF_SUMMARY	AKT1 inhibitors counteract the bacterial manipulation of host signalling processes, thus controlling intracellular growth of bacteria
Literature	GENERIF_SUMMARY	a key regulatory role of the Akt/mTOR pathway in the generation of the effects of As(2)O(3)
Literature	GENERIF_SUMMARY	The results from this investigation show that GBS both requires and activates the PI3K/Akt host-cell signalling pathway during invasion of epithelial cells.
Literature	GENERIF_SUMMARY	No mutations in AKT1 pleckstrin homology domain were identified, making this mutation an unlikely cause of PI3K/AKT pathway activation in acute myeloid leukaemia
Literature	GENERIF_SUMMARY	betaenin can be involved in polyphenol mediated down regulation of AKT1.
Literature	GENERIF_SUMMARY	Chorionic gonadotropin promotes trophoblast invasion and migration through activation of ERK and AKT signaling involving their downstream effector MMP-2.
Literature	GENERIF_SUMMARY	These data provide the first evidence that IL-18 and FN stimulate each other's expression
Literature	GENERIF_SUMMARY	Altogether these results highlight an intertwined role for specific Akt isoforms and XIAP in chemoresistance of uterine cancer cells.
Literature	GENERIF_SUMMARY	Loss of PTEN gene expression is one of the

Main Category	Category	Terms
		important factors of p-Akt high expression in myelodysplastic syndrome.
Literature	GENERIF_SUMMARY	Signal transduction pathways underlying the enhanced cell migration reveal that the IGF-I-IGFBP-VN complex stimulates a transient activation of the ERK/MAPK signaling pathway and a sustained activation of the phosphatidylinositide 3-kinase/AKT pathway.
Literature	GENERIF_SUMMARY	The activation of the protein kinase B (Akt) signaling cascade by carbon nanoparticles was investigated with regard to its relevance for proliferation.
Literature	GENERIF_SUMMARY	support the role of Notch and Akt in breast cancer progression
Literature	GENERIF_SUMMARY	CD28 costimulation activates AKT to phosphorylate NF90 at Ser647 and phosphorylation triggers NF90 to relocate to the cytoplasm and stabilize IL-2 mRNA.
Literature	GENERIF_SUMMARY	Hsp27 inhibits oxidative stress-induced H9c2 damage and inhibition of ROS generation and the augmentation of Akt activation may be involved in the protective signaling.
Literature	GENERIF_SUMMARY	in most primary brain tumors, AKT is activated by molecular mechanisms other than oncogenic G49A mutations of AKT1
Literature	GENERIF_SUMMARY	RIG1 plays a role in IFN-gamma-mediated therapy by downregulating p185 and its downstream PI3K/Akt/mTOR/VEGF-signaling pathway
Literature	GENERIF_SUMMARY	These results suggest that 0.05 Gy of ionizing radiation stimulates cell proliferation through the transient activation of Raf and Akt in CCD-18 Lu cells.
Literature	GENERIF_SUMMARY	The overexpression may be a powerful prognostic marker for predicting disease free and overall survival of breast cancer patients.
Literature	GENERIF_SUMMARY	Rac signaling to cyclin D1 is a crucial pro-proliferative effect of E-cadherin-mediated cell-cell adhesion
Literature	GENERIF_SUMMARY	PI-3K/Akt signaling pathway participates in P2Y receptor-stimulated prostate cancer invasion by enhancing cell motility, rather than up-regulating MMP-2 and MMP-9 activities.
Literature	GENERIF_SUMMARY	Treatment of cells with arsenite also induced significant activation of PI-3K and Akt, which was responsible for the

Main Category	Category	Terms
		anchorage-independent cell growth induced by arsenite exposure.
Literature	GENERIF_SUMMARY	t-Darpp promotes cancer cell survival by up-regulation of Bcl2 through Akt-dependent mechanism.
Literature	GENERIF_SUMMARY	miR-214 induces cell survival and cisplatin resistance primarily through targeting the PTEN/Akt pathway.
Literature	GENERIF_SUMMARY	These results suggest that caffeine has a cytoprotective effect due to the activation of the PI3K/Akt pathways in SH-SY5Y cells.
Literature	GENERIF_SUMMARY	The subcellular localization of DNMT1 can be altered by the addition of IL-6, and this process is greatly enhanced by phosphorylation of the DNMT1 nuclear localization signal (NLS) by PKB/AKT.
Literature	GENERIF_SUMMARY	The phosphorylation pattern and kinase activity of each Akt isoform were compared in primary myotubes from healthy control participants and type 2 diabetic patients.
Literature	GENERIF_SUMMARY	Poliovirus simultaneously activates the phosphatidylinositol 3-kinase (PI3K)/Akt survival signaling pathway in these cells, limiting the extent of JNK activation and thereby cell death.
Literature	GENERIF_SUMMARY	These findings revealed that TNF-alpha induced VCAM-1 expression via multiple signaling pathways.
Literature	GENERIF_SUMMARY	PTEN restores drug sensitivity to cisplatin in human ovarian cancer cell line C13K with multidrug-resistance by decreasing the expression of p-Akt.
Literature	GENERIF_SUMMARY	Human sarcopenia reveals an increase in SOCS-3 and myostatin and a reduced efficiency of Akt phosphorylation.
Literature	GENERIF_SUMMARY	the activities of AMP-activated protein kinase, protein kinase B, and mammalian target of rapamycin by limiting energy availability with 2-deoxyglucose
Literature	GENERIF_SUMMARY	Conclude that PI(3,4,5)P3 and PI(3,4)P2 have distinct roles in determining PKB phosphorylation and activity.
Literature	GENERIF_SUMMARY	These data provide evidence that, although AKT1 mutations are apparently rare in lung cancer (1.9%), the oncogenic properties of E17K-AKT1 may contribute to the development of a fraction of lung carcinoma with squamous histotype (5.5%).
Literature	GENERIF_SUMMARY	The results suggest that CKIepsilon is a new

Main Category	Category	Terms
		positive regulator of the Akt pathway. Here we propose that, rather than inhibiting PTEN function, CKIepsilon positively regulates Akt possibly by inhibiting Protein Phosphatase 2A (PP2A).
Literature	GENERIF_SUMMARY	the Akt/Girdin signalling pathway is essential in VEGF-mediated postneonatal angiogenesis
Literature	GENERIF_SUMMARY	the alpha6beta4 integrin strongly influence Akt phosphorylation through ErbB-3 protein regulation
Literature	GENERIF_SUMMARY	TNF-alpha overrides the G2/M checkpoint in keratinocytes and allows for some cells containing unrepaired cyclobutane pyrimidine dimers to enter the cell cycle. TNF-alpha seems to be dependent on PI3K-Akt activation.
Literature	GENERIF_SUMMARY	The effect of IL-2 on angiogenesis and tube formation was mediated by Reactive Oxygen Species and Akt.
Literature	GENERIF_SUMMARY	Girdin has an important role in tumor progression with aberrant activation of the Akt signaling pathway
Literature	GENERIF_SUMMARY	PKC epsilon mediates TRAIL resistance by Akt-mediated phosphorylation of Hdm2 resulting in suppression of p53 expression and downregulation of Bid in breast cancer cells
Literature	GENERIF_SUMMARY	These findings indicate that v-Crk controls cell migration and membrane dynamics by activating Rac in CAS-deficient mouse embryo fibroblasts.
Literature	GENERIF_SUMMARY	In dermal fibroblasts, Akt has dual profibrotic effects, increasing collagen synthesis and decreasing degradation by MMP1 downregulation. Systemic sclerosis fibroblasts were more sensitive to Akt inhibition, with respect to collagen and MMP1 production.
Literature	GENERIF_SUMMARY	A potential role for increased phosphorylated Akt [in carcinogenesis] may relate to the dose and time of nicotine use [and] an alkaline environment
Literature	GENERIF_SUMMARY	the activities of specific elements of the PI3K signaling pathway, including Akt, are necessary for the synergistic induction of MMP-1 and MMP-13 and the cartilage breakdown stimulated by IL-1 and oncostatin M
Literature	GENERIF_SUMMARY	isosilybin B treatment enhances the formation of complex between Akt, Mdm2

Main Category	Category	Terms
		and AR, which promotes phosphorylation-dependent AR ubiquitination and its degradation by proteasome
Literature	GENERIF_SUMMARY	VprBP depletion abolished the in vivo interaction of Merlin and Roc1-Cullin4A-DDB1, which resulted in Merlin stabilization and inhibited ERK and Rac activation
Literature	GENERIF_SUMMARY	Akt and mTOR activation have distinct functional relevance in mantle cell lymphoma
Literature	GENERIF_SUMMARY	Active AKT and mTOR, were present in all adenomas, carcinomas, and in normal colorectal mucosa.
Literature	GENERIF_SUMMARY	cAMP inhibits cell migration by interfering with Rac-induced lamellipodium formation
Literature	GENERIF_SUMMARY	The Akt isoforms are required for the HGF-stimulated invasive growth of human salivary gland cancer cells.
Literature	GENERIF_SUMMARY	post-translational modification by nitrotyrosylation does not control Akt in systemic sclerosis platelets
Literature	GENERIF_SUMMARY	stabilization of MDMX by Akt may be an alternative mechanism by which Akt up-regulates MDM2 protein levels and exerts its oncogenic effects on p53 in tumor cells
Literature	GENERIF_SUMMARY	Results identify a molecular mechanism by which activated FGFR2 recruits Cbl in raft micro-domains to trigger PI3K ubiquitination and proteasome degradation, and reveal a role for PI3K/Akt in the control of osteoblast survival by FGFR2 signaling.
Literature	GENERIF_SUMMARY	Despite the confirmed oncogenic function of the AKT1 E17K, the rare incidences of the mutation suggest that it may not play a crucial role in the development of the most common types of human cancers.
Literature	GENERIF_SUMMARY	The PI3K/Akt/mTOR signaling pathway is implicated in the development of cervical cancer.
Literature	GENERIF_SUMMARY	The data presented here suggest the involvement of AKT1 in protection of PD through many possible mechanisms involving different signaling pathways that could be potential therapeutic targets in the future.
Literature	GENERIF_SUMMARY	Bortezomib -sensitive and -resistant MCL cell lines showed comparable activation of the AKT pathway
Literature	GENERIF_SUMMARY	Our data indicate a novel role for Akt in

Main Category	Category	Terms
		supporting the progression of human melanomas
Literature	GENERIF_SUMMARY	chlorophyllin exerts antioxidant effect by inducing HO-1 and NQO1 expression mediated by PI3K/Akt and Nrf2
Literature	GENERIF_SUMMARY	Prognostic and/or predictive role of Akt phosphorylation in breast, prostate and non-small cell lung cancer. Review.
Literature	GENERIF_SUMMARY	AKT1 pleckstrin homology domain (E17K) mutations do not play a roles in chronic lymphocytic leukaemia and acute myeloid leukaemia
Literature	GENERIF_SUMMARY	the TSC1-TSC2 complex inhibits mTORC1 and activates mTORC2, which through different mechanisms promotes Akt activation
Literature	GENERIF_SUMMARY	AKT is activated in an ataxia telangiectasia and Rad3-related-dependent manner in response to temozolomide and confers protection against drug-induced cell growth inhibition.
Literature	GENERIF_SUMMARY	in VSMCs exposed to hyperglycemia, IGF-I stimulation of Shc facilitates the transfer of Grb2 to p85 resulting in enhanced PI3K activation and AKT phosphorylation leading to enhanced cell proliferation and migration
Literature	GENERIF_SUMMARY	activation of Akt during preoperative chemotherapy for esophageal cancer correlates with poor prognosis
Literature	GENERIF_SUMMARY	Cav-1 increases the basal and TGF-beta1-induced expression of type I procollagen by regulating two opposite signaling pathways
Literature	GENERIF_SUMMARY	tetrandrine induces G(1) arrest and apoptosis through PI3K/kip1/AKT/GSK3beta pathway
Literature	GENERIF_SUMMARY	Activin A may mediate ovarian oncogenesis by activating Akt and repressing GSK to stimulate cellular proliferation.
Literature	GENERIF_SUMMARY	This study provides a direct link between the growth factor signaling pathways regulated by PI3 kinase/Akt and MAP kinases with Myc-mediated transcription.
Literature	GENERIF_SUMMARY	Transforming growth factor beta induces apoptosis through repressing the phosphoinositide 3-kinase/AKT/survivin pathway in colon cancer cells.
Literature	GENERIF_SUMMARY	The interplay between cancer cell-derived clusterin and IGF-1 may dictate the outcome of cell growth and dormancy during

Main Category	Category	Terms
		tumorigenic progression.
Literature	GENERIF_SUMMARY	Akt and its downstream targets FoxO3a and GSK3 regulate a survival pathway in VSMCs and that their deregulation due to a reduction of IGF1R signaling may promote apoptosis in atherosclerosis.
Literature	GENERIF_SUMMARY	High NG2/MPG expression correlated with multidrug resistance mediated by increased activation of alpha3beta1 integrin/PI3K signaling and their downstream targets, promoting cell survival
Literature	GENERIF_SUMMARY	High expression of PI3K and Akt are involved in proliferation, malignant transformation, invasion and metastasis of cervical carcinoma.
Literature	GENERIF_SUMMARY	ADAM17 mediates the EGFR/AKT/cyclin D1 pathway and cell cycle progression to the S phase induced by UVA radiation
Literature	GENERIF_SUMMARY	deregulated PKB/AKT stabilizes Mcl-1 expression in a mammalian target of rapamycin (mTOR)-dependent pathway
Literature	GENERIF_SUMMARY	Although Akt activation leads to GSK-3beta phosphorylation and the subsequent expression of beta-catenin, activation of this pathway by 1-azakenpaullone was insufficient to stimulate the motile phenotype in cytotrophoblasts.
Literature	GENERIF_SUMMARY	Cdk5 phosphorylates PIKE-A and stimulates its GTPase activity, which activates nuclear Akt and promotes glioblastoma cell migration and invasion.
Literature	GENERIF_SUMMARY	results demonstrate that an insulin or TNF response that uses Akt-dependent signaling activates IKKalpha to induce mTOR activity, whereas a TNF response that does not involve Akt activates IKKbeta to activate mTOR
Literature	GENERIF_SUMMARY	7-ketocholesterol-induced apoptosis has a role in phospholipidosis and down-regulation of the PI3-K/PDK-1/Akt signalling pathway
Literature	GENERIF_SUMMARY	AKT1 has a role in modulating human prefrontal-striatal structure and function and the mechanism of this effect may be coupled to dopaminergic signaling and relevant to the expression of psychosis
Literature	GENERIF_SUMMARY	GILZ is a mediator of glucocorticoid killing, and is regulated by PI3-kinase/AKT.
Literature	GENERIF_SUMMARY	AKT1E17K mutation likely displays a unique tissue-specific pattern in human cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	A signaling cascade for the regulation of p53 in response to ionizing radiation was proposed that involves activation of DNA-PK and Akt/PKB and inactivation of GSK-3 and Mdm2.
Literature	GENERIF_SUMMARY	Shorter overall survival in primary melanoma was associated with the presence of ulceration and BRAF exon 15 mutations, as well as the absence of nuclear activation of Akt and of cytoplasmic activation of ERK.
Literature	GENERIF_SUMMARY	Osteoblast-derived BMP-2 act through Akt and ERK, which in turn activates IKK alpha/beta and NF-kappaB, resulting in the activations of beta1 and beta 3 integrins and contributing the migration of prostate cancer cells.
Literature	GENERIF_SUMMARY	The mTOR-associated protein Raptor is required for the ability of Akt to induce NF-kappaB activity.
Literature	GENERIF_SUMMARY	Akt inactivation in breast cancer causes negative regulation of INK4a/ARF-independent cell cycle progression by Mel-18
Literature	GENERIF_SUMMARY	LMP1 triggers the PI3K/Akt pathway to inactivate FOXO3a and decrease DDB1, which can lead to repression of DNA repair and may contribute to genomic instability in human epithelial cells
Literature	GENERIF_SUMMARY	The percentage of positive cells for TKTL1 and p-Akt was significantly correlated in glioblastoma multiforme (GBM) and other astrocytic gliomas (AGs, grades II and III).
Literature	GENERIF_SUMMARY	strongly activated in HTLV-1 infected T cell lines derived from HAM/TSP patients.
Literature	GENERIF_SUMMARY	Nuclear Akt phosphorylates Aly on threonine-219, which is required for its interaction with Akt.
Literature	GENERIF_SUMMARY	hnRNP A1 has a role in mediating rapamycin-induced alterations of cyclin D1 and c-myc IRES activity in an Akt-dependent manner and provide the first direct link between Akt and the regulation of IRES activity
Literature	GENERIF_SUMMARY	These results indicate that FSH increases the expression of VEGF by upregulating the expression of survivin, which is activated by the PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	Alterations to the Akt signaling pathway and nuclear export determine the stability of AIB1 and nuclear content of this coactivator.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	HDL has endothelial barrier promoting activities, which are attributable to its S1P component and signaling through the S1P1/Akt pathway.
Literature	GENERIF_SUMMARY	CCL2 protects prostate cancer PC3 cells from autophagic death via the phosphatidylinositol 3-kinase/Akt/survivin pathway
Literature	GENERIF_SUMMARY	Akt and AMPK have roles in the pathway of hydrogen peroxide-activated endothelial nitric-oxide synthase phosphorylation and function
Literature	GENERIF_SUMMARY	Akt-mediated phosphorylation modulates eNOS uncoupling and greatly increases O(2)* generation from the enzyme at low Ca(2+) concentrations, and PKCalpha-mediated phosphorylation alters the sensitivity of the enzyme to other negative regulatory signals.
Literature	GENERIF_SUMMARY	Datas show that transforming growth factor beta engages TACE and ErbB3 to activate phosphatidylinositol-3 kinase/Akt in ErbB2-overexpressing breast cancer and desensitizes cells to trastuzumab.
Literature	GENERIF_SUMMARY	AKT1 does not play a significant role in clinical and functional manifestations in patients with schizophrenia who receive risperidone treatment.
Literature	GENERIF_SUMMARY	analysis of Akt-dependent and Akt-independent multiple myeloma subgroups
Literature	GENERIF_SUMMARY	Authors observed a significant increase in AKT phosphorylation in depressed ventral tegmental area postmortem samples from depressed patients, without a change in total AKT levels.
Literature	GENERIF_SUMMARY	the PI3K/Akt pathway is involved in MDR in lymphoma cell lines and PI3K/Akt inhibition correlates down-regulation of NF-kappaB activity and inhibition Pgp function.
Literature	GENERIF_SUMMARY	The effect of AKT1 on DNA-dependent protein kinase catalytic subunit (DNA-PKcs) activity and DNA-dsb repair in irradiated non-small cell lung cancer cell lines A549 and H460, was analysed.
Literature	GENERIF_SUMMARY	eIF4E knockdown is not associated with Akt activation.
Literature	GENERIF_SUMMARY	The lack of Par-4 dramatically enhances Ras-induced lung carcinoma formation in vivo, acting as a negative regulator of Akt activation.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	results show that the mutation E17K/AKT1 was not detected in the pleckstrin homology domain of AKT1 of the investigated cases. We conclude that this mutation is not a major event in B-cell-derived lymphoid leukaemias
Literature	GENERIF_SUMMARY	endocrine hormone insulin contributes to VEGF release in skin wounds through an Akt1-and 4E-BP1- mediated posttranscriptional mechanism in keratinocytes
Literature	GENERIF_SUMMARY	Data demonstrate need for autophagy-mediated degradation of toxic, mutated arginine vasopressin C98X peptides, and suggest that, in the presence of mis-folded proteins, the stimulation of Akt signalling counteracts autophagy and precipitates cell death.
Literature	GENERIF_SUMMARY	PIK3CA mutations, in comparison with PTEN loss and AKT1 mutations, were associated with significantly less and inconsistent activation of AKT and of downstream PI3K/AKT signaling in tumors and cell lines
Literature	GENERIF_SUMMARY	PAK1-specific paxillin phosphorylation at Ser(273) is critically involved in the positive-feedback regulation of the Rac-PAK1 pathway.
Literature	GENERIF_SUMMARY	The role of PI3K/Akt activity and DNA-PKcs on XRCC1 expression/stabilization, was studied.
Literature	GENERIF_SUMMARY	FASN and activated AKT pathway may be a potential target for therapeutic intervention for the treatment of papillary thyroid carcinomas.
Literature	GENERIF_SUMMARY	results suggest that both Transketolase-like enzyme 1 (TKTL1) and p-Akt protein play an important role in the progression of cervical neoplasia
Literature	GENERIF_SUMMARY	isoflavone-induced inhibition of cell proliferation and induction of apoptosis are partly mediated through the regulation of the Akt/FOXO3a/GSK-3beta/AR signaling network.
Literature	GENERIF_SUMMARY	Tissue kallikrein decreased GSK-3beta activity via the phosphatidylinositol 3-kinase-Akt pathway and enhanced VEGF and VEGFR-2 expression in endothelial cells.
Literature	GENERIF_SUMMARY	analysis of differential Akt regulation in plasma membrane microdomains
Literature	GENERIF_SUMMARY	endosomal activation of the PI3K/AKT

Main Category	Category	Terms
		signaling pathway in response to Gbetagamma trafficking, via its interaction with Rab11, is a relevant step in the mechanism controlling cell survival and proliferation
Literature	GENERIF_SUMMARY	Peroxisome proliferator-activated receptor-delta agonist enhances vasculogenesis by regulating endothelial progenitor cells through genomic and nongenomic activations of the phosphatidylinositol 3-kinase/Akt pathway
Literature	GENERIF_SUMMARY	DIDS-sensitive Cl- channels play a key role in the Low-density lipoprotein-induced cell proliferation of human aortic smooth muscle cells via the activation of Erk1/2 and PI3K/Akt and the upregulation of Egr-1.
Literature	GENERIF_SUMMARY	data demonstrate a unique role for Akt in inhibiting TR3 functions that are not related to transcriptional activity but that correlate with the regulation of its mitochondrial association
Literature	GENERIF_SUMMARY	in breast DCIS patients, strong cytoplasmic and/or nuclear immunoexpression was revealed in 17 cases for NRGalpha, 13 cases for p-AKT, and nuclear immunoexpression with p-MAPK was found in 17 cases.
Literature	GENERIF_SUMMARY	Fstl1 is a secreted muscle protein or myokine that can function to promote endothelial cell function and stimulates revascularization in response to ischemic insult through its ability to activate Akt-eNOS signaling
Literature	GENERIF_SUMMARY	These results suggest that the BMP-2 acts through PI3K/Akt, which in turn activates IKKalpha/beta and NF-kappaB, resulting in the activations of beta1 integrin and contributing the migration of human chondrosarcoma cells.
Literature	GENERIF_SUMMARY	These results reveal a primary role of Akt in shaping intracellular Ca2+ homeostasis, that may underlie its protective role against some proapoptotic stimuli.
Literature	GENERIF_SUMMARY	Direct inhibition of Akt may represent a therapeutic strategy for breast and other cancers.
Literature	GENERIF_SUMMARY	rac regulates its effector phospholipase Cgamma2 through interaction with a split pleckstrin homology domain
Literature	GENERIF_SUMMARY	Gate control for cell polarization can explain how Rac can be employed for both PI3K-dependent and -independent signaling pathways coexisting in the same cell.

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Literature	GENERIF_SUMMARY	no aberrant SSCP pattern in exon 2 of the AKT1 gene was found, indicating that there were no E17K mutations of the AKT1 gene in these samples of gastrointestinal and liver cancers in the Korean patients
Literature	GENERIF_SUMMARY	anti-apoptotic functions of cFLIPs may be attributed to inhibit oxaliplatin-induced apoptosis through the sustained XIAP protein level and Akt activation.
Literature	GENERIF_SUMMARY	These findings demonstrate that HAPO induces endothelial cell proliferation through the PI-3K/Akt pathway.
Literature	GENERIF_SUMMARY	Inhibition of PI3-K/Akt induces a 40% decrease of cyclin D1 half-life as a result of accumulation of the dephosphorylated/ active form of GSK-3beta within the nucleus, where it can phosphorylate cyclin D1 on Thr286 thereby promoting its nuclear export.
Literature	GENERIF_SUMMARY	Confocal imaging of fluorescently labeled ZEBOV indicated that inhibition of PI3K, Akt, or Rac1 disrupted normal uptake of virus particles into cells and resulted in aberrant accumulation of virus into a cytosolic compartment
Literature	GENERIF_SUMMARY	Glycine-extended gastrin induced signalling involves a JAK2/PI3-kinase/Akt/NF-kappaB sequence leading to COX-2 transcription.
Literature	GENERIF_SUMMARY	study found significantly increased nuclear but also cytoplasmic p-Akt expression in renal cell carcinoma subtypes
Literature	GENERIF_SUMMARY	there are two parallel cell-survival pathways in prostate cancer cells
Literature	GENERIF_SUMMARY	These findings demonstrate that PI3K/Akt signaling is a key regulator in terminal chondrocyte differentiation in both embryonic and adult chondrogenesis.
Literature	GENERIF_SUMMARY	H. pylori infection of gastric epithelial cells induced phosphorylation of Akt at Ser 473 and Thr 308.
Literature	GENERIF_SUMMARY	AKT1 (E17K) G-->A mutation was not detected in any of the pancreatic cancer tissue samples nor in any cell line DNA; results indicate that the AKT1 (E17K) mutation may not play a significant role in pancreatic cancer
Literature	GENERIF_SUMMARY	COX-2 promotes colon tumor progression, but not initiation, and it does so, in part, by activating EGFR and Akt signaling pathways
Literature	GENERIF_SUMMARY	Suppression of LKB1 expression led to

Main Category	Category	Terms
		apoptosis in three cell lines in which Akt is constitutively active but not in two cell lines without Akt activation.
Literature	GENERIF_SUMMARY	PTH induction of apoptosis is mediated by translocation of mitochondria to the perinuclear region, dephosphorylation of Akt, dephosphorylation of Bad and its movement to the mitochondria, release of cytochrome c and Smac/Diablo, activating casp3.
Literature	GENERIF_SUMMARY	an activating mutation of AKT1 found in prostate cancer
Literature	GENERIF_SUMMARY	populations enriched for cancer stem cells are preferentially sensitive to an inhibitor of Akt, a prominent cell survival and invasion signaling node
Literature	GENERIF_SUMMARY	activation of the Akt-mTOR-p70S6K pathway plays a significant role in HCC progression by promoting neoangiogenesis.
Literature	GENERIF_SUMMARY	Activated Akt seems to characterize well-differentiated invasive squamous laryngeal carcinomas, loss of E-cadherin and activation of beta-catenin correlated with high grade carcinomas.
Literature	GENERIF_SUMMARY	HuR is a direct transcription target of NF-kappaB; its activation in gastric cancer cell lines depends on phosphatidylinositol 3-kinase/AKT signaling. HuR activation by this pathway has proliferative and antiapoptotic effects on gastric cancer cells.
Literature	GENERIF_SUMMARY	AKT was expressed in 83.3% of ameloblastomas, showing high expression of the p-thr(308)AKT and p-ser(473) AKT forms in 93.3% and 56.6% of cases, respectively.
Literature	GENERIF_SUMMARY	nuclear factor kappaB is inhibited through an AKT-independent pathway by liposome-encapsulated curcumin suppresses growth of head and neck squamous cell carcinoma in vitro and in xenografts
Literature	GENERIF_SUMMARY	Complement C1q chemoattracts human dendritic cells and enhances migration of mature dendritic cells to CCL19 via activation of AKT and MAPK pathways.(
Literature	GENERIF_SUMMARY	AKT1 polymorphisms and haplotypes were not significantly associated with TD. However, we detected a significant interaction between rs6275 of DRD2 and rs3730358 of AKT1
Literature	GENERIF_SUMMARY	These results suggest a unique role of AKT in modulating estrogen signaling in ERalpha-

Main Category	Category	Terms
		positive breast cancers and highlights how extracellular signal activated kinases can change the landscape of transcription factor binding to the genome.
Literature	GENERIF_SUMMARY	there was a significant inverse correlation between nuclear pAkt and apoptotic index and between cytoplasmatic and nuclear pAkt and patient survival.
Literature	GENERIF_SUMMARY	These data demonstrate that integrin/EGFR cross-talk is required for expression of Egr-1 through a novel regulatory cascade involving the activation of the PI3K/Akt/Forkhead pathway.
Literature	GENERIF_SUMMARY	SEMA3B is a potential tumor suppressor that induces apoptosis in SEMA3B-inactivated tumor cells through the Np-1 receptor by inactivating the Akt signaling pathway.
Literature	GENERIF_SUMMARY	the Akt signalling pathway is inhibited by NO-sulindac, which inhibits the hypoxia response of PC-3 prostate cancer cells
Literature	GENERIF_SUMMARY	DEP-1 is a positive regulator of VEGF-mediated Src and Akt activation and endothelial cell survival.
Literature	GENERIF_SUMMARY	PTEN upregulation is responsible for Pfn1-dependent attenuation of AKT activation in MDA-MB-231 cells.
Literature	GENERIF_SUMMARY	These data indicate that skeletal muscle is a source of FGF21 and that its expression is regulated by a phosphatidylinosistol 3-kinase (PI3-kinase)/Akt1 signaling pathway-dependent mechanism.
Literature	GENERIF_SUMMARY	Following down-regulation of MR-1, the phosphorylations of MLC2, focal adhesion kinase (FAK), and Akt were dramatically decreased
Literature	GENERIF_SUMMARY	HMGB1 induced the activation of p38MAPK, ERK1/2 and Akt.
Literature	GENERIF_SUMMARY	In vitro studies reveal that the E17K mutation of ATK1 PH domain dramatically increases the affinity for the constitutive plasma membrane lipid PI(4,5)P(2)
Literature	GENERIF_SUMMARY	In human melanoma cells, MFG-E8 knockdown attenuated Akt and Twist signaling and thereby compromised tumor cell survival, EMT, and invasive ability
Literature	GENERIF_SUMMARY	these results indicate the importance of C. burnetii modulation of host signaling and demonstrate a critical role for Akt and Erk1/2 in successful intracellular parasitism

Main Category	Category	Terms
		and maintenance of host cell viability.
Literature	GENERIF_SUMMARY	Scleroderma serum-induced EPC apoptosis is mediated chiefly by the Akt-FOXO3a-Bim pathway, which may account, at least in part, for the decreased circulating EPC levels in scleroderma patients.
Literature	GENERIF_SUMMARY	VacA activates the PI3K/Akt signaling pathway, resulting in phosphorylation and inhibition of GSK3beta, and subsequent translocation ofbeta-catenin to the nucleus, consistent with effects of VacA on beta-catenin-regulated transcriptional activity.
Literature	GENERIF_SUMMARY	Results indicated that the JNK and Akt pathways and inhibition of NF-kappaB activity were key regulators of apoptosis in response to withaferin A in human leukemia U937 cells.
Literature	GENERIF_SUMMARY	AKT1 inhibits homologous recombination by inducing cytoplasmic retention of BRCA1 and RAD51.
Literature	GENERIF_SUMMARY	Altered insulin activation of the PI3K/Akt but not the MAPK pathway precedes and may contribute to development of whole-body insulin resistance and type 2 diabetes in men with low birth weight.
Literature	GENERIF_SUMMARY	NGF induces Mac-2BP expression via the PI3K/Akt/NF-kappaB pathway.
Literature	GENERIF_SUMMARY	RGS13 overexpression inhibited CXCL12-evoked Ca(2+) mobilization, Akt phosphorylation and chemotaxis.
Literature	GENERIF_SUMMARY	mTORC2 and Akt facilitate CD40-inducible expression of VEGF in endothelial cells
Literature	GENERIF_SUMMARY	ILK overexpression during liver oncogenesis and cirrhosis correlates with activation of Akt but not with other conventional ILK targets
Literature	GENERIF_SUMMARY	glioma cells enhance EPC angiogenesis via VEGFR-2, not VEGFR-1, mediated by the MMP-9, Akt and ERK signal pathways
Literature	GENERIF_SUMMARY	The association and altered regulation of IRS proteins by Ad5E1A contribute to the adenovirus-transformed phenotype and modulates viral infection in an Akt-dependent manner.
Literature	GENERIF_SUMMARY	IGFBP-7 can regulate glioma cell migration through the AKT-ERK pathway, thereby playing an important role in glioma growth and migration
Literature	GENERIF_SUMMARY	ZAP70 expression levels in B cells give an

Main Category	Category	Terms
		estimate of the proliferative potential and the associated protein kinase B availability.
Literature	GENERIF_SUMMARY	Akt signals are involved in decoy receptor 3 expression in PANC-1 human pancreatic epithelioid carcinoma cells.
Literature	GENERIF_SUMMARY	AKT1 exerts its effect on verbal learning and memory via neural networks involving prefrontal cortex.
Literature	GENERIF_SUMMARY	Vitamin D3-triggered differentiation of human myeloid leukemia cells depends on downregulation of Akt, which dissociates from Raf1 and activates MAPK signaling.
Literature	GENERIF_SUMMARY	These data demonstrated that cell adhesion-mediated PI3K/Akt activation could be one of the important mechanisms of resistance to genotoxin-induced cell death in differentiated epithelial cells.
Literature	GENERIF_SUMMARY	PI3K/Akt/mdm2 signalling pathway can be activated by doxorubicin and suppress apoptosis by promoting phosphorylation of mdm2 in a gastric cancer cell line.
Literature	GENERIF_SUMMARY	Overexpression of P-AKT and NF-kappaB p65 were involved in the carcinogenesis and metastasis of ovarian cancer.
Literature	GENERIF_SUMMARY	Coexpression with active Akt1 increased the phosphorylation of Ser397 and thereby SGK1 kinase activity. SGK1 activation was further augmented by coexpression with the protein kinase WNK1 (with no lysine kinase 1).
Literature	GENERIF_SUMMARY	The phosphatidylinositol 3-kinase/akt cassette regulates purine nucleotide synthesis.
Literature	GENERIF_SUMMARY	We confirmed the partial responsibility of p-Akt activation in invasion of some bladder cancer cell lines expressing N-cadherin or p-EGFR
Literature	GENERIF_SUMMARY	results suggest that CCL5 acts through PI3K/Akt, which in turn activates IKKalpha/beta and NF-kappaB, resulting in the activation of alphavbeta3 integrin and contributing to the migration of human lung cancer cells
Literature	GENERIF_SUMMARY	an important role of PKC-delta and PI3K/Akt1,2 pathways in activating mTOR as an endogenous modulator to ensure a tight regulation of NF-kappaB signaling of ICAM-1 expression in endothelial cells.
Literature	GENERIF_SUMMARY	the c-Src-Rac1-p38 MAPK pathway is required for activation of Akt in response to

Main Category	Category	Terms
		radiation and plays a cytoprotective role against radiation in human cancer cells
Literature	GENERIF_SUMMARY	Promoter hypermethylation of CST6 is associated with aberrant AKT1 activation in epithelial cells, as well as the disabled INNP4B regulator resulting from the suppression by cancer associated fibroblasts.
Literature	GENERIF_SUMMARY	inhibition of the PI3K-AKT-mTOR pathway potently increases sensitivity of melanoma cells to chemotherapy
Literature	GENERIF_SUMMARY	Id-1 is a novel PTEN inhibitor that could activate the Akt pathway and its downstream effectors, the Wnt/TCF pathway and p27(Kip1) phosphorylation
Literature	GENERIF_SUMMARY	Notch-1 may be mediated through regulating the expression of cell cycle regulatory proteins cyclin D1, CDK2 and p21 and the activity of Akt signaling
Literature	GENERIF_SUMMARY	the prosurvival effect of NADPH oxidase inhibitor parabutoporin depends on activation of protein kinase B/Akt via lipid raft signaling.
Literature	GENERIF_SUMMARY	Akt/mTOR-dependent translation of HIF-1alpha plays a critical role in the postirradiation up-regulation of intratumoral HIF-1 activity in response to radiation-induced alterations of glucose and oxygen availability in a solid tumor.
Literature	GENERIF_SUMMARY	the pathway of apoptin-induced apoptosis and show that it essentially depends on abnormal phosphatidylinositol 3-kinase (PI3-kinase)/Akt activation, resulting in the activation of the cyclin-dependent kinase CDK2
Literature	GENERIF_SUMMARY	KAI1/CD82 decreased the metastatic phenotype of H1299 lung carcinoma cells by down-regulating Rac1 expression through the PI3K/Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	These results suggest the anti-inflammatory activity of platonin is a result of reduced NF-kappaB activity due to inhibition of Akt and IKKbeta.
Literature	GENERIF_SUMMARY	Overexpression of Akt1 inhibits invasive migration of various breast cancer cell lines, but silencing of Akt1 using siRNA enhances migration. In fibroblasts, however, AKT1 enhances migration. Review.
Literature	GENERIF_SUMMARY	Protein kinase B (pAkt) and vascular endothelial growth factor (VEGF) were the insulin-responsive intermediates in

Main Category	Category	Terms
		proliferating endothelial cells (ECs).
Literature	GENERIF_SUMMARY	the PTEN/PI3K/Akt pathways are critical for prostate cancer stem-like cell maintenance
Literature	GENERIF_SUMMARY	ligation of ApoER2 by APC signals via Dab1 phosphorylation and subsequent activation of PI3K and Akt and inactivation of GSK3beta, thereby contributing to APC's beneficial effects on cells.
Literature	GENERIF_SUMMARY	pneumococci exploit the vitronectin-alphavbeta3-integrin complex as a cellular receptor for invasion and this integrin-mediated internalization requires the cooperation between the host signalling molecules ILK, PI3K and Akt.
Literature	GENERIF_SUMMARY	Data suggest that overexpression of Akt1 and Mdm2 may be related to esophageal carcinogenesis, and that Akt expression regulates Mdm2 expression, which may in turn regulate the function of wild-type p53.
Literature	GENERIF_SUMMARY	AKT1 pleckstrin homology domain mutation appear to be rare in Japanese gastrointestinal and liver cancer patients.
Literature	GENERIF_SUMMARY	Tubulin acetylation stimulates the binding and the signaling function of the kinase Akt/PKB.
Literature	GENERIF_SUMMARY	Trends suggested down-modulation of cyclooxygenase-2 and Ki-67 in some tissues, increased pAKT-Ser473 expression, and an inverse relationship between PGE(2) and BCL2 expression.
Literature	GENERIF_SUMMARY	AKT activation has a role in hypoxia-increased prostate epithelial cell resistance to receptor-mediated apoptosis
Literature	GENERIF_SUMMARY	These results suggest a role for PTEN in normal adult brain and that the PTEN/Akt pathway may be involved in neuronal survival or plasticity after ischemic injury.
Literature	GENERIF_SUMMARY	the specificity of DNA-PK action as a Ser-473 kinase for PKB in DNA repair signaling.
Literature	GENERIF_SUMMARY	S100B interacts with Src kinase, thereby stimulating the PI3K/Akt and PI3K/RhoA pathways.
Literature	GENERIF_SUMMARY	PTEN, PTEN/p-AKT and PTEN/p-mTOR have roles in extrahepatic cholangiocarcinoma
Literature	GENERIF_SUMMARY	E-cadh is predominantly expressed at cell-cell contacts and its functionality is necessary and sufficient for the activation of the PI3K/AKT pathway

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the level of phosphorylation on Thr308, but not on Ser473, was significantly correlated with high-risk karyotype in AML
Literature	GENERIF_SUMMARY	the Thyroid hormone beta1 receptor mediates the T3 upregulation of protein synthesis and cell size, together with the cell proliferation and survival, playing a crucial role in the T3 regulation of the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	genetic variations in AKT1 were associated with survival in esophageal cancer
Literature	GENERIF_SUMMARY	Phosphorylation of AKT and mTOR is correlated with anaplastic lymphoma kinase expression in anaplastic large cell lymphoma.
Literature	GENERIF_SUMMARY	EGFR signals to mTOR through PKC and independently of Akt in glioma
Literature	GENERIF_SUMMARY	Phosphorylated AKT as well as activated mTOR and its down-stream effector S6K were strongly expressed in cystic epithelia of autosomal recessive polycystic kidney disease.
Literature	GENERIF_SUMMARY	Endometrial stathmin is linked to HIF-1alpha protein accumulation and VEGF expression through the PI3K/Akt signaling pathway and may be involved in regeneration of the endometrium during the menstrual cycle in human uterine cells.
Literature	GENERIF_SUMMARY	ET-1 and TGF-beta1 independently promote fibroblast resistance to apoptosis through signaling pathways involving PI3K/AKT.
Literature	GENERIF_SUMMARY	the PPARgamma-dependent PI3K/Akt pathway has a role in telmisartan-induced proliferation of human endothelial progenitor cells
Literature	GENERIF_SUMMARY	KiSS1 metastasis suppressor gene product induces suppression of tyrosine kinase receptor signaling to Akt, tumor necrosis factor family ligand expression, and apoptosis.
Literature	GENERIF_SUMMARY	Study identifies PHLDA3 as a p53 target gene that encodes a PH domain-only protein and finds that PHLDA3 competes with the PH domain of Akt for binding of membrane lipids, inhibiting Akt translocation to the cellular membrane and activation.
Literature	GENERIF_SUMMARY	The phosphorylation of Akt induced by the loss of PTEN took part in the tumorigenesis and development of endometrial carcinoma.
Literature	GENERIF_SUMMARY	These results suggest that one mechanism

Main Category	Category	Terms
		of the apoptotic activity of magnolol involves its effect on epidermal growth factor receptor-mediated EGFR/PI3K/Akt signaling transduction pathways.
Literature	GENERIF_SUMMARY	Constitutively active myristoylated Akt overexpression produces an opposite effect to insulin/insulin-like growth factor-1 stimulation by inhibiting secretion of amyloid precursor protein (APP) and APP metabolites in multiple cell lines.
Literature	GENERIF_SUMMARY	beta-arrestin2 inhibits opioid-induced breast cancer cell death through Akt and caspase-8 pathways.
Literature	GENERIF_SUMMARY	VEGF-A signalling acts on tumour cells as a stimulator of the AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	These results show that Akt/PKB overexpression would be suggestive of malignant progression and invasive regrowth of diffuse astrocytoma, and it can serve as a novel prognostic marker for this tumour.
Literature	GENERIF_SUMMARY	Leptin exerts an anti-apoptotic effect on human dendritic cells via the PI3K-Akt signaling pathway.
Literature	GENERIF_SUMMARY	EGFR activation by H pylori infection has an antiapoptotic effect in gastric epithelial cells that appears to involve Akt signaling and Bcl family members.
Literature	GENERIF_SUMMARY	Bcr-Abl represses the expression of PHLPP1 and PHLPP2 and continuously activates Akt1, -2, and -3 via phosphorylation on Ser-473, resulting in the proliferation of CML cells
Literature	GENERIF_SUMMARY	bortezomib overcame TRAIL resistance at clinically achievable concentrations in hepatocellular carcinoma cells, and this effect is mediated at least partly via inhibition of the PI3K/Akt pathway
Literature	GENERIF_SUMMARY	beta-arrestins and c-Src are implicated in the activation of Akt in response to ghrelin through the GHS-R1a
Literature	GENERIF_SUMMARY	cell Ca2(+) plays a fundamental role in wound healing even without PL, p38 and ERK1/2 are essential for PL effects but are also activated by wounding
Literature	GENERIF_SUMMARY	increased phosphorylation of huntingtin via calcineurin inhibition, rather than via Akt induction or activation, is the likely mechanism by which RCAN1-1L may be protective against mutant huntingtin.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Akt/protein kinase B (PKB) interacts with and directly phosphorylates Skp2.
Literature	GENERIF_SUMMARY	Akt1 phosphorylates Skp2 at Ser 72, which is required to disrupt the interaction between Cdh1 and Skp2.
Literature	GENERIF_SUMMARY	loss of PTEN as well as other genetic alterations that result in constitutive activation of Akt/mTORC1 signaling may contribute to the development of sporadic chordomas.
Literature	GENERIF_SUMMARY	PI3K/Akt mediates phosphorylation of the pro-apoptotic Bcl-2 family member Bax
Literature	GENERIF_SUMMARY	Akt-mediated transactivation of the S1P1 receptor in caveolin-enriched microdomains regulates endothelial barrier enhancement by oxidized phospholipids.
Literature	GENERIF_SUMMARY	The results suggest that the PI3K/Akt-dependent signaling pathways serve to regulate hypoxia-induced epithelial-mesenchymal transition of hepatocellular carcinoma cells.
Literature	GENERIF_SUMMARY	Insulin attenuates the insulin-like growth factor-I (IGF-I)-Akt pathway, not IGF-I-extracellularly regulated kinase pathway, in luteinized granulosa cells with an increase in PTEN.
Literature	GENERIF_SUMMARY	Data showed that tumor-associated matrices induced PI3K and beta1-integrin dependent Akt/PKB activity in MDA-MB-231 cells.
Literature	GENERIF_SUMMARY	A functional polymorphism that impairs the activity of PHLPP2 and correlates with elevated Akt phosphorylation and increased PKC levels, was identified.
Literature	GENERIF_SUMMARY	periostin promoted the survival of A549 non-small-cell lung cancer cells under hypoxic microenvironment via activation of Akt/PKB pathway
Literature	GENERIF_SUMMARY	TNFalpha facilitates the reduction of FLIP(L) protein, which is dependent on the phosphatidylinositol 3-kinase/Akt signaling.
Literature	GENERIF_SUMMARY	the AKT1 E17K mutation is rare in prostate adenocarcinomas, esophageal squamous cell carcinomas, laryngeal squamous cell carcinomas, urothelial carcinomas, hepatoblastomas, gastrointestinal stromal tumors and malignant meningiomas
Literature	GENERIF_SUMMARY	AKT activity is downregulated when Tsc1/Tsc2 function is reduced.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	IL-6 functions in a cytoprotective manner, in part, by suppressing Bax translocation and dimerization through PI3K/Akt-mediated Bax phosphorylation.
Literature	GENERIF_SUMMARY	Akt1 activation alone was sufficient to induce G(1)/S checkpoint bypass
Literature	GENERIF_SUMMARY	AKT1 E17K mutation was searched in 26 total RNA samples obtained from 26 patients known to have urothelial carcinoma.
Literature	GENERIF_SUMMARY	The intracellular concentration of TAL1 is regulated by TGF-beta, which triggers its polyubiquitylation & degradation by the proteasome. This is mediated by AKT1, which phosphorylates TAL1 at threonine 90.
Literature	GENERIF_SUMMARY	EP2 and EP4 receptor inhibition induces apoptosis of human endometriotic cells through suppression of ERK1/2, AKT, NFkappaB, and beta-catenin pathways and activation of intrinsic apoptotic mechanisms
Literature	GENERIF_SUMMARY	ILK is inhibited by QLT0267, which, when combined with docetaxel, reduces P-AKT and alters F-actin architecture
Literature	GENERIF_SUMMARY	uMtCK in prostate neoplasm cells (LNCaP) contributes to overproduction of reactive oxygen species, activation of Akt signaling pathway and more aggressive phenotypes including androgen independence development.
Literature	GENERIF_SUMMARY	Results show that high levels of Akt-1, assessed by quantum dots, deconvolution imaging, and image analysis, are associated with a higher risk of biochemical recurrence and prostate cancer-specific death.
Literature	GENERIF_SUMMARY	complete concordance of PIK3CA and AKT1 mutations in matched samples of invasive and in situ tumor indicates that these mutations occur early in breast cancer development
Literature	GENERIF_SUMMARY	Mutation E17K/AKT1 was not detected in the 24 samples of retinoblastoma analyzed. K-RAS mutations were identified in two samples. There were no mutations in any of the other genes analyzed.
Literature	GENERIF_SUMMARY	dysregulation of the phosphoinositide 3-kinase/Akt/PTEN pathway is relevant to the prognosis in node-negative breast carcinoma
Literature	GENERIF_SUMMARY	Our results suggest that up-regulation of AKT1 and 4EBP1 in eutopic endometrium may be associated with the pathogenesis of

Main Category	Category	Terms
		endometriosis.
Literature	GENERIF_SUMMARY	Findings indicate that AP-1 has an important function in pancreatic cancer cells and provide evidence for a previously unknown Akt-mediated mechanism of c-Jun activation.
Literature	GENERIF_SUMMARY	PTEN modulated VEGF-mediated signaling and affected tumor angiogenesis through PI3K/Akt/VEGF/eNOS pathway.
Literature	GENERIF_SUMMARY	Results show that blockade of Akt/HIF-1alpha/PDGF-BB autocrine signaling could enhance the chemosensitivity of liver cancer cells and tumorigenic hepatic progenitor cells under hypoxic conditions and thus provide an effective therapeutic strategy for HCC.
Literature	GENERIF_SUMMARY	A significant difference in the immunoexpression for p-Akt was observed when squamous cell carcinoma samples were compared with normal and dysplasic epithelial groups.
Literature	GENERIF_SUMMARY	Apoptotic body engulfment by hepatic stellate cells promotes their survival by the JAK/STAT and Akt/NF-kappaB-dependent pathways.
Literature	GENERIF_SUMMARY	mutations of PTEN, PI3K, or AKT were identified in 47.7% of 44 cases of T-cell acute lymphoblastic leukemia
Literature	GENERIF_SUMMARY	In glioblastoma multiforme, oncogenic deregulation of the PI3K pathway does not involve somatic mutations in the coding region of AKT1.
Literature	GENERIF_SUMMARY	The constitutive activation of the PI3K-Akt pathway in Fanconi anaemia cells upregulates the expression of IL-1beta.
Literature	GENERIF_SUMMARY	Results show that this unique and specific bacterial entry process is dependent on the cooperation and activation of Rho family GTPase Cdc42, PI3K, and Akt.
Literature	GENERIF_SUMMARY	EGFR activity inhibition abrogates anoikis resistance in cancer cells through a loss of Fak-Src interactions and down-activation of Src/MEK/Erk and PI3-K/Akt-1
Literature	GENERIF_SUMMARY	Results identify PACS-2 as an essential TRAIL effector, and show that Akt cooperates with 14-3-3 to regulate the homeostatic and apoptotic properties of PACS-2 that mediate TRAIL action.
Literature	GENERIF_SUMMARY	ROS-driven Akt dephosphorylation at Ser-473 is involved in 4-HPR-mediated

Main Category	Category	Terms
		apoptosis in NB4 cells.
Literature	GENERIF_SUMMARY	Findings show that PKIB and PKA-C kinase can have critical functions of aggressive phenotype of PCs through Akt phosphorylation and that they should be a promising molecular target for PC treatment.
Literature	GENERIF_SUMMARY	Data show that blockade of endothelial cell-derived IL-6, CXCL8, or EGF by gene silencing or neutralizing antibodies inhibited phosphorylation of STAT3, Akt, and ERK in tumor cells, respectively.
Literature	GENERIF_SUMMARY	Absence of PIK3CA/AKT1 mutations in one specimen may not reflect the status at other sites because these mutations arise during progression.
Literature	GENERIF_SUMMARY	angiogenic effects of A549 conditioned culture media are largely mediated through activation of the PI3K-Akt in endothelial cells, and that the Akt1 is crucial in this process.
Literature	GENERIF_SUMMARY	Activation of the AKT/mTOR signaling pathway plays an important role in the development of diffuse large B-cell lymphoma, and is closely related to the low or non-expression of bcl-6.
Literature	GENERIF_SUMMARY	Two AKT1 (E17K) mutations in the endometrial cancer tissue samples and no mutations in the cell lines.
Literature	GENERIF_SUMMARY	PTEN/Akt/beta-catenin signaling pathway plays an importatn role in regulation of normal and malignant mammary stem/ progenitor cell populations.
Literature	GENERIF_SUMMARY	Resutls show that the IS formation induced antiapoptotic signaling events, including activation of the kinase Akt1 and localization of NF-kappaB and FOXO1 to the nucleus and cytoplasm.
Literature	GENERIF_SUMMARY	a novel role of PI3K/Akt in fibrogenesis of human lung fibroblasts and its regulation by ROS, which could be exploited for the treatment of pulmonary fibrosis and related disorders.
Literature	GENERIF_SUMMARY	Data indicate that PI3K inhibitor, LY294002, almost completely blocked Lf(Fe(3+))-induced Akt activation, and prevented the cytoplasmic localization of p27(kip1).
Literature	GENERIF_SUMMARY	PP2Cgamma is a novel regulator of p21(Cip1/WAF1) protein stability via the Akt signaling pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results provide evidence for the potential application of roscovitine/wormannin combination in clinical cancer treatment by repressing PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Data suggest that decreased habituation of arousal, NRG1-induced AKT phosphorylation, and anxiety are related to delusional ideation in the general population.
Literature	GENERIF_SUMMARY	Studies indicate inhibition of the PI3K/AKT/mTOR pathway is not only a potential first-line therapeutic strategy in ovarian cancer but also could be used in recurrent disease as a means of reversing acquired chemotherapy resistance.
Literature	GENERIF_SUMMARY	loss of stromal Cav-1 is predictive of elevated levels of epithelial Cav-1 and epithelial Akt-activation and is associated with advanced prostate cancer.
Literature	GENERIF_SUMMARY	Data show that the DNA-binding activities of c-Myc to the hTERT promoter were decreased in butein-treated cells, and reatment with butein also suppressed the activation of Akt, thereby inhibiting hTERT phosphorylation and translocation into the nucleus
Literature	GENERIF_SUMMARY	commitment of cell transformation by Aurora-A is determined by at least co-activation of Akt/mTOR pathway
Literature	GENERIF_SUMMARY	Results demonstrated that the inhibition of spontaneous B cell apoptosis by CpG DNA was correlated to up-regulation of Bcl-xL, IAP and down-regulation of p53 and caspase 3 mediated through PI3K/AKT signaling.
Literature	GENERIF_SUMMARY	Results suggested that PTEN/AKT pathway was involved in trichostatin A induced cell growth inhibition and apoptosis of oral squamous cell carcinoma cells.
Literature	GENERIF_SUMMARY	Findings show that PI3K inhibition diminished hepatoblastoma cell growth being accompanied by reduced AKT and GSK-3beta phosphorylation.
Literature	GENERIF_SUMMARY	Data show that IGFBP-3 is regulated by the PI3-kinase/AKT and MAPK/ERK1/2 pathways.
Literature	GENERIF_SUMMARY	Akt, activated downstream from ERK 1/2, was required for TGF-beta1-induced ZEBRA expression and enabled Smad3, a mediator of TGF-beta1 signaling, to be acetylated by direct interaction with the co-activator CREB-binding protein

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Literature	GENERIF_SUMMARY	Interaction of Akt-phosphorylated SRPK2 with 14-3-3 mediates cell cycle and cell death in neurons.
Literature	GENERIF_SUMMARY	Data show that inhibition of TACE activity or expression enhanced the cell surface TbetaRI levels and TGF-beta-induced Smad3 and Akt activation.
Literature	GENERIF_SUMMARY	TLR4 was expressed in colorectal cancer(CRC) cells and LPS treatment increased Akt phosphorylation of TLR4-positive CRCs in a time-dependent manner
Literature	GENERIF_SUMMARY	OPN enhances angiogenesis directly through PI3K/AKT- and ERK-mediated pathways with VEGF acting as a positive feedback signal.
Literature	GENERIF_SUMMARY	Data show that aberrant expression of ErbB2, ErbB3 and ErbB4 in the two resistant cell lines, suggesting the existence of an autocrine loop leading to constitutive activation of MAPK and PI3K/Akt survival pathways.
Literature	GENERIF_SUMMARY	Data suggest an interaction between COX-2 enzymatic activity and PI3Kinase/AKT signaling pathway in EOC.
Literature	GENERIF_SUMMARY	SHIP gene can inhibit K562 cell proliferation and promote cell apoptosis via inactivating PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Results predict that other functions shared by Akt and NHE1, including cell growth and survival, might be regulated by increased H(+) efflux.
Literature	GENERIF_SUMMARY	In THP-1 cells, corticotropin-releasing factor activates the phosphatidylinositol 3-kinase/ Akt and ERK1/2 pathways.
Literature	GENERIF_SUMMARY	ILK mediated pressure-stimulated adhesion through regulating phosphorylation of AKT and FAK.
Literature	GENERIF_SUMMARY	The Akt pathway is involved in peptidoglycan-induced interleukin-6 production in rheumatoid arthritis synovial fibroblasts.
Literature	GENERIF_SUMMARY	Data show that with SF1126 treatment, caspase 3 and PARP proteins were cleaved, survivin was down-regulated, and inhibition of PI3-kinase was confirmed by reduced phosphorylation of Akt.
Literature	GENERIF_SUMMARY	RUNX1 may play a critical role in chemotherapy response in acute megakaryocytic leukemia by regulating the PI3-kinase/Akt pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Our results demonstrated that shRNA targeting COX-2, Akt1 and PIK3R1 downregulates their expression significantly in a sequence-specific manner, exerting proliferation and invasion inhibition effects on SGC7901 and U251 cells
Literature	GENERIF_SUMMARY	suppression of the PI3 kinase/Akt pathway is correlated with the development of BMP2 resistance and invasion in BMP2-induced EMT in colon cancer.
Literature	GENERIF_SUMMARY	Aberrant overexpression of microRNAs activate AKT signaling via down-regulation of tumor suppressors in natural killer-cell lymphoma/leukemia.
Literature	GENERIF_SUMMARY	Chronic moderate hypoxia induces macrophage activation via the Akt and beta-catenin pathways, providing new insight into the pathogenesis of inflammatory diseases.
Literature	GENERIF_SUMMARY	pAkt regulates expression for proliferation, survival and invasion of breast cancer cells
Literature	GENERIF_SUMMARY	The results indicate that serum activates Akt and ERK pathways earlier than XIAP expression.
Literature	GENERIF_SUMMARY	Expression of pAkt, GLUT1 and TKTL1 were higher in breast cancer and DCIS than in normal tissue.
Literature	GENERIF_SUMMARY	Study has shown that EGCG protects A549 cells from apoptosis induced by serum deprivation via Akt activation and this protective effect may limit the clinical use of EGCG in treating and preventing NSCLC.
Literature	GENERIF_SUMMARY	Results demonstrated the augmented adjuvant anti-tumor effects of AC-CF not only through down regulation of MDR-1 expression but also through a COX-2 dependent apoptosis pathway, involving down-regulation of COX-2 and p-AKT and up-regulation of PARP-1.
Literature	GENERIF_SUMMARY	zinc deficiency differentially modulated the Akt-Mdm2-p53 signaling axis in normal prostate cells vs. the Akt-p21 signaling axis in malignant prostate cells.
Literature	GENERIF_SUMMARY	Wortmannin can inhibit the proliferation and induce apoptosis of K562 leukemia cells possibly by down-regulating Akt and NF-kappaB.
Literature	GENERIF_SUMMARY	Data show that p53-independent curcumin-induced apoptosis in ovarian carcinoma cells involves p38 MAPK activation, ablation of Akt signaling, and reduced expression of the antiapoptotic proteins Bcl-2 and survivin.

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Literature	GENERIF_SUMMARY	Studies indicate that drugs are effective in targeting essential molecular pathways of BRAF, PTEN, Akt and mammalian target of rapamycin.
Literature	GENERIF_SUMMARY	YadA and invasin of Yersinia pseudotuberculosis both trigger signal transduction pathways during integrin-mediated phagocytosis into epithelial cells, which lead to the activation of Akt, PLC-gamma1, PKCalpha and -beta downstream of PI3 kinase.
Literature	GENERIF_SUMMARY	This study not only reveals a role of CRT in motility promotion and anoikis resistance in ESCC cells, but also identifies CRT as an upstream regulator in the CRT-STAT3-CTTN-Akt pathway.
Literature	GENERIF_SUMMARY	PI3Kbeta plays an essential role in GPVI-mediated platelet aggregation and Akt activation
Literature	GENERIF_SUMMARY	Chronic lymphocytic leukemia clones consistently contain activated Akt which plays a pivotal role in maintaining cell survival.
Literature	GENERIF_SUMMARY	TRAF6 was found to be direct E3 ligase for Akt & essential for Akt ubiquitination, membrane recruitment & phosphorylation upon growth-factor stimulation; cancer-associated Akt mutant displayed increase Akt ubiquitination
Literature	GENERIF_SUMMARY	Results show that ceramide is a necessary and sufficient intermediate mediating the inhibition of the AKT/eNOS signaling pathway by palmitate in endothelial cells.
Literature	GENERIF_SUMMARY	Results indicate that AMPK is crucial in apoptosis induced by curcumin and further that the pAkt-AMPK-COX-2 cascade or AMPK-pAkt-COX-2 pathway is important in cell proliferation
Literature	GENERIF_SUMMARY	Results suggest that genistein could inhibit cell growth by inhibiting AKT phosphorylation, ERK1/2 activity and activating p38 MAPK.
Literature	GENERIF_SUMMARY	Exposure of cultured human renal proximal tubular cells to high glucose increased Akt and pMAPK phosphorylation coinciding with increased apoptosis.
Literature	GENERIF_SUMMARY	Selective inhibition of AKT renders normally resistant first-trimester trophoblast cells sensitive to FAS-mediated apoptosis by regulating XIAP expression.
Literature	GENERIF_SUMMARY	These results identify nNOS as a target gene

Main Category	Category	Terms
		of a novel RA/PI3K/Akt/DAX1-dependent pathway in human neuroblastoma cells
Literature	GENERIF_SUMMARY	Cell cycle regulators, AKT, p53, and MAPK, as therapeutic targets for MONCPT to induce cell cycle G2/M arrest in the context of anticancer therapy.
Literature	GENERIF_SUMMARY	Data showed that 9 tumor samples had strong expression of P-Akt and 5/9 had P-MAPK, and none of them had EGFR expression.
Literature	GENERIF_SUMMARY	Data show that the N-t-boc-Daidzein induced apoptosis is characterized by caspase activation, XIAP and AKT degradation.
Literature	GENERIF_SUMMARY	No AKT1 E17K mutation was found among 86 Chinese Philadelphia-positive CML patients, suggesting that the pre-existence of this mutation has little role in CML.
Literature	GENERIF_SUMMARY	In the striatum, reduced levels of PHLPP1 can contribute to maintain high levels of activated Akt that may delay cell death and allow the recovery of neuronal viability after mutant huntingtin silencing
Literature	GENERIF_SUMMARY	Inhibition of apoptosis by GIP is mediated via a key pathway involving Akt-dependent inhibition of apoptosis signal-regulating kinase 1, which subsequently prevents the pro-apoptotic actions of p38 MAPK and JNK.
Literature	GENERIF_SUMMARY	Diabetic monocytes show increased migratory capacity in response to GSK-3beta inhibition.
Literature	GENERIF_SUMMARY	These findings demonstrate that excretions/secretions directly induces human microvascular endothelial cells-1 cell migration and this event is partially mediated by the activation of AKT1.
Literature	GENERIF_SUMMARY	The over-expression of AKT accompanied by the loss of PTEN involves carcinogenesis and development of colon carcinoma.
Literature	GENERIF_SUMMARY	Data show that gene silencing of Ob-R with Ob-R siRNA in EOC cells resulted in down regulation of phospho-AKT and its down stream targets.
Literature	GENERIF_SUMMARY	RACK1, partly by inhibiting Src, promotes mitochondrial cell death and blocks Akt-mediated cell survival.
Literature	GENERIF_SUMMARY	Testosterone induces ERK and Akt phosphorylation via the membrane-initiated signaling pathways mediated by ER-alpha36, suggesting a possible involvement of ER-

Main Category	Category	Terms
		alpha 36 in testosterone carcinogenesis.
Literature	GENERIF_SUMMARY	The authors report that M-T5 can bind Akt and the host SCF complex (via Skp1) simultaneously in myxoma virus-infected cells.
Literature	GENERIF_SUMMARY	findings indicate DNA dependent protein kinase as an important enzyme maintaining the quiescent endothelial phenotype by actively inhibiting Akt thus restraining endothelial cell proliferation preventing excessive growth.
Literature	GENERIF_SUMMARY	in multivariate analyses, none of the phospho-Akt levels appeared to be independent predictors of breast cancer recurrence or death
Literature	GENERIF_SUMMARY	demonstrate a key role for beta-TrCP in controlling the level of PHLPP1, and activation of Akt negatively regulates this degradation process.
Literature	GENERIF_SUMMARY	SYUIQ-5 induces autophagy in nasopharyngeal cancer cells which may be related to SYUIQ-5-mediated p-Akt down-regulation and FOXO3a nuclear translocation.
Literature	GENERIF_SUMMARY	Data show that B[a]PDE increases cyclin D1 expression, and activate the PI3-kinase/Akt/MAPK- and p70s6k pathway in Beas-2B bronchial epithelial cells.
Literature	GENERIF_SUMMARY	Data show that stimulation of HIF-1alpha transcription by the ND6 mutation was mediated by overproduction of reactive oxygen species (ROS) and subsequent activation of phosphatidylinositol 3-kinase (PI3K)-Akt and protein kinase C (PKC) signaling pathways.
Literature	GENERIF_SUMMARY	Data show an enhanced activation of AKT1 when E17K and E49K mutations are in tandem suggests that they can co-operate.
Literature	GENERIF_SUMMARY	Results suggest that Akt confers resistance, in part, by modulating CDDP-induced, p53-dependent FLIP ubiquitination.
Literature	GENERIF_SUMMARY	Data show that everolimus/RAD001 significantly inhibits cell viability in a dose- and time-dependent fashion, and diminishes phosphorylation of Akt downstream targets, mTOR and p70S6K, in human medullary thyroid carcinoma cells.
Literature	GENERIF_SUMMARY	LRP1 mediates anti-apoptotic function in neurons by regulating insulin receptor and the Akt survival pathway

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data show that Akt inhibitors inhibited Akt phosphorylation activity.
Literature	GENERIF_SUMMARY	Data show that mutated GPC3 lacking the GPI anchoring domain (sGPC3) significantly inhibited the in vivo growth, blocked Wnt signaling and Erk1/2 and Akt phosphorylation in tumors.
Literature	GENERIF_SUMMARY	EGF receptor up-regulated the expression of MT1-MMP and down-regulated the synthesis of MMP-2 through the mitogen-activated protein kinase/extracellular signal-regulated kinase pathway.
Literature	GENERIF_SUMMARY	induction of EMT is controlled by microRNAs whose abundance depends on the balance between Akt1 and Akt2 rather than on the overall activity of Akt
Literature	GENERIF_SUMMARY	the PDX1 C terminus has a role in mediating the effects of glucose, which modulates PDX1 stability via the AKT-GSK3 axis
Literature	GENERIF_SUMMARY	BMP-2 promotes motility and invasion of gastric cancer cells by activating the PI-3 kinase/Akt signal pathway.
Literature	GENERIF_SUMMARY	Together, these results suggest that TNFalpha induced oxidative stress affects Akt activation to regulate actin organization and growth of glioma cells.
Literature	GENERIF_SUMMARY	OPN promotes prostate cancer cell progression processes by activating the Akt/beta-catenin pathway. The OPN/Akt pathway is required for the stabilization of beta-catenin. Nuclear beta-catenin activates the TCF/LEF family of transcription factors.
Literature	GENERIF_SUMMARY	Data shows the AKT1 E17K mutation in 4% of endometrial carcinomas. The presence of double AKT1/ PTEN mutants is in accord with the hypothesis that in EC more than one hit is required to completely activate the PI3K pathway.
Literature	GENERIF_SUMMARY	Data show that RNA silencing of choline kinase in HeLa cells completely abrogates the high concentration of phosphocholine, which in turn decreases phosphatidylcholine, phosphatidic acid and signaling through the MAPK and PI3K/AKT pathways.
Literature	GENERIF_SUMMARY	The lytic replication of gamma-2 herpesvirus was enhanced by inhibiting the PI3K-Akt pathway.
Literature	GENERIF_SUMMARY	Akt modulates TGF-beta signaling by temporarily adjusting the levels of two TGF-beta pathway negative regulators, Ski and

Main Category	Category	Terms
		Smad7
Literature	GENERIF_SUMMARY	germ line genetic variations in the PI3K-AKT-mTOR pathway may have roles as bladder cancer susceptibility factors
Literature	GENERIF_SUMMARY	baicalein not only inhibited IL-6-mediated phosphorylation of signaling proteins, such as Jak, STAT3, MAPK, and Akt, but also inhibited the expression of their target genes, such as bcl-xl.
Literature	GENERIF_SUMMARY	AKT activation is crucial for PTEN suppression in surviving cells following ultraviolet B irradiation.
Literature	GENERIF_SUMMARY	the Akt phosphorylation and inactivation of its downstream target, BAD may play a role in survival of breast cancer cell
Literature	GENERIF_SUMMARY	anthocyanins isolated from Vitis coignetiae Pulliat induce apoptosis might at least in part through activating p38-MAPK and suppressing Akt in human colon cancer HCT-116 cells.
Literature	GENERIF_SUMMARY	Suppression of Akt1 expression by small interference RNA inhibits SGC7901 cell growth in vitro and in vivo
Literature	GENERIF_SUMMARY	Epigallocatechin gallate decreases ET-1 expression and secretion from endothelial cells, in part, via Akt- and AMPK-stimulated FOXO1 regulation of the ET-1 promoter.
Literature	GENERIF_SUMMARY	Wnt-3A may activate canonical Wnt signaling and PI3K/AKT through distinct receptors.
Literature	GENERIF_SUMMARY	Oscillatory shear stress induces osteoblast-like cell proliferation through activation of alpha(v)beta(3) and beta(1) integrins and synergistic interactions of FAK and Shc with PI3K, leading to the modulation of ERK and Akt/mTOR/p70S6K pathways.
Literature	GENERIF_SUMMARY	increasing Hsp70 levels combined with inhibiting its ATPase function may serve to dramatically reduce Akt levels and facilitate cell death in certain types of cancer
Literature	GENERIF_SUMMARY	Aurora-A, via activating Akt, stimulated nuclear factor-kappaB signaling pathway to promote cancer cell survival.
Literature	GENERIF_SUMMARY	The key signaling molecules in the PI3K signaling cascade, the abnormalities known to occur in thyroid cancer, and the potential for therapeutic targeting of PI3K pathway members are discussed.
Literature	GENERIF_SUMMARY	two-thirds of papillomas are driven by

Main Category	Category	Terms
		mutations in the PI3CA/AKT pathway
Literature	GENERIF_SUMMARY	Data show that DAB2IP is a potent growth inhibitor by inducing G(0)/G(1) cell cycle arrest and is proapoptotic in response to stress, and that DAB2IP can suppress the PI3K-Akt pathway and enhance ASK1 activation leading to cell apoptosis.
Literature	GENERIF_SUMMARY	Elevated expression of p53-R175H mutant may exert gain-of-function activity to activate the EGFR/PI3K/AKT pathway and thus may contribute to the invasive phenotype in endometrial cancer.
Literature	GENERIF_SUMMARY	Data suggest that the targeting of Akt by T. cruzi could be an important mechanism that underlies the long-term survival of infected cells.
Literature	GENERIF_SUMMARY	Data reveal PTPN2, PTPRJ and PTEN as potent regulators of Akt signalling which contribute to ameliorating the consequences of oncogenic K-Ras activity.
Literature	GENERIF_SUMMARY	Rac activation by mechanical forces is rapid, direct and does not depend on Src activation
Literature	GENERIF_SUMMARY	p-Akt and COX-2 are expressed in gastric adenocarcinomas
Literature	GENERIF_SUMMARY	Data show 5-HT can induce activation of two interrelated signaling pathways, Erk1/2 and PI3K/Akt to induce proliferation, migration, and differentiation.
Literature	GENERIF_SUMMARY	The allele, genotype, and haplotype analyses in this study and the meta-analysis do not support an association between AKT1 and schizophrenia.
Literature	GENERIF_SUMMARY	These results established a novel pathway that consists of proto-oncogene c-Akt, COX-2 and Mcl-1 for acquired apoptosis resistance in lung cancer cells.
Literature	GENERIF_SUMMARY	These results indicate that forskolin (FSK) does not activate the Akt-mTOR pathway in human skeletal muscle; however, these results suggest that FSK may inhibit the actions of insulin on this pathway.
Literature	GENERIF_SUMMARY	wtSHIP gene can down-regulate Akt phosphorylation and up-regulate cell cycle related proteins in K562 cells.
Literature	GENERIF_SUMMARY	hypoxia switches glucose depletion-induced necrosis to apoptosis and ERK1/2 and PI3K-Akt exert anti-necrotic and pro-apoptotic activities in the cell death, respectively

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ICAM-3 enhances the migratory and invasive potential of human non-small cell lung cancer cells by inducing MMP-2 and MMP-9 via Akt and CREB
Literature	GENERIF_SUMMARY	Data show that RTX treatment results in a time-dependent inhibition of the BCR-signaling cascade involving Lyn, Syk, PLC gamma 2, Akt, and ERK, and calcium mobilization.
Literature	GENERIF_SUMMARY	Data show that suppression of PUMA through the phosphoinositide 3-kinase (PI3K)/protein kinase B (AKT)/p53 axis in the intestinal stem cells as a novel molecular mechanism of growth factor-mediated intestinal radioprotection.
Literature	GENERIF_SUMMARY	glutamate signaling mediated by AMPA receptor induces U-87MG human malignant glioblastoma cell line proliferation via EGFR-phospho-Akt pathway.
Literature	GENERIF_SUMMARY	Increased expression of Lewis y antigen plays an important role in promoting cell proliferation through activating PI3K/Akt signaling pathway in ovarian carcinoma-derived RMG-I cells.
Literature	GENERIF_SUMMARY	Data suggest that in Caco-2 cells, PP2A and the cAMP pathway act in concert to inactivate the AKT survival pathway in PTH-induced intestinal cell apoptosis.
Literature	GENERIF_SUMMARY	Leptin plays an important role in papillary thyroid carcinoma pathogenesis through PI3K/AKT pathway via Ob-R and is a potential prognostic marker associated with an aggressive phenotype and poor disease-free survival.
Literature	GENERIF_SUMMARY	VEGF increases the phosphorylation and activation of ERK and Akt within CD4+CD45RO+ T cells
Literature	GENERIF_SUMMARY	identified Akt1 as a new signaling component of the amino-Nogo pathway. Akt1 phosphorylation is decreased by amino-Nogo
Literature	GENERIF_SUMMARY	Heat shock protein 70 (Hsc70) plays a significant role in vascular endothelial cells via the phosphatidylinositol 3-kinase/Akt pathway.
Literature	GENERIF_SUMMARY	Akt1 was located in the cytoplasm of human cell lines.
Literature	GENERIF_SUMMARY	findings suggest that CCL2 hyperactivates mTORC1 through simultaneous regulation of both AMPK and Akt pathways and reveals a new network that promotes prostate

Main Category	Category	Terms
		cancer
Literature	GENERIF_SUMMARY	PI3K/AKT mutations may not play a major role in multiple myeloma.
Literature	GENERIF_SUMMARY	Data showed that curcumin decreases HCV gene expression via suppression of the Akt-SREBP-1 activation, not by NF-kappaB pathway.
Literature	GENERIF_SUMMARY	These findings demonstrate the importance of cross-talk between the PI3K/Akt and beta-catenin pathways
Literature	GENERIF_SUMMARY	Studies indicate that the PI3K/Akt/mTOR pathway plays a critical role in breast cancer development and resistance to traditional anti-tumor strategy.
Literature	GENERIF_SUMMARY	TRPV6 was stimulated by protein kinase B PKB/Akt.
Literature	GENERIF_SUMMARY	From our data, we proposed that Choline kinase plays a key role in regulating Akt(ser473) phosphorylation, thereby promoting cell survival and proliferation
Literature	GENERIF_SUMMARY	in human epithelial cells, shrinkage-induced activation of Akt1 is involved in the regulatory volume increase (RVI) process and apoptotic inhibition of RVI is caused by inhibition of Akt activation, which results from ROS-mediated activation of ASK1.
Literature	GENERIF_SUMMARY	Expression of a constitutively activated STAT3 mutant or myristoylated AKT partially blocked the effects of sunitinib in these tumor cells.
Literature	GENERIF_SUMMARY	the integrity of lipid rafts is required for the activity of Akt and cell survival in epidermal cancers
Literature	GENERIF_SUMMARY	the activation of the Akt pathway signalling is closely correlated with shear-induced VSMC reorientation.
Literature	GENERIF_SUMMARY	Patients with ulcerative colitis higher PTEN expression and lower AKT expression than those with colorectal cancer, but lower PTEN and higher Akt expressions than in normal tissue.
Literature	GENERIF_SUMMARY	Data show that PF-2341066 reduced HGF-stimulated phosphorylation of c-Met in the tyrosine kinase domain as well as phosphorylation of the downstream signaling effectors, Akt and Erks.
Literature	GENERIF_SUMMARY	The cancer-associated Akt mutant recently identified in a subset of human cancers displays enhanced Akt ubiquitination, in turn

Main Category	Category	Terms
		contributing to Akt hyperactivation, suggesting a potential role of Akt ubiquitination in cancers.
Literature	GENERIF_SUMMARY	Data show that PI3K activation and PIP3 production lead to recruitment of the PKB/beta-arrestin/PDE4 complex to the membrane via the PKB PH domain, resulting in degradation of the TCR-induced cAMP pool and allowing full T-cell activation to proceed.
Literature	GENERIF_SUMMARY	Data show T117/T384 as Akt phosphorylation sites in MST2, and mutation of these sites inhibited MST2 binding to Raf-1 but enhanced binding to RASSF1A, accentuating downstream c-jun N-Terminal Kinase and p38 MAPK signaling and promoting apoptosis.
Literature	GENERIF_SUMMARY	The PI3K/Akt pathway may play an inhibitory role in inflammatory mediators-induced human vascular smooth muscle cell calcification through regulating alkaline phosphatase expression.
Literature	GENERIF_SUMMARY	PGE(2)-induced CYP1B1 expression is mediated by ligand-independent activation of the ERalpha pathway as a result of the activation of ERK, Akt, and PKA in breast cancer cells.
Literature	GENERIF_SUMMARY	Enhanced MCP-1 release by keloid CD14+ cells augments fibroblast proliferation via Akt pathway in keloids.
Literature	GENERIF_SUMMARY	mutations in critical genes within the PI3-kinase (PI3K) pathway are not functionally equivalent, and that other cooperative genetic events may be necessary to achieve oncogenic PI3K pathway activation in cancers that contain the AKT1 E17K mutation.
Literature	GENERIF_SUMMARY	In breast cancer AKT3 amplifications and AKT1 and AKT2 deletions were seen by FISH; AKT1 amplification was identified in prostatic carcinomas
Literature	GENERIF_SUMMARY	Caffeine confers radiosensitization of PTEN-deficient malignant glioma cells by enhancing ionizing radiation-induced G1 arrest and negatively regulating Akt phosphorylation
Literature	GENERIF_SUMMARY	Phosphorylated Akt expression is augmented in peripheral blood lymphocytes of both Parkinson's and Alzheimer's disease patients.
Literature	GENERIF_SUMMARY	Data show that the expression of tenascin-C is induced in immune myeloid cells, and its

Main Category	Category	Terms
		synthesis is transcriptionally regulated and requires the specific activation of AKT/PI3K and NF-kappaB signaling pathways.
Literature	GENERIF_SUMMARY	Results represent the first indication that Akt isoforms regulate intermediate filament protein expression and support the hypothesis that IFs are involved in PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Data show that distinct phosphatases dephosphorylate conserved AKT motifs within the FOXO family and that PP2A is entwined in a dynamic interplay with AKT and 14-3-3 to directly regulate FOXO3a subcellular localization and transcriptional activation.
Literature	GENERIF_SUMMARY	Adhesion of ST6Gal I-mediated human colon cancer cells to fibronectin contributes to cell survival by integrin beta1-mediated paxillin and AKT activation.
Literature	GENERIF_SUMMARY	Expression of Par-4 sensitizes Caki cells to TRAIL via modulation of multiple targets, including DR5, Bcl-2, Akt, and NF-kappaB.
Literature	GENERIF_SUMMARY	Demographic and clinical characteristics and AKT1 single markers and haplotypes, but not AKTIP polymorphisms or interactions between AKT1 and AKTIP, are associated with increased risk for suicidal behavior in bipolar patients.
Literature	GENERIF_SUMMARY	Data show that AKT1 and AKT2 appeared to regulate growth through FOXO proteins, but not through either GSK3beta or mTOR, and in contrast, inactivation of PDPK1 affected GSK3beta and mTOR activation.
Literature	GENERIF_SUMMARY	Results indicate that AKT1 polymorphisms are associated with susceptibility to pulmonary TB.
Literature	GENERIF_SUMMARY	the PTEN/AKT pathway has a role as a target of CsA in epidermal keratinocytes, suggesting a previously unknown mechanism in CsA-enhanced skin carcinogenesis
Literature	GENERIF_SUMMARY	There is an intensive relationship between pAKT and CD44v6 expression in breast cancer tissues.
Literature	GENERIF_SUMMARY	AKT1-SNP4 A/A genotype seems to be a candidate biomarker of primary resistance in non-small lung cancer patient treated with gefitinib.
Literature	GENERIF_SUMMARY	The authors observed that extracellular signal-regulated kinase (ERK) activation promotes viral fusion via ezrin-mediated

Main Category	Category	Terms
		cytoskeletal rearrangements, whereas AKT1 attenuates fusion by promoting phosphorylation of F protein of Sendai virus.
Literature	GENERIF_SUMMARY	Data from functional molecular imaging demonstrated that beta-parvin plays a regulatory role in the ILK-mediated Akt (also called protein kinase B) signaling cascades, suggesting that beta-parvin might be a crucial modulator of cell survival.
Literature	GENERIF_SUMMARY	Characterizing the expression of PTEN, AKT and pS6 Ser235/236 might be a useful predictive tool for deciding whether to use mTOR inhibitors to treat conjunctiva melanomas.
Literature	GENERIF_SUMMARY	coordinated regulation of IGF-1/AKT/mTOR pathways and the endosomal compartment by the p53 pathway integrates the molecular, cellular, and systemic levels of activities in response to stress [review]
Literature	GENERIF_SUMMARY	PP-1 (PP-1alpha or PP-1beta) acts as a major phosphatase to dephosphorylate AKT1 at Thr-450 and thus modulate its functions in regulating gene expression, cell survival and differentiation.
Literature	GENERIF_SUMMARY	We evaluated the presence of mutations in PIK3CA, AKT1, AKT2, AKT3, PTEN, and PDPK1 genes in 83 papillary thyroid carcinomas
Literature	GENERIF_SUMMARY	In rheumatoid arthritis fibroblast like synovial cells phosphorylation of Akt protects against Fas-induced apoptosis through inhibition of Bid cleavage.
Literature	GENERIF_SUMMARY	Liver X Receptor activation downregulates AKT survival signaling in lipid rafts and induces apoptosis of prostate cancer cells.
Literature	GENERIF_SUMMARY	The radiosensitivity shown in H460 cells may be due to the higher expression of PTEN through p53 signaling pathway.
Literature	GENERIF_SUMMARY	In dermal fibroblasts, CCN2 is required for the induction of MMP1 following Akt inhibition. CCN2 induces MMP1 expression via an Erk1/2 and Ets1-dependent pathway.
Literature	GENERIF_SUMMARY	A sliding window procedure showed a five-SNP haplotype (TCGAG) to be associated with schizophrenia and bipolar disorder and a four-SNP haplotype (TCGA) with the combined sample, showing involvement of AKT1 in both schizophrenia and bipolar disorder.
Literature	GENERIF_SUMMARY	IL-24 has a role in sensitizing melanoma cells to erlotinib through modulation of the

Main Category	Category	Terms
		Apaf-1 and Akt signaling pathways
Literature	GENERIF_SUMMARY	ErbB/PI3K/Akt/NF-kappaB signalling in the progression of prostate cancer
Literature	GENERIF_SUMMARY	subcellular localization of activated AKT is dependent on TCL1B expression in estrogen receptor- and progesterone receptor-expressing breast cancer
Literature	GENERIF_SUMMARY	Hypoxia enhances BMP-2 expression in osteoblasts by an HIF-1alpha-dependent mechanism involving the activation of ILK/Akt and mTOR pathways.
Literature	GENERIF_SUMMARY	Data define a new model of integrin beta4-dependent PKB/Akt and ERK1/2 activation demonstrating the keratin filament as a structure necessary in signal transmission.
Literature	GENERIF_SUMMARY	Data show that PTPN11 mutations causing Leopard syndrome (LS) facilitate EGF-induced PI3K/AKT/GSK-3beta stimulation through impaired GAB1 dephosphorylation, resulting in deregulation of a novel signaling pathway that could be involved in LS pathology.
Literature	GENERIF_SUMMARY	These results suggest that Akt plays a major role in Na(+)/K(+)-ATPase intracellular translocation and thus in alveolar fluid reabsorption.
Literature	GENERIF_SUMMARY	Ack1 mediated AKT/PKB tyrosine 176 phosphorylation regulates its activation
Literature	GENERIF_SUMMARY	the NIC-Akt-Mfn signaling cascade identifies a pathway regulating cell-survival, independent of canonical functions associated with NIC activity
Literature	GENERIF_SUMMARY	Myricetin inhibits Akt survival signaling and induces Bad-mediated apoptosis in a low dose ultraviolet (UV)-B-irradiated HaCaT human immortalized keratinocytes
Literature	GENERIF_SUMMARY	Thermo-chemotherapy has a stronger inhibitory effect than chemotherapy alone in lung tumor cell growth, probably through induction of ROS production and subsequent inhibition of Akt pathway activation and Caspase pathway-induced cancer cell apoptosis.
Literature	GENERIF_SUMMARY	Data support a model by which AKT is a major switch involved in regulating hypoxia-induced cell death.
Literature	GENERIF_SUMMARY	AR mediates EGF- and bFGF-induced colon cancer cell proliferation by activating or expressing G(1)-S phase proteins such as E2F-1, cdks, and cyclins through the reactive

Main Category	Category	Terms
		oxygen species/phosphoinositide 3-kinase/AKT pathway.
Literature	GENERIF_SUMMARY	IRSp53 and spinophilin regulate localized Rac activation by T-lymphocyte invasion and metastasis protein 1
Literature	GENERIF_SUMMARY	PDK1 with AKT1 is sufficient for pathway activation independent of membrane localization and phosphatidylinositol 3 kinase
Literature	GENERIF_SUMMARY	Co-expression of Pc2 and Akt1 results in both phosphorylation and ubiquitylation of CtBP1, thereby targeting CtBP1 for degradation.
Literature	GENERIF_SUMMARY	Syrbactin class proteasome inhibitor-induced apoptosis and autophagy occurs in association with p53 accumulation and Akt/PKB activation in neuroblastoma.
Literature	GENERIF_SUMMARY	Impaired AKT-related intracellular signaling is partly related to the developed illness and cannot fully be explained by the genetic background of schizophrenia.
Literature	GENERIF_SUMMARY	Hepatocyte growth factor protects hepatoblastoma cells from chemotherapy-induced apoptosis by AKT activation
Literature	GENERIF_SUMMARY	NF-kappaB and Akt inhibition by an antibody-avidin fusion protein sensitizes malignant B-cells to cisplatin-induced apoptosis
Literature	GENERIF_SUMMARY	CNK1 promotes oncogenic signalling through Akt in breast cancer cell lines and tumours.
Literature	GENERIF_SUMMARY	HtrA1 has a role in promoting anoikis by attenuating activation of EGFR/AKT pathway that may contribute to its metastasis suppression capacity
Literature	GENERIF_SUMMARY	In HER2-positive breast cancer, basal co-expression of PTEN and p-Akt might identify those patients who are more likely to benefit from trastuzumab-based therapies.
Literature	GENERIF_SUMMARY	Data describe the role of the IGF-IR in bladder cancer and show that it required the activation of the Akt and MAPK pathways as well as IGF-I-induced Akt- and MAPK-dependent phosphorylation of paxillin.
Literature	GENERIF_SUMMARY	Studies indicate that three different isoforms Akt1, Akt2, and Akt3 have distinct expression patterns and functions.
Literature	GENERIF_SUMMARY	PKB phosphorylates one of the anti-apoptotic proteins--transcription factor

Main Category	Category	Terms
		Twist-1 at Ser42.
Literature	GENERIF_SUMMARY	Ovarian cancer ascites induces FAK and Akt activation in an alphavbeta5 integrin-dependent pathway, which confers protection from TRAIL-induced cell death and caspase activation.
Literature	GENERIF_SUMMARY	The E17K substitution in AKT1 is rare in prostate cancer.
Literature	GENERIF_SUMMARY	pAkt may have a role in progression of glioblastoma
Literature	GENERIF_SUMMARY	FoxOs inhibit mTORC1 and activate Akt by inducing the expression of Sestrin3 and Rictor
Literature	GENERIF_SUMMARY	Single-nucleotide polymorphisms, including Asp54Gly, of the ADCYAP1 gene are unlikely to play a sizeable role in the genetic susceptibility to schizophrenia.
Literature	GENERIF_SUMMARY	Inhibition of Akt may have beneficial therapeutic effects in malignant breast cancers.
Literature	GENERIF_SUMMARY	inactivation of functional PTEN and the consequent activation of the Akt pathway prevents TRAIL-induced apoptosis, leading to TRAIL resistance
Literature	GENERIF_SUMMARY	The present results provide further evidence to support the AKT1 association with schizophrenia.
Literature	GENERIF_SUMMARY	Recent advances in understanding the physiological functions of Ack1 signaling in normal cells and the consequences of its hyperactivation in various cancers, are discussed.
Literature	GENERIF_SUMMARY	Angiotensin II increased Akt phosphorylation and PI3K activity in cultured hypertrophic scar fibroblasts in a dose- and time-dependent manner.
Literature	GENERIF_SUMMARY	Overexpression of ss-arrestin 2 decreased the percentage of apoptosis and caspase-3 activation and attenuated resveratrol-reduced levels of p-Akt and p-GSK3ss.
Literature	GENERIF_SUMMARY	These findings suggest that angiotensin II plays a role in the growth of angiotensin II type I receptor-positive breast cancer cells through PI3-kinase/Akt pathway activation.
Literature	GENERIF_SUMMARY	The better prognosis of patients with MS neuroblastomas compared with classical neuroblastomas was secondary to spontaneous tumour regression mediated by higher levels of apoptosis associated

Main Category	Category	Terms
		with low activation of AKT.
Literature	GENERIF_SUMMARY	Akt1 phosphorylates palladin, inducing cytoskeletal reorganization and inhibiting the migration of breast cancer cells.
Literature	GENERIF_SUMMARY	Endogenous MADD is phosphorylated at three highly conserved sites by Akt, and only the phosphorylated MADD can directly interact with the TRAIL receptor DR4 thereby preventing Fas-associated death domain recruitment.
Literature	GENERIF_SUMMARY	PI3K-PKB-mTOR signaling plays a crucial role in the development of functional CD34-derived myeloid Dendritic cells
Literature	GENERIF_SUMMARY	significantly decreased cell viability and increased apoptosis in human neuroblastoma cell line in association with down-regulation of p-ERK1/2, p-Akt, p-STAT3 survival pathways.
Literature	GENERIF_SUMMARY	These results suggest that PI3K/Akt inhibits endoplasmic reticulum stress-induced apoptosis in HEK293 cells, at least in part, by promoting GRP78 protein stability.
Literature	GENERIF_SUMMARY	Direct sequencing analysis of endometrial DNA analysis did not reveal any activating somatic mutations in either PI3KCA or AKT1 gene, indicating that these mutations are rarely associated with endometriosis in South Indian women.
Literature	GENERIF_SUMMARY	these studies suggest that FUT4 regulates A431 cell growth through controlling cell cycle progression via MAPK and PI3K/Akt signaling pathways.
Literature	GENERIF_SUMMARY	Data demonstrate that PI3K/Akt pathway regulates the expression of DNMT3B at transcriptional and post-transcriptional levels, which is particularly important to understand the effects of PI3K/Akt and DNMT3B on hepatocarcinogenesis.
Literature	GENERIF_SUMMARY	The inhibitory effect of the GHRH antagonist JMR-132 on proliferation is due, in part, to an interference with the EGFR-Akt pathway in ovarian cancer cells.
Literature	GENERIF_SUMMARY	Data suggest that the induction of apoptosis by tanshinone I in leukemia cells is mainly related to the disruption of Deltapsim, inactivation of PI3k/Akt signaling pathway, upregulation of Bax expression, and the activation of caspase-3.
Literature	GENERIF_SUMMARY	Glutamine, leucine and proline increase Akt phosphorylation in HepG2 cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Decreased expression of PTEN together with overexpression of PI3K and AKT contribute to gastric cancer progression.
Literature	GENERIF_SUMMARY	NFKB/p65 is a downstream target of Akt.
Literature	GENERIF_SUMMARY	E prostanoid (EP) 2 and EP3 have roles in cAMP/protein kinase A- and PI3-K/Akt-dependent NF-kappaB activation during shear-induced interleukin-6 synthesis in chondrocytes
Literature	GENERIF_SUMMARY	Genetic and functional interactions between COMT and AKT1 may provide novel insights into pathogenesis of schizophrenia and other ErbB-associated human diseases such as cancer.
Literature	GENERIF_SUMMARY	miR-22 acts to fine-tune the dynamics of PTEN/AKT/FoxO1 pathway.
Literature	GENERIF_SUMMARY	Role of germ line genetic variations in the PI3K-AKT-mTOR pathway genes as predictors of bladder cancer metastasis and invasiveness.
Literature	GENERIF_SUMMARY	Nifedipine inhibits vascular smooth muscle dedifferentiation by suppressing Akt signaling, thereby preventing neointimal thickening.
Literature	GENERIF_SUMMARY	AKT pathways are involved in the tumorigenesis of CD133 positive colon cancer cells.
Literature	GENERIF_SUMMARY	IQGAP1 is a scaffold that facilitates mTOR and Akt interaction.
Literature	GENERIF_SUMMARY	PTEN/Akt and MAP kinase signaling pathways are regulated by the ubiquitin ligase activators Ndfip1 and Ndfip2
Literature	GENERIF_SUMMARY	These findings cannot exclude the hypothesis of involvement of Akt and MAPK signalling pathways in pathogenesis of idiopathic pulmonary fibrosis.
Literature	GENERIF_SUMMARY	TNFalpha promotes the interaction of N-terminally truncated RXRalpha with the p85alpha subunit of PI3K, activating PI3K/AKT signaling in cancer cell lines and primary tumors.
Literature	GENERIF_SUMMARY	Impaired cell cycle regulation by AKT contributes to the distinct growth phenotype of preadipocytes in visceral fat of obese subjects.
Literature	GENERIF_SUMMARY	Activation of the PI3K/AKT signalling pathway in non-melanoma skin cancer is not mediated by oncogenic PIK3CA and AKT1 hotspot mutations.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	PDCD4 regulates TRAIL sensitivity in gastric cancer cells by inhibiting the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	Akt has a pivotal role in EGFR-mediated cell migration
Literature	GENERIF_SUMMARY	PKB activity was found to be required for induction of adhesion to bone marrow-derived stromal cells and detrimental for migration of umbilical cord blood-derived CD34(+) hematopoietic progenitors.
Literature	GENERIF_SUMMARY	Aberrantly expressed UCH-L1 boosts signaling through the Akt pathway by downregulating the antagonistic phosphatase PHLPP1, an event that requires its de-ubiquitinase activity
Literature	GENERIF_SUMMARY	The activated PI3K/Akt/COX-2 signal transduction pathway was found to be the main cause for decline in radiosensitivity in HeLa cells.
Literature	GENERIF_SUMMARY	These results demonstrate that down-regulation of PI3K/Akt pathway with specific cleavage of HSP90 to 45 KDa modulates the pro-apoptotic effects of deoxycholate in gastric cells.
Literature	GENERIF_SUMMARY	analysis of a novel Gi, P2Y-independent signaling pathway mediating Akt phosphorylation in response to thrombin receptors
Literature	GENERIF_SUMMARY	M2PK has a role in preventing, and pAkt has a role in enhancing progression of breast cancer
Literature	GENERIF_SUMMARY	P21-activated kinase 1 stimulates colon cancer cell growth and migration/invasion via ERK- and AKT-dependent pathways.
Literature	GENERIF_SUMMARY	Phosphorylation of deleted in liver cancer 1 (DLC1) and DLC2 by Akt at the conserved residue points to a common regulatory mechanism of the DLC tumor suppressor family.
Literature	GENERIF_SUMMARY	mechanisms behind the P2X7-dependent depletion of nuclear phosphorylated Akt
Literature	GENERIF_SUMMARY	Akt activation is a common event in pediatric malignant gliomas.
Literature	GENERIF_SUMMARY	Collectively, these data suggested that, in oral epithelial cells, the herpes simplex virus 1-induced PI3K/Akt activation was involved in the regulation of apoptosis blockage and viral gene expression.
Literature	GENERIF_SUMMARY	Studies indicate that targeting the

Main Category	Category	Terms
		deregulated PI3K/PTEN-Akt signaling axis has currently emerged as one of the major tenets in anticancer drug development.
Literature	GENERIF_SUMMARY	Our findings not only provided the first evidence that Akt1 is a direct target of miRNA but also demonstrated that miR-149* is a pro-apoptotic miRNA by repressing the expression of Akt1 and E2F1.
Literature	GENERIF_SUMMARY	Findings reveal a novel mechanism for recruitment of PDK1 to the SHPS-1 signaling complex, which is required for IGF-I-stimulated AKT Thr(308) phosphorylation and inhibition of apoptosis.
Literature	GENERIF_SUMMARY	Down-regulation of the brain-derived neurotrophic factor (BDNF)-Akt-Bcl2 antiapoptotic signaling pathway in the autistic brain could be one of the underlying mechanisms responsible for the pathogenesis of autism.
Literature	GENERIF_SUMMARY	Nuclear p-AKT expression together with stage and grade constitute independent prognostic factors for tumour progression and cancer-specific survival.
Literature	GENERIF_SUMMARY	Data indicate that stathmin is a target of the PI3K/Akt pathway in cervical cancer cells.
Literature	GENERIF_SUMMARY	FOXO3a is in a constant inactive state due to its cytoplasmic localization, and that neither PI3K/Akt nor extra-signal-regulated kinase-specific inhibition resulted in its nuclear translocation.
Literature	GENERIF_SUMMARY	demonstrate that alphaVbeta3 ligation results in sustained increases of PI3 kinase/Akt-dependent binding of transcription factor NF-kappaB to DNA.
Literature	GENERIF_SUMMARY	ADAM17-mediated activation of Lyn/Akt maintains the viability of U937 cells and that suppression of the pathway is responsible for PILP-1-induced apoptosis.
Literature	GENERIF_SUMMARY	AKT activation controls cell survival to ionizing radiation by phosphorylating CLK2, revealing an important regulatory mechanism required for promoting cell survival.
Literature	GENERIF_SUMMARY	SGK3 is not required for insulin-induced PFK-2 activation and that this effect is likely mediated by PKBalpha.
Literature	GENERIF_SUMMARY	Overexpression of Akt1 inhibits cell proliferation and motility in MDA-MB-231 cells through isoform-specific mechanisms.
Literature	GENERIF_SUMMARY	Data show that formylpeptides induce

Main Category	Category	Terms
		sequential activation of AKT, ERK1/2, and PLD, which represents a novel signaling pathway.
Literature	GENERIF_SUMMARY	Phosphorylated Akt1 (protein kinase B) expression is associated with a favourable prognosis in pancreatic cancer
Literature	GENERIF_SUMMARY	Data show that DLBS1425 significantly reduced phosphoinositide-3 (PI3)-kinase/ protein kinase B (AKT) signalling by reducing PI3K transcript level and subsequent reduction in AKT phosphorylation.
Literature	GENERIF_SUMMARY	Celastrol inhibits the proliferation and induced the apoptosis of HL-60 cells which may be related with down-regulation of p-Act and cyclin D1 expressions.
Literature	GENERIF_SUMMARY	Data show that blockade of amphiregulin suppressed the activities of EGFR, HER3, and Akt and the expression of amphiregulin itself.
Literature	GENERIF_SUMMARY	Data indicate that high ErbB2 expression enhances cholangiocarcinoma invasion, motility and proliferation via the AKT/ p70S6K pathway.
Literature	GENERIF_SUMMARY	Vascular endothelial growth factor receptor 1 contributes to Escherichia coli K1 invasion of human brain microvascular endothelial cells through the phosphatidylinositol 3-kinase/Akt signaling pathway.
Literature	GENERIF_SUMMARY	lymph node metastasis and poor survival in invasive ductal breast carcinoma are linked to an uncoupling of the Akt-FOXO3a signaling axis
Literature	GENERIF_SUMMARY	Rhabdomyosarcoma pathogenesis involves increased IGF1R expression that enhances AKT and Bcl-x(L)-mediated cell survival.
Literature	GENERIF_SUMMARY	The levels of total Akt and phosphorylated Akt (active Akt) are increased in E6-AP overexpressing prostate gland and LNCaP cells suggesting that E6-AP regulates the PI3K-Akt signaling pathway.
Literature	GENERIF_SUMMARY	Rac activation by dedicator of cytokinesis (DOCK)180 regulates cardiovascular development by acting downstream of chemokine receptor CXCR4.
Literature	GENERIF_SUMMARY	Data show that exposure of postmitotic neurons to caffeine leads to changes in tau phosphorylation concomitantly with downregulation of Akt signaling.
Literature	GENERIF_SUMMARY	the present study shows total p-AKT

Main Category	Category	Terms
		overexpression to be more frequent in metastases than in primary renal cell carcinomas
Literature	GENERIF_SUMMARY	COX-2 is constitutively expressed in osteoblasts in the dynamic bone growth area, which facilitates osteoblast proliferation via PTEN/Akt/p27(Kip1) signaling.
Literature	GENERIF_SUMMARY	AKT1 activation induces soft-tissue sarcoma cell motility and invasiveness at least partially through a novel interaction with the intermediate filament vimentin
Literature	GENERIF_SUMMARY	C. jejuni activates a PI3K/Akt-dependent anti-inflammatory pathway in human intestinal epithelial cells.
Literature	GENERIF_SUMMARY	PTEN selectively inhibits expression of VCAM-1 but not ICAM-1 through modulation of PI3K/Akt/GSK-3beta/GATA-6 signaling cascade in TNF-alpha-treated endothelial cells
Literature	GENERIF_SUMMARY	Functional characterization of a haplotype in the AKT1 gene associated with glucose homeostasis and metabolic syndrome.(
Literature	GENERIF_SUMMARY	The interactions between the pleckstrin-homology and kinase domains mediate binding of the inhibitor to AKT1.
Literature	GENERIF_SUMMARY	TSH suppression does not affect Akt phosphorylation in benign human thyroid nodules.
Literature	GENERIF_SUMMARY	We observed epistasis between NRG1 (rs10503929; Thr286/289/294Met) and its receptor ERBB4 (rs1026882; likelihood ratio test P = .035); a 3-way interaction with these 2 SNPs and AKT1 (rs2494734) was also observed.
Literature	GENERIF_SUMMARY	This perspective highlights recent advances using in vivo studies that reveal how AKT signaling supports tumor formation.
Literature	GENERIF_SUMMARY	annexin 2 regulates endothelial morphogenesis through an adherens junction-mediated pathway upstream of Akt.
Literature	GENERIF_SUMMARY	we confirmed the important roles of Akt and Mre11 during the process of DNA double-strand break repair in cancer cells treated with ionizing radiation
Literature	GENERIF_SUMMARY	Akt signaling pathway activity increased diffusively in skeletal muscle of myotonic dystrophy patients.
Literature	GENERIF_SUMMARY	expression of a pseudophosphorylated NEP

Main Category	Category	Terms
		variant (Ser6Asp) abrogates the inhibitory effect of NEP on insulin/insulin-like growth factor-1 (IGF-1) stimulated activation of Akt
Literature	GENERIF_SUMMARY	cIAP2 upregulated by E6 via EGFR/PI3K/AKT cascades may contribute to cisplatin resistance, revealing that the EGFR or PI3K inhibitor combined with cisplatin may improve the chemotherapeutic efficacy in HPV-infected lung cancer.
Literature	GENERIF_SUMMARY	Alkylation of PTEN by alpha,beta-enals/enones and interference with its restraint of cellular PKB/Akt signaling may accentuate hyperplastic and neoplastic disorders associated with chronic inflammation, oxidative stress, or aging.
Literature	GENERIF_SUMMARY	DynAP promotes activation of Akt in cancer cells.
Literature	GENERIF_SUMMARY	VSV M protein inactivates the cellular Akt signaling pathway.
Literature	GENERIF_SUMMARY	the transforming AKT1 E17K mutation may be present at a lower frequency than initially suggested, even in NSCLC with squamous cell carcinoma histology
Literature	GENERIF_SUMMARY	the PI3K/Akt signaling pathway might originate in part with Tat-induced activation of NADPH oxidase and long terminal repeat transactivation
Literature	GENERIF_SUMMARY	Data suggest that the PI3K/Akt/mTOR pathway may contribute to maintenance of quiescence of cancer stem cells.
Literature	GENERIF_SUMMARY	Constitutive activation of EGFR/Akt/mTOR pathway was present in defined subset of nonsmall cell lung carcinomas.
Literature	GENERIF_SUMMARY	There is a mutual suppressive interaction between AMPK and Akt.
Literature	GENERIF_SUMMARY	the PI3K/AKT signaling pathway is aberrantly active and plays a critical role for the cell cycle progression in human pleural mesothelioma cells.
Literature	GENERIF_SUMMARY	Data suggest that DLX5 plays a significant role in the pathogenesis of some ovarian cancers by enhancing IRS-2-AKT signaling.
Literature	GENERIF_SUMMARY	Findings show that oncogenic factors activating the phosphoinositide 3-kinase/Akt pathway can regulate alternative splicing of Casp9 via a coordinated mechanism involving the phosphorylation of SRp30a.
Literature	GENERIF_SUMMARY	these results implicate the Akt1 isoform in

Main Category	Category	Terms
		regulating hippocampal neuroplasticity and cognition and in contributing to the etiology of schizophrenia
Literature	GENERIF_SUMMARY	Plexin-B1 expression correlates with malignant phenotypes of serous ovarian tumors, probably via phosphorylation of AKT at Ser473
Literature	GENERIF_SUMMARY	Study shows that the H1 12 kb AKT1 haplotype is significantly associated with higher fasting serum glucose levels, and an increased risk for developing metabolic syndrome.
Literature	GENERIF_SUMMARY	Akt and autophagy cooperate to promote survival of drug-resistant glioma.
Literature	GENERIF_SUMMARY	Dopamine acting through its D(2) receptor, inhibits IGF-I-induced proliferation of gastric adenocarcinoma cells by up-regulating KLF4, a negative regulator of the cell cycle through down regulation of IGF-IR and AKT phosphorylation.
Literature	GENERIF_SUMMARY	COX-2 activation is associated with Akt phosphorylation and poor survival in ER-negative, HER2-positive breast cancer
Literature	GENERIF_SUMMARY	Omentin inhibits osteoblast differentiation of calcifying vascular smooth muscle cells through the PI3K/Akt pathway
Literature	GENERIF_SUMMARY	Calcium-induced apoptosis in bladder cancer cells is delayed by HER1 receptor activation involving the Akt and PLCgamma signalling pathways.
Literature	GENERIF_SUMMARY	Expressions of Akt1, Akt2 and PI3K were decreased in ovarian epithelial cancer cells transduced with PTEN.
Literature	GENERIF_SUMMARY	Common genetic variation in PTEN, PIK3CA, AKT1, MLH1, or MSH2 was not statistically significantly associated with endometrial cancer.
Literature	GENERIF_SUMMARY	Akt play an important role in cross-linking IL-4 priming, SCF signaling, and IgE-dependent activation of mature human mast cells.
Literature	GENERIF_SUMMARY	Recent development of anticancer therapeutics targeting Akt.
Literature	GENERIF_SUMMARY	Activation of the PI3K/AKT in addition to MAPK/ERK signaling pathways may underlie biological aggressiveness in pilocytic astrocytoma (PA). Specifically, it may mediate the increased proliferative activity observed in histologically anaplastic PA.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Up-regulation of p22(phox) represents a novel pro-survival mechanism through which growth factors and Akt inhibit apoptosis in PaCa cells.
Literature	GENERIF_SUMMARY	Confocal microscopy analysis revealed that menin regulates AKT1 in part by reducing the translocation of AKT1 from the cytoplasm to the plasma membrane.
Literature	GENERIF_SUMMARY	INPP4B functions as a tumor suppressor by negatively regulating normal and malignant mammary epithelial cell proliferation through regulation of the PI3K/Akt signaling pathway
Literature	GENERIF_SUMMARY	Lower expression of PTEN protein and higher expression of P-AKT protein may play an important role in leukemia pathogenesis.
Literature	GENERIF_SUMMARY	Amino acids activate mammalian target of rapamycin complex 2 (mTORC2) via PI3K/ Akt signaling.
Literature	GENERIF_SUMMARY	Findings support a role for the AKT-dependent regulation of AIP4/Itch activity in mediating the differential cyclin D1 and c-MYC transcriptional responses to rapamycin.
Literature	GENERIF_SUMMARY	Mcl-1 and pAKT scores were directly correlated during various stages of colon carcinogenesis.
Literature	GENERIF_SUMMARY	Akt requires glucose metabolism to suppress puma expression and prevent apoptosis of leukemic T cells.
Literature	GENERIF_SUMMARY	Data show that the combined use of resveratrol and rapamycin resulted in modest additive inhibitory effects on the growth of breast cancer cells, mainly through suppressing rapamycin-induced AKT activation.
Literature	GENERIF_SUMMARY	Among many putative PTEN- and PIP(3)-dependent events, phosphorylation of PKB substrates is the key downstream regulator of cell polarity
Literature	GENERIF_SUMMARY	Phosphorylation of Akt at Ser-473 was enhanced by overexpression of PINK1, and the Akt activation was crucial for protection of SH-SY5Y cells from various cytotoxic agents, including oxidative stress
Literature	GENERIF_SUMMARY	Heat shock protein 90 maintains the tumour-like character of rheumatoid synovial cells by stabilizing integrin-linked kinase, extracellular signal-regulated kinase and protein kinase B.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results suggest that low magnitude stretch can induce cPLA(2) phosphorylation through the MEK/ERK and PI3K-Akt pathways, independently.
Literature	GENERIF_SUMMARY	Activity of the Akt pathway is greatly decreased following downregulation of urokinase plasminogen activator receptor, leading to induction of papillary thyroid carcinoma cell senescence.
Literature	GENERIF_SUMMARY	A novel binding interaction for Akt1 and focal adhesion kinase (FAK) makes them therapeutic targets for prevention of cancer cell adhesion and, eventually, cancer metastasis.
Literature	GENERIF_SUMMARY	the Ang1/Tie2 signal potentiates basal Notch signal controlling vascular quiescence by up-regulating Dll4 through AKT-mediated activation of beta-catenin.
Literature	GENERIF_SUMMARY	RACK1 regulates VEGF/Flt1-mediated cell migration via activation of a PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Here, the authors show that VP11/12 is required for virus-induced activation of PI3K-Akt signaling in herpes simplex virus 1-infected Jurkat T cells and primary fibroblasts.
Literature	GENERIF_SUMMARY	Studies indicate that SKP2 works in concert with Akt to induce FOXO1 ubiquitination and proteasome degradation.
Literature	GENERIF_SUMMARY	Results indicate that activation of Akt is involved in Brca1-deficiency mediated tumorigenesis and that the mTOR pathway can be used as a novel target for treatment of Brca1-deficient cancers.
Literature	GENERIF_SUMMARY	PPAR-gamma ligands inhibit human myofibroblast differentiation of normal and idiopathic pulmonary fibrotic (IPF) fibroblasts, by blocking Akt phosphorylation at Ser473 by a PPAR-gamma-independent mechanism.
Literature	GENERIF_SUMMARY	we detected a significant correlation between p28(GANK) expression and p-AKT levels in a cohort of hepatocellular carcinoma biopsies, and the combination of these two parameters is a more powerful predictor of poor prognosis.
Literature	GENERIF_SUMMARY	Data show that the effects of PI3K-Akt signalling on IL-10 responses were mediated at least in part by GSK3.
Literature	GENERIF_SUMMARY	AKT is critical for transmitting the double strand break signal to ERK.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Autophagy and Akt promote survival in glioma.
Literature	GENERIF_SUMMARY	Akt-dependent mechanisms of interference with motility/invasion of human glioblastoma cells
Literature	GENERIF_SUMMARY	Data show that activation of beta-catenin and Akt pathways are required for the sustention of EMT-associated stem cell-like traits.
Literature	GENERIF_SUMMARY	studies showed that phosphatidylinositol-3 kinase (PI3K)/Akt/nuclear factor-kappaB signalling pathway was involved in chronic myeloid leukaemia pathogenesis
Literature	GENERIF_SUMMARY	For long-lasting T cell immunity, after ligation by OX40 ligand, OX40 assembles a signaling complex containing the adapter tumor necrosis factor receptor-associated factor 2 (TRAF2) as well as PKB and its upstream activator, phosphoinositide 3-kinase.
Literature	GENERIF_SUMMARY	Dependence of Evi1-expressing leukemic cells on AKT/mTOR signaling provides the first example of targeted therapeutic modalities that suppress the leukemogenic activity of Evi1.
Literature	GENERIF_SUMMARY	PI3K/AKT/mTOR signalling pathway activation is a common event in high-risk neuroblastoma.
Literature	GENERIF_SUMMARY	Leucine deprivation was associated with Akt phosphorylation in breast cancer cell lines.
Literature	GENERIF_SUMMARY	The interaction between ERalpha and Akt plays a crucial role in the regulation of proliferative activity in postmenopausal endometrium.
Literature	GENERIF_SUMMARY	the loss of E-cadherin itself may contribute to dysregulated PI3K/Akt signaling through its effects on PTEN.
Literature	GENERIF_SUMMARY	Results demonstrate that PP1-mediated inhibition of the key anti-apoptotic protein, Akt, plays an important role in SPH-mediated apoptosis in Jurkat cells.
Literature	GENERIF_SUMMARY	The phosphatidylinositol 3-kinase/Akt signaling pathway is activated by VEGF and plays a critical role in the angiogenesis of chronic subdural hematoma.
Literature	GENERIF_SUMMARY	Studies indicate that CNK1-driven proliferation relies on Akt-dependent phosphorylation and inactivation of FoxO proteins.

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Literature	GENERIF_SUMMARY	mTOR complex 2 targets Akt for proteasomal degradation via phosphorylation at the hydrophobic motif
Literature	GENERIF_SUMMARY	The regulation of AKT1 would represent a fine transitory system for controlling homologous recombination and maintaining genomic integrity.
Literature	GENERIF_SUMMARY	These findings indicated that up-regulation of pAkt contributed to elevated AT1R expression in castration resistant prostate cancer.
Literature	GENERIF_SUMMARY	Loss of Desmocollin2 promotes cell proliferation and enables tumor growth in vivo through the activation of Akt/beta-catenin signaling.
Literature	GENERIF_SUMMARY	Data suggest that irradiation causes RFP2 overexpression, which enhances ionizing radiation-induced apoptosis by increasing p53 stability and decreasing AKT kinase activity through MDM2 and AKT degradation.
Literature	GENERIF_SUMMARY	the basis for enhanced activity of mTOR inhibitors in non-Hodgkin's lymphomas may be the lack of compensatory activation of AKT and eIF4E phosphorylation in lymphoma cells
Literature	GENERIF_SUMMARY	Selective activation of Akt1 through mTORC2 and P-Rex1 regulates cancer cell migration, invasion and metastasis.
Literature	GENERIF_SUMMARY	in response to cellular stress, GSK-3beta restrains mTORC2-Akt signaling by specifically phosphorylating rictor, thereby balancing the activities of GSK-3beta and Akt, two opposing players in glucose metabolism
Literature	GENERIF_SUMMARY	PI3K-activating hormones inhibit ROMK by enhancing its endocytosis via a mechanism that involves phosphorylation of WNK1 by Akt1 and SGK1.
Literature	GENERIF_SUMMARY	Expression of AKT was associated with lymph node metastasis of gastric carcinoma. Overall 5-year survival rate was significantly higher in patients with AKT expression.
Literature	GENERIF_SUMMARY	Ghrelin activated and phosphorylate extracellular signal-regulated protein kinase (ERK)1/2 and Akt.
Literature	GENERIF_SUMMARY	Data indicate that Akt inhibitor may potentiate the apoptotic effect of trichostatin A on ovarian carcinoma cell lines by increasing the activation of the

Main Category	Category	Terms
		caspase-8-dependent pathway.
Literature	GENERIF_SUMMARY	Data show that PrLZ elevated the phosphorylation of Akt and Stat3 and upregulated Bcl-2 expression.
Literature	GENERIF_SUMMARY	Knockdown of SIRT1 abolished GR-induced p16 repression as well as Akt/p70S6K1 activation implying that SIRT1 may affect p16 repression through direct deacetylation effects and indirect regulation of Akt/p70S6K1 signaling.
Literature	GENERIF_SUMMARY	Wnt/beta-catenin signaling regulates MAPK and Akt1 expression and growth of hepatocellular carcinoma cells.
Literature	GENERIF_SUMMARY	Co-occurrence of MYC amplification and PI3K-pathway alteration was observed, raising the possibility that these two lesions cooperate in prostate cancer progression.
Literature	GENERIF_SUMMARY	Studies indicate that Akt phosphorylates acetylated-FoxO and then phosphorylated FoxO interacts with 14-3-3 proteins in the nucleus, which in turn results in cytoplasmic retention of FoxO.
Literature	GENERIF_SUMMARY	Temozolomide induced c-Myc-mediated apoptosis via Akt signalling in MGMT expressing glioblastoma cells
Literature	GENERIF_SUMMARY	Results show the mechanism by which EZH2 regulates BRCA1 expression and genomic stability mediated by the PI3K/Akt-1 pathway.
Literature	GENERIF_SUMMARY	148 Penile squamous cell carcinoma were microarrayed and immunostained for phosphorylated EGFR (pEGFR), HER2, HER3, HER4, phosphorylated Akt (pAkt), Akt1 and PTEN proteins.
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in AKT gene is not associated with colorectal cancer.
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in AKT1 is associated with prostate cancer.
Literature	GENERIF_SUMMARY	analysis of the roles of Akt1 and Akt2 in cancer [review]
Literature	GENERIF_SUMMARY	these results indicate that pharbin is an important negative regulator of IGF-1/PI3K/Akt signalling and protein synthesis.
Literature	GENERIF_SUMMARY	Regrowth of axons within the adult nigrostriatal projection, when prominently affected in Parkinson's disease, can be induced by activation of Akt transgene signaling in surviving endogenous

Main Category	Category	Terms
		dopaminergic neurons by viral vector transduction.
Literature	GENERIF_SUMMARY	Studies indicate that FoxOs have emerged as an important effector arm of PI3K/Akt signaling by driving multiple apoptotic gene expression.
Literature	GENERIF_SUMMARY	Studies indicate that the inhibition of mTORC1 by FOXO could lead to the activation of Akt via the inhibition of the negative feedback loop driven by mTORC1 and S6K1.
Literature	GENERIF_SUMMARY	Data suggest that the PI3K/Akt signal transduction pathway is involved in the regulation and release of pro-inflammatory cytokines such as TNF-alpha and plays an important role in the development and progression of ulcerative colitis.
Literature	GENERIF_SUMMARY	PRMT1-mediated methylation of BAD inhibits phosphorylation by Akt.
Literature	GENERIF_SUMMARY	An in-situ proximity ligation assay revealed a physical interaction between GRP78 and AKT at the plasma membrane of cells following induction of ER stress
Literature	GENERIF_SUMMARY	CCL5 signaling induces GLI2 through a PI3K-AKT-IkappaBalpha-p65 pathway and requires GLI2 transcriptional activity to modulate IL-6 expression and Ig secretion in vitro and in vivo
Literature	GENERIF_SUMMARY	the phosphorylation status of hnRNP A1 serine 199 regulates the AKT-dependent sensitivity of cells to rapamycin and functionally links IRES-transacting factor annealing activity to cellular responses to mTOR complex 1 inhibition.
Literature	GENERIF_SUMMARY	Mislocalization of the E3 ligase, beta-transducin repeat-containing protein 1 (beta-TrCP1), in the pleckstrin homology domain leucine-rich repeat protein phosphatase 1 (PHLPP1) and Akt.
Literature	GENERIF_SUMMARY	Age-dependent accumulation of soluble amyloid beta oligomers reverses the neuroprotective effect of soluble amyloid precursor protein-alpha by modulating phosphatidylinositol 3-kinase/Akt-GSK-3beta pathway in Alzheimer mouse model.
Literature	GENERIF_SUMMARY	IkappaB kinase epsilon and TANK-binding kinase 1 activate AKT by direct phosphorylation.
Literature	GENERIF_SUMMARY	Akt ubiquitination could be chemically exploited to effectively facilitate its

Main Category	Category	Terms
		deactivation, thus identifying an avenue for pharmacological intervention in Akt signaling.
Literature	GENERIF_SUMMARY	findings suggest that relative to the normal myometrium, there is indeed dysregulation of the P13K-Akt-mTOR pathway in uterine smooth muscle tumors
Literature	GENERIF_SUMMARY	The knockdown of STEAP4 inhibits insulin-stimulated glucose transport and GLUT4 translocation via the attenuated phosphorylation of Akt, independent of the effects of EEA1.
Literature	GENERIF_SUMMARY	capsaicin-mediated IL-6 increase in prostate cancer PC-3 cells is regulated at least in part by TNF-alpha secretion and signaling pathway involving Akt, ERK and PKC-alpha activat
Literature	GENERIF_SUMMARY	Parkinson's disease, induced by alpha-synuclein mutation, is evoked by deregulation of the AKT-signaling cascade.
Literature	GENERIF_SUMMARY	Myotubularin regulates Akt-dependent survival signaling via phosphatidylinositol 3-phosphate.
Literature	GENERIF_SUMMARY	Data show that the Ras/PI3K/Akt pathway can account for Fyn over-expression in cancers.
Literature	GENERIF_SUMMARY	A detachment-induced and PI3K/Akt-mediated resistance to apoptotic effects of TRAIL, is demonstrated.
Literature	GENERIF_SUMMARY	Nuclear translocation of epidermal growth factor receptor by Akt-dependent phosphorylation enhances breast cancer-resistant protein expression in gefitinib-resistant cells.
Literature	GENERIF_SUMMARY	The raft-associated but not total membrane Akt determines its cellular functions.
Literature	GENERIF_SUMMARY	Data show that hepatocyte growth factor inhibits anoikis of pancreatic carcinoma cells through phosphatidylinositol 3-kinase pathway in which activation of Akt may be involved.
Literature	GENERIF_SUMMARY	Moreover, sustained PI3K/Akt signalling in Aspergillus fumigatus infected cells is an important determinant to resist apoptosis.
Literature	GENERIF_SUMMARY	loss of Akt1 or Akt2 decreased proliferation of Pten wild-type astrocytes, whereas combined loss of multiple isoforms was needed to inhibit proliferation of Pten-null astrocytes

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results suggest that prosurvival PTEN-Akt signalling is impaired in the diseased diabetic myocardium.
Literature	GENERIF_SUMMARY	knockdown of mTOR did not alter or only slightly reduced Akt (ser473) phosphorylation in phosphatase and tensin homolog deleted from chromosome 10 (PTEN)-null (LNCaP and MDA-MB-468 cells) and PTEN-positive (DU 145 and MDA-MB-231) cells
Literature	GENERIF_SUMMARY	elevated Id1 expression was observed in lung cancer cell lines as well as lung cancer tissues. Id1 overexpression increased cell proliferation while Id1 knockdown decreased cell proliferation, mostly through Akt-related pathway
Literature	GENERIF_SUMMARY	Studies indicate that in addition to the direct control of FOXO by PKB, a number of proteins are controlled separately by PKB and FOXO giving rise to antagonistic control.
Literature	GENERIF_SUMMARY	Findings show that the oncogenic PI3K/Akt signalling pathway participates in maintaining cells in a type II phenotype by excluding CD95 from lipid rafts.
Literature	GENERIF_SUMMARY	An Akt-dependent increase in canonical Wnt signaling and a decrease in sclerostin protein levels are involved in strontium ranelate-induced osteogenic effects in human osteoblasts.
Literature	GENERIF_SUMMARY	Glucose deprivation resulted in activation of AMPK and inhibition of Akt phosphorylation
Literature	GENERIF_SUMMARY	Transcriptional activation of the Akt1/PKB pathway indicates that it is involved in lumbar disc degeneration.
Literature	GENERIF_SUMMARY	COX2-dependent and independent activation of CK2alpha-Akt/uPA signal is mainly involved in urothelial carcinoma cell survival
Literature	GENERIF_SUMMARY	mTORC2 is the kinase that phosphorylates Akt Ser(473) in human platelets but that this phosphorylation is dispensable for Thr(308) phosphorylation and Akt1 activity.
Literature	GENERIF_SUMMARY	Acute manipulation of eccentric exercise velocity does not seem to differently influence intracellular hypertrophy signaling through the Akt/mTORCI/p70S6K pathway.
Literature	GENERIF_SUMMARY	three AKT phosphorylation-site-recognition motifs present on FoxO3, which are required for FoxO3 phosphorylation, are dispensable for AKT binding, suggesting that AKT has a docking point on FoxO3 distinct from the

Main Category	Category	Terms
		phosphorylation-recognition motifs
Literature	GENERIF_SUMMARY	Data demonstrate that DNA double-strand breaks activate signaling that promotes a PI3K-independent pathway of AKT phosphorylation that is dependent on MRE11-ATM-RNF168 signaling.
Literature	GENERIF_SUMMARY	Insulin inhibits amyloid-beta42-induced intracellular reactive oxygen species generation via Akt activation.
Literature	GENERIF_SUMMARY	mindin serves as a novel mediator that protects against cardiac hypertrophy and the transition to heart failure by blocking AKT/GSK3beta and TGF-beta1-Smad signalling.
Literature	GENERIF_SUMMARY	AKT signaling helps ABCA12 deficient keratinocytes to survive during the keratinization process.
Literature	GENERIF_SUMMARY	Insulin signaling plays a significant role in protecting against Toll-like receptor (TLR)3-mediated apoptosis of human bronchial epithelial cells via phosphatidylinositol 3-kinase/Akt and ERK-MAP kinase pathways.
Literature	GENERIF_SUMMARY	Results identify unique sites of PTEN and Akt regulation by means of S-nitrosylation, resulting in an ""on-off"" pattern of control of Akt/PI3K signaling."
Literature	GENERIF_SUMMARY	Data show that LRRK2 directly phosphorylates Akt1, a central molecule involved in signal transduction for cell survival and prevention of apoptosis.
Literature	GENERIF_SUMMARY	Data show that THBS1 was highly expressed at RNA and protein levels in LBW-ECFCs, and silencing THBS1 restored the angiogenic properties of LBW-ECFCs by increasing AKT phosphorylation.
Literature	GENERIF_SUMMARY	findings suggest that decreased B55alpha expression in AML is at least partially responsible for increased AKT signaling in AML and suggests that therapeutic targeting of PP2A could counteract this
Literature	GENERIF_SUMMARY	A cell penetrating peptide derived from azurin inhibits angiogenesis and tumor growth by inhibiting phosphorylation of VEGFR-2, FAK and Akt.
Literature	GENERIF_SUMMARY	a potential role for curcumin in treating tendon inflammation through modulation of NF-kappaB signaling, which involves PI-3K/Akt and the tendon-specific transcription factor scleraxis in tenocytes.
Literature	GENERIF_SUMMARY	In the presence of glucose, activated Akt

Main Category	Category	Terms
		prevented Loss of Mcl-1 expression & protected cells from growth factor deprivation-induced apoptosis. Mcl-1 associated with & inhibited the proapoptotic protein Bim, contributing to cell survival.
Literature	GENERIF_SUMMARY	Studies indicate that PI3K can phosphorylate and convert phosphatidylinositol 4,5-bisphosphate (PIP2) to phosphatidylinositol 3,4,5-trisphosphate (PIP3) that leads to activation of AKT.
Literature	GENERIF_SUMMARY	autophagy contributes to the anti-apoptotic function of the PML-RARalpha protein through inhibiting the Akt/mTOR pathway
Literature	GENERIF_SUMMARY	Angiopoietin-2, an angiogenic regulator, promotes initial growth and survival of breast cancer metastases to the lung through the integrin-linked kinase (ILK)-AKT1,2-B cell lymphoma 2 (Bcl-2) pathway.
Literature	GENERIF_SUMMARY	SPARC has a cell-autonomous survival activity, which requires Akt-dependent regulation of p53
Literature	GENERIF_SUMMARY	Metastases in most subcellular compartments showed comparable and stronger expression for pAKT, pmTOR, and PTEN than primary renal cell carcinomas (PRCCs).
Literature	GENERIF_SUMMARY	Knockdown of Scribble (Scrib) results in redistribution of PHLPP1 from the membrane to the cytoplasm and an increase in Akt phosphorylation.
Literature	GENERIF_SUMMARY	Dual function of protein kinase C (PKC) in TPA-induced MnSOD expression
Literature	GENERIF_SUMMARY	Activation of the PI3K/Akt/mTOR pathway correlates with tumour progression and reduced survival in patients with urothelial carcinoma of the urinary bladder.
Literature	GENERIF_SUMMARY	Studies indictet that the mammalian FoxO family consists of FoxO1, 3, 4 and 6 and are regulated by by AKT and 14-3-3 proteins.
Literature	GENERIF_SUMMARY	Administration of a Toll-like receptor (TLR)-2 ligand attenuates focal cerebral perfusion injury via a transgenic TLR2- and phosphatidylinositol 3-kinase/Akt-dependent mechanism.
Literature	GENERIF_SUMMARY	TGF-beta1 undergo a switch into mesenchymal cells and PI3K/Akt and MAPK/Erk1/2 signaling pathways serve to regulate TGF-beta1-induced epithelial-to-mesenchymal transition of A549 cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	mTOR/AKT inhibition by temsirolimus decreased endometriotic cell proliferation both in vitro and in vivo in a mouse model of deep infiltrating endometriosis
Literature	GENERIF_SUMMARY	beta-catenin/Tcf-4 binds to the potential binding sites in the gene promoter of AKT1.
Literature	GENERIF_SUMMARY	AKT1 and AKT2 both contribute to cell survival, albeit via different mechanisms; the effects on cell growth and migration are predominantly regulated by AKT1
Literature	GENERIF_SUMMARY	These results suggest that 18beta-glycyrrhetinic acid induces apoptosis in human breast carcinoma MCF-7 cells via caspase activation and modulation of Akt/FOXO3a/Bim pathway.
Literature	GENERIF_SUMMARY	In contrast to previously published results, our data do not support the hypothesis of genetic variants in AKT1 confering protection against Parkinson's disease.
Literature	GENERIF_SUMMARY	Exogenous H(2)O(2) triggers two parallel independent pathways, one leading to autophagy and autophagy-dependent apoptosis, the other to transient Akt phosphorylation.
Literature	GENERIF_SUMMARY	Clusterin induces matrix metalloproteinase-9 expression via ERK1/2 and PI3K/Akt/NF-kappaB pathways in monocytes/macrophages.
Literature	GENERIF_SUMMARY	Matrix-dependent regulation of AKT in Hepsin-overexpressing PC3 prostate cancer cells.
Literature	GENERIF_SUMMARY	Data suggest that cancer cells appear to continuously flux between symmetric and asymmetric division depending on the precise state of their AKT signaling network.
Literature	GENERIF_SUMMARY	Akt phosphorylation at Thr308 and Ser473 is required for CHIP-mediated ubiquitination of the kinase.
Literature	GENERIF_SUMMARY	The correlation between expression of pEGFR and pAKT is indicative of activation of the PI3-K/AKT pathway through phosphorylation of EGFR in head and neck squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Data indicate that co-administration perifosine with paclitaxel in ovarian cancer cells led to the inhibition of AKT, and a striking increase in the activation of pro-apoptosis pathways, including caspase 3, JNK1 and AMPKalpha1.
Literature	GENERIF_SUMMARY	The deacetylase SIRT1 promotes membrane

Main Category	Category	Terms
		localization and activation of Akt and PDK1 during tumorigenesis and cardiac hypertrophy.
Literature	GENERIF_SUMMARY	Monoamine oxidase A (MAO A) is induced following valproic acid (VPA) administration and MAO A is a novel target for VPA via Akt/FoxO1 signaling pathway.
Literature	GENERIF_SUMMARY	Cannabis use before onset of psychosis may have long-lasting effects on measures of sustained attention, even in the absence of current use, contingent on AKT1 rs2494732 genotype.
Literature	GENERIF_SUMMARY	Akt activity enhances the osteogenic function of Osterix, at least in part, through protein stabilization and that BMP-2 regulates the osteogenic function of Osterix, at least in part, through Akt.
Literature	GENERIF_SUMMARY	Our data demonstrate that Akt1 and notably Akt3 regulate proliferation, survival, migration and epidermal growth factor-mediated signal transduction in non small cell lung cancer-derived disseminated tumor cells
Literature	GENERIF_SUMMARY	cigarette smoke induces cytotoxicity, partly through Akt degradation via the ubiquitin-proteasome system, in which TTC3 acts as a ubiquitin ligase for active Akt.
Literature	GENERIF_SUMMARY	This study suggested that the AKT1 affects risk for schizophrenia and accompanying cognitive deficits, at least in part through specific genetic interactions related to brain neuroplasticity and development
Literature	GENERIF_SUMMARY	The Proteus syndrome is caused by a somatic activating mutation in AKT1, proving the hypothesis of somatic mosaicism and implicating activation of the PI3K-AKT pathway in the characteristic clinical findings of this disorder.
Literature	GENERIF_SUMMARY	Paracrine factors from normal endometrial stromal cells can inhibit hormone-stimulated cell proliferation in Ishikawa cells by regulating cell survival and apoptosis through PI3K/AKt/Survivin signaling
Literature	GENERIF_SUMMARY	SHP-1 antagonizes the action of receptor and non-receptor-tyrosine kinases on GIV and down-regulates the phospho-GIV-PI3K-Akt axis of signaling.
Literature	GENERIF_SUMMARY	our results suggest that ULK1 may act as a major node for regulation by multiple kinases including AMPK and Akt that play both stimulatory and inhibitory roles in regulating autophagy.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Overexpression of DJ-1 decreased the expression of PTEN, promoted the activation of Akt.
Literature	GENERIF_SUMMARY	Were associated with an increase in AKT.
Literature	GENERIF_SUMMARY	absence of mutations in AKT1, PTEN, and RAS genes in the present study is in accordance with previous studies confirming that these genes are rarely mutated in oral squamouos cell carcinoma
Literature	GENERIF_SUMMARY	mda-9/syntenin, a positive regulator of cancer metastasis, regulates the activation of Akt (also known as protein kinase B) by facilitating ILK adaptor function during adhesion to type I collagen (COL-I) in human breast cancer cells.
Literature	GENERIF_SUMMARY	For a Treg enhancing strategy to be successful in the treatment of autoimmune inflammation, resistance because of PKB/c-akt hyperactivation should be targeted as well.
Literature	GENERIF_SUMMARY	activation of Akt may cause apoptosis resistance, contributing to the overall mechanism of nickel carcinogenesis.
Literature	GENERIF_SUMMARY	Immunoblotting showed a high degree of Akt and mTOR phosphorylation in tumor samples compared with that in non-neoplastic tissue
Literature	GENERIF_SUMMARY	our results indicate that ERK1/2, Akt, and cav-1 are involved in the regulatory mechanisms of PPAR-mediated protection against HIV-1-induced MMP-9 expression in brain endothelial cells
Literature	GENERIF_SUMMARY	AKT1 polymorphisms are associated with early stage non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Akt phosphorylation event occurs at a residue (Thr33) located at the interface where the norovirus RNA-dependent RNA polymerase finger and thumb domains interact and decreases de novo activity of the polymerase.
Literature	GENERIF_SUMMARY	The N-cadherin in prostate cancer cell mediates cell-cell adhesion and regulates MCP-1 expression via the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	our study reveals a novel link between NF-kappaB and PI3K/Akt and establishes c-FLIP as an important regulator of FasL-mediated cell death.
Literature	GENERIF_SUMMARY	Targeting cap-dependent translation blocks converging survival signals by AKT and PIM

Main Category	Category	Terms
		1, 2 kinases in lymphoma
Literature	GENERIF_SUMMARY	Akt fine-tunes NF-kappaB-dependent gene expression during T cell activation.
Literature	GENERIF_SUMMARY	we here show that PD-1 signaling in cord blood T cells facilitates their differentiation into induced functional FOXP3+ Treg cells through a mechanism involving reduced PKB signaling.
Literature	GENERIF_SUMMARY	Ectopic Akt-1 expression induces resistance of MCF-7 cells to tamoxifen.
Literature	GENERIF_SUMMARY	CREG plays a critical role in protecting the vascular endothelium from apoptosis, and the protective effort of CREG against ECs apoptosis is through the activation of the VEGF/PI3K/AKT signaling pathway
Literature	GENERIF_SUMMARY	essential role of membrane microdomain compartmentalization in enabling PI3K/Akt signaling.
Literature	GENERIF_SUMMARY	The E-box binding factors Max/Mnt, MITF, and USF1 act coordinately with FoxO to regulate expression of proapoptotic and cell cycle control genes by phosphatidylinositol 3-kinase/Akt/glycogen synthase kinase 3 signaling.
Literature	GENERIF_SUMMARY	Aberrant expressions of p-AKT1 and p-MTOR in intrahepatic cholangiocarcinoma were associated with a favorable prognosis, possibly in a PTEN-independent manner.
Literature	GENERIF_SUMMARY	Chemokine CXCL12 uses CXCR4 and a signaling core formed by bifunctional Akt, ERK1/2, and mTORC1 proteins to control chemotaxis and survival simultaneously in mature dendritic cells.
Literature	GENERIF_SUMMARY	show that the kinase AKT regulates ribosome biogenesis at multiple levels to promote ribosomal RNA (rRNA) synthesis
Literature	GENERIF_SUMMARY	Protein kinase B is shown to modulate PGC1alpha activity and, hence, hepatitis B virus transcription and replication in a cell line-specific manner.
Literature	GENERIF_SUMMARY	Report cooperative effects of Akt-1 and Raf-1 on the induction of cellular senescence in doxorubicin or tamoxifen treated breast cancer cells.
Literature	GENERIF_SUMMARY	Study found that low levels of AKT activity associated with elevated levels of FOXOs are required to maintain the function and immature state of leukemia-initiating cells (LICs).

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	FGF2 and EGF affect the motor neuron fate decision in human fetal neural stem cells differently through a fine tuning of the PI3K/AKT/GSK3beta pathway.
Literature	GENERIF_SUMMARY	Complex I defects could promote tumorigenesis through induction of oxidative stress and activation of AKT pathway.
Literature	GENERIF_SUMMARY	Beta-HPV 5 and 8 E6 degrade p300 in a proteasomal-dependent, E6AP-independent manner. E6 competes with AKT for binding to p300.
Literature	GENERIF_SUMMARY	a model in which Akt can be phosphorylated at Ser-473 and activated in the absence of mTORC2 by mechanisms that depend on removal of the PH domain from the kinase domain.
Literature	GENERIF_SUMMARY	Syk drives EBV+ B cell lymphoma survival through PI3K/Akt activation, which prevents the HtrA2-dependent loss of XIAP. Syk, Akt, and XIAP antagonists may present potential new therapeutic strategies for PTLD
Literature	GENERIF_SUMMARY	Phospho-ERK and AKT status, but not KRAS mutation status, are associated with outcomes in rectal cancer treated with chemoradiotherapy
Literature	GENERIF_SUMMARY	findings demonstrate that Akt is a downstream signalling component of TGF-beta-mediated biglycan core protein synthesis but not glycosaminoglycan chain hyperelongation in vascular smooth muscle
Literature	GENERIF_SUMMARY	dissociation of PRAS40 from insulin-stimulated 4E-BP protein binding to mTORC1 and enhanced mTORC1 substrate binding results from Akt and mTORC1 activation and makes little or no contribution to mTORC1 signaling
Literature	GENERIF_SUMMARY	We sought to determine if ezrin, KBA.62, p-Akt, CD166, and nestin, may be helpful in distinguishing melanoma from nevi and atypical melanocytic lesions
Literature	GENERIF_SUMMARY	Data illustrated the importance of Akt1 in mammary epithelial cell proliferation, as well as in the regulation of Btn1a1 and subsequent expression of ss-casein.
Literature	GENERIF_SUMMARY	Loss of Akt1 evokes epithelial-mesenchymal transition by autocrine regulation of transforming growth factor-beta1
Literature	GENERIF_SUMMARY	CREG induces endothelial cell migration by activating the integrin-linked kinase/AKT/mTOR/VEGF(165) signaling pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	PIK3CA mutations in pancreatic Intraductal tubulopapillary neoplasms were associated with strong expression of phosphorylated AKT
Literature	GENERIF_SUMMARY	These findings, representing the first genomewide search for variants influencing human protein phosphorylation, provide useful information about the PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	Akt activation plays a critical role in enabling Arsenic-transformed human bronchial epithelial cell migration and invasion by promoting ZEB1 and ZEB2 expression.
Literature	GENERIF_SUMMARY	Overexpression of MUC1 enhances proangiogenic activity of non-small-cell lung cancer cells through activation of Akt and extracellular signal-regulated kinase pathways.
Literature	GENERIF_SUMMARY	EGFR-AKT-Smad signaling promotes formation of glioma stem-like cells and tumor angiogenesis by ID3-driven cytokine induction.
Literature	GENERIF_SUMMARY	findings establish that Shh signaling promotes the metastasis of gastric cancer through activation of the PI3K/Akt pathway, which leads to mesenchymal transition and MMP-9 activation
Literature	GENERIF_SUMMARY	High and low phospho-ERK and phospho-AKT expression shows a trend toward worse overall survival in patients with high expression of phospho-AKT.
Literature	GENERIF_SUMMARY	Akt/mTOR might be effective targets to overcome the resistance to IGF-1R mAbs in head and neck squamous cell carcinoma and non-small cell lung carcinomas.
Literature	GENERIF_SUMMARY	IKKbeta induced AKT activity, whereas IKKbeta-driven NF-kappaB transcription was required for GLUT1 surface localization downstream of AKT.
Literature	GENERIF_SUMMARY	Findings suggest that large HBsAg (LHBs) promotes tumorigenesis of hepatoma cells by triggering a PKCalpha/Raf1 to Src/PI3K/Akt signaling pathway, revealing novel insights into the underlying mechanisms of HBV-associated hepatocarcinogenesis.
Literature	GENERIF_SUMMARY	Fascin may regulate functions of human hepatic stellate cells (HSCs) through the FAK-phosphoinositide 3-kinase-Akt pathway.
Literature	GENERIF_SUMMARY	Endothelial cell-induced Akt phosphorylation is abrogated in primary but not metastatic human head and neck squamous cell

Main Category	Category	Terms
		carcinoma stem cells.
Literature	GENERIF_SUMMARY	Discoverery of a novel reverse regulation within the traditional PTEN/AKT signaling pathway, whereby AKT induces GATA2 with consequent decreased PTEN transcription in breast cancer.
Literature	GENERIF_SUMMARY	AMPK, Akt and mTOR pathways have roles in resveratrol-enhanced prostate cancer cell response to ionizing radiation
Literature	GENERIF_SUMMARY	Data present a novel pathway for the regulation of Ca(2+) signaling and PASMC proliferation involving activation of Akt/mTOR and upregulation of STIM1/Orai1.
Literature	GENERIF_SUMMARY	analysis of resistance of Akt kinase and PP2A to dephosphorylation through ATP-dependent conformational plasticity
Literature	GENERIF_SUMMARY	Study confirms the contribution of loss of PTEN expression in Akt phosphorylation and spontaneous apoptosis suppression in the specimens of oral cancer.
Literature	GENERIF_SUMMARY	spheroid dispersion was at least partly dependent upon AKT activity
Literature	GENERIF_SUMMARY	These results suggest that the Nrf2/ARE/Jun/Akt/Erk pathway plays an important role in the regulation of icariside-mediated antioxidant effects in HepG2 cells.
Literature	GENERIF_SUMMARY	CXCL12, on binding CXCR7, induced phosphorylation of extra cellular regulated protein kinases (ERK 1/2) and Akt.
Literature	GENERIF_SUMMARY	In endometriotic stromal cells, overactivation of the phosphatidylinositol 3-kinase/AKT signaling pathway contributes to the reduced expression of the decidua-specific gene, IGFBP1, potentially through reduced levels of nuclear FOXO1.
Literature	GENERIF_SUMMARY	multiple-site acetylation of Rictor signals for increased activation of mTORC2, providing a critical link between nutrient-sensitive deacetylases and mTORC2 signaling to Akt
Literature	GENERIF_SUMMARY	Notch activation could stimulate beta-catenin and NF-kappaB signaling through AKT activation in glioma cells. Notch activation-stimulated beta-catenin and NF-kappaB signaling synergistically promote the migratory and invasive properties of glioma cells.
Literature	GENERIF_SUMMARY	results provide a first rationale and justification for targeting the ETV6/RUNX1 fusion gene and the PI3K/AKT/mTOR pathway therapeutically

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	analysis of AKT signaling pathway activated by HIN-1 methylation in non-small cell lung cancer
Literature	GENERIF_SUMMARY	Proliferation of human IVD cells is regulated by exogenous and autocrine growth factors mainly via the MEK/ERK and PI-3K/Akt pathways.
Literature	GENERIF_SUMMARY	The prognostic impact of Akt (Akt1) phosphorylated at threonine308 and serine473, Akt2, Akt3, PI3K and PTEN, alone and in coexpression with ER and PgR in non-gastrointestinal stromal tumor soft tissue sarcomas.
Literature	GENERIF_SUMMARY	inhibition of miR-17 approximately 92 expression suppressed the PI3K/AKT pathway and inhibited tumor growth in a xenograft MCL mouse model
Literature	GENERIF_SUMMARY	Radiotherapy can increase the levels of AKT in nasopharyngeal carcinoma (NPC) cells that are associated with NPC metastasis and increase in mortality.
Literature	GENERIF_SUMMARY	Epidermal growth factor receptor (EGFR)-mediated positive feedback of protein-tyrosine phosphatase epsilon (PTPepsilon) on ERK1/2 and AKT protein pathways is required for survival of human breast cancer cells.
Literature	GENERIF_SUMMARY	DNA-PK inhibition might prevent platinum-induced AKT activation without interfering with normal glucose homeostasis, an unwanted toxicity of direct AKT inhibitors.
Literature	GENERIF_SUMMARY	The addition of RTK inhibitors can prevent this reactivation of AKT signaling and cause profound cell death and tumor regression in vivo.
Literature	GENERIF_SUMMARY	BDNF promotes EGF-induced proliferation and migration of neural stem/progenitor cells through the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	The AKT1 rs2498801 genotype significantly increased risk of endometrial cancer.
Literature	GENERIF_SUMMARY	Genetic variant of AKT1 might be associated with attentional deficits and brain morphological vulnerability in patients with schizophrenia.
Literature	GENERIF_SUMMARY	The elevation of ENTPD5 activity therefore protects AKT-active cancer cells from protein-overloading-induced endoplasmic reticulum stress and the resulting growth arrest and apoptosis.
Literature	GENERIF_SUMMARY	Using knockdown of AKT & STAT3, studied

Main Category	Category	Terms
		the effects of IL-17/AKT signaling in HCC, resulting in IL-6 production, which in turn activated JAK2/STAT3 signaling and subsequently up-regulated its downstream targets IL-8, MMP2, and VEGF.
Literature	GENERIF_SUMMARY	cell survival role for CaM KK operating through AKT and BAD in LNCaP prostate cancer cells
Literature	GENERIF_SUMMARY	VEGF-A stimulation of lymphatic endothelial cells leads to the phosphorylation of VEGFR2 (Tyr 951, 1054, 1059, 1175, and 1214) which subsequently triggers PKC dependent phosphorylation of ERK1/2 and PI3-K dependent phosphorylation of Akt.
Literature	GENERIF_SUMMARY	Akt activation is a mechanism of chemotherapeutic resistance in B-pre ALL and suggest that Akt can be a therapeutic target for the treatment of relapsed or refractory pediatric B-pre ALL.
Literature	GENERIF_SUMMARY	unrecognized cell-autonomous function of TORC1 and TORC2 in the up-regulation of PTEN, which prevents phosphorylation of Akt and may shield against the development of malignancy in TSC patients
Literature	GENERIF_SUMMARY	Data suggest that wild-type PTEN gene transfection might increase drug sensitivity or reverse drug resistance via inhibiting the PI3K/Akt pathway and regulating downstream molecules of the cell signaling transduction pathway in K562/ADM cells.
Literature	GENERIF_SUMMARY	Analysis of Akt1 gene TRR methylation showed decreased methylation level in bladder cancer than normal.
Literature	GENERIF_SUMMARY	14-3-3sigma regulation by p53 mediates a chemotherapy response to 5-fluorouracil in MCF-7 breast cancer cells via Akt inactivation.
Literature	GENERIF_SUMMARY	Results describe the selective loss of Akt activity, in vivo, due to oxidative modification of Akt and provide mechanistic insight into oxidative stress-induced down-regulation of cell survival pathway in mouse midbrain following exposure to MPTP.
Literature	GENERIF_SUMMARY	RET mutations may have a role in medullary thyroid carcinoma, while BRAF, AKT1, and CTNNB1 do not; the role of HRAS, KRAS, and NRAS mutations are not determined
Literature	GENERIF_SUMMARY	AKT protein kinase has a role in gastric cancer [review]
Literature	GENERIF_SUMMARY	Results show a novel anti-inflammatory role for the PI3K/Akt signaling pathway in

Main Category	Category	Terms
		microglia, and show that Ad-IRF3 suppressed proinflammatory genes and enhanced anti-inflammatory genes.
Literature	GENERIF_SUMMARY	VE-cadherin outside-in signaling controls locally Rac activity stimulated by barrier protective agonists.
Literature	GENERIF_SUMMARY	AIB1 plays an important role in proliferation and chemoresistance of cholangiocarcinoma through simultaneous activation of Akt and Nrf2 pathways.
Literature	GENERIF_SUMMARY	The results showed that loss of RUNX3 expression enhanced the Akt1-mediated signaling pathway. The RUNX3-mediated inhibition of Akt1 caused beta-catenin protein degradation and then cyclin D1 downregulation.
Literature	GENERIF_SUMMARY	miR-7 functions as a tumor suppressor and plays a substantial role in inhibiting the tumorigenesis and reversing the metastasis of hepatocellular carcinoma through the PI3K/Akt/mTOR-signaling pathway in vitro and in vivo.
Literature	GENERIF_SUMMARY	miR-122 as an amplifier of the antiapoptotic Akt/p53 circuit
Literature	GENERIF_SUMMARY	identification of AKT as being one important downstream component of activated BMP signalling on epithelial ovarian cancer spheroid pathobiology, which may have important implications on the metastatic potential of this malignancy
Literature	GENERIF_SUMMARY	NAMPT rs1319501 minor allele associates with increased MI risk in young women. In young men a protective effect of the AKT1 rs3730358 minor allele was suggested, possibly related to an attenuated inflammation
Literature	GENERIF_SUMMARY	Epstein-Barr virus latent membrane protein-1 protects B-cell lymphoma from rituximab-induced apoptosis through miR-155-mediated Akt activation and up-regulation of Mcl-1
Literature	GENERIF_SUMMARY	Leptin-induced epithelial-mesenchymal transition in breast cancer cells requires beta-catenin activation via Akt/GSK3- and MTA1/Wnt1 protein-dependent pathways.
Literature	GENERIF_SUMMARY	Cyclin G1 overexpression enhanced Akt activation through interaction with p85 (regulatory subunit of phosphoinositide 3-kinase)
Literature	GENERIF_SUMMARY	The findings of this study supported the view that AKT1 genetic variants are shared

Main Category	Category	Terms
		by both BPD and SCZ.
Literature	GENERIF_SUMMARY	Protein Kinase B/Akt is a target of alpha1-adrenoceptor-blocker therapy, which has been unknown to date.
Literature	GENERIF_SUMMARY	UCA1 regulated cell cycle through CREB via PI3K-AKT dependent pathway in bladder cancer.
Literature	GENERIF_SUMMARY	pAKT expression, as examined by immunohistochemistry, is an independent prognostic factor for peripheral T-cell lymphoma (PTCL).
Literature	GENERIF_SUMMARY	data suggest that lamin A/C proteins are positively involved in malignant behavior of PC cells through the PI3K/AKT/PTEN pathway. Lamin A/C may represent a new oncogenic factor and a novel therapeutic target for PC
Literature	GENERIF_SUMMARY	Results suggest that oncogenic co-operation between HBx and Akt may be important for cell proliferation, abrogation of apoptosis and tumorigenic transformation of cells.
Literature	GENERIF_SUMMARY	EphB3 suppresses non-small-cell lung cancer metastasis via a PP2A/RACK1/Akt signalling complex
Literature	GENERIF_SUMMARY	Akt1 regulates the interferon response through phosphorylation of the transcriptional repressor EMSY
Literature	GENERIF_SUMMARY	A subset of ovarian cancers exhibits AKT pathway activation and is sensitive to selective AKT inhibition.
Literature	GENERIF_SUMMARY	melatonin inhibits cell proliferation and induces apoptosis in MDA-MB-361 breast cancer cells in vitro by simultaneously suppressing the COX-2/PGE2, p300/NF-kappaB, and PI3K/Akt/signaling and activating the Apaf-1/caspase-dependent apoptotic pathway.
Literature	GENERIF_SUMMARY	Endometrial carcinoma patients with loss of PTEN and p-AKT positivity have a favorable prognosis.
Literature	GENERIF_SUMMARY	synemin positively regulates glioblastoma cell proliferation by helping sequester PP2A away from Akt, thereby favoring Akt activation.
Literature	GENERIF_SUMMARY	Akt/mTOR is a key signaling component through which c-Met protects multiple myeloma cells from chemotherapy-induced growth inhibition and apoptosis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Studies suggest that the effects of PHLPP1 deletion on prostate carcinogenesis may be explained by the strong dependence of prostate cancer on phosphoinositide 3-kinase (PI3K)/Akt signalling.
Literature	GENERIF_SUMMARY	Overexpression of mu-opioid receptor has a possible direct affect on Akt and mTOR activation and lung cancer progression.
Literature	GENERIF_SUMMARY	API-1 reduces c-FLIP and enhances TRAIL-induced apoptosis independent of its Akt-inhibitory activity
Literature	GENERIF_SUMMARY	Data indicate that the mechanisms of action of ON 01910.Na invloved PI3K/AKT inhibition and induction of oxidative stress response.
Literature	GENERIF_SUMMARY	This review summarizes the regulation and potential function(s) of PRAS40 in the complex Akt- and mTOR-signaling network in health and disease
Literature	GENERIF_SUMMARY	PI3K/AKT pathway increased the activity of cystathionine gamma-lyase gene promoter via Sp1.
Literature	GENERIF_SUMMARY	The relationship between expression and genetic alterations of the components of the PI3K/AKT pathway in non-small cell lung cancer, was investigated.
Literature	GENERIF_SUMMARY	Merkel cell carcinoma samples were analyzed for activating PIK3CA and AKT1 mutations.
Literature	GENERIF_SUMMARY	These findings reveal a previously unknown regulatory mechanism for nuclear import of hTERT through a bipartite nuclear localization signal mediated by Akt phosphorylation.
Literature	GENERIF_SUMMARY	PTEN protein phosphatase activity correlates with control of gene expression and invasion, a tumor-suppressing phenotype, but not with AKT activity.
Literature	GENERIF_SUMMARY	GUCY2C opposes systemic genotoxic tumorigenesis by regulating AKT-dependent intestinal barrier integrity
Literature	GENERIF_SUMMARY	analysis of expression of EGFR, pAkt, NF-kappaB and MIC-1 in prostate cancer stem cells and their progenies
Literature	GENERIF_SUMMARY	analysis of the role of RICTOR in beta1 integrin-mediated cell survival and how receptor-specific mechanisms regulate phosphorylation of AKT at Ser473
Literature	GENERIF_SUMMARY	Differential expression of MUC5AC involves

Main Category	Category	Terms
		crosstalk between interleukin-1beta and Akt, whereby the trefoil factor family peptide TFF3 activates Akt by phosphorylation of EGFR.
Literature	GENERIF_SUMMARY	Flavopiridol induces phosphorylation of AKT-Ser473 in human T98G glioblastoma cell line.
Literature	GENERIF_SUMMARY	USP46-mediated stabilization of PHLPP and the subsequent inhibition of Akt.
Literature	GENERIF_SUMMARY	decreased PBMC AKT1 expression is a stable trait in recent onset, male schizophrenia patients
Literature	GENERIF_SUMMARY	JNK-potentiated Akt/FoxO3a and JNK-mediated c-Jun pathways co-operatively trigger Puma expression, which determines the threshold for overcoming chemoresistance in ovarian cancer cells.
Literature	GENERIF_SUMMARY	PDPN expression in human glioblastoma multiforme is caused by loss of PTEN function and activation of the PI3K-AKT-AP-1 signaling pathway, accompanied by epigenetic regulation of PDPN promoter activity
Literature	GENERIF_SUMMARY	Signaling through the PI3K (phosphatidylinositol 3-kinase)-Akt-mTOR (mammalian target of rapamycin) pathway was necessary and sufficient for the increase in p57Kip2
Literature	GENERIF_SUMMARY	Akt was found to directly interact with MULAN and to be ubiquitinated by MULAN in vitro and in vivo.
Literature	GENERIF_SUMMARY	we have analyzed the prevalence of somatic mutations in the FGFR3, PIK3CA, AKT1, KRAS, HRAS, and BRAF genes in bladder cancers
Literature	GENERIF_SUMMARY	a novel USP1-PHLPP1-Akt signaling axis, and decreased USP1 level in lung cancer cells may play an important role in lung cancer progress.
Literature	GENERIF_SUMMARY	Akt1 drives progression in early breast cancers.
Literature	GENERIF_SUMMARY	We have demonstrated that vincristine resistance mediated by CD40 activation was induced by an increased expression of MRP1 by AKT signaling in human multiple myeloma cell lines.
Literature	GENERIF_SUMMARY	Overexpression of hyaluronan-binding protein 1 (HABP1/p32/gC1qR) in HepG2 cells leads to increased hyaluronan synthesis and cell proliferation by up-

Main Category	Category	Terms
		regulation of cyclin D1 in AKT-dependent pathway.
Literature	GENERIF_SUMMARY	Positive correlations are found between p-AKT, p-GSK3beta and beta-catenin in epithelial ovarian carcinoma. The activation of beta-catenin is possibly correlated with inactivation of p-GSK3beta that binds to p-AKT.
Literature	GENERIF_SUMMARY	TLR3 induces the apoptotic cell death of salivary gland epithelial cells (SGECs) via the PI3K-Akt signaling pathway.
Literature	GENERIF_SUMMARY	Data indicate a functional link between the Notch1 and PI3K/Akt pathways in clear cell renal cell carcinoma (CCRCC).
Literature	GENERIF_SUMMARY	miR-93 can directly target PTEN, and participates in the regulation of the AKT signaling pathway
Literature	GENERIF_SUMMARY	Targeting PI3 kinase/AKT/mTOR signaling in cancer.
Literature	GENERIF_SUMMARY	study concludes that Akt2 is indispensable for the regulation of preadipocyte and adipocyte number, whereas Akt1 and Akt2 are equally important for the regulation of insulin-stimulated metabolic pathways in adipocytes.
Literature	GENERIF_SUMMARY	The Akt signaling is as an inhibitor of Oct4 and Nanog expression and an activator for PHF19 expression
Literature	GENERIF_SUMMARY	The PI3K/AKT pathway can be an important signaling pathway for the survival of BRCA1-defective breast cancer cells.
Literature	GENERIF_SUMMARY	CYLD negatively regulates TGFb1 signalling via deubiquitinating Akt.
Literature	GENERIF_SUMMARY	AKT mutations are associated with pediatric T-cell acute lymphoblastic leukemia.
Literature	GENERIF_SUMMARY	Findings suggest that dysadherin might contribute to breast cancer progression through AKT activation.
Literature	GENERIF_SUMMARY	Integrin beta1 mediates vaccinia virus entry through activation of PI3K/Akt signaling.
Literature	GENERIF_SUMMARY	Data demonstrated that in myoepithelial cells, expression of E17K inhibited growth, migration, and protein synthesis compared with wild-type AKT1. In luminal cells, E17K enhanced cell survival and migration, possibly offering a selective advantage.
Literature	GENERIF_SUMMARY	PARG knockdown, concomitant with inhibition of PARP, suppressed the

Main Category	Category	Terms
		metastatic potency of colon carcinoma cells by activation of PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	The results of this study genetic modulation of DRD2-AKT1-related prefrontal-subcortical circuits could at least in part influence cognitive dysfunction in psychosis and its treatment.
Literature	GENERIF_SUMMARY	The result indicated that Protein kinase B was involved in knee osteoarthritic progression, along with the increase of cell apoptosis.
Literature	GENERIF_SUMMARY	AKT1-GLI3-VMP1 pathway mediates KRAS oncogene-induced autophagy in cancer cells.
Literature	GENERIF_SUMMARY	Changes in the proportion of circulating VEGFR2- and pAkt- positive cells may be a potential pharmacodynamic marker of the efficacy of antiangiogenic agents, and could prove valuable in determining drug dosage and administration schedule
Literature	GENERIF_SUMMARY	Cysteine-rich protein Cyr61 activates interleukin (IL)-6 production via the integrin alphavbeta5/Akt/NF-kappaB signaling pathway in rheumatoid arthritis.
Literature	GENERIF_SUMMARY	The existence of a specific activity of Akt on IP3R type III, was leading to channel inhibition, diminished Ca2+ transfer to mitochondria and protection from apoptosis.
Literature	GENERIF_SUMMARY	Mutation of this amino acid led to increased ERBB receptor activation and upregulation of the ERBB3/PI3K/AKT signaling pathway, which was no longer responsive to MEK inhibition
Literature	GENERIF_SUMMARY	Data indicate that Akt expression was up-regulated with high glucose and insulin in both cell lines, whereas PLCgamma expression was enhanced in colon cancer cells only.
Literature	GENERIF_SUMMARY	Expression of pAkt and mTOR expression is more significantly associated with adenocarcinoma than squamous cell carcinomaof the lung
Literature	GENERIF_SUMMARY	CXCL8 and CXCR1 are involved in the pathogenesis of endometriosis by up-regulating proliferation and growth and restricting apoptosis in endometrial stromal cells by activating the PTEN/Akt pathway and mediating the expression of survivin and Bcl-2
Literature	GENERIF_SUMMARY	Protein kinase G and focal adhesion kinase

Main Category	Category	Terms
		converge on Src/Akt/beta-catenin signaling module in osteoblast mechanotransduction.
Literature	GENERIF_SUMMARY	PTPN22 promotes survival of antigen-stimulated CLL cells by selectively activating AKT.
Literature	GENERIF_SUMMARY	CD133 negatively regulates tumorigenicity via AKT pathway in synovial sarcoma.
Literature	GENERIF_SUMMARY	We further identify the activation of the PI3K/AKT signal pathway by T3 as an underlying mechanism for the enhanced conversion to cell pluripotency in this model.
Literature	GENERIF_SUMMARY	This study identified one functionally meaningful genetic variant within the Akt signaling pathway that is associated with both lymphocyte count and composition alterations during natalizumab treatment in Multiple sclerosis
Literature	GENERIF_SUMMARY	AKT activation plays a crucial role in neutrophil-like differentiated HL-60 cell polarization
Literature	GENERIF_SUMMARY	FKBP5 as a selection biomarker for gemcitabine and Akt inhibitors in treatment of pancreatic cancer.
Literature	GENERIF_SUMMARY	Src, Akt, NF-kappaB, BCL-2 and c-IAP1 may be involved in an anti-apoptotic effect in patients with BCR-ABL positive and BCR-ABL negative acute lymphoblastic leukemia.
Literature	GENERIF_SUMMARY	Chronic modulation of AMP-Kinase, Akt and mTOR pathways by ionizing radiation.
Literature	GENERIF_SUMMARY	miR-26b and miR-128 as central regulators of pituitary somatomammotrophic tumor growth through activation of the PTEN-AKT pathway.
Literature	GENERIF_SUMMARY	Data suggest that phosphoinositide-3-kinase/Akt and mammalian target of rapamycin (mTOR) pathway signaling differentially regulate phospho-Mst1-Thr-120/Thr-183.
Literature	GENERIF_SUMMARY	Overexpression of human Akt1 enhances adipogenesis and leads to lipoma formation in zebrafish.
Literature	GENERIF_SUMMARY	integrin maintains pro-survival signaling via continuous activation of AKT
Literature	GENERIF_SUMMARY	LEDGF gene silencing impairs the tumorigenicity of prostate cancer DU145 cells by abating the expression of Hsp27 and activation of the Akt/ERK signaling pathway

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Literature	GENERIF_SUMMARY	Angiosarcoma (AS) is characterized by a complex pattern of alterations of the TP53 and PIK3CA/AKT/mTOR pathways that does not depend on MYC amplification status.
Literature	GENERIF_SUMMARY	This study demonstrates a central role for HuR in Akt signaling and reveals a mechanism by which modest changes in HuR levels below or above normal may be amplified, potentially resulting in cell death or cellular transformation.
Literature	GENERIF_SUMMARY	The interaction between HGF and c-Met increases OPN expression in human osteoblasts via the PI3K, Akt, c-Src, c-Jun, and AP-1 signaling pathway.
Literature	GENERIF_SUMMARY	expression levels of EGFR, pAkt, and PTEN differ between oropharyngeal and oral cavity cancer and it may be attributed to HPV-related molecular pathogenesis
Literature	GENERIF_SUMMARY	levels of EphB4, mTOR and Akt were distinctly lower than those in other groups. It was concluded that suppression of EphB4 may inhibit the growth of ovarian cancer cells by downregulation of the PI3K/Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	Data indicate that maintained AKT1 expression correlated with low copy number, an increased frequency of integration and increased HPV16 E7 expression.
Literature	GENERIF_SUMMARY	analysis of regulation of AKT and GSK-3beta by O-GlcNAcylation in various types of cells
Literature	GENERIF_SUMMARY	Mucinous breast carcinomas lack AKT1 mutations.
Literature	GENERIF_SUMMARY	Blocking individual branches in epidermal growth factor (EGF) signaling leads to Akt activation and plays a major role in EGF-induced alternative splicing.
Literature	GENERIF_SUMMARY	the regulation of HT-29 proliferation is mediated by cPLA(2)alpha-dependent PGE(2) production. PGE(2)via EP induces CREB phosphorylation by the PKA pathway and regulates beta-catenin and cyclin D1 cellular localization by PKB/Akt pathway.
Literature	GENERIF_SUMMARY	Selective effects of PD-1 on Akt and Ras pathways regulate molecular components of the cell cycle and inhibit T cell proliferation.
Literature	GENERIF_SUMMARY	amplification of AKT1 and/or AKT2 and high-level polysomy were found in 16% of total lung carcinoma cases, and this defined subset was characterized by the overexpression/activation of Akt, reciprocal

Main Category	Category	Terms
		to EGFR aberrations.
Literature	GENERIF_SUMMARY	Data indicate that erufosine caused dose-dependent decreases in the phosphorylation of PI3K (p85), Akt (PKB) at Thr 308 and cRaf in both MCF-7 and MDA-MB 231 cell lines.
Literature	GENERIF_SUMMARY	The resultant cells with concurrent NF-kappaB and Akt blockade were significantly more sensitive to cisplatin-induced cell death in vitro.
Literature	GENERIF_SUMMARY	These data support a strong biological-link between NFkB and the PI3-kinase/AKT pathway in the modulation of anti-apoptotic effects in primary effusion lymphoma cells.
Literature	GENERIF_SUMMARY	AKT regulation of the mesenchymal stem cell differentiation
Literature	GENERIF_SUMMARY	24 h of exposure to simulated microgravity promotes angiogenesis among HUVEC-C and this process is mediated through the PI3K-Akt-eNOS signal pathway
Literature	GENERIF_SUMMARY	these data define an inhibitory role for both AKT1 and AKT2 in prostate cancer migration and invasion and highlight the cell type-specific actions of AKT kinases in the regulation of cell motility.
Literature	GENERIF_SUMMARY	The association between single nucleotide polymorphisms (SNPs) of PTEN/AKT/mTOR genes and prostate cancer (PCa) in Chinese population, was examined.
Literature	GENERIF_SUMMARY	AKT1 rs2494732 contributes to susceptibility to the psychotogenic effect of cannabis.
Literature	GENERIF_SUMMARY	Which includes inhibition of tyrosine kinase activity of AKT and ERK1/2, suppression of NF-kappaB protein expression.
Literature	GENERIF_SUMMARY	Epstein-Barr virus latent membrane protein-2A induces ITAM/Syk- and Akt-dependent epithelial migration through alphav-integrin membrane translocation
Literature	GENERIF_SUMMARY	Data provide information for developing new therapeutics targeting PI3K/Akt/mTOR signaling pathway in autoimmune diseases like psoriasis and rheumatoid arthritis.
Literature	GENERIF_SUMMARY	This study is the first to prove the significance of Akt kinase isoform, activity and compartmentalization for the prediction of response to trastuzumab-based therapy in patients with HER2-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	Data suggest that targeting Hedgehog (Hh)

Main Category	Category	Terms
		pathway alone or in combination with PI3K/AKT pathway could be a novel therapeutic option in treating endocrine-resistant breast cancer.
Literature	GENERIF_SUMMARY	AZ7328 inhibited proliferation and AKT substrate phosphorylation in a concentration-dependent manner.
Literature	GENERIF_SUMMARY	MLS-2438 inhibits viability and induces apoptosis of human melanoma cells associated with inhibition of STAT3 and Akt signaling.
Literature	GENERIF_SUMMARY	RhoC GTPase is a substrate for Akt1 and its phosphorylation is absolutely essential for inflammatory breast cancer cell invasion.
Literature	GENERIF_SUMMARY	Akt1 plays a key role in monocrotophos-induced apoptosis in human umbilical cord blood stem cells.
Literature	GENERIF_SUMMARY	AKT1 rs2494752 polymorphism is associated with treatment response in advanced non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Grape proanthocyanidin inhibit pancreatic cancer cell growth in vitro and in vivo through induction of apoptosis and by targeting the PI3K/Akt pathway
Literature	GENERIF_SUMMARY	association of the transmembrane domain of CD154 in lipid rafts is necessary for CD154-mediated Akt and p38 activation with consequent enhancement of IL-2 production
Literature	GENERIF_SUMMARY	ADAMTS9 acts as a functional tumor suppressor in gastric cancer through inhibiting oncogenic AKT/mTOR signaling pathway
Literature	GENERIF_SUMMARY	Hyperphosphorylation of Tau induced by naturally secreted amyloid-beta at nanomolar concentrations is modulated by insulin-dependent Akt-GSK3beta signaling pathway.
Literature	GENERIF_SUMMARY	These observations demonstrate ERK and Akt activation induced by Dyrk1A in the brain of hyperhomocysteinemic mice.
Literature	GENERIF_SUMMARY	XB130 is a substrate of multiple protein tyrosine kinases, and it can regulate cell proliferation and survival through modulating selected down-stream signals of PI3K/Akt pathway
Literature	GENERIF_SUMMARY	AKT1 fails to replicate as a longevity-associated gene in Danish and German nonagenarians and centenarians.
Literature	GENERIF_SUMMARY	ASPP 049 from C. comosa induced

Main Category	Category	Terms
		osteoblastic cell proliferation and differentiation through ERalpha-, Akt-, and GSK-3beta-dependent activation of beta-catenin signaling.
Literature	GENERIF_SUMMARY	Data indicate that KRAS, BRAF, PIK3CA, and AKT1 mutations can be rapidly and accurately detected for cancer diagnosis.
Literature	GENERIF_SUMMARY	Lunasin inhibited alphaVbeta3 integrin-mediated proinflammatory markers and downregulated Akt-mediated NF-kappaB pathways through interaction with alphaVbeta3 integrin.
Literature	GENERIF_SUMMARY	Inhibition of EGFR-AKT axis results in the suppression of ovarian tumors in vitro and in preclinical mouse model
Literature	GENERIF_SUMMARY	Akt phosphorylation was down-regulated in OA chondrocytes after the strain, and was up-regulated after transfection of p53R2.
Literature	GENERIF_SUMMARY	tumors with high TIMP-1 and NOS2 behave more aggressively by mechanisms that favor Akt pathway activation
Literature	GENERIF_SUMMARY	These results indicated that moscatilin inhibited MDA-MB-231 cell migration via Akt- and Twist-dependent pathways; this finding was consistent with moscatilin's antimetastatic activity in vivo.
Literature	GENERIF_SUMMARY	Results indicate that CIB1 is uniquely positioned to regulate PI3K/AKT and MEK/ERK signaling and that simultaneous disruption of these pathways synergistically induces a nuclear GAPDH-dependent cell death.
Literature	GENERIF_SUMMARY	Data suggest that sustained hyperinsulinemia (prolonged infusion of insulin) without amino acid supply does not stimulate synthesis of mitochondrial proteins or ATP production in skeletal muscle; phosphorylation of c-Akt (but not mTOR) is stimulated.
Literature	GENERIF_SUMMARY	PKB-mediated PHF20 phosphorylation led to an inhibition of p53 induction following UV treatment, leading to the reduction of p21 transcriptional activity.
Literature	GENERIF_SUMMARY	CXCL12/CXCR4 protein signaling axis induces sonic hedgehog expression in pancreatic cancer cells via extracellular regulated kinase- and Akt kinase-mediated activation of nuclear factor kappaB
Literature	GENERIF_SUMMARY	EGFR and downstream genetic alterations in KRAS/BRAF and PI3K/AKT pathways have roles in colorectal cancer and treatment

Main Category	Category	Terms
		[review]
Literature	GENERIF_SUMMARY	The results suggest that inactivation of IKKalpha, followed by Akt and FOXO1 phosphorylation and caspase-3 activation, contributes to zerumbone-induced GBM cell apoptosis.
Literature	GENERIF_SUMMARY	Data suggest that parthenolide may be mediated not only by the inhibition of NF-kappaB but also by the inhibition of Akt signalling and the activation of apoptotic proteins.
Literature	GENERIF_SUMMARY	in embryonal carcinoma cells, unphosphorylated Oct4 bound to the AKT1 promoter and repressed its transcription. Phosphorylation of Oct4 by Akt resulted in dissociation of Oct4 from the AKT1 promoter, which activated AKT1 transcription and promoted cell survival
Literature	GENERIF_SUMMARY	Glycogen synthase kinase-3beta (GSK-3beta) is inactivated by recombinant human Hsp70-induced Akt activation.
Literature	GENERIF_SUMMARY	Results suggest that AKT is activated by hypoxia in an EGFR-independent way in tumors.
Literature	GENERIF_SUMMARY	AKT activation is associated with the pathogenesis of myelodysplastic syndromes with excess of blasts and evolution to leukaemia.
Literature	GENERIF_SUMMARY	These findings suggest that Rac is an important regulator in establishing the inflammatory microenvironment that is initiated in the human airway upon exposure to rhinovirus.
Literature	GENERIF_SUMMARY	Substantial decrease of heat shock protein (Hsp)90 client Akt signaling pathway is observed after Hsp72/73 depletion in cultured myeloma cells.
Literature	GENERIF_SUMMARY	Lactotransferrin acts as a tumor suppressor in nasopharyngeal carcinoma by repressing AKT through multiple mechanisms.
Literature	GENERIF_SUMMARY	PI3K/Akt signaling is utilized by alphaIIb-mediated outside-in signaling to activate platelets.
Literature	GENERIF_SUMMARY	Data suggest that insulin secretion is controlled by two different mTOR complexes (mTORC1; mTORC2) acting through c-Akt; data provide evidence for antagonistic effects of raptor and rictor on insulin secretion by modulating activity of c-Akt.
Literature	GENERIF_SUMMARY	Akt phosphorylates the transcriptional

Main Category	Category	Terms
		repressor bmi1 to block its effects on the tumor-suppressing ink4a-arf locus.
Literature	GENERIF_SUMMARY	HCV transiently activates the PI3K-AKT pathway to facilitate its entry
Literature	GENERIF_SUMMARY	Expression of survivin modulated by sodium valproate is under the regulation of AKT.
Literature	GENERIF_SUMMARY	PI3K/AKT/mTOR signalling components appear to be differentially implicated in urothelial tumorigenesis and, with the exception of p85aPI3K, are unrelated to the PIK3CA or AKT1 mutational status.
Literature	GENERIF_SUMMARY	Akt-mediated phosphorylation of Beclin 1 functions in autophagy inhibition, oncogenesis, and the formation of an autophagy-inhibitory Beclin 1/14-3-3/ vimentin intermediate filament complex.
Literature	GENERIF_SUMMARY	AKT1 and MAPK1 expressions were upregulated in SLE patients and correlated with Th17, Treg, TGFB)-2, and Th2-(interleukin (IL)-5)-related genes.
Literature	GENERIF_SUMMARY	Study indicates that FLT3-CBL interaction sites and the AKT pathway as critical mediators of transformation by oncogenic CBL mutants.
Literature	GENERIF_SUMMARY	Phospho-Akt immunoreactivity in prostate cancer shows a relationship to disease severity and outcome, Ki67 and phosphorylated EGFR expression
Literature	GENERIF_SUMMARY	The AKT1 somatic mutants are constitutively active, leading to oncogenic signaling.
Literature	GENERIF_SUMMARY	In cancer cells in vitro or in vivo, RhoB functions as a tumor suppressor that restricts EGF receptor (EGFR) cell surface occupancy as well as Akt signaling
Literature	GENERIF_SUMMARY	Data show that miR-7 inhibits the effects of TLR9 signaling on lung cancer cells through regulation of the PIK3R3/Akt pathway.
Literature	GENERIF_SUMMARY	FAIM modulates IGF-1-induced Akt activation and IRF4 expression and has a role in multiple myeloma cell survival
Literature	GENERIF_SUMMARY	Patients with steatosis had decreased phosphorylation of the insulin kinase AKT1, mediating insulin receptor signaling, and the transcription factor FOXO1.
Literature	GENERIF_SUMMARY	Analyzed the effects of dual inhibition of mTOR and AKT on the proliferation of HCC cell lines & measured feedback activation of each AKT isoforms after mTOR inhibition in

Main Category	Category	Terms
		HCC cell lines and enzymatic activity in primary samples from HCC patients.
Literature	GENERIF_SUMMARY	These data suggested hepatitis C virus inhibited the AKT-TSC-MTORC1 pathway via endoplasmic reticulum stress, resulting in autophagy.
Literature	GENERIF_SUMMARY	Also induced phosphorylation of Akt.
Literature	GENERIF_SUMMARY	Data show that a photodynamic therapy (PDT) dose-dependent upregulation of CRP gene, as well as of PTX3 and ficolin 1 genes in lung tumor A549 cells, and indicate critical role played by PI3K/Akt/AP-1 pathway.
Literature	GENERIF_SUMMARY	A new cell cycle regulatory pathway was identified ain which the miR-302-367 cluster directly down-regulated both cyclin D1 and AKT1 and indirectly up-regulated p27Kip1 and p21Cip1, leading to the suppression of cervical cancer cell proliferation.
Literature	GENERIF_SUMMARY	the present observations show that PKB in conjunction with PIKfyve activates Kir2.1 channels.
Literature	GENERIF_SUMMARY	Activation of GABA(B) receptors significantly inhibits Akt/GSK-3 signaling in a beta-arrestin-dependent pathway.
Literature	GENERIF_SUMMARY	In conclusion, authors have found that the varicella-zoster virus ORF12 protein activates the PI3K/Akt pathway to regulate cell cycle progression.
Literature	GENERIF_SUMMARY	Akt1 may inhibit autophagy by decreasing UVRAG expression, which also sensitizes cancer cells to UV irradiation.
Literature	GENERIF_SUMMARY	Chronic low-dose radiation increases the levels of AKT and acinus proteins via NF-kappaB activation
Literature	GENERIF_SUMMARY	Complexing of beta1 integrin the 70-kDa with CDCP1 fragment induced intracellular phosphorylation signaling, involving focal adhesion kinase-1 (FAK) and PI3 kinase (PI3K)-dependent Akt activation
Literature	GENERIF_SUMMARY	Report shows that ROS levels in PC3 cells are constitutively maintained by DUOX 1 and 2 enzymes, and these ROS positively regulate AKT signalling through inactivating phosphatases, leading to increased resistance to apoptosis.
Literature	GENERIF_SUMMARY	results demonstrate that MTA1 plays an important role in controlling the malignant transformation of prostate cancer cells through the p-AKT/E-cadherin pathway

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	GPR30 mediates the nontranscriptional effect of estrogen on the activation of PI3K/Akt pathway in endometrial cancer cells.
Literature	GENERIF_SUMMARY	pAkt expression significantly correlates with HER2 expression and with a poor prognosis in gastric cancer.
Literature	GENERIF_SUMMARY	Findings indicate driver mutations in AKT1, and extend the number and type of mutations that activate the PI3-kinase pathway in human breast cancers.
Literature	GENERIF_SUMMARY	Our observations suggest that PIK3CA and AKT1 are CS susceptibility genes.
Literature	GENERIF_SUMMARY	Data indicate that the PTEN loss and activation of AKT signaling pathway contributed to erlotinib resistance in EGFR-mutant non-small cell lung cancer (NSCLC) cell line H1650.
Literature	GENERIF_SUMMARY	a new mechanism by which eNOS translocation to mitochondria is regulated by the phosphorylation of eNOS at Ser(617) and Ser(1179) by Akt1 and that this is enhanced when Akt1 becomes nitrated at Tyr(350).
Literature	GENERIF_SUMMARY	Linoleic acid induces opening of connexin26 hemichannels through a PI3K/Akt/Ca(2+)-dependent pathway
Literature	GENERIF_SUMMARY	Coincided with the activation of the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Inhibition of Akt prevented the antiautophagic effect of long exposure to smoking compounds, indicating that smoking-induced late activation of Akt prevents autophagy.
Literature	GENERIF_SUMMARY	resveratrol's anti-tumor actions in prostate cancer could be explained, in part, through inhibition of Akt/miR-21 signaling pathway
Literature	GENERIF_SUMMARY	Inhibition of AKT activity by MK-2206 blocks GRP78 expression in endometrial cancer (EC) cells. SiRNA studies also revealed that knockdown of GRP78 reduces but does not abrogate AKT activity, demonstrating that GRP78 is required for optimal AKT activity.
Literature	GENERIF_SUMMARY	Data suggest mechanisms of carcinogenesis by heavy metals involve increased activation of signaling pathways involving phosphatidylinositol 3-kinase/c-AKT, mTOR (mechanistic target of rapamycin), and EGRF (epidermal growth factor receptor). [REVIEW]
Literature	GENERIF_SUMMARY	Cytoplasmic Skp2 is overexpressed with p-

Main Category	Category	Terms
		Akt1 in invasive breast carcinomas.
Literature	GENERIF_SUMMARY	AKT isoforms 1 and 3 promote basal as well as EGF-induced trophoblast migration.
Literature	GENERIF_SUMMARY	SLIT2 suppresses colon tumor metastasis, and it exerts its suppressive activity against colorectal cancer metastasis by restraining AKT-GSK3beta signaling.
Literature	GENERIF_SUMMARY	In the context of oxidative stress, p-AKT facilitated apoptosis by inducing pVHL function in prostate cells.
Literature	GENERIF_SUMMARY	sEcad contributes to skin carcinogenesis via association with the HER/IGF-1R-family of receptors and subsequent activation of the MAPK and PI3K/Akt/mTOR pathways.
Literature	GENERIF_SUMMARY	We found that AKT activation in serum-deprived conditions is not common among lung cancer cell lines and correlates with activating mutation in either EGFR or PIK3CA, loss of PTEN, or amplification of ERBB2
Literature	GENERIF_SUMMARY	Phosphorylated Akt and VEGF-A are involved in angiogenesis of gastric adenocarcinoma, and Akt activation may contribute to angiogenesis via VEGF-A upregulation.
Literature	GENERIF_SUMMARY	Glioma is formed by active Akt1 alone and promoted by active Rac1 in transgenic zebrafish.
Literature	GENERIF_SUMMARY	A subset of meningiomas lacking NF2 alterations harbored recurrent oncogenic mutations in AKT1 (p.Glu17Lys) and SMO (p.Trp535Leu) and exhibited immunohistochemical evidence of activation of these pathways.
Literature	GENERIF_SUMMARY	the present study demonstrated that Src plays a biologically significant role in tumor proliferation and apoptosis and enhances the cytotoxic effect of temozolomide through AKT supression in glioma.
Literature	GENERIF_SUMMARY	The stimulatory effect of follicle stimulating hormone on Cyp19 and AKT depends on IGF-I and on the expression and activation of the IGF-IR.
Literature	GENERIF_SUMMARY	The expression of phosphorylated AKT was determined by ScanScope XT and ACIS III and compared with the manual analysis by two observers.
Literature	GENERIF_SUMMARY	Meningioma mutations in TRAF7 commonly occurred with a recurrent mutation (K409Q) in KLF4, a transcription factor known for its

Main Category	Category	Terms
		role in inducing pluripotency, or with AKT1, a mutation known to activate the PI3K pathway.
Literature	GENERIF_SUMMARY	The rate of PIK3CA mutations in this series of micropapillary carcinomas is similar to invasive ductal carcinomas; however, there may be an enrichment of AKT1 mutations.
Literature	GENERIF_SUMMARY	kinase-independent function of PI3Kgamma could directly inhibit GSK-3 function by preventing the PP2A-PPMT-1 interaction and that this inhibition of GSK-3 was independent of Akt.
Literature	GENERIF_SUMMARY	the phosphorylation/activation of Akt was significantly suppressed by EESB treatment, whereas that of p53 was enhanced. These results suggest that EESB could effectively induce G1/S arrest in human colon carcinoma cells
Literature	GENERIF_SUMMARY	Studies indicate that DISC1 and AKT were invlved in the regulation of GSK3beta and Wnt signaling paathway.
Literature	GENERIF_SUMMARY	found that Gata1 mutations protect MKs from activated AKT-induced apoptosis
Literature	GENERIF_SUMMARY	Increased endothelial microparticles in mitral valve disease impair mitral valve endothelial cell function by inhibiting the Akt/eNOS-HSP90 signaling pathway.
Literature	GENERIF_SUMMARY	14-3-3 negatively regulates the RGC downstream of the PI3-kinase/Akt signaling pathway
Literature	GENERIF_SUMMARY	Runx2 is a novel and important downstream mediator of the phosphatidylinositol 3-kinase (PI3K)/Akt pathway that is linked to metastatic properties of breast cancer cells.
Literature	GENERIF_SUMMARY	down regulation of fibroblast growth factor receptor 3 (FGFR3) and PI3K/Akt signaling mechanisms leading to inhibition of cell proliferation and induction of molecular mechanisms of apopto
Literature	GENERIF_SUMMARY	neuroprotective effects of angiogenin
Literature	GENERIF_SUMMARY	Studies indicate tht Akt dysregulation is associated with several diseases including infertility and ovarian cancer.
Literature	GENERIF_SUMMARY	Results indicate that the induction of MCP-2/CCL8 by mycobacteria is dependent on the activation of TLR2/PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	RFP-mediated ubiquitination of PTEN modulates its effect on AKT activation.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	These data suggest that HER2/Akt is an important negative regulator of p57 (Kip2), and that p57 restoration in HER2-overexpressing cells can reduce breast tumor growth.
Literature	GENERIF_SUMMARY	astrocytes coexpressing c-Myc and Akt were tumorigenic when implanted into the brain of immunocompetent C57BL/6 mice
Literature	GENERIF_SUMMARY	The correlation between T stage of gastric cancer and the higher expression of CD40, VEGF, AKT, and PI3K, along with lower S100 expression in DC, may provide insights into future targets for more effective immunotherapy for cancer.
Literature	GENERIF_SUMMARY	Akt phosphorylation at S404 targets the precursor prelamin A for degradation. Akt also regulates Lmna transcription.
Literature	GENERIF_SUMMARY	Results show that Akt regulates acetyltransferase activity through CBP phosphorylation, which may contribute to tumorigenesis.
Literature	GENERIF_SUMMARY	In conclusion, our study highlights the potential importance of Akt as a signaling factor in leukemia survival
Literature	GENERIF_SUMMARY	We further showed that SGT1 could regulate Akt signaling pathway by modulating Akt ser473 phosphorylation status.
Literature	GENERIF_SUMMARY	This study supports immune dysfunction and pathways involving AKT1 in the aetiopathophysiology of schizophrenia and their response to antipsychotic medication.
Literature	GENERIF_SUMMARY	Data indicate that tumor regression was seen in three of five patients with BRAF-mutated, low pAKT melanomas.
Literature	GENERIF_SUMMARY	the activation of the calcium/CaM/Akt pathway resulting from stimulation of overexpressed PDGF receptor-beta may contribute to the survival and tumorigenicity of MPNST cells.
Literature	GENERIF_SUMMARY	amino acids stimulate Fru-2,6-P2 synthesis by Akt-dependent PFKFB2 phosphorylation and activation and show how signaling and metabolism are inextricably linked.
Literature	GENERIF_SUMMARY	These results imply that, in glioblastoma cells, constitutive Akt signaling drives RIO kinase overexpression, which creates a feedforward loop that promotes and maintains oncogenic Akt activity through stimulation of mTor signaling
Literature	GENERIF_SUMMARY	Expression of phosphorylated Akt in

Main Category	Category	Terms
		cholesteatoma was significantly increased compared to normal external ear canal. In cholesteatoma, a significant positive association was observed between p-EGFR and p-Akt expression, and between p-Akt and NF-kappaB.
Literature	GENERIF_SUMMARY	sorafenib reduced the stability of the Mcl-1 protein by enhancing its ubiquitination and degradation by the proteasome via the AKT/GSK3beta and the ERK pathways.
Literature	GENERIF_SUMMARY	In a manner that depended on the level of phosphorylated AKT1/2 protein.
Literature	GENERIF_SUMMARY	15-LO-1/15-HETE pathway prevents hepatocellular carcinoma cells from apoptosis and promotes hepatocellular carcinoma progression via a specific intracellular signaling pathway centered by the interaction of Akt with heat shock protein 90
Literature	GENERIF_SUMMARY	The activated Akt is the upstream signaling regulating hPMS2 expression, stability and nuclear localization.
Literature	GENERIF_SUMMARY	Characterization of heparanase-induced phosphatidylinositol 3-kinase-AKT activation and its integrin dependence.
Literature	GENERIF_SUMMARY	Findings indicate a role for Akt in viral entry, link Akt and calcium signaling, and suggest a new target for HSV treatment and suppression.
Literature	GENERIF_SUMMARY	Results suggest that AIB1 promotes BC cell proliferation through the AKT pathway and E2F1. Furthermore, AIB1 overexpression predicts tumour progression in patients with non-muscle-invasive BC.
Literature	GENERIF_SUMMARY	Data indicate that binding affinities for the PH domains of Akt and PHLPP1 were greater than for other PH domain-containing proteins, which may underlie the preferential recruitment of these proteins to membranes containing tocopherols.
Literature	GENERIF_SUMMARY	By activating the Akt pathway, glutathione may prevent free fatty acids-induced human brain vascular endothelial cells (HBVECs)damage, oxidative stress, and apoptosis.
Literature	GENERIF_SUMMARY	Data indicate that CHI3L1 expression was positively correlated with high level of p-Akt, but inversely correlated with expression of E-cadherin in hepatocellular carcinoma (HCC) tissues.
Literature	GENERIF_SUMMARY	hyperactivated ERK and/or AKT signaling

Main Category	Category	Terms
		pathways promoted MED1 overexpression in prostate cancer cells.
Literature	GENERIF_SUMMARY	These data support a potential role for AKT overexpression and the JAK2/STAT3 pathway in the development of cisplatin drug resistance in human gastric cancer cells.
Literature	GENERIF_SUMMARY	miRNA-200c inhibited Akt signaling through its effects on E-cadherin and PTEN, resulting in the inhibition of doxorubicin resistance in breast cancer cells.
Literature	GENERIF_SUMMARY	Data suggest that 17 beta-estradiol enhances the the glucose uptake, activates PI3K/Akt signaling pathway, leading to translocation of glucose transporter 4 to the plasma membrane in an ERalpha-dependent manner.
Literature	GENERIF_SUMMARY	AIF-1 can protect rheumatoid arthritis fibroblast-like synoviocytes from apoptosis induced by NO by upregulating the expression of p-Akt and p-BAD.
Literature	GENERIF_SUMMARY	AKT is essential for enabling EMT in response to upregulation of Snail and Slug.
Literature	GENERIF_SUMMARY	The hypoxia-related activation of AKT could explain the resistance to various treatments including chemotherapy and radiotherapy.
Literature	GENERIF_SUMMARY	These results implied that nicotinamide might be a potential therapeutic agent for human pancreatic cancer treatment through downregulating SIRT1, K-Ras and P-Akt expression.
Literature	GENERIF_SUMMARY	APG dramatically reduced Nrf2 expression at both the messenger RNA and protein levels through downregulation of PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Deptor is induced by the Baf60c-Six4 transcriptional complex and mediates activation of Akt and glycolytic metabolism by Baf60c in a cell-autonomous manner
Literature	GENERIF_SUMMARY	AKT1 and AKT2 have non-redundant roles in the regulation of prostate cancer cell proliferation and migration.
Literature	GENERIF_SUMMARY	The findings presented in this study demonstrate a novel role for PTEN and Akt in controlling centrosome composition and integrity during mitosis.
Literature	GENERIF_SUMMARY	GLI1 contributes to the survival of diffuse large B-cell lymphoma (DLBCL) cells and that this effect occurs in part through promotion of the transcription of AKT

Main Category	Category	Terms
		genes.
Literature	GENERIF_SUMMARY	Data indicate that the chimeric protein is a direct inhibitor of GSK3beta in a Akt phosphorylation-dependent manner.
Literature	GENERIF_SUMMARY	It is a key player in cell-growth signaling in a number of lymphoid malignancies
Literature	GENERIF_SUMMARY	we showed that the synergism between HIV-1 Nef and Kaposi's sarcoma-associated herpesvirus IL-6 in the promotion of angiogenesis and oncogenesis depends on the AKT signaling pathway
Literature	GENERIF_SUMMARY	Data indicate that fucoxanthin and siphonaxanthin down-regulate the phosphorylation of FGF-2-mediated intracellular signaling proteins ERK1/2 and Akt.
Literature	GENERIF_SUMMARY	results suggest that c-Src suppression by PP2 or siRNAs may regulate the progression of cell cycle through AKT/GSK3beta and ERK1/2 pathways
Literature	GENERIF_SUMMARY	Results show that stromal cell-derived factor-1 (SDF-1) enhanced ovarian cancer cell invasion through alphavbeta6 integrin-mediated urokinase-type plasminogen activator (uPA) expression via the p38 MAPK and PI3 K/Akt pathway.
Literature	GENERIF_SUMMARY	Data indicate that the PI3K/Akt pathway plays an important role in the regulation of telomerase reverse transcriptase (hTERT) activity in response to histone deacetylases (HDACs) inhibitor.
Literature	GENERIF_SUMMARY	Results indicate that ADAM17 is upregulated by hypoxia and contributes to hypoxia-induced cisplatin resistance via EGFR/PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	the PI3K/AKT/mTOR signaling pathway, which was overactivated in APA and IHA compared with normal zona glomerulosa, may mediate aldosterone hypersecretion and participate in the development of PA.
Literature	GENERIF_SUMMARY	Akt inhibition promoted glucose uptake in both cell lines, suggesting that Akt inhibition may increase glucose uptake by detaching HK2 from the mitochondria.
Literature	GENERIF_SUMMARY	The AKT/S6K signaling pathway participates in atrial fibrosis and atrial fibrillation in patients with rheumatic mitral valve disease.
Literature	GENERIF_SUMMARY	SNPs within the AKT1 gene are not associated with increased risk for testicular

Main Category	Category	Terms
		germ cell tumors.
Literature	GENERIF_SUMMARY	our results suggest that CAPE suppresses phosphoinositide 3-kinase/AKT/XIAP pathway leading to apoptosis in melanoma tumor cells in vitro and in vivo.
Literature	GENERIF_SUMMARY	Data show that macrophages on gastric and colorectal cancer cells migration through epidermal growth factor receptor and phosphorylation of Akt, c-Src and ERK1/2, and led to an increase of RhoA and Cdc42 activity.
Literature	GENERIF_SUMMARY	Akt1 enhances CA916798 expression through mTOR pathway.
Literature	GENERIF_SUMMARY	AIMP1 peptide promotes the proliferation of BMMSCs by activating the beta-catenin/TCF complex via FGFR2-mediated activation of Akt, which leads to an increase in mesenchymal stem cells in peripheral blood.
Literature	GENERIF_SUMMARY	These findings indicate that bFGF could rapidly activate the protein kinase B activities, enhance the expression of survivin and the proliferation of hepatocellular carcinoma cells via the PI3K pathway.
Literature	GENERIF_SUMMARY	Pre-treatment of the system with Akt inhibitor IV, Bisindolylmaleimide (PKC inhibitor) and src inhibitor I resulted in a significant decrease of the GLP-1 induced angiogenesis
Literature	GENERIF_SUMMARY	These findings for the first time define H2O2 as a signaling molecule that modulate Smad3 linker phosphorylation.
Literature	GENERIF_SUMMARY	Signalling pathways regulating muscle mass in ageing skeletal muscle involve roles in the IGF1-Akt-mTOR-FoxO pathway.
Literature	GENERIF_SUMMARY	Ttreatment with Wit A resulted in inhibition of TGFbetainduced phosphorylation of Akt.
Literature	GENERIF_SUMMARY	These results strongly suggest that DEX protects podocytes by stabilizing the expression and subcellular distribution of CD2AP and by maintaining the expression of phosphor-activated Akt and GSK3beta
Literature	GENERIF_SUMMARY	The existence of an autocrine loop in the IGF-1R/Akt pathway.
Literature	GENERIF_SUMMARY	Data reveal that Erbin is a negative regulator of AKT activation and suggest that Erbin may play a role in breast cancer progression.
Literature	GENERIF_SUMMARY	activation of AKT plays a role in contactin-1-

Main Category	Category	Terms
		mediated downregulation of E-cadherin.
Literature	GENERIF_SUMMARY	A downregulation of FASN effectively inhibits the activity of ""HER2-PI3K/Akt axis"" and alters the malignant phenotype in colorectal cancer cells."
Literature	GENERIF_SUMMARY	report that Mps1/AKT and B-Raf(WT)/ERK signaling form an auto-regulatory negative feedback loop in melanoma cells
Literature	GENERIF_SUMMARY	We also examined 61 of our cases for the presence of PIK3CA, AKT1, PTEN and K-RAS mutations.
Literature	GENERIF_SUMMARY	the latter evident by a reduction in PI3K subunit p100beta levels and decreased phosphorylated Akt
Literature	GENERIF_SUMMARY	Cancer associated E17K mutation causes rapid conformational drift in AKT1 pleckstrin homology (PH) domain.
Literature	GENERIF_SUMMARY	Specific inhibition of Akt activation increased HIV-1 Env-dependent cell death of CCR5+ CD4 T cells.
Literature	GENERIF_SUMMARY	The PI3K/Akt inhibitor LY294002 can enhance chemosensitivity of human gastric cancer to VCR.
Literature	GENERIF_SUMMARY	This study reveled that no association between AKT1 gene variants and schizophrenia.
Literature	GENERIF_SUMMARY	The activated serine/threonine kinase Akt/protein kinase B phosphorylated BTK on two sites prior to 14-3-3 zeta binding.
Literature	GENERIF_SUMMARY	Data indicate that AKT promotes EGFR recycling by phosphorylating and activating PIKfyve.
Literature	GENERIF_SUMMARY	results demonstrated that VEGF enhanced the expression of MRP1, and the PI3K/Akt signaling pathway and SP1 may be involved in this modulation
Literature	GENERIF_SUMMARY	Data suggest over-expression of microRNA-133a in breast cancer cells results in suppression of phosphorylation of Akt (p-Akt) and inhibition of p-Akt nuclear translocation; this involves down-regulation of epidermal growth factor receptor expression.
Literature	GENERIF_SUMMARY	results suggest that Notch1 can regulate PP2A dephosphorylation of critical cellular regulators including AKT, AMPKalpha, and p70S6K
Literature	GENERIF_SUMMARY	IL-6 upregulates PON1 gene expression

Main Category	Category	Terms
		through an AKT/NF-kappaB signaling axis in human hepatocyte-derived HepG2 cell line.
Literature	GENERIF_SUMMARY	Our results integrated TGF-beta1 signals to SKP2 via Akt1 and c-Myc during epithelial-mesenchymal transition.
Literature	GENERIF_SUMMARY	Increased AKT activation is associated with cervical clear cell carcinoma.
Literature	GENERIF_SUMMARY	Etoposide-induced multinucleation in HepG2 cells is dependent on the catalytic activity of Akt. The Akt/PTEN cellular axis is a major determinant of etoposide resistance in these cells.
Literature	GENERIF_SUMMARY	These results suggest that the phosphatidylinositol 3-kinase/Akt signaling pathway plays an important role in beta1 integrin-mediated internalization of Staphylococcus aureus by alveolar epithelial cells.
Literature	GENERIF_SUMMARY	The model showed agreement at several key nodes, involving scaffolding proteins Gab1, Gab2 and their complexes with Shp2. VEGFR2 recruitment of Gab1 is greater in magnitude, slower, and more sustained than that of Gab2. As Gab2 binds VEGFR2 complexes more transiently than Gab1, VEGFR2 complexes can recycle and continue to participate in other signaling pathways.
Literature	GENERIF_SUMMARY	Results show that Endostar may inhibit the endothelial progenitor cells (EPCs) proliferation by reducing the expression of p-protein kinase B.
Literature	GENERIF_SUMMARY	Data indicate that Erk1/2 and Akt activation were increased in CCA tissues when compared with their normal adjacent tissue.
Literature	GENERIF_SUMMARY	Expression of cytoplasmic survivin with activation of the PI3K/Akt pathway identifies a group of acute myeloid leukemia patients with excellent outcome.
Literature	GENERIF_SUMMARY	findings demonstrate that manipulation of the host PI3K/Akt signaling pathway and Nox4 gene expression is a novel mechanism involved in T. gondii survival and proliferation
Literature	GENERIF_SUMMARY	Data indicate that increased activation of PI3K/AKT signaling was observed in the cholangiocarcinoma (CCA) tissues, and suggest that inhibition of PI3K and mTOR activity by the inhibitor NVP-BEZ235 has anti-cancer activity against CCA.
Literature	GENERIF_SUMMARY	Data indicate that nicotinamide N-

Main Category	Category	Terms
		methyltransferase (NNMT) positively correlated with protein kinase B (pAkt) expression and was independent adverse prognosticators of patient survival.
Literature	GENERIF_SUMMARY	results indicate that PTEN loss defines a PI3K/AKT-dependent GCB DLBCL subtype that is addicted to PI3K and MYC signaling and suggest that pharmacologic inhibition of PI3K might represent a promising therapeutic approach in these lymphomas
Literature	GENERIF_SUMMARY	these findings present a novel mechanism of ROS-regulated apoptosis and autophagy that involves Akt-mTOR signaling in prostate cancer cells with mutant p53 exposed to DHA.
Literature	GENERIF_SUMMARY	inhibition of protein biosynthesis induced phosphorylation/activation of AKT and led to phosphorylation of AKT target substrates.
Literature	GENERIF_SUMMARY	mTOR complex 2-Akt signaling at mitochondria-associated endoplasmic reticulum membranes (MAM) regulates mitochondrial physiology.
Literature	GENERIF_SUMMARY	frequent PIK3CA mutations in patients with high-risk HNSCC confined predominantly to the oropharyngeal and sinonasal subsites; for the first time, mutation in AKT1 has been identified in HNSCC
Literature	GENERIF_SUMMARY	Telomere damage and reduced TPP1 dimerization as a result of Akt inhibition was also accompanied by diminished recruitment of TPP1 and POT1 to the telomeres.
Literature	GENERIF_SUMMARY	evidence suggests that many processes attributed to the Akt pathway in myelination depend, at least partly, on mTOR signalling
Literature	GENERIF_SUMMARY	This study shows that PI3K inhibitors only transiently inhibit AKT which limits their antitumor activities.
Literature	GENERIF_SUMMARY	Increases in phosphate AKT1 levels in the spheroid cells.
Literature	GENERIF_SUMMARY	Polyethylene glycated leukemia inhibitory factor antagonist inhibits human blastocyst implantation and triggers apoptosis by down-regulating embryonic AKT1.
Literature	GENERIF_SUMMARY	In Proteus syndrome, cerebriform connective tissue nevi in dermis and epidermal nevi in epidermis are associated with AKT1 mutation.
Literature	GENERIF_SUMMARY	Among 30 breast tumors, 10(33%) had BCSCs with AKT1, HRAS, or PIK3CA

Main Category	Category	Terms
		mutations. A single mutation was detected in AKT1.Tumors in which BCSCs have defects in PI3K/Akt signaling are significantly more likely to manifest nodal metastases.
Literature	GENERIF_SUMMARY	SUMOylation of Akt is associated with cell growth and tumorigenesis
Literature	GENERIF_SUMMARY	Increase in the expression of AKT was associated with colorectal carcinoma.
Literature	GENERIF_SUMMARY	AKT1 E17K is a bona fide oncogene in a human luminal breast cancer context. Distinct PI3K pathway mutations confer differential sensitivity to drugs targeting the pathway at different points and by distinct mechanisms.
Literature	GENERIF_SUMMARY	epigenetic downregulation of RUNX3 can induce docetaxel resistance in human lung adenocarcinoma cells by activating AKT signaling.
Literature	GENERIF_SUMMARY	Hyperactivation of the Akt pathway in Teff cells from children with lupus nephritis is associated with reduced induction of TRAF6 and up-regulation of OX40, which may cause Teff cell resistance to Treg cell-mediated suppression.
Literature	GENERIF_SUMMARY	These findings suggest that oxidative stress might be associated with AKT activation and hTERT induction and might confer a high-risk status for gastric carcinogenesis.
Literature	GENERIF_SUMMARY	Junctional localization of Akt-1 correlates with regions of high unidirectional shear stress whereas a perinuclear localization correlates with areas of disturbed flow in endothelial cells.
Literature	GENERIF_SUMMARY	activation of RAF1-MEK1-ERK/AKT axis may determine the resistance of non-small cell lung cancer cell lines bearing wild type EGFR to erlotinib
Literature	GENERIF_SUMMARY	detected and measured all three AKT isoforms 1, 2 and 3 to enable the study of the multiple and variable roles that these isoforms play in AKT breast tumorigenesis
Literature	GENERIF_SUMMARY	Bcr-Abl activates AURKA and AURKB in chronic myeloid leukemia cells via AKT signaling.
Literature	GENERIF_SUMMARY	Data suggest that a combination of gamma-secretase inhibitor GSI application and 532 nm low-power laser irradiation (LLI) can increase cell proliferation via Akt activation.
Literature	GENERIF_SUMMARY	concurrent expression of AKT and hTERT at

Main Category	Category	Terms
		high levels in gastric canceris associated with a poor prognosis.
Literature	GENERIF_SUMMARY	Adiponectin inhibits neutrophil apoptosis via activation of AMP kinase, PKB and ERK 1/2 MAP kinase.
Literature	GENERIF_SUMMARY	Data indicate that ZNF703 overexpression in MCF-7 breast cancer cells activated the Akt/mTOR signaling pathway, downregulated ERalpha, and reduced the antitumor effect of tamoxifen.
Literature	GENERIF_SUMMARY	Protumorigenic actions of S100A2 involve regulation of PI3/Akt signaling and functional interaction with Smad3.
Literature	GENERIF_SUMMARY	Activation of AKT signaling promotes epithelial-mesenchymal transition and tumor growth in colorectal cancer cells.
Literature	GENERIF_SUMMARY	Data indicate that PYK2 N-terminal domain interacting receptor 1 (Nir1) could induce epithelial-mesenchymal transition by stabilising Snail via the PI3K/Akt/GSK3beta/Snail signalling pathway through binding to CCL18.
Literature	GENERIF_SUMMARY	a new role of AKT in the MM promotion of osteoclast formation and bone osteolysis through, at least in part, the ATF4-dependent up-regulation of RANK expression in osteoclast precursors.
Literature	GENERIF_SUMMARY	The role of Akt1 as a regulator of alternative splicing and cell cycle progression found that SUMOylation of Akt is required for the activity of this kinase in these different cellular processes.
Literature	GENERIF_SUMMARY	PCN-mediated oxidative stress activates the EGFR-PI3K-AKT/MEK1/2-ERK1/2 MAP kinase signaling pathway, leading to nuclear NRF2 translocation and ARE responsiveness in pulmonary epithelial cells.
Literature	GENERIF_SUMMARY	results suggest that variation at tagSNP rs3730358 in the AKT1 gene conferred an increased association to develop late onset depression.
Literature	GENERIF_SUMMARY	Continuous FLX treatment after neurogenesis elevated the survival rate of differentiated neural cells probably by enhanced induction of Akt1 phosphorylation
Literature	GENERIF_SUMMARY	overexpression of 4EBP1, p70S6K, Akt1 or Akt2 could promote the Coxsackievirus B3-induced apoptosis.
Literature	GENERIF_SUMMARY	AKT1 is cardioprotective to the heart by supporting its physiological growth and

Main Category	Category	Terms
		function.
Literature	GENERIF_SUMMARY	Studied the cytotoxic effect of NO donors on ovarian cancer cell lines, as well as their effect on posttranslational modification of STAT3 and AKT proteins in these cells.
Literature	GENERIF_SUMMARY	Orai3 constitutes a native store-operated calcium entry pathway in non small cell lung adenocarcinoma that controls cell proliferation and cell cycle progression, likely via Akt pathway.
Literature	GENERIF_SUMMARY	Loss of cavin-3 promotes Akt signaling through suppression of EGR1 and PTEN
Literature	GENERIF_SUMMARY	suggesting that the changes in AKT/BCL2 signaling molecules might mediate growth and anti-apoptosis processes at least in part, thus increasing the survival potential of Congenital Giant Nevi melanocytes and maintaining their integrity
Literature	GENERIF_SUMMARY	Akt can switch the TopBP1 function from checkpoint activation to transcriptional regulation by regulating its quaternary structure.
Literature	GENERIF_SUMMARY	The Notch1 is not only down regulated the Shh pathway, but also inhibited recombinant Shh mediated erythropoiesis.
Literature	GENERIF_SUMMARY	We observed strong up-regulation of SFRP1 expression in all meningiomas with AKT1E17K mutation
Literature	GENERIF_SUMMARY	angiotensin II signaling in human preadipose cells involves an ERK1,2-dependent attenuation of Akt activity
Literature	GENERIF_SUMMARY	The AKT cascade is induced in breast epithelial and tumor cells following exposure to heavy metals in cigarette smoke.
Literature	GENERIF_SUMMARY	The expression of miR-33a on TGF-beta1-induced hepatic stellate cell activation may be modulated via the activation of PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Induced apoptosis mainly via the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	The AKT/mTOR signaling pathway is highly activated in ALK + ALCL patients and targeting the AKT/mTOR signaling pathway might confer a great therapeutic potential in ALCL.
Literature	GENERIF_SUMMARY	A novel AKT1/NF-kappaB/Notch1/PTEN axis has an important role in the development of chemoresistance in gastric

Main Category	Category	Terms
		cancer.
Literature	GENERIF_SUMMARY	Its signal plays a crucial role in tumor genesis and tumor progression by promoting cell proliferation and inhibiting apoptosis.
Literature	GENERIF_SUMMARY	DHA not only displaced several raft-associated onco-proteins, including EGFR, Hsp90, Akt, and Src, from the rafts but also decreased total levels of those proteins via multiple pathways, including the proteasomal and lysosomal pathways
Literature	GENERIF_SUMMARY	Increased APN may contribute to hepatocellular carcinoma (HCC) at least in part through its activation of AKT signalling and may serve as a prognostic factor in HCC
Literature	GENERIF_SUMMARY	Found progressive loss of PTEN and increase in EGFR, TGF-alpha, P-AKT expression from benign samples to non small cell lung cancer. Changes were correlated to differentiation extent of cancer tissue, metastasis of lymph nodes and histological classification.
Literature	GENERIF_SUMMARY	This study identified a set of biomarkers of Akt inhibition that can be used in the clinical setting to assess target engagement.
Literature	GENERIF_SUMMARY	Results found that DRR expression leads to elevated AKT activation by recruiting AKT to focal adhesios in a manner dependent on SRC-family kinases (SFKs) and cell adhesion.
Literature	GENERIF_SUMMARY	Inhibition of AKT decreased the phosphorylation level of MDM2.
Literature	GENERIF_SUMMARY	These findings suggested that inhibition of VCP could suppress osteosarcoma cells invasion and migration through down-regulating AKT/NF-kB signaling pathway.
Literature	GENERIF_SUMMARY	TRAF4 attenuation impaired glucose metabolism by inhibiting expression of Glut1 and HK2 mediated by the Akt pathway.
Literature	GENERIF_SUMMARY	Data show that modin directly inhibited TWIST1 expression, upregulated E-cadherin, and inhibited TWIST1-induced epithelial-mesenchymal transition (EMT) by inhibiting the beta-catenin and Akt pathways.
Literature	GENERIF_SUMMARY	ROR2 is an epigenetically inactivated tumor suppressor gene that antagonizes both beta-catenin and AKT signaling in multiple various carcinomas.
Literature	GENERIF_SUMMARY	we investigated the effect of S(+)-ibuprofen on neuroblastoma cell lines, focusing on the

Main Category	Category	Terms
		expression of the MYCN, MYC, AKT, p53 proteins and the favorable neuroblastoma genes in vitro as biomarkers of malignancy
Literature	GENERIF_SUMMARY	In a Chinese cohort, no significant difference was found between osteosarcoma patients and controls in the genotype of AKT rs6973569. However, after stratified analysis, the genotype AA of AKT rs6973569 carried a higher risk of osteosarcoma metastasis.
Literature	GENERIF_SUMMARY	These results imply that curcumin inhibits Ab-induced tau hyperphosphorylation involving PTEN/Akt/GSK-3b pathway.
Literature	GENERIF_SUMMARY	FOXO1 inhibits cisplatin-induced apoptosis in gastric cancer cells via activating PI3K/Akt pathway
Literature	GENERIF_SUMMARY	These findings suggest that Bcl-xl overexpression exerts an anti-apoptotic and protective effect on EC function. The Akt/eNOS signaling pathway is probably involved in these processes.
Literature	GENERIF_SUMMARY	Data indicate that miltefosine and erufosine induced dephosphorylation of Akt and phosphorylation of JNK in cutaneous T-cell lymphoma cell line.
Literature	GENERIF_SUMMARY	ERK, AKT, and GSK-3beta have roles in boldine-induced cell cycle arrest and apoptosis in T24 human bladder cancer cell line
Literature	GENERIF_SUMMARY	HIV-1 Tat-mediated induction of CCL5 in astrocytes involves NF-kappaB, AP-1, C/EBPalpha and C/EBPgamma transcription factors and JAK, PI3K/Akt and p38 MAPK signaling pathways
Literature	GENERIF_SUMMARY	Cur-NPs upregulated the protein expression levels of Bad and downregulated the protein expression level of p-Akt in U2OS cells
Literature	GENERIF_SUMMARY	Experimental validation indicate that AKT1, AKT2 and AKT3 proteins may all be novel unfavorable prognostic factors for patients with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Inhibition of miR-223 was found to maintain the undifferentiated state of hESCs, while addition of miR-223 induced differentiation. Furthermore, these effects were found to be likely dependent on IGF-1R/Akt signaling
Literature	GENERIF_SUMMARY	Data indicate that phospholipase D2 (PLD2) promotes autophagy through regulation of Akt in glioblastoma cells.
Literature	GENERIF_SUMMARY	Estrogen receptor and PI3K/Akt signaling

Main Category	Category	Terms
		pathway involvement in S-(-)equol-induced activation of Nrf2/ARE in endothelial cells.
Literature	GENERIF_SUMMARY	Akt activation by ODC+COX-2 over-expression.
Literature	GENERIF_SUMMARY	Chromogranin A and its fragments regulate metastatic small intestinal neuroendocrine neoplasm proliferation via the AKT pathway.
Literature	GENERIF_SUMMARY	A novel AKT1 mutant amplifies an adaptive melanoma response to BRAF inhibition.
Literature	GENERIF_SUMMARY	Findings show that PPP1R1B-STARD3 fusion transcript has a key role in subsets of gastric cancers through the activation of PI3K/AKT signaling.
Literature	GENERIF_SUMMARY	The PDGFRbeta-AKT pathway plays a critical role in the development of cisplatin resistance in testicular tumoral cells.
Literature	GENERIF_SUMMARY	The activation level of the EEF1A2/PI3K/AKT/mTOR/MDM4 axis significantly influences the survival probability of hepatocellular carcinoma patients.
Literature	GENERIF_SUMMARY	Expression of AKT, 4EBP1, PTEN, mTOR was assessed in 19 gastrointestinal Kaposi sarcoma biopsies by immunohistochemistry (17 patients).
Literature	GENERIF_SUMMARY	downregulation of Rap2a promoted glioma migration and invasion, and raised the phosphorylation level of AKT
Literature	GENERIF_SUMMARY	miR-221/222 confers radioresistance in glioblastoma cells through activating Akt independent of PTEN status.
Literature	GENERIF_SUMMARY	These data suggest a dual role for Akt1 in tumor cell migration and invasion and highlight the cell type-specific actions of Akt1 kinases in the regulation of cell motility.
Literature	GENERIF_SUMMARY	These data support a model in which activated Akt enhances rRNA synthesis both by preventing TIF-IA degradation and phosphorylating CK2alpha, which in turn phosphorylates TIF-IA.
Literature	GENERIF_SUMMARY	Data suggest that AKT1 phosphorylation in response to insulin signaling is upregulated in isolated monocytes (but not in isolated T-cells) and is independent of insulin resistance of subject (that is, same in normal, prediabetes, or diabetes type 2).
Literature	GENERIF_SUMMARY	Although Akt supports survival of melanocytes, autophagy inhibition specifically targeted the metastatic

Main Category	Category	Terms
		melanoma cells, thus indicating a stage-specific requirement for Akt and autophagic flux, throughout melanoma progression.
Literature	GENERIF_SUMMARY	Dengue virus 2 subgenomic RNA induces apoptosis through the Bcl-2-mediated PI3k/Akt signaling pathway.
Literature	GENERIF_SUMMARY	in a defined subset of bone and soft tissue tumors, including benign tumors, Akt was frequently overexpressed and activated, and AKT1/2 copy number was increased
Literature	GENERIF_SUMMARY	Data suggest that the ribosomal S6 kinase
Literature	GENERIF_SUMMARY	AKT/beta-catenin/Snail signaling pathway is mechanistically associated with cancer stem cell-like properties and epithelial-mesenchymal transition features of A549/CDDP cells, and thus, this pathway could be a novel target for the treatment of NSCLC.
Literature	GENERIF_SUMMARY	Akt signaling controls growth and chemical-induced apoptosis in Hepatocellular carcinoma and p-AKT may be a potential target for therapeutic interventions in Hepatocellular carcinoma patients.
Literature	GENERIF_SUMMARY	Endogenous or exogenous PGD2 induces CRTH2-dependent apoptosis in human differentiated osteoclasts; beta-arrestin-1, ERK1/2, and Akt, but not IKK2/NF-kappaB are probably implicated in the signaling pathways.
Literature	GENERIF_SUMMARY	Our data suggest that Akt/Ezrin Tyr353/NF-kappaB pathway regulates EGF-induced EMT and metastasis inTSCC, and Ezrin may serve as a therapeutic target to reverse EMT in tongue cancers and prevent TSCC progression
Literature	GENERIF_SUMMARY	blocking PI3K/Akt signaling pathway by LY294002 or Akt siRNA could remarkably inhibit the PAK1 activation and cell invasion
Literature	GENERIF_SUMMARY	Akt/mTOR was activated in 50% of the cases of Solitary fibrous tumors and was associated with receptor tyrosine kinases, which were phosphorylated at different rates. The Akt-mTOR pathway may be involved in the tumorigenesis of Solitary fibrous tumors.
Literature	GENERIF_SUMMARY	Ghrelin promotes intestinal epithelial cell proliferation through PI3K/Akt pathway and EGFR trans-activation both converging to ERK 1/2 phosphorylation.
Literature	GENERIF_SUMMARY	The findings from our study suggest the existence of a positive feedback regulation between Akt phosphorylation and FASN

Main Category	Category	Terms
		expression and that this loop may play an important role in the malignant phenotype of osteosarcoma cells.
Literature	GENERIF_SUMMARY	of mTORC1 signaling to 4E-BP1 requires the coordinated activity of two tumor suppressors, p53 and p63. In contrast, suppression of S6K1 and ribosomal protein S6 phosphorylation by DNA damage is Akt-dependent.
Literature	GENERIF_SUMMARY	CHIP-modulated AKT/FoxO/Bim signaling pathway induces apoptosis resistance in breast cancer.
Literature	GENERIF_SUMMARY	augmentation of proinflammatory chemokines CXCL1/2, by potentiating NF-kappaB activation through EGFR-transactivated Akt, contributes to CXCR2-driven ovarian cancer progression
Literature	GENERIF_SUMMARY	Data indicate that AKT inhibitor MK-2206 plus carboplatin and paclitaxel, docetaxel, or erlotinib was well-tolerated, with early evidence of antitumor activity.
Literature	GENERIF_SUMMARY	1) NHE3 basal activity is regulated by a signaling complex that is controlled by sequential effects of two kinases, Akt and GSK-3, which act on a Ser cluster in the same NHE3 C-terminal domain that binds ezrin
Literature	GENERIF_SUMMARY	The suppression of K-ras expression by siRNA interfered with this synergism and inhibited both EGFR and Akt activity in A549 cells.
Literature	GENERIF_SUMMARY	EGCG inhibited growth by affecting the cell cycle and induced apoptosis in different HCC cells by downregulating PI3K/AKT activity
Literature	GENERIF_SUMMARY	These findings establish that lysosomal accumulation of Akt and Phafin2 is a critical step in the induction of autophagy via an interaction with phosphatidylinositol 3 phosphate.
Literature	GENERIF_SUMMARY	Platelet-derived growth factor-C (PDGF-C) induces anti-apoptotic effects on macrophages through Akt and Bad phosphorylation.
Literature	GENERIF_SUMMARY	these findings suggest that MHY-449 induces apoptosis via downregulation of the Akt/FoxO1 and activation of ERK in androgen-independent, p53-null and PTEN-negative PC3 human prostate cancer cells.
Literature	GENERIF_SUMMARY	AFP can promote the proliferation of hepatoma cells via activation of PI3K/Akt

Main Category	Category	Terms
		signaling pathway.
Literature	GENERIF_SUMMARY	Data reveal a mechanism by which BRAF(V600E) and PI3K signaling cooperate to regulate melanoma proliferation through AKT-independent effects on protein translation.
Literature	GENERIF_SUMMARY	Data show that sangivamycin induced the apoptosis of primary effusion lymphoma (PEL) cells by activating caspase-7 and -9 and suppressed the phosphorylation of Erk1/2 and Akt, thus inhibiting activation of the proteins.
Literature	GENERIF_SUMMARY	AKT also associated with a nuclear sirtuin, Sirt1; however, inhibition of PI3K resulted in dissociation from Sirt1 and increased association with Sirt2.
Literature	GENERIF_SUMMARY	Akt is a molecular switch that regulates the NETosis-apoptosis axis
Literature	GENERIF_SUMMARY	IWS1, an RNA processing regulator, is phosphorylated by Akt3 and Akt1 at Ser720/Thr721 in lung cancer
Literature	GENERIF_SUMMARY	mutations may confer resistance to epidermal growth factor receptor-tyrosine kinase inhibitors in lung adenocarcinoma patients
Literature	GENERIF_SUMMARY	molecular or pharmacologic inhibition of the Lyn-PI3K/AKT pathway, markedly increased the sensitivity of the otherwise chemoresistant Cbl mutant-JMML cells to chemotherapeutic agents currently used in the treatment of JMML patients
Literature	GENERIF_SUMMARY	Altogether, our study uncovered the existence of a novel Akt-HSF-1 signaling axis that leads to Slug upregulation and EMT, and potentially contributes to progression of HER2-positive breast cancer
Literature	GENERIF_SUMMARY	miR-375 negatively regulates the expression of 3-phosphoinositide-dependent protein kinase 1 (PDK1) by directly targeting the 3'UTR of the PDK1 transcript, throught Akt signaling pathway.
Literature	GENERIF_SUMMARY	the results show that EphA2 endows invasiveness of glioma stem cells in vivo in cooperation with Akt and regulates glioma stem cell properties
Literature	GENERIF_SUMMARY	When we tested the effects of a panel of signal transduction modifiers on the signal transduction pathways of MDA in oxLDL-treated HCAECs, we found that MDA-induced cytotoxicity was mediated partly through the Akt pathway.

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Literature	GENERIF_SUMMARY	Studied the expression of phosphatidylinositol 3 kinase (PI3-K), protein kinase B (PKB) and glycogen synthase kinase 3 beta (GSK-3 beta) in skeletal muscle tissue of gestational diabetes mellitus patients.
Literature	GENERIF_SUMMARY	AKT-induced uptake of [(18)F]FDG was found to be HIF-1alpha independent in breast cancer cell lines.
Literature	GENERIF_SUMMARY	Modulation of AKT by type II TGFbeta receptor interacting protein (TRIP)-1-mediated enhancement of AKT phosphorylation is the implicated pathway.
Literature	GENERIF_SUMMARY	Data indicate that phosphorylation of Carma1 down-stream of TCR/CD28-dependent NF-kappaB induction is regulated in part by Akt.
Literature	GENERIF_SUMMARY	Mechanistically, we found that protein kinase B (PKB, also known as Akt) signaling pathway was a functional link in TrkB-induced anoikis suppression in CRC cells.
Literature	GENERIF_SUMMARY	conclude that downregulation of BIM represents a key survival signal downstream of oncogenic MYC and PI3K-AKT signaling in treatment-resistantT-cell acute lymphoblastic leukemia
Literature	GENERIF_SUMMARY	results of the experiments showed that hCGbeta action leads to the increase in ERK and AKT kinases phosphorylation in cancer cells and indicate that these biological effects can be achieved independently of human chorionic gonadotropin presence
Literature	GENERIF_SUMMARY	Reduces phosphorylation of AKT.
Literature	GENERIF_SUMMARY	Data show the expression of N-myc downstream-regulated gene 2 (NDRG2) is frequently downregulated in adult T-cell leukaemia-lymphoma, resulting in enhanced phosphorylation of PTEN and activation of the phosphatidylinositol 3-kinase (PI3K)-AKT pathway.
Literature	GENERIF_SUMMARY	Selective inhibition of mTORC1 leads to hyperactivation of the AKT/GSK-3beta pathway, inducing E-cadherin repressor complexes and epithelial mesenchymal transition.
Literature	GENERIF_SUMMARY	suggest that BMP-2 exerts an inhibitory effect on the growth and migration of hepatocellular carcinoma cells, possibly via a blockade of PI3K/AKT signaling
Literature	GENERIF_SUMMARY	Phosphorylated Akt plays an important role in regulating the expression level of cyclin

Main Category	Category	Terms
		B1 by interacting with AR and increasing the transcriptional activity of AR.
Literature	GENERIF_SUMMARY	Data indicate that MK-2206 at maximum tolerated doses only modest decreases in proto-oncogene c-Akt (AKT).
Literature	GENERIF_SUMMARY	The Akt/mTOR pathway is activated by TGFbetaRII through Kv2.1 regulation by GDF15 in cerebellar granule cells.
Literature	GENERIF_SUMMARY	Authors showed that the PI3K/Akt pathway negatively regulated the expression of DAF on the epithelial cell surface and thus inhibited the adhesion of Dr(+) E. coli to epithelial cells.
Literature	GENERIF_SUMMARY	results provide a significant insight into the potential anticarcinogenic action of DNAzyme against AKT1 which might be used as a valuable therapy to leukemia
Literature	GENERIF_SUMMARY	Data suggest that the expression of ATP-binding cassette sub-family G member 2 (ABCG2) and the side population may be regulated by phosphatase and tensin homologue deleted on chromosome -10 (PTEN) through the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Data indicate the potential of miR-21 and the PTEN/Akt pathway as therapeutic targets in intervertebral disc degeneration.
Literature	GENERIF_SUMMARY	Data suggest that activation/ phosphorylation/post-translational protein processing of AKT plays activ role in establishment of ectopic endometrial tissues as in endometriosis; phosphorylation of AKT at serine-473 appears to be involved.
Literature	GENERIF_SUMMARY	Our data indicate that activation of the PI3K/ AKT pathway evidenced on the protein level might be a valuable prognostic marker to stratify patients for their risk of tumour recurrence.
Literature	GENERIF_SUMMARY	Suggest the combination of AKT, mTORC1 and/or MEK inhibitors would be a promising therapeutic strategy for liver cancer treatment.
Literature	GENERIF_SUMMARY	Data shows that the existence of an alternative promoter driving the expression of the unique AKT1m transcript may provide a mechanism by which the levels of AKT1 can be temporally and spatially regulated at particular physiological states such as cancer.
Literature	GENERIF_SUMMARY	In resistance to oxidative stress cells, Akt expression is decreased but its

Main Category	Category	Terms
		phosphorylation enhanced compared with MCF-7 breast cancer cells.
Literature	GENERIF_SUMMARY	P2Y2 receptor fails to repress IL-6 expression by valve interstitial cells through Akt in calcific aortic valve disease.
Literature	GENERIF_SUMMARY	genetic variants in the PTEN, AKT1, MDM2 and p53 tumor suppressor-oncoprotein network may play roles in mediating the susceptibility to NPC in Chinese populations
Literature	GENERIF_SUMMARY	these data provide evidence in support of a significant role for HGF-induced c-Met/PI3k/Akt signaling and NADPH oxidase activation in lamellipodia formation and motility of lung endothelial cells.
Literature	GENERIF_SUMMARY	Low expression of phospho-AKT is associated with renal carcinoma.
Literature	GENERIF_SUMMARY	The present experiments demonstrated a critical role for the GPRC6A-ERK1/2 and PI3K/Akt signaling pathway in arginine-mediated fibroblast survival.
Literature	GENERIF_SUMMARY	The data indicate that PI3Kgamma has a critical role in cAMP-mediated inflammatory hypernociception and analgesic signalling via mu-opioid receptors and PI3K/Akt in neuronal cells.
Literature	GENERIF_SUMMARY	miR-9500 repressed tumorigenesis and metastasis by targeting Akt1.
Literature	GENERIF_SUMMARY	Low protein expression of p-Akt308 was associated with improved DFS and OS among patients treated with hormonal therapy following adjuvant chemotherapy. Coexpression of EGFR and p-mTOR was associated with worse OS.
Literature	GENERIF_SUMMARY	Our results showed that although PKB/Akt was not associated with survival in NSCLC patients, it may be a potential therapeutic target for NSCLC
Literature	GENERIF_SUMMARY	In this review, we discuss the emerging evidence for a functional relationship between the IGF-I/PI3K/AKT/mTOR pathway and ASDs, as well as a possible role of this signaling pathway in the diagnosis and treatment of ASDs. [review]
Literature	GENERIF_SUMMARY	CaMK4-dependent activation of AKT/mTOR and CREM-alpha underlies autoimmunity-associated Th17 imbalance.
Literature	GENERIF_SUMMARY	AKT activation upregulates GATA-1 and FOXO3 transcriptional pathways with resulting inhibition of apoptosis, modulation of cell cycle, and enhanced maturation of

Main Category	Category	Terms
		erythroid cells.
Literature	GENERIF_SUMMARY	the ErbB3/4 ligand heregulin (HRG) reduced both ErbB2 and ErbB3 mRNA and protein levels in human breast cancer cell lines.
Literature	GENERIF_SUMMARY	these results suggest that betaarrestin1 regulates rasgrf2 gene expression and Rac activation to affect membrane protrusion and cell migration and invasion.
Literature	GENERIF_SUMMARY	These data suggest that embelin can inhibit pancreatic cancer growth, angiogenesis and metastasis by suppressing Akt and Shh pathways, and can be developed for the treatment and/or prevention of pancreatic cancer.
Literature	GENERIF_SUMMARY	We examined the response of 80 samples of primary cells from acute myeloid leukemia patients to selective inhibitors of the phosphatidylinositol 3 kinase (PI3K)/Akt/mammalian target of rapamycin (mTOR) axis.
Literature	GENERIF_SUMMARY	Pretreatment of T24 cells with the antioxidant N-acetylcysteine (NAC) significantly inhibited activation of caspase-3 and MAPK-p38 and prevented inactivation of Akt and Bcl-2
Literature	GENERIF_SUMMARY	Data show that the crosstalk between the Rb-AKT-signaling pathways forms this switch by controlling the overlapping functions of FoxO3a and FoxM1 transcription factors in cultured fibroblasts.
Literature	GENERIF_SUMMARY	SUMO induces the activation of AKT, whereas, conversely, down-modulation of the SUMO machinery diminishes AKT activation and cell proliferation.
Literature	GENERIF_SUMMARY	Findings indicate that dual phosphorylation of PRAS40 by Akt and mTORC1 promotes formation of a nuclear-specific PRAS40- and RPL11-containing complex distinct from mTORC1 that inhibits the RPL11-HDM2-p53 pathway.
Literature	GENERIF_SUMMARY	The results have provided evidence for clinical investigation of GDC0068, a novel ATP-competitive pan-Akt inhibitor, as the second-line treatment after the failure of sorafenib-medicated molecular targeted therapy for advanced HCC
Literature	GENERIF_SUMMARY	Data indicate PI3K/Akt signaling as an essential pathway regulating the distribution of nucleolin.
Literature	GENERIF_SUMMARY	This effect correlated with depletion of principle proteins of the Akt/mTOR and

Main Category	Category	Terms
		MAPK survival and proliferation pathways.
Literature	GENERIF_SUMMARY	The survival of CSF1R(pos) cells requires active AKT (v-akt murine thymoma viral oncogene homolog 1) signaling, which contributed to increased levels of nuclear, transcriptionally competent beta-catenin.
Literature	GENERIF_SUMMARY	PAR4-P2Y12 association supports arrestin-mediated sustained signaling to Akt.
Literature	GENERIF_SUMMARY	Protein expression of PI3K, Akt and survivin were significantly associated with the development, progression and metastasis of gastric cancer and may have value as diagnostic and prognostic markers in gastric cancer
Literature	GENERIF_SUMMARY	AKTi combined with BRAFi-based therapy may benefit patients with tumors harboring BRAF mutations and particularly PTEN deletions or AKT mutations.
Literature	GENERIF_SUMMARY	Aberrant activation of the PI3K/Akt/mTOR axis acts in parallel to RCAN1 in phosphorylating tau, in DS and DS/AD.
Literature	GENERIF_SUMMARY	The GG allele in the 3' side of AKT1 is closely related to clinical outcomes of gastric cancer.
Literature	GENERIF_SUMMARY	proteomic view to characterize the effect of chitosan nanoparticle to hepatic cells
Literature	GENERIF_SUMMARY	TEIF is a downstream effector in EGF/PI3K/Akt signaling.
Literature	GENERIF_SUMMARY	CK2-dependent phosphorylation is therefore a crucial event which, discriminating between Akt1 and Akt2, can account for different substrate specificities.
Literature	GENERIF_SUMMARY	Fundamental role of the PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	Data show that salvianolic acid B protects endothelial progenitor cells against oxidative stress by modulating Akt/mTOR/4EBP1, p38 MAPK/ATF2, and ERK1/2 signaling pathways.
Literature	GENERIF_SUMMARY	QSN-10c dose-dependently suppressed the phosphorylation of AKT.
Literature	GENERIF_SUMMARY	Data demonstrate that p34SEI-1 induces the activation of either AKT or ILK signaling on HER2/neu expression status.
Literature	GENERIF_SUMMARY	Furthermore, we confirmed that these phenotypic changes are associated with the inactivation of AKT and ERK. Our findings suggest that PTK7 has different oncogenic

Main Category	Category	Terms
		roles in organs and target tumors.
Literature	GENERIF_SUMMARY	Results suggest that cyclin A2 overexpression may drive cancer by promoting AKT-dependent proliferation and survival.
Literature	GENERIF_SUMMARY	Hyperactivation of Akt/mTOR and deficiency in tuberin increased the oxidative DNA damage in kidney cancer patients with diabetes.
Literature	GENERIF_SUMMARY	A high-fat medium containing 100 muM of palmitate stimulates proliferation in PNT1A cells by decreasing the activation of AMPK and increasing activation of AKT after longer exposure time.
Literature	GENERIF_SUMMARY	After overexpression and knockdown of XIAP, the AKT, pAKT, JNK2 and GSK3beta levels were determined in periodontal ligament cells.
Literature	GENERIF_SUMMARY	Silencing miR-21 in radioresistant NSCLC A549 cells sensitized them to IR by inhibiting cell proliferation and enhancing cell apoptosis through inhibition of PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	FAM3A plays crucial roles in the regulation of glucose and lipid metabolism in the liver, where it activates the PI3K-Akt signaling pathway by way of a Ca(2+) /CaM-dependent mechanism
Literature	GENERIF_SUMMARY	phenotype variations that were believed to have the highest impact in AKT1 gene
Literature	GENERIF_SUMMARY	AKT-mTOR pathway is a novel component of oncogene induced DNA damage response in immortalized 'normal-like' breast cells and its suppression may contribute to growth arrest and arrest of the breast tumorigenesis
Literature	GENERIF_SUMMARY	Results of Western blotting demonstrated that the combined treatment-induced A375 cells growth inhibition was achieved by triggering reactive oxygen species-mediated DNA damage involving inactivation of AKT and MAPKs.
Literature	GENERIF_SUMMARY	besides HIF-1alpha, several other pathways, including PI3K/Akt, may be involved in the response to hypoxia of immature and, more specifically, of mature DC to sustain their trafficking and functions within hypoxic microenvironments.
Literature	GENERIF_SUMMARY	During free radical stress, hUCBC paracrine factors activate myocyte Akt, which increases myocyte viability by decreasing

Main Category	Category	Terms
		activation of death-promoting protein kinases JNK and p38.
Literature	GENERIF_SUMMARY	Dual MAPK and PI3K/Akt inhibition was detected.
Literature	GENERIF_SUMMARY	demonstrate that ShcA-dependent activation of AKT, but not the RAS/MAPK pathway, induces VEGF production by bolstering VEGF mRNA translation
Literature	GENERIF_SUMMARY	The effect of N-(3-oxododecanoyl)-L-homoserine lactone on mucin 3 and Akt expression in differentiated and undifferentiated Caco 2 cells is reported.
Literature	GENERIF_SUMMARY	Correlation between PDZK1, Cdc37, Akt and breast cancer malignancy
Literature	GENERIF_SUMMARY	the important role of Akt during the process of invasion and metastasis in esophageal squamous cell cancer
Literature	GENERIF_SUMMARY	MCP-1 was shown to have a role in activation of the ER1 by phosphorylating Ser167 via the PI3K/Akt/mTORC1 signaling pathway
Literature	GENERIF_SUMMARY	IL-10 apparently inhibits fibrosis by activating AKT and STAT3 phosphorylation downstream of the IL-10 receptor
Literature	GENERIF_SUMMARY	Further experiments demonstrated that phosphatidylinositide 3-kinases/protein kinase B/mammalian target of rapamycin/p70S6 kinase (PI3K/AKT/mTOR/p70S6K) and extracellular regulated protein kinases (ERK).
Literature	GENERIF_SUMMARY	Data indicate that the CD166/AKT axis modulates tumorigenesis via promotion of phosphorylation and facilitation of degradation, ubiquitination and cytosolic accumulation of FOXO proteins in liver cancer cells.
Literature	GENERIF_SUMMARY	Data indicate that upregulation of TRPM cation channel TRPM8 promoted epithelial-mesenchymal transition of breast cancer cells via activating c-akt protein/glycogen synthase kinase 3 beta (GSK-3beta) pathway.
Literature	GENERIF_SUMMARY	Data indicate that Src kinase participates in the regulation of apoptosis through Src/PI-3 K/Akt signaling pathway in Hep-2 cell line.
Literature	GENERIF_SUMMARY	Down-regulate NF-kappaB, Akt and MAPK/JNK/p38/ERK pathways.
Literature	GENERIF_SUMMARY	By reducing MMP-2/-9 expression via AKT/mTOR and ROS/ERK1/2 pathways.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Curcumin inhibits the invasion of thyroid cancer cells via down-regulation of PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	miR-185 inhibits Hepatocellular carcinoma cell growth by targeting DNMT1, leading to PTEN induction and Akt inhibition.
Literature	GENERIF_SUMMARY	signaling via the Akt/eNOS axis is causally linked with impaired outgrowth endothelial cells-mediated vascular repair in South Asian men.
Literature	GENERIF_SUMMARY	our results demonstrate that activation of the PI3K/Akt/mTOR/p70S6K signaling pathway is involved in S100A4-induced viability, migration, upregulation of VEGF and downregulation of E-cadherin in CRC cells.
Literature	GENERIF_SUMMARY	Findings suggest that inhibition of H19 long non-coding RNA (LncRNAH19) and miR-675 expression can promote migration and invasion of hepatocellular carcinoma (HCC) cells via AKT/GSK-3beta/Cdc25A signaling pathway.
Literature	GENERIF_SUMMARY	EGFL7 enhances EGFR-AKT signaling, epithelial-mesenchymal transition, and metastasis of gastric cancer cells.
Literature	GENERIF_SUMMARY	miR-29c plays an important role in the progression of bladder cancer and induces anti-proliferative effects at least in part by inhibiting the AKT-PI3K pathway.
Literature	GENERIF_SUMMARY	Data show that doxazosin inhibited the phosphorylation of VEGF receptor-2 (VEGFR-2) Akt and mammalian target of rapamycin (mTOR), and hypoxia-inducible factor 1 (HIF-1alpha) expression.
Literature	GENERIF_SUMMARY	Data indicate that malic enzyme 2 knockdown impacts phosphatidylinositol 3-kinases/proto-oncogene protein akt (PI3K/AKT) signaling.
Literature	GENERIF_SUMMARY	These results indicate that constitutive Akt1 activity disrupts the normal blood vessel growth and maturation.
Literature	GENERIF_SUMMARY	PI4KIIIbeta likely plays a role in breast oncogenesis and that cooperation between Rab11a and PI4KIIIbeta represents a novel Akt activation pathway.
Literature	GENERIF_SUMMARY	Data indicate that AKT control TRAIL resistance of cancer cells through downregulation of interferon stimulated gene 12a (ISG12a) by MicroRNA-942 (miR-942).

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Akt1 phosphorylates TKT on Thr382, markedly enhancing enzyme activity.
Literature	GENERIF_SUMMARY	data strongly suggest that increased PI3K/AKT-mediated metastatic invasiveness in CaP is associated with FOXO4 loss, and that mechanisms to induce FOXO4 re-expression might suppress CaP metastatic aggressiveness.
Literature	GENERIF_SUMMARY	Combined IGF-1R/HER family and IGF-1R/Src family inhibition may have therapeutic potential in HNSCC. AKT may be a node of convergence between IGF-1R signaling and pathways that compensate for IGF-1R inhibition.
Literature	GENERIF_SUMMARY	study demonstrated that AuNPs significantly inhibited VEGF165-induced HUVEC migration and tube formation by affecting the cell surface ultrastructure, cytoskeleton and might have inhibited angiogenesis via the Akt pathway.
Literature	GENERIF_SUMMARY	Data show that downstream of Akt protein, IkappaB kinase alpha (IKKalpha) directly phosphorylates mammalian target of rapamycin (mTOR) to drive mTORC1 activation.
Literature	GENERIF_SUMMARY	OSU-53 activated Foxo3a through two Akt-dependent pathways
Literature	GENERIF_SUMMARY	Different targeted therapies may be necessary for Triple-negative breast cancer that involves activation of PIK3CA and AKT1 .
Literature	GENERIF_SUMMARY	naringenin is capable of inhibiting bladder cancer cell migration through the downregulation of the AKT and MMP2 pathways
Literature	GENERIF_SUMMARY	Rapamycin inhibition of mTOR results in feedback activation of AKT.
Literature	GENERIF_SUMMARY	results reveal a correlation between p-Akt and p-caspase-9 expression in gastric and colorectal cancers
Literature	GENERIF_SUMMARY	the role of Akt in the regulation of MUC5AC production
Literature	GENERIF_SUMMARY	Results show that phosphorylation of Sox2 by AKT1 at T118 blocks K119 methylation by Set7 and stabilizes Sox2.
Literature	GENERIF_SUMMARY	Studies indicate the importance of the EGFR/PI3K/PTEN/Akt/mTORC1/GSK-3 pathway in breast cancer.
Literature	GENERIF_SUMMARY	the results demonstrate that REDD1 acts not

Main Category	Category	Terms
		only as a repressor of mTORC1 but also as a constant modulator of the phosphorylation of Akt in response to growth factors and nutrients.
Literature	GENERIF_SUMMARY	results indicated rs1130214 and rs3803300 were related to OSCC susceptibility in Chinese Han Population. rs3730358 might be associated with progression-free survival time of OSCC patients
Literature	GENERIF_SUMMARY	mode of action for calycopterin in our study provides strong support that inhibition of PI3K/Akt and activation of MAPKs are pivotal in G2/M cell cycle arrest
Literature	GENERIF_SUMMARY	Vulnerability to acute psychomotor impairments induced by cannabis depends on variation in AKT1.
Literature	GENERIF_SUMMARY	sensitivity of lung cancer cells to miR-153 is determined by its endogenous miR-153 level
Literature	GENERIF_SUMMARY	we have found that the PI3Ka/b isoforms and AKT are involved in the downstream mechanisms leading to actin polymerization, ATP secretion and aggregation of human platelets stimulated by 2-AG.
Literature	GENERIF_SUMMARY	MDM2-mediated degradation of SIRT6 phosphorylated by AKT1 has a role in promoting tumorigenesis and trastuzumab resistance in breast cancer
Literature	GENERIF_SUMMARY	activating effect of PI3K inhibitor on IL-1beta release was mediated mainly by the enhancement of the intracellular IL-1beta and caspase-1 content release to the extracellular medium and not by increasing the corresponding mRNA and protein expression levels
Literature	GENERIF_SUMMARY	this is the first study to demonstrate that bFGF can induce EMT via AKT/GSK-3beta/ Snail signaling pathway in prostate cancer cells.
Literature	GENERIF_SUMMARY	Germline SNPs in the PI3 K-AKT-mTOR pathway were associated with breast cancer.
Literature	GENERIF_SUMMARY	Studied conditions that increase the sensitivity of cancer cells to MK-2206. and found reduction by salinomycin of Akt and downregulation of pAkt, pGSk3beta, pTSC2, and p4EBP1 by cotreatment with MK-2206.
Literature	GENERIF_SUMMARY	Collectively, these data reveal that activation of the PI3K/Akt pathway limits JNK-mediated apoptosis by phosphorylating and inactivating ASK1 during human enterovirus

Main Category	Category	Terms
		71 infection.
Literature	GENERIF_SUMMARY	IK-induced ROS generation regulates cell growth inhibition and it induces apoptosis through caspasedependent and -independent pathways via modulation of PI3K/AKT signaling in SK-MEL-2 cells
Literature	GENERIF_SUMMARY	It is anticipated that through an improved understanding of the biology of the PI3K/Akt pathway in prostate cancer, relevant biomarkers and rationale combination therapies will optimize targeting of this pathway
Literature	GENERIF_SUMMARY	miR-29b functions as a tumor suppressor in TSCC, and the miR-29b/Sp1/PTEN/AKT axis might represent a potential therapeutic target for TSCC intervention
Literature	GENERIF_SUMMARY	Due to the Akt activation, PTEN loss may play an important role in prostate cancer by promoting cancer stemness through a mechanism that involves enhanced NF-kappaB signaling.
Literature	GENERIF_SUMMARY	Studies indicate that the PI3K/PTEN/Akt/mTOR pathway is in the pathogenesis of malignancies and development of resistance to anticancer therapies, and inhibitors are a promising therapeutic option.
Literature	GENERIF_SUMMARY	Data indicate that microRNA miR-181b could activate HSCs, at least in part, via tensin homologs deleted on chromosome 10 (PTEN)/Akt proto-oncogene protein pathway.
Literature	GENERIF_SUMMARY	ANXA2 is involved in gastric cancer multi-drug-resistance (MDR) through regulating p38MAPK and AKT pathways as well as certain MDR factors
Literature	GENERIF_SUMMARY	RBP2 promotes HIF-1alpha-VEGF-induced angiogenesis of non-small cell lung cancer via the Akt pathway.
Literature	GENERIF_SUMMARY	MT1-MMP/Akt signaling axis is a key modifier for TNF-alpha-induced signaling pathways for modulation of procoagulant activity and apoptosis of endothelial cells.
Literature	GENERIF_SUMMARY	AKT isoform-specific knock down combined with quantitative phosphoproteomics provided a powerful strategy to unravel AKT isoform-specific signaling
Literature	GENERIF_SUMMARY	Immunohistochemistry analyses show that the cell signalling proteins AKT1 and ERK1/2 shuffled from the nucleus to the cytoplasm with tumour progression in

Main Category	Category	Terms
		breast cancer.
Literature	GENERIF_SUMMARY	These observations support the notion that loss of TRIB3 is associated with a more aggressive phenotype in various types of tumors by enhancing the activity of the mTORC2/AKT/FOXO axis.
Literature	GENERIF_SUMMARY	Data indicate that proto-oncogene c-Akt (AKT1) is directly involved in miR-99a-mediated tumor suppression.
Literature	GENERIF_SUMMARY	Authors show that MNK regulates SRPK via mTOR and AKT.
Literature	GENERIF_SUMMARY	Data show that IL-27 on IFN-gamma mediated autophagy perpetuates by simultaneous induction of kinases mTOR via JAK/PI3 K/Akt cascade and increment in myeloid cell leukemia sequence 1 (Mcl-1) level through PI3 K pathway.
Literature	GENERIF_SUMMARY	gastric circulating tumor cells exhibited remarkable epithelial-mesenchymal transition process, and p-Akt/miR200s signaling regulates
Literature	GENERIF_SUMMARY	In this review we present a state of art review on various classes of small molecule inhibitors selectively involving PI3K/Akt/mTOR pathway and their clinical potential in this disease
Literature	GENERIF_SUMMARY	The OHPg/PR-B dependent induction of PTEN causes the down-regulation of PI3K/AKT signal, switching on the autophagy process
Literature	GENERIF_SUMMARY	Thymosin beta4 induces invasion and migration of human colorectal cancer cells through the ILK/AKT/beta-catenin signaling pathway.
Literature	GENERIF_SUMMARY	In conclusion, the present study indicates that C5a may promote the proliferation of breast cancer cells through Akt1 activation of the RGC-32 gene.
Literature	GENERIF_SUMMARY	Data suggest that three isoforms of the serine/threonine protein kinase Akt (Akt1, Akt2, Akt3) regulate cell survival, cell growth, cell proliferation, and cell metabolism in breast cancer cells. [REVIEW]
Literature	GENERIF_SUMMARY	Akt1 polymorphisms are associated with cachexia development in pancreatic cancer patients.
Literature	GENERIF_SUMMARY	AKT will be destabilized by administration of the HSP90 inhibitor 17-allylamino-geldanamycin.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with lymph node metastasis of non-small cell lung cancer.
Literature	GENERIF_SUMMARY	The individual contribution of each Akt isoform in p120 RasGAP fragment N-mediated cell protection against Fas ligand induced cell death, was investigated.
Literature	GENERIF_SUMMARY	Phosphorylation of cortactin by AKT1 is important for mutant PI3K-enhanced cell migration and invasion.
Literature	GENERIF_SUMMARY	Akt1 and Akt2 are involved in albumin endocytosis, and phosphorylation of Dab2 by Akt induces albumin endocytosis in proximal tubule epithelial cells.
Literature	GENERIF_SUMMARY	Impaired EPCs proliferation could be reversed by BBR via the PI3K/AKT/eNOS signaling pathway.
Literature	GENERIF_SUMMARY	Increased ACK1 tyrosine phosphorylation correlated with upregulated PDGFR-beta activity and AKT activation.
Literature	GENERIF_SUMMARY	RI represents an important predictor of progression in patients with CRC and suppresses proliferation and metastasis in CRC cells through inhibiting PI3K/AkT pathway.
Literature	GENERIF_SUMMARY	MicroRNA-141 and microRNA-146b-5p have a role in inhibiting the prometastatic mesenchymal characteristics through the RNA-binding protein AUF1 targeting the transcription factor ZEB1 and the protein kinase AKT
Literature	GENERIF_SUMMARY	In this review, a basic overview of the role of AKT in normal and malignant tissues is provided, along with an in-depth discussion of its role in gastric cancer and its potential as a target for therapy.
Literature	GENERIF_SUMMARY	Data indicate that lanthionine synthetase C-like 2 (LanCL2) depletion sensitizes cells to apoptosis through down-regulating serine/threonine protein kinase Akt phosphorylation.
Literature	GENERIF_SUMMARY	High AKT is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Combination of ibrutinib with ABT-199
Literature	GENERIF_SUMMARY	PI3Kgamma-induced VEGF expression was reduced when the human tocopherol-associated protein 1 (hTAP1/SEC14L2) was overexpressed suggesting formation of an inactive PI3Kgamma/hTAP1 heterodimer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Most notably, WWP1 downregulation both inactivated PTEN-Akt signaling pathway in MKN-45 and AGS cells. our findings suggest that WWP1 acts as an oncogenic factor and should be considered as a novel interfering molecular target for gastric carcinoma
Literature	GENERIF_SUMMARY	RSV induced apoptosis in ovarian cancer cells by impairing glucose uptake, a process involving Akt-regulated plasma membrane GLUT1 trafficking.
Literature	GENERIF_SUMMARY	CrkL knockdown markedly suppressed the phosphorylated ERK (p-ERK) as well as the phosphorylated AKT (p-AKT) (p
Literature	GENERIF_SUMMARY	The phosphoinositide 3-kinase/Akt signaling pathway was activated in cells overexpressing Sox2.
Literature	GENERIF_SUMMARY	Data show that the induction of resveratrol-dependent autophagy is mediated via the mechanistic target of Akt proto-oncogene protein/rapamycin complex 1 (mTORC1)/S6 kinase 1 (S6K1) signaling pathway.
Literature	GENERIF_SUMMARY	Data indicate that quercetin induces apoptosis via phosphatidylinositol 3-kinase PI3k/Akt pathways, suggesting quercetin has potential to be used as an anti-tumor agent against human cervix cancer.
Literature	GENERIF_SUMMARY	we showed that ovarian cancer cells activate Akt phosphorylation in endothelial cells inducing resistance to bevacizumab leading to an autocrine loop based on FGF2 secretion.
Literature	GENERIF_SUMMARY	Results indicate that orexin-A significantly enhanced the expression of 3beta-hydroxysteroid dehydrogenase (3beta-HSD) and the production of cortisol, and increased the phosphorylation of AKT protein in the NCI-H295R adrenocortical cells.
Literature	GENERIF_SUMMARY	Importantly, the AKT1/MYC status was independent of Gleason grade and pathologic staging. Our findings show how prostate tumors undergo a metabolic reprogramming that reflects their molecular phenotypes
Literature	GENERIF_SUMMARY	Results indicate that proto-oncogene protein c-Akt inhibition contributed to cell death.
Literature	GENERIF_SUMMARY	Children with lupus nephritis have enhanced Akt-dependent migratory capacity of Th17 cells.
Literature	GENERIF_SUMMARY	Peptide-based TCL1-interphase mimics

Main Category	Category	Terms
		were potent in steric AKT antagonization.
Literature	GENERIF_SUMMARY	Authors show that the transcription factor hepatocyte nuclear factor 4alpha (HNF4alpha) is a target of HBx-regulated AKT, and link HNF4alpha to HBx-regulated AKT modulation of hepatitis B virus transcription and replication.
Literature	GENERIF_SUMMARY	CXCR7 siRNA efficaciously suppressed CCL19-induced AKT.
Literature	GENERIF_SUMMARY	We conclude that cPLA2alpha is required for sustaining AKT phosphorylation at Ser473 and cell proliferation in colorectal cancer cells with PI3K mutation
Literature	GENERIF_SUMMARY	A decrease in the expression of Akt.
Literature	GENERIF_SUMMARY	Results suggest that Mst1 regulates proliferation of glioma cells via AKT/mTOR signaling pathway
Literature	GENERIF_SUMMARY	In colorectal cancer patient the AKT gene expression is up regulated and its a good indicator for the prognosis
Literature	GENERIF_SUMMARY	These results suggest that the PKC pathway and PDGFR/EGFR transactivation pathway play important roles in HCA2-mediated Akt activation.
Literature	GENERIF_SUMMARY	Mechanistic investigations uncovered opposing functions for different Akt isoforms in this regulation, where Akt1 promotes and Akt3 inhibits vascular tumor growth.
Literature	GENERIF_SUMMARY	Data indicate that phosphorylation status of proto-oncogene c-AKT(Ser473) serves as a biomarker to identify advanced head and neck squamous cell carcinoma (HNSCC) patients at high risk for radiotherapy treatment failure.
Literature	GENERIF_SUMMARY	Finally, a strong association between the expression of EEF1A2, phosphorylated AKT and MDM4 was observed in human HCC samples. Strong activation of the EEF1A2/PI3K/AKT/mTOR/MDM4 signaling pathway was observed in HCC patients.
Literature	GENERIF_SUMMARY	Nck1 depletion induces activation of the PI3K/Akt pathway by attenuating PTP1B protein expression
Literature	GENERIF_SUMMARY	The PTEN/AKT signaling axis was defined as a critical pathway regulated by betaIII-tubulin in NSCLC cells.
Literature	GENERIF_SUMMARY	Akt, through its downstream targets, mTORC1 and hence autophagy, negatively

Main Category	Category	Terms
		regulates cholesterol efflux to apoA-I.
Literature	GENERIF_SUMMARY	The results have demonstrated that ATO synergized with sorafenib to inhibit the proliferation and promote the apoptosis of HCC cells by diminishing the increased activation of Akt by sorafenib.
Literature	GENERIF_SUMMARY	The activity of wedelolactone on osteoclasts and breast cancer-mediated osteoclastogenesis was associated with the inhibition of Akt/mammalian target of the rapamycin signaling pathway (mTOR).
Literature	GENERIF_SUMMARY	Re-expression of miR-149 significantly inhibited HCC cell proliferation and tumorigenicity by regulating AKT1/mTOR pathway.
Literature	GENERIF_SUMMARY	MALAT1 might suppress the tumor growth and metastasis via PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	These results indicated that 1-BT inhibited gastric cancer cells proliferation and migration through the Akt pathway, which has the potential clinical significance in the prevention and treatment of gastric cancer
Literature	GENERIF_SUMMARY	Centipedegrass extract induces apoptosis through the activation of caspases and the downregulation of PI3K/Akt and MAPK phosphorylation in leukemia cells.
Literature	GENERIF_SUMMARY	CD138 myeloma cell-directed cross-talk with BMSCs (bone marrow stromal cells) and reveal that CD138 myeloma cells regulate multiple myeloma-BMSC stiffness through SDF-1/CXCR4/AKT signaling.
Literature	GENERIF_SUMMARY	certain Akt/mTOR/4E-BP1 pathway signals could be novel therapeutic targets for Merkel cell carcinomas regardless of Merkel cell polyomavirus infection status
Literature	GENERIF_SUMMARY	We also report that the extra-genomic effects elicited by both ligands, leading to recruitment of active Akt to chromatin, are essential for phosphorylation of serine 10 in histone H3 by IKKa
Literature	GENERIF_SUMMARY	the role of PLC and PI3K/Akt/mTOR signal transduction pathways in pathophysiology.
Literature	GENERIF_SUMMARY	pharmacologic activation of Akt in Tregs can induce Glut1, overcoming the effects of Foxp3. Together, these results illustrate the molecular basis behind differential glucose metabolism in Tregs.
Literature	GENERIF_SUMMARY	Data indicate that ibrutinib significantly reduced PI3K-Akt activity in cardiac cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Upregulation of the AKT-pathway seems to be one reason for poor radioresponse
Literature	GENERIF_SUMMARY	PCAF-mediated Akt1 acetylation enhanced Akt1 phosphorylation.
Literature	GENERIF_SUMMARY	GLUT1 may be part of an AKT1-dependent mechanism allowing cells to survive in low levels of glucose
Literature	GENERIF_SUMMARY	signalings can reciprocally and positively regulate each other, leading to enhanced non-small cell lung cancer (NSCLC) cell proliferation and survival.
Literature	GENERIF_SUMMARY	Through the inhibition of AKT/ NF-kappaB/ mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Data reveal that AKT is frequently activated in ovarian clear cell carcinoma.
Literature	GENERIF_SUMMARY	functional studies reveal that the Akt-mTOR signaling pathway, a major negative regulator of autophagy, is inhibited by oxamate.
Literature	GENERIF_SUMMARY	Findings establish that the convergence of 2 distinct Ras effector pathways on mammalian target of rapamycin signaling maintains neurofibromatosis type 1 mouse optic glioma growth.
Literature	GENERIF_SUMMARY	self-renewal versus differentiation of human ESCs (hESCs) in response to Wnt signaling is predominantly determined by a two-layer regulatory circuit involving beta-catenin, E-cadherin, PI3K/Akt, and Slug in a time-dependent manner.
Literature	GENERIF_SUMMARY	AKT1 polymorphisms correlate with overall survival in patients with head and neck squamous cell cancer treated with radiotherapy.
Literature	GENERIF_SUMMARY	GRIM-19 overexpression suppressed hepatocellular carcinoma (HCC) growth and downregulated AKT1 expression, suggesting that GRIM-19 might play a crucial role in hepatocarcinogenesis through negatively regulating the PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	results broaden the current view of AKT function and have important implications for the development of AKT inhibitors for cancer
Literature	GENERIF_SUMMARY	Trimetazidine protects human UC-MSCs from H/SD-induced apoptosis via the Akt pathway and may therefore be a potentially useful therapeutic adjunct for transplanting MSCs into damaged heart after myocardial

Main Category	Category	Terms
		infarction
Literature	GENERIF_SUMMARY	Attenuation of the phosphatidylinositol 3-kinase/Akt signaling pathway by Porphyromonas gingivalis gingipains RgpA, RgpB, and Kgp.
Literature	GENERIF_SUMMARY	BTG3 binds and suppresses AKT, a kinase frequently deregulated in cancers.
Literature	GENERIF_SUMMARY	the expression of Aven is regulated by the Akt signaling pathway through cathepsin D activity, which contributes to the sensitivity of cancer cells to chemotherapeutic agents.
Literature	GENERIF_SUMMARY	Phosphorylated Akt overexpression is an adverse prognostic factor for breast cancer.
Literature	GENERIF_SUMMARY	Activation of the Akt/mTOR signaling pathway induces cardiac fibrosis and hypertrophy.
Literature	GENERIF_SUMMARY	eIF2alphaP acts as a molecular switch that dictates either cell survival or death by activated Akt in response to oxidative stress.
Literature	GENERIF_SUMMARY	Treatment of LNCaP cells with obacunone and OG resulted in dose-dependent changes in expression of proteins responsible for the induction of programmed cell death through the intrinsic pathway and down-regulation of Akt
Literature	GENERIF_SUMMARY	Suppressed expression of miR-637 and increased Akt1 protein levels were correlated with unfavorable progression and poor prognosis, respectively, and a negative relationship between the miR-637 expression and Akt1 protein levels was observed in gliomas
Literature	GENERIF_SUMMARY	we will summarize the current knowledge about the role of PAM (PI3K/AKT/mTOR) signaling in medulloblastoma and discuss the strategies that are currently being evaluated with targeted agents against this pathway.
Literature	GENERIF_SUMMARY	BMCC1 negatively regulates phosphorylation pathway of AKT, resulted in apoptosis and the BNIP2 homology region of BMCC1 interacts with BCL2.
Literature	GENERIF_SUMMARY	Emerging concepts such as the role of the PI3K/AKT/TWIST pathway in the regulation of the E-cadherin expression will be highlighted, since it seems to be consistently involved in cells EMT
Literature	GENERIF_SUMMARY	knockdown of PTX3 suppressed tumor proliferation and invasion of lung

Main Category	Category	Terms
		adenocarcinoma cells, followed by decreased expression of p-AKT, p-NF-kappa B, PCNA, and MMP-9
Literature	GENERIF_SUMMARY	Aktmediated phosphorylation of Oct4A or its homolog protein was associated with the proliferation of stemlike cancer cells.
Literature	GENERIF_SUMMARY	This study demonstrated down-regulation of the Akt/mTOR pathway, specifically via p70S6K/eIF4B, in idiopathic autism.
Literature	GENERIF_SUMMARY	LGR4 promotes tumorigenesis of prostate cancer via PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	our results unveil a functional cross-talk between YB-1, DeltaNp63alpha and the PI3K/AKT pathway critically governing survival of squamous carcinoma cells.
Literature	GENERIF_SUMMARY	Pharmacological and genetic analyzes using receptor subtype blocking and RNA interference approaches revealed that this effect selectively depended on b2-AR signaling involving the regulation of AKT
Literature	GENERIF_SUMMARY	we made the first demonstration that dysregulation of PAFR and the positive regulatory loop between PAFR and pAKT contribute to malignant progression of esophageal squamous cell carcinoma
Literature	GENERIF_SUMMARY	Our findings afford new insights into the role played by YB-1 in the TGF-beta1 signaling pathway.
Literature	GENERIF_SUMMARY	Data show that serine/threonine kinase LKB1 and proto-oncogene protein Akt are targets of neddylation, suggesting the post-translational modification (PTM) as a potential therapeutic target during liver tumor management and treatment.
Literature	GENERIF_SUMMARY	identified the PI3K/Akt pathway, the cell-cycle regulator Wee1 kinase, and protein kinase C (PKC) as prospective regulatory nodes of neuronal excitability through modulation of the FGF14
Literature	GENERIF_SUMMARY	MACC1 overexpression predicted worse prognosis after liver resection, which was attributed to the repression of HCC cell apoptosis via a molecular mechanism in which MACC1 accelerated the activation of the HGF/c-MET/PI3K/AKT pathway
Literature	GENERIF_SUMMARY	miR-21 is a key regulator of fibrogenic epithelial-mesenchymal transition in hepatocytes via PTEN/Akt pathway.
Literature	GENERIF_SUMMARY	this study is the first to demonstrate that HG treatment of A549 human lung epithelial

Main Category	Category	Terms
		cells promotes tumor cell invasion and increases metastasis-associated protein expression by up-regulating HO-1 expression via ROS or the TGF-1/PI3K/Akt signaling
Literature	GENERIF_SUMMARY	miR-494-3p inhibitor could prevent migration, invasion, proliferation, and promote apotosis in gliomas through PTEN/AKT pathway.
Literature	GENERIF_SUMMARY	the AKT/miR-199a-5p/CAV1 pathway is identified as a regulator of innate immunity, which is dysfunctional in cystic fibrosis macrophages contributing to lung hyper-inflammation.
Literature	GENERIF_SUMMARY	Akt and phosphoinositide 3-kinase pathway activation plays role in suppressing apoptosis of non-small cell lung cancer cells.
Literature	GENERIF_SUMMARY	SOCS3 negatively regulates proliferation and angiogenesis of NCI-H446 cells and that HIF-1alpha is required in this process. The results also suggested a suppressive role of SOCS3 in Akt signaling, but not STAT3 signaling to block HIF-1alpha expression
Literature	GENERIF_SUMMARY	C23 protein meditates bone morphogenetic protein-2-mediated epithelial-to-mesenchymal transition via up-regulation of Erk1/Erk2 and Akt in gastric cancer.
Literature	GENERIF_SUMMARY	Our results suggest that celecoxib-mediated upregulation of Cbl-b is responsible, at least in part, for the additive antitumor effect of celecoxib and rapamycin via inhibition of rapamycin-induced Akt activation.
Literature	GENERIF_SUMMARY	RASSF4 promotes EV71 replication and the production of viral progeny to accelerate the inhibition of the phosphorylation of AKT.
Literature	GENERIF_SUMMARY	Breast cancer patients with co-heightened EGFR/AKT1 gene copy numbers had a worse outcome than those with only high EGFR gene copy numbers.
Literature	GENERIF_SUMMARY	REIVEW
Literature	GENERIF_SUMMARY	AKT1 E17K mutation in advanced colorectal carcinoma is associated with mucinous morphology, PIK3CA wild-type status, and concurrent RAS/RAF mutations with similar pattern to PIK3CA exon 21 mutants.
Literature	GENERIF_SUMMARY	Imp2 regulates the activity of IGF2, which further activates PI3K/Akt signaling.
Literature	GENERIF_SUMMARY	The results of this study showed that AKT1 and mTOR in the pathophysiology of Bipolar

Main Category	Category	Terms
		disorder and support the relevance of blood mRNA expression.
Literature	GENERIF_SUMMARY	The schweinfurthin class of compounds as a novel approach to modulate oncogenic mTOR/AKT signaling for cancer treatment.
Literature	GENERIF_SUMMARY	Experimental data showed upregulation of the selected EGFR cascade(Akt, nuclear factor kappa B p50) stromal expression of EGFR, and nuclear localization of COX-2 in metaplastic gland cells of laryngeal cancer tissue sample
Literature	GENERIF_SUMMARY	Results show that a new mechanism of oncogenesis is attributed to p68 by upregulation of AKT and consequent nuclear exclusion and degradation of tumor suppressor FOXO3a.
Literature	GENERIF_SUMMARY	Forced activation of Akt and knockdown of FOXO3, p15(INK4b) and p27(KIP1) effectively inhibited hSKP senescence.
Literature	GENERIF_SUMMARY	AKT is activated after imatinib secondary resistance in gastrointestinal stromal tumors.
Literature	GENERIF_SUMMARY	the results of the present study demonstrated that SET was associated with paclitaxel resistance in MCF-7/PTX cells, and that paeonol reversed paclitaxel resistance in MCF-7/PTX cells by downregulating the activity of the SET/PP2A/Akt pathway.
Literature	GENERIF_SUMMARY	mTORC1 regulates cell adhesion through S6K1 and 4E-BP1 pathways, but mTORC2 regulates cell adhesion via Akt-independent mechanism
Literature	GENERIF_SUMMARY	These results demonstrate that endometrial carcinoma stromal cells stimulate epithelial cell proliferation via the HGF/c-Met/Akt signaling pathway.
Literature	GENERIF_SUMMARY	Overexpression of NDRG1 inhibits human glioma proliferation and invasion via phosphoinositide 3-kinase/AKT pathways.
Literature	GENERIF_SUMMARY	A higher SUVmax on 18F-FDG PET is associated with elevated tumor levels of pAkt and pS6 protein and with aggressive behavior and metastatic potential of RCC, as well as with early relapse following radical nephrectomy and shorter overall survival.
Literature	GENERIF_SUMMARY	Interferon-alpha inhibits CD4 T cell responses to interleukin-7 and interleukin-2 and selectively interferes with Akt signaling.
Literature	GENERIF_SUMMARY	TG-interacting factor transcriptionally

Main Category	Category	Terms
		induced by AKT/FOXO3A is a negative regulator that antagonizes arsenic trioxide-induced cancer cell apoptosis.
Literature	GENERIF_SUMMARY	Study did not observe significant association of SNPs rs1130214 from AKT1 gene with schizophrenia in Pakistani samples
Literature	GENERIF_SUMMARY	the results of the present study demonstrated that the acquired taxol resistance of ovarian cancer cells was associated with ROS-dependent upregulation in the expression of Tyro3 RTK and the subsequent activation of Akt.
Literature	GENERIF_SUMMARY	AKT mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	IRS2 phosphorylation is cell cycle-regulated and that Plk1 phosphorylation of IRS2 prevents premature mitotic exit via AKT inactivation.
Literature	GENERIF_SUMMARY	Together, these data suggest that GRh2 may suppress GBM migration through inhibiting Akt-mediated MMP13 activation.
Literature	GENERIF_SUMMARY	In summary, our results indicated that wogonin induced apoptosis and ER stress in HL-60 cells, which was mediated by the inhibition of the PI3K-AKT signaling pathway.
Literature	GENERIF_SUMMARY	IL-32 stimulation promotes the invasion and motility of osteosarcoma cells, possibly via the activation of AKT and the upregulation of MMP-13 expression.
Literature	GENERIF_SUMMARY	in a high number of HNSCC tumours, PI3K/Akt signalling is activated by different molecules or by the combination of these molecules.
Literature	GENERIF_SUMMARY	suppression of Akt1 expression in human RCC ACHN cells using shRNA technology significantly enhanced the sensitivity of these cells to sorafenib both in vitro and in vivo through the regulation of molecules associated with apoptosis.
Literature	GENERIF_SUMMARY	Survivin, via AKT and VEGF, seems to play an important role in oral melanoma and could represent an important prognostic marker of melanoma progression.
Literature	GENERIF_SUMMARY	Our findings support the de-regulation of the PI3K/Akt/mTOR and MAPK signaling pathways in type I endometrial carcinomas suggesting involvement of these pivotal pathways in endometrial carcinogenesis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	PTCL patients with high p-AKT expression showed aggressive clinical courses with significantly worse OS and PFS and a poor chemotherapy response rate
Literature	GENERIF_SUMMARY	we demonstrated here the inhibitory effect of CACP on the growth of H22 cells in vitro and in vivo, which may be through, at least partly, repression of PTTG1 and then followed by the inactivation of P13/Akt and activation of p38 MARK signaling pathways
Literature	GENERIF_SUMMARY	Overexpression of Cav-1 is related to poor prognosis in NPC patients, which correlated with Akt signaling pathway. Abrogation of Akt signaling by shRNA-mediated knockdown of Cav-1 decreased malignant properties of tumor cells
Literature	GENERIF_SUMMARY	adult human Sertoli cells assumed similar morphological features, stable global gene expression profiles and numerous proteins, and activation of AKT and SMAD1/5 during long-period culture.
Literature	GENERIF_SUMMARY	We find that at least 34 % of the downregulated AKT module genes are either mediators of apoptosis or have tumor suppressor functions
Literature	GENERIF_SUMMARY	inhibition of AKT/FoxO3a signaling may contribute to H1-mediated PUMA induction, suggesting that inhibition of AKT/FoxO3a signaling result in PUMA expression in response to p53-independent cytotoxic effects of H1.
Literature	GENERIF_SUMMARY	To increased E-cadherin expression and decreased Akt phosphorylation.
Literature	GENERIF_SUMMARY	KLF5 loss enhances tumor angiogenesis by attenuating PI3K/AKT signaling and subsequent accumulation of HIF1alpha in PTEN deficient prostate tumors.
Literature	GENERIF_SUMMARY	RGS1 suppresses CXCL12-mediated migration and AKT activation in cultured human plasmacytoma cells and plasmablasts.
Literature	GENERIF_SUMMARY	Data show that microRNA miR-21 plays an oncogenic role in diffuse large B-cell lymphoma (DLBCL) by modulating the PI3-Kinase/AKT protein/mTOR protein/forkhead box O1 protein (FOXO1) pathway at multiple levels resulting in strong prognostic implication.
Literature	GENERIF_SUMMARY	Data show that microRNA miR-451 target tuberous sclerosis 1 (TSC1) gene and activates the PI3-kinase/Akt protein/mTOR protein signaling in multiple myeloma (MM)

Main Category	Category	Terms
		Side population (SP) cells.
Literature	GENERIF_SUMMARY	MicroRNA-302a suppresses tumor cell proliferation by inhibiting AKT in prostate cancer.
Literature	GENERIF_SUMMARY	SMYD2-mediated methylation negatively regulates PTEN tumor suppressor activity and results in activation of the phosphatidylinositol 3-kinase-AKT pathway.
Literature	GENERIF_SUMMARY	Myristoylated AKT1 was incompatible with mouse hematopoietic stem cell proliferation driven by both STAT5 and Fms-related tyrosine kinase 3-internal tandem duplication, which triggered cell cycle block and apoptosis.
Literature	GENERIF_SUMMARY	Data indicate that combining the CDK inhibitor dinaciclib with the pan-AKT inhibitor MK-2206 showed promising therapeutic results in pancreatic cancer murine model.
Literature	GENERIF_SUMMARY	Low expression of PTEN in clearcell renal cell carcinoma contributes to chemoresistance through activating the Akt/HDM2 signaling pathway.
Literature	GENERIF_SUMMARY	These results demonstrated that miR-133a improves the cardiac function and fibrosis through inhibiting Akt in heart failure rats.
Literature	GENERIF_SUMMARY	these results identify mediators of hypoxia-induced p53-dependent apoptosis and suggest AKT inhibition may improve radiotherapy response in p53-deficient tumors.
Literature	GENERIF_SUMMARY	treatment with butein significantly increased reactive oxygen species (ROS) generation and reduced the phosphorylation of PI3K, AKT and mTOR expression, which contributes to the inhibition of the tumor growth of cervical cancer
Literature	GENERIF_SUMMARY	ACh-induced activation of EGFR/PI3K/AKT pathway and subsequent IL-8 upregulation may be one of the important mechanisms of M3R function.
Literature	GENERIF_SUMMARY	These results show that insulin attenuates arsenic-induced neurite outgrowth impairment possibly via activation of PI3K/Akt/SIRT1 signaling, and arsenic may exert neurite outgrowth inhibition through a mechanism involving reduction of the same molecules.
Literature	GENERIF_SUMMARY	STAT3 (signal transducer and activator of transcription 3)-induced control of AKT1 transcription was functional in ALK+ ALCL

Main Category	Category	Terms
		and blocking of STAT3 and AP-1 signaling synergistically affected cell proliferation
Literature	GENERIF_SUMMARY	ET-1 may induce fibrosis in dermal fibroblasts through Akt signals.
Literature	GENERIF_SUMMARY	Akt1 phosphorylates NCT at Ser437, significantly reducing NCT stability. Akt1 functions as a negative regulator of the gamma-secretase activity through phosphorylation and degradation of NCT.
Literature	GENERIF_SUMMARY	a model that CD166 regulates MCAM through a signaling flow from activation of PI3K/AKT and c-Raf/MEK/ERK signaling to the inhibition of potential MCAM ubiquitin E3 ligases, betaTrCP and Smurf1.
Literature	GENERIF_SUMMARY	results demonstrate that XBP1 mRNA splicing plays an important role in maintaining the function of bone marrow-derived macrophages and provide new insight into the study and treatment of atherosclerosis
Literature	GENERIF_SUMMARY	Expression of active Akt1 in endothelial cells inhibited angiogenesis; hemiphosphorylated Akt1 with reduced kinase activity had intermediate inhibitory effect. Suppressed Akt activity increased endothelial MAP kinase signaling and angiogenesis
Literature	GENERIF_SUMMARY	The overexpression of MCP1 in HNC cells may partially induce EMT through the AKT pathway. A high cellular expression of MCP1 was associated with pathological neck metastases.
Literature	GENERIF_SUMMARY	The intracellular localization of Akt plays an important role in breast cancer stem-like cells.
Literature	GENERIF_SUMMARY	this study provides the evidence of Rab31 overexpression in HCC, and Rab31 is potentially used as a novel biomarker of poor prognosis in patients with HCC. PI3K/AKT/Bcl-2/BAX axis was involved in Rab31-promoting HCC progression.
Literature	GENERIF_SUMMARY	DAPK2-induced apoptosis is negatively regulated by Akt and 14-3-3 proteins.
Literature	GENERIF_SUMMARY	IQGAP1 and its two isoforms regulate the actin-cytoskeleton alone and with their binding partners, Rac and Cdc42, thereby controlling diverse cellular processes, such as cell migration and adhesion. (Review)
Literature	GENERIF_SUMMARY	Data show that cytoplasmic cyclin-dependent kinase inhibitor p27 is preferentially observed in primary non-small cell lung cancer (NSCLC) with activated

Main Category	Category	Terms
		proto-oncogene proteins c-akt1 and predicts poor survival.
Literature	GENERIF_SUMMARY	p53 attenuates AKT signaling by modulating membrane phospholipid composition
Literature	GENERIF_SUMMARY	Loss of PIPKIgamma or its focal adhesion-targeting variant, PIPKIgammai2, impaired PI3K/Akt activation upon stimulation with growth factors or extracellular matrix proteins in different tumor cells.
Literature	GENERIF_SUMMARY	a correlation between TRAP1 and AKT expression is found in vivo in human colorectal tumours. These results provide new insights into TRAP1 role in the regulation of cell migration in cancer cells, tumour progression and metastatic mechanisms.
Literature	GENERIF_SUMMARY	These results showed that deguelin possessed antitumor effect by targeting Akt in dual axis such as EGFR and IGF1R signaling pathways and suggested that it provides an applicable therapeutic strategy for HNSCC patients.
Literature	GENERIF_SUMMARY	Berberine reverses drug resistance by sensitizing human ovarian cancer cells to cisplatin through miR-93/PTEN/Akt signaling pathway.
Literature	GENERIF_SUMMARY	SMAD3 and AKT phosphorylation was enhanced during the culture of human SSCs
Literature	GENERIF_SUMMARY	MicroRNA-374b Suppresses Proliferation and Promotes Apoptosis in T-cell Lymphoblastic Lymphoma by Repressing AKT1 and Wnt-16
Literature	GENERIF_SUMMARY	Our data show that combined targeting of MEK and PI3K-AKT with mTOR is a better option than single agents alone for KRAS mutant NSCLC, thus opening the possibility of a beneficial treatment strategy in the future.
Literature	GENERIF_SUMMARY	p120RasGAP shields Akt from deactivating phosphatases in FGF1 signaling, but loses this ability once cleaved by caspase-3.
Literature	GENERIF_SUMMARY	Data show that activation of EphB2 receptor kinase arrests tau protein hyperphosphorylation through phosphatidylinositol 3-kinase (PI3K)/Akt protein-mediated glycogen synthase kinase-3beta (GSK-3beta) inhibition.
Literature	GENERIF_SUMMARY	SHIP2 inhibition was accompanied by an increased Akt and FOXO-1 phosphorylation, whereas overexpression of the wild-type SHIP2 gene had the opposite effects

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data show that significantly better overall survival was observed in patients with coexistence of high epidermal growth factor receptor (EGFR) expression and low p-Akt proto-oncogene protein expression.
Literature	GENERIF_SUMMARY	Doxo-induced apoptosis has a role in inhibiting hepatocellular carcinoma by targeting Madcam1 and AKT and blocking protein translation initiation
Literature	GENERIF_SUMMARY	Our results establish that Fyn can arrest SW-induced apoptosis via the activity of Akt and its effective phosphorylation in 293T cells.
Literature	GENERIF_SUMMARY	These data suggest that the rs2498794 polymorphism of the AKT1 gene is associated with a long smoking duration and may be involved in the predisposition to cancer when the smoking duration is short or the cigarettes per day is rate low.
Literature	GENERIF_SUMMARY	In-frame duplications in the pleckstrin-homology domain (PHD) found in more than 60% of Juvenile Granulosa Cell Tumors (JGCTs)
Literature	GENERIF_SUMMARY	This study incriminates somatic mutations of AKT1 as a major event in the pathogenesis of Juvenile Granulosa Cell Tumors.
Literature	GENERIF_SUMMARY	CRISPR-Cas9n-mediated knockout and reconstitution studies establish that dynamin-1 is activated by Akt/GSK3beta signaling in H1299 non-small lung cancer cells.
Literature	GENERIF_SUMMARY	The AKT1 isoform plays a dominant role in the survival and chemoresistance of chronic lymphocytic leukaemia cells.
Literature	GENERIF_SUMMARY	Genetic deletion and pharmacological inhibition of Akt1 isoform attenuates bladder cancer cell proliferation, motility and invasion
Literature	GENERIF_SUMMARY	Data show that immunoglobulin-like transcript 4 (ILT4) increases the expression of the co-inhibitory molecule B7-H3 through PI3K/AKT/mTOR signalling.
Literature	GENERIF_SUMMARY	These observations suggest that not only ERK1/2 but also Akt activity is essential to maintain Ets-1 in an active state.
Literature	GENERIF_SUMMARY	Girdin regulates the migration and invasion of glioma cells via the PI3K-Akt signaling pathway.
Literature	GENERIF_SUMMARY	8bromo7methoxychrysin induces apoptosis and reverses drug resistance by regulating

Main Category	Category	Terms
		Akt/FOXO3a pathway in cisplatinsensitive and resistant ovarian cancer cells.
Literature	GENERIF_SUMMARY	Data show that hypoxia-inducible factor 1, alpha subunit (HIF-1alpha) is activated via an proto-oncogene protein Akt/mTOR protein signaling pathway in LCC2 and LCC9 breast cancer cells.
Literature	GENERIF_SUMMARY	Akt/mTORC1 signaling and glucose metabolism is inhibited by halofuginone treatment of colorectal cancer
Literature	GENERIF_SUMMARY	Pygo2 is a common node downstream of oncogenic Wnt and Akt signaling pathways.
Literature	GENERIF_SUMMARY	A3G inactivated PTEN by binding to the C2 tensin-type and PDZ domains, thereby inducing anoikis resistance through Akt activation
Literature	GENERIF_SUMMARY	Study suggested that AKT1 rs2498786 polymorphism in insulin signaling pathway may be associated with Alzheimer's disease risk and different genotypes may affect levels of protein expression
Literature	GENERIF_SUMMARY	Membrane attack complexes activate noncanonical NF-kappaB by forming a novel Akt(+)NIK(+) signalosome on Rab5(+) endosomes.
Literature	GENERIF_SUMMARY	Reciprocal positive regulation between Cx26 and PI3K/Akt pathway confers acquired gefitinib resistance in non-small-cell lung carcinoma cells by promoting epithelial mesenchymal transition via a gap-junctional communication-independent manner.
Literature	GENERIF_SUMMARY	Loss of protein tyrosine phosphatase, non-receptor type 2 is associated with activation of AKT and tamoxifen resistance in breast cancer
Literature	GENERIF_SUMMARY	We show that the decrease in PI(4,5)P2 level under non-stimulated conditions inhibits PTEN activity leading to the aberrant activation of the oncoprotein Akt. As well as defining a novel mechanism of Akt phosphorylation with important therapeutic consequences, we also demonstrate that differential expression levels of FGFR2, Plc11 and Grb2 correlate with patient survival
Literature	GENERIF_SUMMARY	These findings indicate that rhTIMP-1 promotes clonogenic expansion and survival in human progenitors via the activation of the CD63/PI3K/pAkt signaling pathway
Literature	GENERIF_SUMMARY	These data suggest that the p53(R248)

Main Category	Category	Terms
		mutant endows ovarian cancer cells with increased adhesiveness and that integrin beta4 and Akt signalling are associated with the mutation-enhanced ovarian cancer-mesothelial cell adhesion.
Literature	GENERIF_SUMMARY	Inactivated Sendai virus induces apoptosis and autophagy via the PI3K/Akt/mTOR/p70S6K pathway in human non-small cell lung cancer cells.
Literature	GENERIF_SUMMARY	Aschantin inhibited epidermal growth factor (EGF)-induced full activation of Akt by phosphorylation at Ser473/Thr308, resulting in inhibition of the mTORC2/Akt and Akt/mTORC1/p70S6K signaling pathways and activation of GSK3b
Literature	GENERIF_SUMMARY	Molecular dynamic simulation results of AKT1 in different states unraveled the mechanism of allosteric regulation of Akt1 phosphorylation by ATP on the conformational level, and elucidated the function of a disease-related mutant and its rescue.
Literature	GENERIF_SUMMARY	EBP1 participates in the regulation of intestinal inflammation via mediating Akt signaling pathway.
Literature	GENERIF_SUMMARY	CD82 regulated BCL2L12 expression via STAT5A and AKT signaling and stimulated proliferation and engrafting of leukemia cells.
Literature	GENERIF_SUMMARY	we demonstrated that the inactivation of the RHOC/AKT/MAPK pathway was involved in the tumor-suppressive functions of HOXD10 in cholangiocellular carcinoma .
Literature	GENERIF_SUMMARY	Combined PI3K/Akt and Hsp90 targeting synergistically suppresses essential functions of alloreactive T cells and increases Tregs.
Literature	GENERIF_SUMMARY	Findings indicate inositol polyphosphate 5-phosphatase PIPP (INPP5J) as a suppressor of oncogenic phosphoinositide 3-kinase (PI3K)/AKT signaling in breast cancer.
Literature	GENERIF_SUMMARY	the present study indicates the potential value of SNP rs712829 in the promoter region of the EGFR gene and SNP rs1130214 in the AKT1 gene as novel prognostic markers in NSCLC patients treated with EGFR-TKIs.
Literature	GENERIF_SUMMARY	LF triggered CREB1 phosphorylation in IL-1beta-induced Human articular chondrocytes apoptosis through AKT1 signaling.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	results uncover Akt as a key regulator of NPC1 degradation and link NPC1 to cancer cell proliferation and migration.
Literature	GENERIF_SUMMARY	miR-542-3p acts as a negative regulator in astrocytoma progression and that miR-542-3p down-regulation contributes to aberrant activation of AKT signaling
Literature	GENERIF_SUMMARY	Data show that small molecule NSC156529 inhibits the interaction of endogenous serine/threonine kinase AKT (AKT1) and 3-phosphoinositide dependent protein kinase-1 (PDPK1) proteins.
Literature	GENERIF_SUMMARY	Over-expressed TWIST associates with markers of epithelial mesenchymal transition and predicts poor prognosis in breast cancers via ERK and AKT activation.
Literature	GENERIF_SUMMARY	Difference between alleles frequency of SNP1, SNP2, SNP3 of patients with endometrial cancer and controls was not significant. SNPs (rs72715985), (rs2494750), and (rs74090038) of Akt1 gene are not associated with endometrial cancer in Iranian subjects.
Literature	GENERIF_SUMMARY	Data show that leukemia inhibitory factor (LIF) signaling promote chemoresistance in cholangiocarcinoma by up-regulating myeloid cell factor-1 (Mcl-1) via phosphatidylinositol 3-kinase (PI3K)/c-akt protein (AKT)-dependent pathway.
Literature	GENERIF_SUMMARY	MicroRNA-31 inhibits lung adenocarcinoma stem-like cells via down-regulation of MET-PI3K-Akt signaling pathway
Literature	GENERIF_SUMMARY	DNAJB6a reduces AKT signaling, and DNAJB6 expression in cancer cells reduces their proliferation and growth of xenograft esophageal squamous cell tumors in mice.
Literature	GENERIF_SUMMARY	Data suggest that a dietary component, EGCG (epigallocatechin-3-gallate) found in green tea, activates c-Akt and MAPK signaling in colon cancer cells.
Literature	GENERIF_SUMMARY	Data show that interleukin-8 (IL8) was a direct target of microRNA miR-203, and reduced miR-203 promoted nasopharyngeal carcinoma (NPC) cell radioresistance by activating IL8/AKT (akt proto-oncogene protein) signaling.
Literature	GENERIF_SUMMARY	These data suggest that the AKT and ERK/p38 MAPK pathways are involved in the synergistic effects of butein and cisplatin.
Literature	GENERIF_SUMMARY	These results suggest that Akt1 functions as a critical oncogene in human oral

Main Category	Category	Terms
		squamous cell carcinoma
Literature	GENERIF_SUMMARY	AKT rs1130233 polymorphisms were associated with increased gastric cancer risk in males and increased atrophic gastritis risk.
Literature	GENERIF_SUMMARY	AKT3 has a role in prostate cancer proliferation through regulation of Akt, B-Raf, and TSC1/TSC2
Literature	GENERIF_SUMMARY	AKT1 but not AKT2 protects islet cells from apoptosis and drives proliferation.
Literature	GENERIF_SUMMARY	O-GlcNAcylation enhanced oncogenic phenotypes possibly partially involving PI3K/Akt signaling pathway
Literature	GENERIF_SUMMARY	Activation of AKT1 is associated with poor prognosis in esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Furthermore, we demonstrated that high-LET carbon ions stimulated the unfolded protein response (UPR) and mediated autophagy via the UPR-eIF2alpha-CHOP-Akt signaling axis.
Literature	GENERIF_SUMMARY	these data suggest that DHA-induced apoptosis and autophagy in non-small cell lung cancer cells are associated with AMPK activation and PI3K/Akt inhibition, which in turn lead to suppression of mTOR
Literature	GENERIF_SUMMARY	Data show that tumors with proto-oncogene protein c-akt (AKT1 E17K) mutations are rational therapeutic targets for AKT inhibitors.
Literature	GENERIF_SUMMARY	Data show that rapamycin-insensitive companion of mTOR protein (RICTOR) plays a central role in phosphatidylinositol 3-kinase (PI3K)/proto-oncogene protein c-akt (AKT) pathway in melanocytes and its deregulation could be involved in melanoma development.
Literature	GENERIF_SUMMARY	that Abeta1-42 oligomers could cause disturbances in insulin/Akt/EAAT signaling in astrocytes
Literature	GENERIF_SUMMARY	Differentially expressed genes involved in cytokine and hormone signaling and cell division-related processes were found in juvenile granulosa cell tumors bearing AKT1 mutations.
Literature	GENERIF_SUMMARY	Transcriptomic dysregulations might be mediated by a limited set of transcription factors perturbed by AKT1 hyperactivation in JGCTs

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	gankyrin regulates HIF-1alpha protein stability and cyclin D1 expression, ultimately mediating FSH-driven ovarian cancer cell proliferation
Literature	GENERIF_SUMMARY	Data show that long ncRNAs MALAT1 silencing downregulated the expression of the microRNA miR-22-3p target gene CXCR2 and AKT pathway.
Literature	GENERIF_SUMMARY	PP-1 and Akt have roles in AR-v7 protein expression and activities when AR is functionally blocked
Literature	GENERIF_SUMMARY	GCN5 Potentiates Glioma Proliferation and Invasion via STAT3 and AKT Signaling Pathways
Literature	GENERIF_SUMMARY	hepatic insulin resistance in human obesity is
Literature	GENERIF_SUMMARY	Overexpression of PI3K-p110alpha is associated with progression of uterine cervical neoplasia, and the expression of pAkt and DJ-1 is positively correlated with PI3K-p110alpha expression in this process.
Literature	GENERIF_SUMMARY	Down-regulation of AKT is associated with osteoporosis.
Literature	GENERIF_SUMMARY	The CaM-binding interface in Akt(pleckstrin homology domain) was mapped to two loops adjacent to the PI(3,4,5)P3 binding site, which represents a CaM-binding motif and suggests a synergistic relationship between CaM and PI(3,4,5)P3 upon Akt activation.
Literature	GENERIF_SUMMARY	29b lowers postoperative IOP and sustains the function of filtering bleb. It inhibits the proliferation of fibroblasts and reduces collagen deposition by repressing the PI3K/Akt/Sp1 pathway in rabbits subjected to GFS.
Literature	GENERIF_SUMMARY	These results suggested that fucosterol executed its protective effects against CoCl2 induced cytotoxicity and inflammation by the inhibition of hypoxia inducible factor through PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	our findings indicate that TSN inhibits growth and induces apoptosis in CRC cells through suppression of AKT/GSK-3b/b-catenin pathway, suggesting that TSN may have potential for use in CRC treatment
Literature	GENERIF_SUMMARY	activated Akt could directly bind to PMS2 and cause degradation of PMS2 in Epithelial ovarian cancer (EOC) cells.
Literature	GENERIF_SUMMARY	Data show that both serine/threonine

Main Category	Category	Terms
		phosphatases PHLPP and dephosphorylated the physiological substrates of Akt1 and Akt3 with similar efficiencies.
Literature	GENERIF_SUMMARY	Akt and PKA phosphorylated KLHL3 at S433, and phosphorylation of KLHL3 by PKA inhibited WNK4 degradation.
Literature	GENERIF_SUMMARY	critical pathophysiologic event in the evolution of HER2-amplified cancers is the loss of the input signals that normally drive TORC2 signaling, repositioning it under Akt dependency, and fundamentally altering the role of HER3
Literature	GENERIF_SUMMARY	Lower hydrogen sulfide is associated with cardiovascular mortality, which involves PKCBII/Akt pathway in chronic hemodialysis patients.
Literature	GENERIF_SUMMARY	Data show that rab5 GTP-Binding Protein (Rab5a) is overexpressed in human hepatocellular carcinoma (HCC) and contributes to cancer cell proliferation and invasion through regulation of FAK and AKT signaling.
Literature	GENERIF_SUMMARY	Data show that apoptotic cell death by esculetin was associated with inhibition of the EGFR/PI3K/Akt signaling pathway and the decreased expression of nucleophosmin (NPM).
Literature	GENERIF_SUMMARY	Co-administration of the mTORC1/TORC2 inhibitor INK128 and the Bcl-2/Bcl-xL antagonist ABT-737 kills human myeloid leukemia cells through Mcl-1 down-regulation and AKT inactivation.
Literature	GENERIF_SUMMARY	RICTOR/MTORC2-AKT can integrate convergent hormonal and metabolic signals to provide coordinated and sensitive regulation of hepatic FOXO1-target gene expression.
Literature	GENERIF_SUMMARY	etoposidestimulated Akt is translocated into the mitochondria, thereby possibly enhancing its interaction with Smac and promoting apoptosis in HeLa cells.
Literature	GENERIF_SUMMARY	Data show that RNA silencing of CCN family member 1 protein (CCN1) inhibits umbilical vein endothelial cell (HUVEC) proliferation under hypoxic conditions by inhibiting phosphoinositide 3-kinase (PI3K)/AKT protein signaling.
Literature	GENERIF_SUMMARY	Results indicate that resveratrol could increase the anti-proliferative activity of bestatin through downregulating P-gp expression via suppressing the PI3K/Akt/

Main Category	Category	Terms
		mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Data show that synergistic apoptotic effect of the combination treatment is correlated with the block of the PI3K/Akt signal pathway in breast cancer cells.
Literature	GENERIF_SUMMARY	c-kit promotes the survival/growth and migration of human CPCs cultured ex vivo via the activation of PI3K and MAPK pathways
Literature	GENERIF_SUMMARY	Data show that microRNA miR-221 targets PTEN protein, leading to activation of the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Dephosphorylation of PDK-1 and the resulting changes to Akt phosphorylation is one of the mechanisms by which infection with Helicobacter pylori alter the balance between apoptosis and cell proliferation.
Literature	GENERIF_SUMMARY	This review focuses on recent advances in understanding of the roles of Akt and SGK1 in the regulation of renal tubular transport.
Literature	GENERIF_SUMMARY	Our results demonstrated that GRP94 is a key molecule in Hepatocellular carcinoma (HCC) progression that modulates the AKT pathway and eNOS levels
Literature	GENERIF_SUMMARY	suggest the AKT/SOX2 molecular axis as a regulator of breast carcinoma (BC) clonogenicity and AKT inhibitors as promising drugs for the treatment of SOX2-positive BC
Literature	GENERIF_SUMMARY	Data show that puquitinib mesylate (XC-302) induced autophagy in nasopharyngeal cancer cells CNE-2 by promoting cell autophagy and inhibits the phosphatidylinositol 3-Kinase (PI3K)/AKT protein/TOR serine-threonine kinases (mTOR) signaling pathway.
Literature	GENERIF_SUMMARY	PI3K-Akt-GSK3beta signaling pathway is required for SIRT1 induction by ER stress.
Literature	GENERIF_SUMMARY	Data suggest that the overexpression of microRNA miR16 inhibits the proliferation, invasion and metastasis of hepatocellular carcinoma cells associated with the
Literature	GENERIF_SUMMARY	Results indicate that p110alpha primarily mediates PI3K/AKT signaling in HER2+ PTEN deficient breast tumors whereas p110beta conditionally mediates HER2/PI3K signaling in the absence of p110alpha.
Literature	GENERIF_SUMMARY	Together, these results demonstrate that escin protects RPE cells from oxidative stress possibly through activating AKT-Nrf2

Main Category	Category	Terms
		signaling.
Literature	GENERIF_SUMMARY	AKT1 gene polymorphism rs1130214 is associated with antidepressant treatment response in patients with depressive disorders.
Literature	GENERIF_SUMMARY	Data indicate the role of tropomyosin-related kinase-B (TrkB) in activation of the interleukin-6/Janus kinase 2/STAT3 transcription factor and PI3 kinase/c-AKT proto-oncogene protein pathway in breast cancer.
Literature	GENERIF_SUMMARY	High AKT expression is associated with cancer.
Literature	GENERIF_SUMMARY	Akt-mediated phosphorylation of USP14 at Ser432, which normally blocks its catalytic site in the inactive conformation, activates its deubiquitinating activity in vitro and in cells.
Literature	GENERIF_SUMMARY	Findings indicate the role of CMTM7 protein in the regulation of epidermal growth factor receptor (EGFR)-AKT proto-oncogene protein signaling in tumor cells, and as a molecule related to Rab5 GTP-binding protein activation.
Literature	GENERIF_SUMMARY	new insight concerning the intracellular networks of the PI3K/AKT and MAPK signaling cascades for regulating T. gondii-induced IL-23 and IL-12 secretion in human monocytic cells.
Literature	GENERIF_SUMMARY	These data suggest that targeting the IL-8/AKT1 signaling pathway and DNMT1 may provide a potential therapeutic approach for blocking NPC metastasis.
Literature	GENERIF_SUMMARY	Down-regulation of microRNA-155 attenuates retinal neovascularization via the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Cytokines IL-2/7/9 stimulation activates PI3K/Akt pathways downstream of Ras in RasGRP1 T-cell acute lymphoblastic leukemia (T-ALL).
Literature	GENERIF_SUMMARY	Data suggest that primary cancer, hypoxic microenvironment generates positive feedback loops which hyperactivate proto-oncogene protein AKT, forming extreme robustness that forcing cancer cells to migrate to the distant site.
Literature	GENERIF_SUMMARY	Findings indicate that Epstein-Barr virus(EBV)-encoded latent membrane protein 1 (LMP1), PI3K/AKT, miR-21 and PTEN constitute a positive feedback loop and have a key role in LMP1-induced cancer

Main Category	Category	Terms
		stem cells (CSCs) in nasopharyngeal carcinoma (NPC).
Literature	GENERIF_SUMMARY	Data show that downregulating proto-oncogene protein Akt (p-Akt) by inhibiting PP2A inhibitor SET-mediated protein phosphatase 2A (PP2A) inactivation determined the pro-apoptotic effects of EMQA and paclitaxel combination treatment.
Literature	GENERIF_SUMMARY	In human colon cancer cells, Akt1 binds focal adhesion kinase via the Akt1 kinase domain independently of the pleckstrin homology domain
Literature	GENERIF_SUMMARY	Notch 1, pAkt and Bcl-2 proteins have roles in radioresistance of tumor cells
Literature	GENERIF_SUMMARY	Thus, GWL is a human oncoprotein that promotes the hyperactivation of AKT via the degradation of its phosphatase, PHLPP, in human malignancies.
Literature	GENERIF_SUMMARY	Further investigations on the molecular mechanisms revealed that total Akt protein level was decreased in a caspase-dependent way after treatment with 13OD
Literature	GENERIF_SUMMARY	MiR-185 acts as a tumor suppressor by targeting AKT1 in non-small cell lung cancer cells.
Literature	GENERIF_SUMMARY	we found that CXCR4-CXCL12/AKT axis regulates RhoA, Rac-1, and Cdc42 to modulate cell invasion and tumor metastasis. Together, these results demonstrate a role for CXCR4 in ESCC metastasis and progression and suggest potential targets for therapeutic intervention.
Literature	GENERIF_SUMMARY	Our data collectively indicate that miR-409-3p functions as a tumor suppressor in BC through downregulating Akt1.
Literature	GENERIF_SUMMARY	miR-23b acts as a suppressor of epithelial-to-mesenchymal transition in diabetic nephropathy through repressing PI3K-AKT signaling pathway by targeting HMGA2 and thus alleviating the disease.
Literature	GENERIF_SUMMARY	Data show that CD133 overexpression resulted in decreased EGF and EGFR expression, increased telomerase reverse transcriptase (TERT) expression, and increased Akt phosphorylation.
Literature	GENERIF_SUMMARY	we discovered a novel mechanism mediated by Smad4 to trigger 5-FU chemosensitivity through cell cycle arrest by inhibiting the

Main Category	Category	Terms
		PI3K/Akt/CDC2/survivin cascade.
Literature	GENERIF_SUMMARY	Activated Akt signalling pathway mediates Sema4A-PlexinD1- enhanced collagen contraction in human lung fibroblasts.
Literature	GENERIF_SUMMARY	One of the mechanisms was TF3 inactivated Akt/mTOR/p70S6K/4E-BP1 pathway and Akt/c-Myc pathway
Literature	GENERIF_SUMMARY	Acetaldehyde treatment led to a significant decrease in the levels of activated Akt and cyclic AMP-responsive element binding protein (CREB).
Literature	GENERIF_SUMMARY	Upregulation of AKT is associated with breast cancer.
Literature	GENERIF_SUMMARY	Overexpression of AKT is associated with osteosarcoma.
Literature	GENERIF_SUMMARY	The underlying mechanisms involve reduced phosphorylation of Akt and nuclear factor-kappaB (NF-kappaB) p65. We noted that as with EtP, EtOH reduced the inflammatory response in lung epithelial cells under acute inflammatory conditions
Literature	GENERIF_SUMMARY	The transcription activity of PI3K/AKT was markedly inhibited by CA, as well as IGF-1 which functions as an anti-apoptotic factor
Literature	GENERIF_SUMMARY	this study demonstrated that JC induced the apoptosis of hepatocellular carcinoma (HCC) cells by activating Akt/Foxo signaling pathway and increasing intracellular ROS levels.
Literature	GENERIF_SUMMARY	Data show one single nucleotide polymorphism (SNP) in Fox transcription factor FOXO3 (FOXO3) was significantly associated with longevity, and the six SNPs in proto-oncogene protein AKT1 (AKT1) gene and in IGF-2 Receptor (IGF-2R) gene are not.
Literature	GENERIF_SUMMARY	Novel pharmacodynamic biomarkers for MYCN protein and PI3K/AKT/mTOR pathway signaling in children with neuroblastoma.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with prostate cancer.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with neuroblastoma.
Literature	GENERIF_SUMMARY	The cross-talk between SP1 and p65, and the positive feedback regulatory loop of PI3-K/Akt signaling by EP4 contribute to the overall responses of solamargine in this process

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Akt activation can provide useful prognostic information and that tobacco represents a serious risk factor for recurrence in a cohort of Tunisian patients
Literature	GENERIF_SUMMARY	AKT mutations are associated with cancers.
Literature	GENERIF_SUMMARY	By inhibiting mTOR and MnSOD, we confirmed that rapamycin functioned through the mTOR/MnSOD/reactive oxygen species (ROS) signaling pathway, and the existence of Akt governed the rapamycininduced asymmetric division
Literature	GENERIF_SUMMARY	Akt2, but not Akt1, colocalizes in the nucleus with lamin B in the nuclear envelope.
Literature	GENERIF_SUMMARY	High expression of AKT is associated with resistance to sorafenib in liver cancer.
Literature	GENERIF_SUMMARY	Taken together, in current study, we found a novel tumor suppressor, DDX10, is epigenetic silenced by miR-155-5p in ovarian cancer, and the down-regulated expression pattern of DDX10 promotes ovarian cancer proliferation through Akt/NF-kappaB pathway.
Literature	GENERIF_SUMMARY	These data suggest that CBX8 modulates SIPS through the RB-E2F1 pathway in CML cells and provide important insight into its application in CML treatment.
Literature	GENERIF_SUMMARY	Data show that 14-3-3beta protein augmented the expression of matrix metalloproteinasea MMP2 and MMP9 through PI3 kinase/Akt protein/NF-kappaappa B pathway, thereby enhancing the invasiveness of hepatocellular carcinoma (HCC) cells.
Literature	GENERIF_SUMMARY	Following transepithelial migration, neutrophils adhesion to ICAM-1 resulted in activation of Akt and beta-catenin signaling, increased epithelial-cell proliferation, and wound healing.
Literature	GENERIF_SUMMARY	Taken together, these results show that CHAF1A contributes to the proliferation of glioblastoma cells and may be developed as a de novo drug target and prognosis biomarker of glioblastoma.
Literature	GENERIF_SUMMARY	our results indicate that miR-181A expression is associated with CRC patients with late liver metastases through PTEN/AKT signaling.
Literature	GENERIF_SUMMARY	YY1 promotes mTORC2-mediated AKT activation

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data suggest that regulation of Bax stability and transport between cytosol and mitochondria is more tightly regulated by Bcl-xL/BCL2L1 when Bax has been phosphorylated at Ser184 by AKT1; experiments involved expression in haploid yeast strains.
Literature	GENERIF_SUMMARY	Our study demonstrates that CSE/H2S system is regulated by miR-216a, and regulates ABCA1-mediated cholesterol efflux and cholesterol levels through the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	The data provide the first evidence that SJP protects SH-SY5Y cells against 6-OHDA toxicity possibly by inhibiting MAPK and NF-kappaB and activating PI3K/Akt signaling pathways.
Literature	GENERIF_SUMMARY	Our results demonstrated the pro-inflammatory and angiogenic effects of PlGF on HIMECs in IBD through activation of PI3K/Akt signaling pathway. PlGF/PI3K/Akt signaling may serve as a potential therapeutic target for IBD.
Literature	GENERIF_SUMMARY	Low AKT1 expression is associated with osteosarcoma.
Literature	GENERIF_SUMMARY	Mefloquine acts on gastric cancer cells via suppressing PI3K/Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	High expression of AKT1 is associated with Colorectal cancer.
Literature	GENERIF_SUMMARY	results give a clue to account for an underlying cause of HBx-mediated hepatocarcinogenesis. It appears that Ser31 phosphorylation of HBx by Akt plays an important role. The current study provides an example of association of HBV genome variations with risks of HCC and chronic hepatitis.
Literature	GENERIF_SUMMARY	Our data suggested a potential role of Rab11-FIP2 in tumor progression and provided novel insights into the mechanism of how Rab11-FIP2 positively regulated cell migration and invasion in CRC cells.
Literature	GENERIF_SUMMARY	effects. Further studies indicated that PRKCI knockdown-mediated autophagy was associated with the inactivation of phosphatidylinositol 3-kinase alpha/AKT-mammalian target of rapamycin (PIK3CA/AKT-MTOR) signaling.
Literature	GENERIF_SUMMARY	Therefore, we identified a tumorigenic role of Id-1 in OS and suggested a potential therapeutic target for OS patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Based on the collective findings, we suggest that vIRF2 acts as an activator in PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	These results suggested that MiR-486 mediated the hypoxia-induced angiogenic activity and promoted the proliferation and survival of BM-MSCs through regulating PTEN-PI3K/AKT signaling.
Literature	GENERIF_SUMMARY	the potentially functional AKT1 rs2494752 SNP may affect Gastric Adenocarcinoma susceptibility, likely by modulating the AKT1 promoter transcriptional activity
Literature	GENERIF_SUMMARY	Findings reveal that miR-302 regulates OCT4 by suppressing AKT1, which provides hPSCs two characteristics related to their potential for clinical applications
Literature	GENERIF_SUMMARY	KCNN4 inhibition differentially regulates migration of intestinal epithelial cells in inflamed vs. non-inflamed conditions in a PI3K/Akt-mediated manner.
Literature	GENERIF_SUMMARY	The findings indicate that succinate-GPR91 signaling may be involved in right ventricular hypertrophy via PI3K/Akt signaling in vivo and in vitro.
Literature	GENERIF_SUMMARY	Three AKT1 SNPs might confer protection against oesophageal squamous cell carcinoma risk; nevertheless, these effects may be dependent on other risk factors.
Literature	GENERIF_SUMMARY	AIM2 contributes to the maintenance of intestinal integrity via Akt and protects against Salmonella mucosal infection.
Literature	GENERIF_SUMMARY	miR-137 suppresses the phosphorylation of AKT.
Literature	GENERIF_SUMMARY	High p-AKT expression is associated with Chromosomal Instability in Gastric Cancer.
Literature	GENERIF_SUMMARY	Results indicate that eIF4B integrates the signals from Pim and PI3K/Akt/mTOR pathways in Abl-expressing leukemic cells.
Literature	GENERIF_SUMMARY	Akt1 and Akt2 activated both SREBP-1 and SREBP-2, whereas Akt3 upregulated SREBP-1 to enhance hepatitis C virus translation.
Literature	GENERIF_SUMMARY	AKT1E17K induces hyperplasia of mouse lung epithelium in vivo and cooperates with urethane to induce the fully malignant phenotype
Literature	GENERIF_SUMMARY	MiR-205 promotes proliferation, migration and invasion of nasopharyngeal carcinoma cells by activation of AKT signaling and

Main Category	Category	Terms
		down-regulation of PTEN.
Literature	GENERIF_SUMMARY	The AKT1 gene codes for a protein kinase that forms part of the striatal dopamine receptor signalling cascade.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with malignant pleural mesothelioma.
Literature	GENERIF_SUMMARY	expression of Akt and mTOR protein was enhanced in the endurance exercise group, whereas only the level of mTOR was elevated following Resistance training.
Literature	GENERIF_SUMMARY	Data suggest that a dietary component/ supplement, L-carnitine, protects hepatocytes against oxidative stress/cell damage (induced here by hydrogen peroxide) through AKT- (proto-oncogene protein c-akt)-mediated activation of NRF2 (nuclear factor [erythroid-derived 2]-like 2 protein) signaling pathway.
Literature	GENERIF_SUMMARY	These findings demonstrated that Akt is related to NF-kappaB and Bad signaling pathway possibly playing a direct role in the progression of liver cancer. Thus, Akt might be an important and potential treatment choice for the clinical diagnosis and treatment in the future.
Literature	GENERIF_SUMMARY	p53 R273H mutation unleashes the inhibition effect of miR-30a on IGF-1R expression, thus leading to elevated activation of IGF-1R-AKT signaling cascade in tumor cells.
Literature	GENERIF_SUMMARY	Stimulation with H2O2 decreased the phosphorylation of Akt and FoxO3a, and induced nuclear localization of FoxO3a and apoptosis of H9c2 cells. These effects of H2O2 were abrogated by pretreatment with FGF-2
Literature	GENERIF_SUMMARY	Akt phosphorylates MAZ at Thr385, and the phosphorylated MAZ is released from the p53 promoter, leading to transcriptional activation of p53.
Literature	GENERIF_SUMMARY	AKT protects cancer cells from autophagic cell death during nutritional starvation.AKT-p53 axis play a significant role in cell survival through regulating autophagy during nutritional deprivation.
Literature	GENERIF_SUMMARY	P-AKT and TTF-1 using IHC had statistically significant correlation with EGFR mutation with high negative predictive value
Literature	GENERIF_SUMMARY	These results suggest that exposure to nicotine might negatively impact the apoptotic potential of chemotherapeutic

Main Category	Category	Terms
		drugs and that alpha5-nAChR/AKT signaling plays a key role in the anti-apoptotic activity of nicotine induced by cisplatin.
Literature	GENERIF_SUMMARY	The expression of PI3K, AKT, and phospho-AKT proteins was significantly lower in the thin endometrium group than in the normal endometrium group.
Literature	GENERIF_SUMMARY	Down-regulation of CD133 inhibits migration of gallbladder carcinoma cells through reducing Akt phosphorylation.
Literature	GENERIF_SUMMARY	results suggest that Akt and JNK pathways are involved in the regulation of B-cell maturation antigen(BCMA)
Literature	GENERIF_SUMMARY	GHRH and GHRH-R loops are involved in placental choriocarcinoma cell line viability and apoptosis through Akt and eIF2a pathways.
Literature	GENERIF_SUMMARY	Data suggest that claudin-18 suppresses the abnormal proliferation and motility of lung epithelial cells mediated by inhibition of phosphorylation of pyruvate dehydrogenase kinase isoform 1 (PDK1) and proto-oncogene protein c-akt (Akt).
Literature	GENERIF_SUMMARY	mediates mitophagy thus contributing to alveolar macrophage apoptosis resistance and pulmonary fibrosis
Literature	GENERIF_SUMMARY	results reveal that the AKT signaling pathway plays a critical role in the malignancy of adult T-cell leukemia-derived cells
Literature	GENERIF_SUMMARY	Studies indicate key feedback loops involved in the cross-talk among the insulin-proto-oncogene protein c-akt (AKT) and MAPK/ERK signaling pathways.
Literature	GENERIF_SUMMARY	The Asian-American variant of human papillomavirus type 16 exhibits higher activation of MAPK and PI3K/AKT signaling pathways, transformation, migration and invasion of primary human keratinocytes.
Literature	GENERIF_SUMMARY	AKT (Tyr308/Ser473) is upregulated by Notch1 activation in human breast carcinoma cells.
Literature	GENERIF_SUMMARY	In the conditon of miR- 128b over-expression, we also observed spontaneous inactivation of the Akt/NF-kappaB signalling, implying PDK1 was a potential regulator of this pathway. In conclusion, our study shed some novel light on miR-128b-PDK1/Akt/NF-kappaB axis onGastric cancer (GC) progression

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Studies show that AKT1 regulates apoptosis in limiting cytokine concentrations and promotes angiogenesis mainly via phosphorylation and controlling important downstream effectors which promote varied aspects of angiogenic signaling. [review]
Literature	GENERIF_SUMMARY	results suggest that IL10-mediated inhibition of autophagy is facilitated by the cross talk between STAT3, AKT, and mTOR; in other words, the IL10-IL10R-STAT3 and IL10-AKT-mTOR pathways.
Literature	GENERIF_SUMMARY	High AKT expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	The Akt pathway controls endothelial Jagged1 expression and, thereby, Notch signalling regulating VSMC maintenance.
Literature	GENERIF_SUMMARY	phosphorylation of STAT3 and AKT was highly induced by treatment with IL-22 via IL-22R1. IL-22R1 was also consistently overexpressed in recurrent NSCLC tissues
Literature	GENERIF_SUMMARY	AKT overexpression is associated with cervical squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Moreover, mechanistic study reveals that PD166866 induces autophagy through repressing Akt/mTOR signaling pathway. Together, the present study provides new insights into the molecular mechanisms underlying the anti-tumor activities of FGFR antagonists, and may further assist the FGFRs-based drug discovery.
Literature	GENERIF_SUMMARY	lower tumor expression of nuclear pAKT was associated with higher rate of response to sorafenib. This observation justifies evaluation of combination therapy with sorafenib and an inhibitor of the PI3K/AKT signaling pathway in RAIR-DTC.
Literature	GENERIF_SUMMARY	ITGB4 is overexpressed in hepatocellular carcinoma tissues and promotes metastases of HCC by conferring anchorage independence through EGFR-dependent FAK-AKT activation.
Literature	GENERIF_SUMMARY	we found that combinatorial MK-2206+progesterone treatments decreased angiogenesis and proliferation in the Pten(d/d) conditional mouse model of endometrial cancer. Taken together, these findings suggest that a combinatorial therapeutic approach utilizing Akt inhibitors with progestins may improve the efficacy of progestin therapy for the treatment of endometrial cancer.
Literature	GENERIF_SUMMARY	AKT1(E17K) prevents HER2-driven

Main Category	Category	Terms
		mammary tumor formation, in part through negative feedback inhibition of RTK signaling.
Literature	GENERIF_SUMMARY	lps to elucidate the p53-independent regulatory function of MDM2 in Akt signaling, offering a novel view of the mechanism by which triptolide functions as an anticancer agent
Literature	GENERIF_SUMMARY	Pregnancy-associated plasma protein-A induced TF expression through the activation of Akt/NF-kappaB axis
Literature	GENERIF_SUMMARY	findings demonstrate that the BRD7/miR-141/PTEN/AKT axis has critical roles in the progression of NPC and provide some promising targets for the diagnosis and treatment of NPC.
Literature	GENERIF_SUMMARY	Conclusively, GRP78 plays a vital role in gemcitabine resistance and clinical strategy to improve gemcitabine efficacy in breast cancer by manipulating GRP78 should be explored.
Literature	GENERIF_SUMMARY	Data show that the elevated activities of PI3Ks, AKt and GSK-3beta were effectively suppressed by exogenous carbon monoxide (CO), leading to the improvement of platelet function.
Literature	GENERIF_SUMMARY	The results suggest that KCa3.1 activation contributes to dysfunctional tubular autophagy in diabetic nephropathy through PI3K/Akt/mTOR signaling pathways.
Literature	GENERIF_SUMMARY	periostin enhanced invasion and migration abilities of EECs and facilitated the epithelial-mesenchymal transition of endometrial epithelial cell through ILK-Akt signaling pathway
Literature	GENERIF_SUMMARY	The results of the present study may indicate that NE activates the phosphoinositide-3 kinase/Akt signaling pathway in leukemia cells to inhibit apoptosis and enhance cell proliferation
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with cervical cancer.
Literature	GENERIF_SUMMARY	Insulin treatment caused sustained Akt activity, whereas EGF or PDGF-AA promoted transient signaling; PDGF-BB produced sustained responses at higher concentrations.Transient responses to EGF were caused by negative feedback at the receptor level, as a second treatment yielded minimal responses, whereas parallel exposure to IGF-I caused full Akt activation

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Our study demonstrates that the IGF1R/p110b/AKT/mTOR axis confers resistance to BYL719 in PIK3CA mutant breast cancers.
Literature	GENERIF_SUMMARY	this study highlights the molecular basis for posttranslational modifications of Gata3 that control the regulation of IFNgamma expression in memory Th2 cells.
Literature	GENERIF_SUMMARY	CCL21 can facilitate chemoresistance and stem cell property of colorectal cancer cells via the upregulation of P-gp, Bmi-1, Nanog, and OCT-4 through AKT/GSK-3beta/Snail signals.
Literature	GENERIF_SUMMARY	Mutations of PIK3CA were found in our cohort in 21 cases (28 %), 10 (13 %) in exon 9 and 11(15 %) in exon 20. Mutation frequency of AKT1 and PTEN genes was 4 and 3 %, respectively
Literature	GENERIF_SUMMARY	rs2498804 not associated with the need for neonatal resuscitation
Literature	GENERIF_SUMMARY	results show that the activation of the ezrin-pAkt signaling axis is associated with the more aggressive clinicopathological features of PPA compared with LPA
Literature	GENERIF_SUMMARY	our results suggest that the mutational spectrum of pediatric AF is more complex than in adult AF, being rich in AKT1 and BRAF, as well as CTNNB1 gene mutations.
Literature	GENERIF_SUMMARY	In conclusion, exenatide significantly improves coronary endothelial function in patients with newly diagnosed type 2 diabetes. The effect may be mediated through activation of AMPK/PI3K-Akt/eNOS pathway via a GLP-1R/cAMP-dependent mechanism.
Literature	GENERIF_SUMMARY	miR-143-3p inhibits hypertrophic scarring by regulating the proliferation and apoptosis of human HSFs, inhibiting ECM production-associated protein expression by targeting CTGF, and restraining the Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	The authors found that Akt interacts with HIV-1 Nef protein. In primary T cells treated with exogenous Nef or acutely infected with Nef-expressing HIV-1 in vitro, Akt became phosphorylated on serine(473) and threonine(308).
Literature	GENERIF_SUMMARY	Collectively, these findings demonstrate that stimulation of G12 proteins is capable of promoting invasion through RhoA/ROCK-JNK activation.

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Literature	GENERIF_SUMMARY	depletion of FANCI, but not FANCD2 or USP1, results in increased phosphorylation and activation of Akt.
Literature	GENERIF_SUMMARY	Akt has a role in several key steps of peripheral nervous system myelination, including axon wrapping and myelin sheath thickness.
Literature	GENERIF_SUMMARY	Data indicate that proto-oncogene protein c-akt (Akt) phosphorylates distinct sites than SRPK1 protein within the arginine-serine (RS) domain of Lamin B Receptor (LBR).
Literature	GENERIF_SUMMARY	These results suggest that COMP-Ang1 enhances survival and proliferation of human PLFs through the activation of Tie2-mediated signaling, where PI3K/Akt and MAPK-c-Jun signaling pathways act as downstream effectors.
Literature	GENERIF_SUMMARY	endothelial-barrier protective role of VEGF in the long term and the importance of Akt1-FoxO signaling on tight-junction stabilization and prevention of vascular leakage through claudin expression, is reported.
Literature	GENERIF_SUMMARY	Melatonin inhibits tumorigenicity of glioblastoma stem-like cells via the AKT-EZH2-STAT3 signaling axis.
Literature	GENERIF_SUMMARY	This review describes the mechanisms by which PI3-K/AKT pathway causes radioresistance and the role of HIV protease inhibitors especially nelfinavir as a potential candidate drug to target the AKT pathway for overcoming radioresistance and its use in various clinical trials for different malignancies
Literature	GENERIF_SUMMARY	Atypical Activation of Akt is associated with Human Cytomegalovirus Infection.
Literature	GENERIF_SUMMARY	we have found that SNG inhibits Akt, a key anti-apoptotic protein kinase by dephosphorylating it at Ser(473), leading to the dephosphorylation of its downstream targets, GSK3beta and mTOR
Literature	GENERIF_SUMMARY	These findings identify phospho-AKT in the primary tumor and miR-200c later during tumor progression as prognostic molecules.
Literature	GENERIF_SUMMARY	Data indicate that proto-oncogene c-Akt (AKT) is important in cell cycle regulation in the control of chemosensitivity in ovarian cancer cells (OVCAs) and suggest an alternate regulatory pathway for the development of rationale therapy for cisplatin-resistant ovarian cancer.
Literature	GENERIF_SUMMARY	Results indicate that des-gamma-carboxy

Main Category	Category	Terms
		prothrombin (DCP) antagonizes the inhibitory effects of Sorafenib on hepatocellular carcinoma (HCC) through activation of the Raf/MEK/ERK and PI3K/Akt/mTOR signaling pathways.
Literature	GENERIF_SUMMARY	This study showed that the induction level of IL-32 was increased in chronic rhinosinusitis with nasal polyps compared to normal nasal mucosa and that LPS-induced IL-32 expression in nasal polyp-derived fibroblasts was regulated via the TLR4/JNK/AKT/CREB signaling pathway.
Literature	GENERIF_SUMMARY	PI3Kdelta and PI3Kgamma inhibition with IPI-145 has anti-proliferative activity in primary AML cells by inhibiting the activity of AKT and MAPK
Literature	GENERIF_SUMMARY	elevated Th17 cells secrete IL-17A by which promotes the proliferation and resistance to daunorubicin in B-cell acute lymphoblastic leukemia cells through activation of Akt signaling.
Literature	GENERIF_SUMMARY	TUSC3 regulates proliferation and invasion of glioblastoma cells by inhibiting the activity of the Akt signaling pathway.
Literature	GENERIF_SUMMARY	IL-10 stimulation activated the JAK/Stat-3 and PI3K/Akt signaling pathways. Moreover, IL-10 treatment increased translocation of p65 NF-kappaB into the nuclear compartment, and up-regulated expression of the pro-survival proteins Bcl-2 and Bcl-xL.
Literature	GENERIF_SUMMARY	TRAP-induced activation of Akt in addition to p38 MAP kinase positively regulates the release of phosphorylated-HSP27 from human platelets, which is closely related to the platelet hyper-aggregation in type 2 diabetes mellitus patients.
Literature	GENERIF_SUMMARY	MiR-21 targets key proteins in PTEN/PI3K/AKT signal pathway, promoting proliferation, migration, invasion, and cell cycle, and inhibiting apoptosis of human esophageal cancer cells.
Literature	GENERIF_SUMMARY	Report expression of microRNAs in lung neoplasms with increased AKt1/2 gene number.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with breast Cancers.
Literature	GENERIF_SUMMARY	miR-21 confers decreased autophagy in cervical cancer cells after IR via the Akt-mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Akt phosphorylates inversin at amino acids 864-866 that are required not only for Akt

Main Category	Category	Terms
		interaction, but also for inversin dimerization.
Literature	GENERIF_SUMMARY	The present study suggests that ADAM17-siRNA inhibits MCF-7 breast cancer and is activated through the EGFR-PI3K-AKT signaling pathway
Literature	GENERIF_SUMMARY	NK2 and HGF drive motogenic and anti-apoptotic signaling but only HGF drives cell proliferation by activating AKT-pathway signaling.
Literature	GENERIF_SUMMARY	Risk of recurrence correlated with cytoplasmic expression of ODAM and pAKT, whereas nuclear expression did not predict recurrence of gastrointestinal stromal tumors.
Literature	GENERIF_SUMMARY	Akt mediates the proliferative effect of miR-221 on Capan-2 pancreatic ductal adenocarcinoma cells.
Literature	GENERIF_SUMMARY	Data show that overexpression of S100 calcium binding protein A16 protein (S100A16) activated AKT protein and ERK signaling pathways.
Literature	GENERIF_SUMMARY	High AKT1 expression is Associated with Endometrial Cancer.
Literature	GENERIF_SUMMARY	Results show that macrophages in the renal cell carcinoma (RCC) tumor microenvironment could increase RCC metastasis via activation of the AKT/mTOR signals.
Literature	GENERIF_SUMMARY	Musashi-1 has a role in regulating AKT-derived IL-6 autocrinal/paracrinal malignancy and chemoresistance in glioblastoma
Literature	GENERIF_SUMMARY	Transforming growth factor-beta1 upregulation triggers pulmonary artery smooth muscle cell proliferation and apoptosis imbalance in rats with hypoxic pulmonary hypertension via the PTEN/AKT pathways
Literature	GENERIF_SUMMARY	Our results show a miRNA editing activity trend aligning with cellular pathways closely associated to hypoxia, such as the VEGF and PI3K/Akt pathways, providing important novel insights on this poorly elucidated phenomenon.
Literature	GENERIF_SUMMARY	We revealed that activation of the PI3K/Akt pathway was the main cause of impaired autophagy in ELK3 KD. Our results suggest that targeting ELK3 may be a potential approach to overcome doxorubicin resistance in breast cancer therapeutics.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Akt1 SNPs were not associated with schizophrenia age of onset.
Literature	GENERIF_SUMMARY	The present study indicated a possibility that mitochondrial reactive oxygen species release through mitochondrial permeability transition pores and subsequent Akt activation were necessary for the anti-apoptotic effect of 50-Hz magnetic field on human amniotic (FL) cells .
Literature	GENERIF_SUMMARY	miR-495 may act as a tumor suppressor by targeting Akt1 in esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	GKN2 suppressed epithelial mesenchymal transition of gastric cancer cells by downregulation of snail through PI3K/AKT/GSK3beta signaling pathway.
Literature	GENERIF_SUMMARY	AQP9 overexpression decreased the protein levels of phosphatidylinositol-3-kinase (PI3K), leading to reduced phosphorylation of Akt, and subsequently the protein levels of forkhead box protein O1 (FOXO1) were increased.
Literature	GENERIF_SUMMARY	These results demonstrate that NLRC5 promotes hepatocellular carcinoma progression via the AKT/VEGF-A signaling pathway
Literature	GENERIF_SUMMARY	NEDD4L negatively regulates PIK3CA protein levels via ubiquitination and is required for the maintenance of PI3K-AKT signaling pathway.
Literature	GENERIF_SUMMARY	results suggest that AKT1, FTO, and AKTIP polymorphisms were not associated with obesity/overweight in Brazilians children.
Literature	GENERIF_SUMMARY	the RRM2/AKT/NF-kappaB signaling pathway is essential for tumor invasiveness in gastric cancer cells
Literature	GENERIF_SUMMARY	Results suggest that polymorphism of AKT1 locus is associated with risk of Parkinson's disease and that the G allele at rs2498799 may decrease the risk of Parkinson's disease in the North-eastern part of Han Chinese female population.
Literature	GENERIF_SUMMARY	activation appears to be a key survival mechanism for HDAC6 inhibitor treatment.
Literature	GENERIF_SUMMARY	Data show that interleukin-12 (IL-12) treatment decreased the expressions of p-AKT protein, p-mTOR protein and p-STAT3 protein.
Literature	GENERIF_SUMMARY	AKT and 14-3-3 proteins down-regulate the activity of several BCR-associated

Main Category	Category	Terms
		components, including BTK, BLNK and SYK and also inhibit SYK's interaction with Importin 7
Literature	GENERIF_SUMMARY	RAS-driven tumors induce PI3K/AKT-dependent beta-catenin activation
Literature	GENERIF_SUMMARY	Mutations in PIK3CA, AKT1, and PTEN also were less frequently found in cancers with mutations in the DNA guardian, TP53. This observation might seem counterintuitive because increased genome instability might facilitate development and selection of advantageous mutations.
Literature	GENERIF_SUMMARY	PIK3CA, PTEN, and AKT1 mutations occurred more frequently in the presence of hormone receptor overexpression (androgen, progesterone, or estrogen receptor). PIK3CA mutations were also more common in the HER2-positive than in the HER2-negative group; the opposite pattern was seen for PTEN mutation or PTEN loss
Literature	GENERIF_SUMMARY	reactivation of ERK and Akt associated respectively with the activity of mutant BRAF and CDC37 renders mutant BRAF colon cancer cells resistant to AUY922
Literature	GENERIF_SUMMARY	Mutational activation of Kit-, Ras/Raf/Erk- and Akt- pathways indicate the biological importance of these pathways and their components as potential targets for therapy.
Literature	GENERIF_SUMMARY	MLN0128 selectively targeted and functionally inhibited acute myeloid leukemia (AML) stem/progenitor cells with high AKT/mTOR kinase signaling activity.
Literature	GENERIF_SUMMARY	Taken together, these results revealed that Lp-PLA2 silencing protected against ox-LDL-induced oxidative stress and cell apoptosis via Akt/mTOR signaling pathway in human THP1 macrophages.
Literature	GENERIF_SUMMARY	Data suggest that phosphatase and tensin homolog deleted on chromosome 10 (PTEN) mutations might not be correlated with activated protein kinase B (Akt).
Literature	GENERIF_SUMMARY	FHL2 facilitates ovarian granulosa cell tumor progression via controlling AKT1 transcription.
Literature	GENERIF_SUMMARY	Mutations in AKT1 gene is associated with ductal adenomas.
Literature	GENERIF_SUMMARY	In acute schizophrenia patients, the mRNA expression levels of DRD2 and PI3KCB were significantly lower than those in the healthy controls, while the AKT1 mRNA levels were

Main Category	Category	Terms
		significantly higher than those in the healthy controls.
Literature	GENERIF_SUMMARY	Data show that zinc finger protein 143 (ZNF143) promotes epithelial-mesenchymal transition (EMT) through regulating PI3K/AKT pathway in gastric cancer (GC) cells.
Literature	GENERIF_SUMMARY	Study of four cases confirms BRAF V600E mutation as a probable driver in a subset of ciliated muconodular papillary tumors (CMPT) of the lung, along with AKT1 mutation, which further supports that CMPT are indolent pulmonary neoplasms.
Literature	GENERIF_SUMMARY	High AKT expression is associated with Recurrence of Prostate Cancer.
Literature	GENERIF_SUMMARY	Overexpression of AKT is associated with pituitary neoplasms.
Literature	GENERIF_SUMMARY	ID1 down-regulation induced parallel changes in the IGF and AKT pathways. The crosstalk of these pathways may enhance malignant phenotypes in salivary gland cancer.
Literature	GENERIF_SUMMARY	Low AKT1 expression is associated with malignant phenotype and temozolomide chemoresistance in glioblastomas.
Literature	GENERIF_SUMMARY	Data suggest that various buttermilk samples exhibit anticancer activity in colon cancer cell line; this antitumor activity appears to involve down-regulation of Wnt signaling, down-regulation of MAP kinase signaling, and down-regulation of AKT1 phosphorylation.
Literature	GENERIF_SUMMARY	This study demonstrated that increased expression phosphorylated Akt1 in the prefrontal cortex in patient with schizophrenia.
Literature	GENERIF_SUMMARY	Results indicate that hypoxia increases IL-11 secretion in anaplastic thyroid carcinoma (ATC) cells via HIF-1alpha induction and that IL-11 then induces epithelial-mesenchymal transition (EMT) in these cells via the PI3K/Akt/GSK3beta pathway.
Literature	GENERIF_SUMMARY	HPV16 E6-E7 promotes cancer stem-like cells phenotype in esophageal squamous cell carcinoma cells through the activation of PI3K/Akt signaling pathway
Literature	GENERIF_SUMMARY	GHRH agonist MR-409 increase phosphorylation of AKT and ERK1/2 in dermal fibroblasts.
Literature	GENERIF_SUMMARY	Our results demonstrated that NC inhibits development of glioblastoma multiforme

Main Category	Category	Terms
		(GBM) by targeting the PI3K/Akt/mTOR signaling pathway and provides a potential therapeutic agent for the treatment of GBM.
Literature	GENERIF_SUMMARY	enhanced NEK2 expression in hepatocellular carcinoma (HCC) promotes HCC progression and drug resistance by promoting PP1/Akt and Wnt pathway activation, which may represent a new therapeutic target for HCC.
Literature	GENERIF_SUMMARY	T cells from patients with acute or chronic B cell leukemia, which were also metabolically exhausted had defective Akt/mTORC1 signaling, reduced expression of Glut1 and hexokinase 2, and decreased glucose metabolism.
Literature	GENERIF_SUMMARY	AKT1 (E17K) is the most likely disease driver in certain breast cancer patients.
Literature	GENERIF_SUMMARY	Data show that SC79 activated Akt-dependent NF-E2-related factor 2 (Nrf2) signaling and inhibited UV-induced oxidative stress in retinal pigment epithelium (RPE) cells.
Literature	GENERIF_SUMMARY	Mechanistically, G-Rg1 promoted the phosphorylation of Akt and FoxO3a and led to the cytoplasmic translocation of FoxO3a, which in turn suppressed FoxO3a-modulated expression of proapoptotic Bim and elevated the ratio of Bcl-2 to Bax.
Literature	GENERIF_SUMMARY	PTEN and GSK3beta are targeted by miR-29a, and miR-29a may contribute to ADR resistance through inhibition of the PTEN/AKT/GSK3beta pathway in breast cancer cells. Thus, miR-29a may be a potential target for the patients who acquired ADR-resistance during the treatment of breast cancer.
Literature	GENERIF_SUMMARY	NS4A and NS4B, cooperatively suppress the Akt-mTOR pathway and lead to cellular dysregulation.
Literature	GENERIF_SUMMARY	Bone morphogenetic protein (BMP)9 and BMP10 are high affinity ligands for activin receptor-like kinase 1 (ALK1).
Literature	GENERIF_SUMMARY	B55alpha-PP2A mutations in acute myeloid leukemia have roles in leukemogenesis by promoting AKT T308 phosphorylation and sensitivity to AKT inhibitor-induced growth arrest
Literature	GENERIF_SUMMARY	These data suggest that there is minimal redundancy between the Akt isoforms in ovarian cancer progression. These findings have important implications in the design of Akt inhibitors for the effective treatment of

Main Category	Category	Terms
		ovarian cancer.
Literature	GENERIF_SUMMARY	Inhibition of protooncogenic microRNA-181a may suppress proliferation and invasion and promote apoptosis of cervical cancer cells through the PTEN/Akt/FOXO1 pathway.
Literature	GENERIF_SUMMARY	GSK3b may up-regulate BK channels, an effect disrupted by lithium or additional expression of AKT and possibly participating in the regulation of cell volume and excitability.
Literature	GENERIF_SUMMARY	We hypothesize that sporadic meningothelial and transitional subtype meningiomas are a forme fruste or microform of Proteus syndrome, and activation of the AKT/PI3K pathway drives hyperostosis in both non-syndromic, and Proteus-related meningiomas.
Literature	GENERIF_SUMMARY	Study found that cannabis use is likely to be associated with a lower Body Mass Index (BMI) in patients with a psychotic disorder. Moreover, AKT1 risk alleles may increase the incidence of cannabis use in patients with a psychotic disorder, but AKT1 does not appear to mediate the effect of cannabis on BMI.
Literature	GENERIF_SUMMARY	High expression of AKT1 is correlated with metastatic triple negative breast cancer.
Literature	GENERIF_SUMMARY	Phosphorylation level of Akt declined from 138.45+/-13.8 to 69.9+/-29.7% in SGC-7901, and from 115.5+/-26.6 to 49.07+/-27% in MGC-803 and phosphorylation level of mTOR also significantly decreased.While apoptosis of gastric cancer(GA) cells increased which we verified with apoptosis proteins and staining analysis. Our data showed that RLIP76 plays a significant oncogenic role in GC and it maybe a potential target in GC
Literature	GENERIF_SUMMARY	Findings indicate that PTEN and its downstream PI3K/AKT signaling targets were modulated by JQ1 compound.
Literature	GENERIF_SUMMARY	Our study suggested that CXCR7 plays an important proangiogenic role in hepatocellular carcinoma (HCC). via activation of the AKT pathway. So CXCR7 may be a potential target for antiangiogenic therapy in HCC
Literature	GENERIF_SUMMARY	Our studies highlighted the top differentially expressed gene AKT1 and its correlation with the estrogen signaling and RAP1 signaling pathways to alter the proliferation

Main Category	Category	Terms
		and apoptosis of melanoma cells.
Literature	GENERIF_SUMMARY	JARID1B directly bound to PI3K/AKT signaling inhibitor SHIP1 gene promoter and decreased SHIP1 gene expression.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	AEP promotes activation of the PI3K-AKT signaling pathway in prostate cancer cells.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with T-cell leukemia.
Literature	GENERIF_SUMMARY	our results highlight that the Nox1/AKT signaling pathway plays an important role in cell survival in oral squamous cell carcinoma (OSCC) cells.
Literature	GENERIF_SUMMARY	Our study revealed genomic differences between Pulmonary sclerosing hemangioma (PSH) and lung adenocarcinomas, including a high rate of AKT1 mutation in PSHs. These genomic features of PSH identified in the present study provide clues to understanding the biology of PSH and for differential genomic diagnosis of lung tumors
Literature	GENERIF_SUMMARY	IL-24 inhibits AKT via regulating the HMGA1/miR-222 signaling node in human lung cancer cells and acts as an effective tumor suppressor.
Literature	GENERIF_SUMMARY	Omentin protects against lipopysaccharides-mediated acute respiratory distress syndrome through promoting endothelial barrier via an Akt/NOS3-dependent mechanism.
Literature	GENERIF_SUMMARY	In summary, our study demonstrates that IL-7/IL-7R axis promotes the invasion and migration of prostate cancer cells, through activation of AKT/NF-kappaB pathway and upregulation of MMP-3 and MMP-7 expression
Literature	GENERIF_SUMMARY	Bis(acetylacetonato)-oxidovanadium(IV) and sodium metavanadate induced the activation of both PI3K/AKT and MAPK/ERK signaling pathways dose- and time-dependently.
Literature	GENERIF_SUMMARY	This study for the first time demonstrated that the PI3K/Akt/mTOR/S6K1 signal pathway plays an important role in regulating replicative senescence of human VSMCs.
Literature	GENERIF_SUMMARY	Studies suggest to maximize therapeutic benefit, drug combinations and schedules

Main Category	Category	Terms
		for Inhibition of the PI3K/AKT/mTOR Pathway must be explored to identify those with the highest efficacy and lowest toxicity overlap.
Literature	GENERIF_SUMMARY	ITGA6 might be involved in a mechanism that underlies radiation resistance and that ITGA6 could be a potential target for therapies aimed at overcoming radiation resistance in breast cancer.
Literature	GENERIF_SUMMARY	SMYD3-mediated methylation of AKT1 at lysine 14 is essential for AKT1 activation and that SMYD3-mediated AKT1 methylation appears to be a good target for development of anti-cancer therapy.
Literature	GENERIF_SUMMARY	Platelet-derived growth factor (PDGF) activates the PI3K/Akt/mTOR pathway, and suppresses apoptosis.
Literature	GENERIF_SUMMARY	the phosphorylation levels of ErbB2, ErbB3 and Akt and the protein levels of cyclin D1 were decreased by lapatinib treatment of HSC3, HSC4 and Ca9-22 cells
Literature	GENERIF_SUMMARY	CB2 activation with sub-micromolar doses of agonists, which could be more similar to endogenous levels of cannabinoids, promote colon cancer progression in a process that involves AKT and GSK3beta.
Literature	GENERIF_SUMMARY	treatment of paclitaxel (PTX) after down-regulating Akt gene expression using RNAi reduces the cell proliferation and increases apoptosis. We propose that the sequential treatment may exhibit more effective approach rather than traditional combination therapy
Literature	GENERIF_SUMMARY	results suggest that the anthracycline therapies, such as Dox or AD198, can be more effective for treatment of oral squamous cell carcinomas when combined with inhibitors of the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Intra-tumour activation of the PTEN/Akt/COX-2 pathway modulates colorectal cancer (CRC) progression and invasive capacity.
Literature	GENERIF_SUMMARY	Spheroids showed relative lower activities in the AKT, mammalian target of rapamycin (mTOR) and S6K (also known as RPS6KB1) signaling pathway compared to cells cultured in two dimensions.
Literature	GENERIF_SUMMARY	CCN1 RNAi reduced the mRNA and protein expression levels of CCN1, Akt and VEGF in the HUVECs (P
Literature	GENERIF_SUMMARY	IL-1beta induced apoptosis and the expression of catabolic mediators by

Main Category	Category	Terms
		inducing autophagy, and the autophagy in part was mediated through the activation of AKT/mTOR/P70S6K signaling pathway in human osteoarthritis chondrocytes.
Literature	GENERIF_SUMMARY	High expression of AKT1 is associated with invasion in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	High expression of AKT1 is associated with glioma.
Literature	GENERIF_SUMMARY	In vitro study using human endothelial cells showed that prostaglandin E2 (PGE2) significantly increased MMP-2 activity as well as tube formation. Inhibition of COX-2 and/or phosphorylated AKT suppressed MMP-2 activity and endothelial tube formation suggesting involvement of PGE2 in regulation of MMP-2 activity during angiogenesis.
Literature	GENERIF_SUMMARY	Binding of PP2A and Akt increased in response to cAMP or phosphatidic acid (PA), suggesting that their binding is directly responsible for the inactivation of Akt during decidualization.
Literature	GENERIF_SUMMARY	High expression of AKT1 is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	High expression of AKT1 is associated with Adriamycin resistance in breast cancer.
Literature	GENERIF_SUMMARY	found that DIXDC1 could promote cell proliferation via modulating cell cycle progression and PI3K/AKT signaling pathway in NHLs
Literature	GENERIF_SUMMARY	Phosphorylation of MITF by AKT affects its downstream targets and causes TP53-dependent cell senescence
Literature	GENERIF_SUMMARY	Ik6, the upstream factor of Akt-FoxO1 pathway, can protect acute lymphoblastic leukemia cells against daunorubicin-induced apoptosis
Literature	GENERIF_SUMMARY	miR-130a could alleviate human coronary artery endothelial cells (HCAECs) injury and inflammatory responses by down-regulating PTEN and activating PI3K/Akt/eNOS signaling pathway.
Literature	GENERIF_SUMMARY	The impact of AKT1 on glucocorticoid receptor (GR)-induced transcriptional activity in cooperation with phospho-serine/threonine-binding protein 14-3-3, was examined.
Literature	GENERIF_SUMMARY	PIK3CA and AKT1 are frequently mutated in Hidradenoma papilliferum (HP) tumours (29% and 14%, respectively). PIK3CA/AKT1

Main Category	Category	Terms
		pathway alterations in HP further support the hypothesis that HP is the vulvar (anogenital mammary-like gland) analogue of breast intraductal papilloma.
Literature	GENERIF_SUMMARY	PTEN/Akt/FOXO1 pathway plays an important role in adriamycin-resistance of breast cancer.
Literature	GENERIF_SUMMARY	The overexpression of CSN6 reduced the effect of quercetin treatment on HT29 cells, suggesting that quercetininduced apoptosis may involve the AktCSN6Myc signaling axis in HT29 cells.
Literature	GENERIF_SUMMARY	We found statistically significant DNA hypomethylation of the promoter regions of DRD3 (P = 0.032), DRD4 (P = 0.05), MB-COMT (P = 0.009), and AKT1 (P = 0.0008) associated with increased expression of the corresponding genes in patients with methamphetamine psychosis (P = 0.022, P = 0.034, P = 0.035, P = 0.038, respectively)
Literature	GENERIF_SUMMARY	In conclusion, the authors demonstrated that AKT activation prevents apoptosis, partly through inhibition of GSK3beta, resulting in pluripotent stem cells survival.
Literature	GENERIF_SUMMARY	IGF-II siRNA inactivates the FAK/PI3K/Akt signaling pathway, and further reduces cell proliferation, N-ras and C-myc levels in SMMC-7721 cells.
Literature	GENERIF_SUMMARY	The findings reveal SAM domain-dependent oligomerization by AKT as a switch for CNK1 activation.
Literature	GENERIF_SUMMARY	Levels of p21 and p27 were decreased in TACO or pAKT overexpressing HCC due to SKP2 upregulation.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with glioblastoma multiforme.
Literature	GENERIF_SUMMARY	conclusion, serum leptin was elevated in CKD patients and it might contribute to endothelial dysfunction by disarrangement of f-actin cytoskeleton via a mechanism involving the AKT/GSK3beta and beta-catenin pathway.
Literature	GENERIF_SUMMARY	These results point to an increase in total and phosphorylated Akt in high-grade gliomas and to a possible role of RICTOR in proliferations of high-grade glioblastomas cells.
Literature	GENERIF_SUMMARY	our studies demonstrated that AMPK was involved in the VES-induced autophagy. Crosstalk exists between AMPK and the Akt/mTOR axis. The results elucidated the

Main Category	Category	Terms
		mechanism of VES-induced autophagy in human gastric cancer cells.
Literature	GENERIF_SUMMARY	Data show that sulfatase 1 (hSulf-1) overexpression in melanoma cells can inhibit cell proliferation and induce cell cycle arrest and apoptosis by decreasing the protein kinase B (AKT) phosphorylation and limiting cyclin dependent kinase 4 (CDK4) nuclear import.
Literature	GENERIF_SUMMARY	These results establish an important role for CCAR2 in cancer cells proliferation and could shed new light on novel therapeutic strategies against cancer, devoid of detrimental side effects.
Literature	GENERIF_SUMMARY	PTEN downregulates the PI3K/AKT/hTERT pathway, thereby suppressing the growth of lung adenocarcinoma cells.
Literature	GENERIF_SUMMARY	this study shows that PTEN dephosphorylates AKT to prevent the expression of GLUT1 on plasmamembrane and to limit glucose consumption in cancer cells
Literature	GENERIF_SUMMARY	Gli1 promotes colorectal cancer cells metastasis in a Foxm1-dependent manner by activating EMT and PI3K-AKT signaling.
Literature	GENERIF_SUMMARY	Data indicate that overexpression of nitric oxide synthase 1 adaptor protein short form (CAPON-S) led to the inactivation of the proto-oncogene protein Akt (Akt) signaling pathway.
Literature	GENERIF_SUMMARY	CYT-Rx20 significantly reduced tumor growth of the implanted esophageal cancer cells accompanied by decreased Ki-67, phospho-AKT, and phospho-STAT3 expression.
Literature	GENERIF_SUMMARY	TWIST1, in part via GAS6 and L1CAM, led to higher expression and activation of Akt upon cisplatin treatment, and inhibition of Akt activation sensitized cells to cisplatin.
Literature	GENERIF_SUMMARY	This was further verified in gene expression analyses, showing downregulation of genes involved in glucose metabolism. Additionally, both AKT1 KO and AKT2 KO demonstrated an impaired fatty acid metabolism. However, genes were upregulated in the Wnt and cell proliferation pathways, which could oppose this effect. AKT inhibition should therefore be combined with other effectors to attain the best effect.
Literature	GENERIF_SUMMARY	In conclusion, baicalin and DDP were synergistic at inhibiting proliferation and invasion of human lung cancer cells at

Main Category	Category	Terms
		appropriate dosages and incubation time in the presence or absence of DDP resistance. The attenuation of DDP resistance was associated with downregulation of MARK2 and p-Akt.
Literature	GENERIF_SUMMARY	Phosphorylated and activated DCK can inhibit radiation-induced cell death including apoptosis and mitotic catastrophe, and promote radiation-induced autophagy through PI3K/Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	suggest that the reduction of CLDN5, 7, and 18 expression loses the suppressive ability of interaction between PDK1 and Akt and causes sustained phosphorylation of Akt, resulting in the disordered proliferation in lung squamous carcinoma cells
Literature	GENERIF_SUMMARY	overexpression of KIAA0247 is able to inhibit phosphorylation of AKT and Stat3 in glioma cells, resulting in inactivation of the AKT and Stat3 signaling pathways, this ultimately decreases the expression of PCNA, CyclinD1, Bcl2 and VEGF.
Literature	GENERIF_SUMMARY	These results indicated that lncRNA ANRIL might play an important role in cervical cancer progression and could serve as a novel prognostic biomarker and therapeutic target in cervical cancer.
Literature	GENERIF_SUMMARY	Data indicate connector enhancer of kinase suppressor of Ras 1 protein (CNK1) as a molecular platform that controls c-raf protein (RAF) and c-akt protein (AKT) signalling and determines cell fate decisions in a cell type- and cell stage-dependent manner.
Literature	GENERIF_SUMMARY	Two different mechanisms
Literature	GENERIF_SUMMARY	SNORD126 activates the PI3K-AKT pathway by upregulation of FGFR2.
Literature	GENERIF_SUMMARY	Overexpression of PHLPP2 without its 3'UTR attenuated the effects of miR-181a on cell proliferation and apoptosis in keloid fibroblast cells.
Literature	GENERIF_SUMMARY	Data show that acid sensing ion channel 1a (ASIC1a) mediated calcium influx, which resulted in the activation of PI3K/AKT signaling and increased drug resistance, suggesting that ASIC1a/Ca(2+)/PI3K/AKT signaling represents a pathway that regulates drug resistance, thus offering a potential target for chemotherapy of hepatocellular carcinoma (HCC).
Literature	GENERIF_SUMMARY	these results indicate that PAK4 confers CDDP resistance via the activation of MEK/

Main Category	Category	Terms
		ERK and PI3K/Akt pathways. PAK4 and PI3K/Akt pathways can reciprocally activate each other.
Literature	GENERIF_SUMMARY	NecroX-5 decreased intracellular calcium concentration by blocking Ca2+ influx, which mediated the inhibition of cell migration, AKT downregulation and the reduction of mitochondrial ROS levels. However, the reduction of mitochondrial ROS was not associated with supressed migration and AKT downregulation
Literature	GENERIF_SUMMARY	FBXL18 plays an oncogenic role through promoting lysine 63-linked ubiquitination of Akt in glioma.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with liver cancer.
Literature	GENERIF_SUMMARY	AKT expression is regulated by PRSS8 in glioma.
Literature	GENERIF_SUMMARY	Data suggest a role for akt proto-oncogene protein (Akt) pathway in the pathogenesis of rheumatoid arthritis (RA).
Literature	GENERIF_SUMMARY	study demonstrates the presence of PIWI-like proteins in somatic cells and the possible role of HIWI2 in preserving the functional integrity of epithelial cells probably by modulating the phosphorylation status of Akt.
Literature	GENERIF_SUMMARY	Thus, nimbolide exerted anticancer effects in vitro by representing the PI3K-Akt-mTOR pathway in PC-3 cells. Thereby, it acts as a potent anticancer drug for prostate cancer.
Literature	GENERIF_SUMMARY	matrine inhibited the phosphorylation of Akt, mTOR and their downstream substrates p70S6K and 4EBP1, which led to the occurrence of autophagy
Literature	GENERIF_SUMMARY	RanBP9/TSSC3 complex cooperatively suppress metastasis via downregulation of Src-dependent Akt pathway to expedite mitochondrial-associated anoikis.
Literature	GENERIF_SUMMARY	CD133 facilitates cancer stem cell-like properties by stabilizing EGFR-AKT signaling in HCC.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with chronic myelogenous leukemia.
Literature	GENERIF_SUMMARY	wortmannin and rapamycin inhibit B7-H4-mediated tumor immunoresistance through regulating B7-H4 subcellular distribution. Taken together, these results suggest that PI3K/Akt/mTOR inhibitors might be used for adjuvant therapy aimed at inhibition of

Main Category	Category	Terms
		immune evasion.
Literature	GENERIF_SUMMARY	HULC regulates PI3K/AKT signaling pathway in chronic myeloid leukemia.
Literature	GENERIF_SUMMARY	Cross-talk between Akt1 and Src regulates endothelial-barrier function.
Literature	GENERIF_SUMMARY	AKT signaling pathway is involved in breast cancer cells apoptosis.
Literature	GENERIF_SUMMARY	our results showed that miR-375 and miR-205 regulated the invasion and migration of LSCC via AKT-mediated Epithelial-mesenchymal transition (EMT) synergistically.
Literature	GENERIF_SUMMARY	the preclinical results suggest that AT7867 inhibits colorectal cancer cells via AKT-dependent and -independent mechanisms.
Literature	GENERIF_SUMMARY	AKT1 mutation is associated with olfactory groove meningiomas.
Literature	GENERIF_SUMMARY	Data show that spleen tyrosine kinase (SYK) and proto-oncogene protein c-akt (AKT1) proteins were increased in the cytoplasm of the cells forming Mallory-Denk bodies.
Literature	GENERIF_SUMMARY	that the presence of NRAS, PTEN and AKT1 gene mutations may have an influence on the occurrence of CNS metastases in patients with SqCC
Literature	GENERIF_SUMMARY	Upregulation of AKT1 is associated with thyroid cancer.
Literature	GENERIF_SUMMARY	Disruption of CK1alpha function in myeloma cells resulted in decreased Mdm2, increased p53 and p21 and reduced expression of beta-catenin and AKT.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with hepatoma.
Literature	GENERIF_SUMMARY	In vitro analysis of EEC4 and AN3CA cells showed a dose-dependent decrease in p(Ser473)-AKT and p(Thr308)-AKT with MK2206. Invasion of EEC4 and AN3CA cells also significantly decreased after 36h and 72h of MK2206 treatment.
Literature	GENERIF_SUMMARY	Together, our results suggest that Gab3 expression in CRC cells is important for Akt-Erk activation and cell proliferation.
Literature	GENERIF_SUMMARY	We demonstrated trophoblast cytoprotection by intervention with supraphysiological concentrations of relaxin, a process in part mediated through the PI3-kinase-Akt/PKB cell survival pathway. These results provide further

Main Category	Category	Terms
		rationale for clinical investigation of relaxin as a potential therapeutic in preeclampsia.
Literature	GENERIF_SUMMARY	MicroRNA miR-155-5p promoted proliferation, invasion and migration, but inhibited apoptosis in hepatocellular carcinoma (HCC) by directly targeting the 3'-UTR of PTEN through the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Data indicate three biomarkers mitogen-activated protein kinase 3 (MAPK3), BCL2 apoptosis regulator (BCL2) and proto-oncogene c-Akt (AKT1) as potential predictors of neurological outcome following cardiac arrest (CA).
Literature	GENERIF_SUMMARY	Extracellular polyamines induced proliferation and cancer cell migration by inducing ODC and SSAT expression, and the Akt1-mediated pathway.
Literature	GENERIF_SUMMARY	Akt controls nuclear translocation and activity of the master lysosomal transcription factor TFEB by phosphorylating TFEB at Ser467.
Literature	GENERIF_SUMMARY	High expression of AKT1 is associated with drug resistance and proliferation of breast cancer.
Literature	GENERIF_SUMMARY	AKT/GSK-3beta-mediated stabilization of SP1 is required for TGF-beta induced up-regulation of NKG2DLs.
Literature	GENERIF_SUMMARY	Derlin-1 is overexpressed in bladder cancer and promotes malignant phenotype through ERK/MMP-2/9 and PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	Integrin beta1 appears to serve as a partner of Stathmin induction of ERK and Akt signaling by inhibiting apoptosis in the cholangiocarcinoma cell.
Literature	GENERIF_SUMMARY	KRAS mutations and AKT activation are present in Wilms tumors (WT) and may represent novel therapeutic targets for this tumor.
Literature	GENERIF_SUMMARY	our results revealed that As2S2 induced G2/M phase arrest, apoptosis, and autophagy via activing ROS/JNK and blocking Akt/mTOR signaling pathway in human osteosarcoma cells. Arsenic sulfide may be a potential clinical antitumor drugs targeting osteosarcoma.
Literature	GENERIF_SUMMARY	biased CD4(+) T cell differentiation was independent of any crosstalk between CD4(+) T cells and dendritic cells and was mediated via direct exposure of CD4(+) T cells to palmitate, leading to increased

Main Category	Category	Terms
		activation of a PI3K p110delta-Akt-dependent pathway upon priming.
Literature	GENERIF_SUMMARY	Data show that growth differentiation factor 15 (GDF15) knockdown significantly inhibited the growth and metastasis of cancer stem-like sphere cells (SCs) through AKT/GSK-3beta/beta-catenin pathway suppression.
Literature	GENERIF_SUMMARY	Activation of stromal MYC was critical for this reprogramming and for the sustained cellular responses elicited by LO, both in vitro and in vivo in an AKT1-dependent manner. Inhibition of LO internalization prevented activation of MYC and impaired the tumor-supporting properties of fibroblasts.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with lung cancer.
Literature	GENERIF_SUMMARY	Activating Akt1 mutations alter DNA double strand break repair and radiosensitivity.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with gallbladder cancer.
Literature	GENERIF_SUMMARY	LINK-A-dependent AKT hyperactivation leads to tumorigenesis and resistance to AKT inhibitors.
Literature	GENERIF_SUMMARY	Here the authors identified a direct interaction of both MEK1 and MEK2 with AKT. The interaction between MEK and AKT affects cell migration and adhesion, but not proliferation. The specific mechanism of action of the MEK-AKT complex involves phosphorylation of the migration-related transcription factor FoxO1.
Literature	GENERIF_SUMMARY	Findings demonstrate a new role of microRNA 185 as a key regulator of hepatocellular carcinoma via autophagy by dysregulation of AKT1 pathway.
Literature	GENERIF_SUMMARY	Data show that cancer-associated fibroblasts (CAFs)-derived hepatocyte growth factor (HGF) or recombinant HGF activated c-Met/phosphoinositide 3-kinase (PI3K)/Akt and glucose-regulated protein 78 (GRP78) signalling pathways in ovarian cancer cells.
Literature	GENERIF_SUMMARY	findings suggest that c-Rel might play a role in promoting the invasion of choriocarcinoma cells through PI3K/AKT signaling
Literature	GENERIF_SUMMARY	In conclusion, overexpression of CXCL8 induced cell proliferation, migration and invasion of colon cancer LoVo cells. CXCL8

Main Category	Category	Terms
		may act through induction of EMT via the PI3K/AKT/NF-kappaB signaling axis
Literature	GENERIF_SUMMARY	Study demonstrates that ROS-mediated oxidative modification of Akt1 contributes to synaptic dysfunction in Alzheimer's disease, seen as loss of activity-dependent protein translation that is essential for synaptic plasticity and maintenance.
Literature	GENERIF_SUMMARY	Polydatin inhibited cell proliferation and induced apoptosis in laryngeal cancer and HeLa cells via inactivation of the PDGF/AKT signaling pathway.
Literature	GENERIF_SUMMARY	High expressions of AKT1 and AKT2 through possible relation with androgen may cause granulosa cells dysfunction in the +HA PCOS patients.
Literature	GENERIF_SUMMARY	after AKT-mediated phosphorylation at serine 319, FOXO1 binds to IQGAP1, a hub for activation of the MAPK pathway, and impedes IQGAP1-dependent phosphorylation of ERK1/2 (pERK1/2).
Literature	GENERIF_SUMMARY	Data show that knockdown of ubiquitin specific peptidase 39 (USP39), a direct target of microRNA microRNA-133a, induced cell apoptosis through inhibition of proto-oncogene proteins c-akt (AKT) signaling pathway in pancreatic cancer (PC) cells.
Literature	GENERIF_SUMMARY	Data show while overexpression of AKT serine/threonine kinase 1 (AKT1) promoted local tumor growth, downregulation of AKT1 or overexpression of AKT serine/threonine kinase 2 (AKT2) promoted peritumoral invasion and lung metastasis.
Literature	GENERIF_SUMMARY	we conclude that Gab3 overexpression in human glioma mediates Akt activation and cancer cell proliferation
Literature	GENERIF_SUMMARY	pAkt1 expression is increased in perineural invasion and metastasis as compared with primary prostate tumors. pAkt1 is associated with increased proliferation and decreased apoptosis.
Literature	GENERIF_SUMMARY	results suggested that the promoter mutation, methylation and gene expression of AKT1 may play distinct roles in tumorgenesis of breast cancer and the integrated analysis of methylation and expression of AKT1 might serve as potential biomarkers for diagnosis and classification of breast cancer
Literature	GENERIF_SUMMARY	CONCLUSION

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Literature	GENERIF_SUMMARY	IGF1 and insulin more likely (at lower concentrations) enhance tumour cell survival than proliferation, via activation and maintenance of phosphatidylinositol 3-kinase activity and p-Akt/PKB.
Literature	GENERIF_SUMMARY	Data suggest that AKT inhibitor MK2206 has potential synergistic efficacy to the current standard of care for glioma patients.
Literature	GENERIF_SUMMARY	IRS-1 and IRS-2 signaling interaction with the microtubule cytoskeleton and its response to AKT determines the response to microtubule disruption in breast carcinoma cells
Literature	GENERIF_SUMMARY	findings shed new light on the STAT1/miR-181a/PTEN pathway in colorectal cancer and add new insight regarding the carcinogenesis of colorectal cancer.
Literature	GENERIF_SUMMARY	Akt contributes to thyroxine-mediated MDM2 phosphorylation in sperm.
Literature	GENERIF_SUMMARY	No significant differences of p-Akt and survivin expression were found between PTC patients with type 2 diabetes mellitus and papillary thyroid carcinoma patients without type 2 diabetes mellitus
Literature	GENERIF_SUMMARY	Inhibition of Akt signaling during ex vivo priming and expansion gives rise to CD19CAR T cell populations that display comparatively higher antitumor activity
Literature	GENERIF_SUMMARY	findings shed new light on the miR-223/PAX6 pathway in glioma and this pathway might modulate the sensitivity of glioma to TMZ via regulating PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	The findings demonstrate that HMGB1 may be implicated in the formation of hemangioma through upregulation of AKT pathway.
Literature	GENERIF_SUMMARY	Moreover, inhibition on miR-374b relieved the suppressed AKT1 protein, and reduced cell apoptosis induced by BLM. These data depict the p53/miR-374b/AKT1 signaling that may regulate BLM-induced apoptosis in Colorectal cancer (CRC)cells, thus facilitating to improve the outcome of chemotherapy in CRC.
Literature	GENERIF_SUMMARY	The role of PI3K/Akt signaling pathway in osteosarcoma
Literature	GENERIF_SUMMARY	we supposed that EphA4 interacted with CDK5 and promoted its expression which in turn enhanced p-AKT expression and promoted cell adhesion-mediated drug

Main Category	Category	Terms
		resistance in multiple myeloma.
Literature	GENERIF_SUMMARY	our study found that intracellular Ca(2+) could promote radioresistance of non-small cell lung cancer cells through phosphorylating of Akt on the 473 site, which contributes to a better understanding on the non-small cell lung cancer radioresistance, and may provide a new target for radioresistance management
Literature	GENERIF_SUMMARY	Tetraarsenic hexoxide (As4O6) induced G2/M arrest, apoptosis and autophagic cell death through PI3K/Akt and p38 MAPK pathways alteration in SW620 colon cancer cells.
Literature	GENERIF_SUMMARY	NT3 upregulates cellular proliferation, extracellular matrix protein production, and collagen deposition in human aortic valve interstitial cells through the Trk-Akt-cyclin D1 cascade.
Literature	GENERIF_SUMMARY	PI3K/AKT/FoxO3a signaling pathway role in melanoma cells
Literature	GENERIF_SUMMARY	This study investigated the associations between three polymorphisms (rs1130214, rs3730358, and rs2494732) in AKT1 and the risk of development of prostate cancer. Compared to the CC genotype of the rs3730358 polymorphism, the CT genotype of the same polymorphism was strongly associated with a decreased risk of prostate cancer. The genotype CT of rs3730358 of AKT1 was associated with a decreased risk of prostate cancer.
Literature	GENERIF_SUMMARY	Mechanistically, USP49 deubiquitinates and stabilizes FKBP51, which in turn enhances PHLPP's capability to dephosphorylate AKT.
Literature	GENERIF_SUMMARY	K2 and AKT display a high degree of cross-regulation of their respective functions, both directly, through physical interaction and phosphorylation, and indirectly, through an intense cross-talk of key downstream effectors, ultimately leading to sustained AKT activation. (Review)
Literature	GENERIF_SUMMARY	Data indicate YAP1 as a candidate marker to predict cell lines that were most sensitive to MSC2363318A, suggesting clinical development of a dual AKT/P70S6K inhibitor.
Literature	GENERIF_SUMMARY	Further study found that knockdown of cPLA2alpha in MDA-MB-231 cells inhibited TGF-beta-induced EMT through the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	results suggest that Kctd20 impacts

Main Category	Category	Terms
		proliferation and invasion of non-small cell lung cancer through enhancing Fak and Akt phosphorylation
Literature	GENERIF_SUMMARY	Here, using PET with [(11)C]raclopride, we identified in the AKT1 gene a new variable number tandem repeat (VNTR) marker associated with baseline striatal dopamine D2/D3 receptor availability and with methylphenidate-induced striatal dopamine increases in healthy volunteers.
Literature	GENERIF_SUMMARY	High Expressions of AKT1 is associated with metastasis of esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Low AKT1 expression is associated with progression of hepatic fibrosis.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with ovarian cancer.
Literature	GENERIF_SUMMARY	PTEN rs701848 and AKT1 rs2494752 polymorphisms might be a candidate pharmacogenomic factor to assess the susceptibility of and response and prognosis prediction for interindividualized CE(A)F chemotherapy in breast cancer patients.
Literature	GENERIF_SUMMARY	Collectively, these results indicate that COX-1/PGE2/EP4 upregulates the beta-arr1 mediated Akt signaling pathway to provide mucosal protection in colitis.
Literature	GENERIF_SUMMARY	p-PTEN, Akt, p-PDK1, and p-GSK-3beta were significantly downregulated after the knockdown of CK2alpha expression.
Literature	GENERIF_SUMMARY	Likewise, inhibiting the PI3K-AKT-mTOR pathway with the ATP-competitive mTOR inhibitor PP242 reduced CD44 protein in SNU-423 and SNU-449 cells without altering CD44 mRNA levels.
Literature	GENERIF_SUMMARY	Data show that activated proto-oncogene protein Akt (AKT) directly phosphorylates Fas associated factor 1 (FAF1) reduces FAF1 at the plasma membrane and results in an increase in TGF-beta type II receptor (TbetaRII) at the cell surface.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with oncogenicity and chemoresistance in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	inhibition of miR21 suppressed cell proliferation and colony formation through regulating the PTEN/AKT pathway, and

Main Category	Category	Terms
		improved PTX sensitivity in cervical cancer cells.
Literature	GENERIF_SUMMARY	ong non-coding RNA ROR may affect the sensitivity of lung adenocarcinoma cells to DDP by targeting PI3K/Akt/mTOR signaling pathway
Literature	GENERIF_SUMMARY	results showed that upregulated lncARSR promotes doxorubicin resistance in HCC via modulating PTEN-PI3K/Akt pathway, and implied that lncARSR may serve as a promising prognostic biomarker and therapeutic target for HCC chemo-resistance.
Literature	GENERIF_SUMMARY	Data indicate that RACK1 increases interactions between Akt and MCM7 and promotes Akt-dependent MCM7 phosphorylation, which in turn increases MCM7 binding to chromatin and miniature chromosome maintenance (MCM) complex formation.
Literature	GENERIF_SUMMARY	analysis performance of the OncoBEAMtrade mark test was evaluated using a novel AKT1 E17K ctDNA reference standard consisting of sheared genomic DNA spiked into human plasma. Both assays are employed at centralized testing laboratories operating according to quality standards for prospective identification of the AKT1 E17K mutation in ER+ breast cancer patients in the context of a clinical trial evaluating the AKT inhi...
Literature	GENERIF_SUMMARY	Patients with skull base meningiomas with AKT1E17K mutation might benefit from additional treatment targeting the mTOR
Literature	GENERIF_SUMMARY	Our results suggested that activated Akt/mTOR-autophagy may have a role in the local T cell-mediated immunoregulatory mechanism of oral lichen planus (OLP). LC3B might be a valuable marker to monitor the disease severity of OLP.
Literature	GENERIF_SUMMARY	The results demonstrate that in renal cells, NHE1 is associated with several regulatory proteins including Hsp90, and that Hsp90 affects its function possibly through altered phosphorylation of the protein via the AKT kinase.
Literature	GENERIF_SUMMARY	AKT inhibition blocks miR-124 silencing-induced AKT1/2, glucose transporter 1, hexokinase II activation, cell proliferation, and glycolytic or energy metabolism changes. In summary, this study demonstrated that miR-124 is able to inhibit proliferation, glycolysis, and energy

Main Category	Category	Terms
		metabolism, potentially by targeting AKT1/2-glucose transporter 1/hexokinase II in non-small cell lung cancer cells.
Literature	GENERIF_SUMMARY	This study provides the first clinical data that AKT1 E17K is a therapeutic target in human cancer. The genomic context of the AKT1 E17K mutation further conditioned response to AZD5363.
Literature	GENERIF_SUMMARY	data suggested that MK2206 significantly enhanced the cytocidal effects of bufalin in multiple myeloma cells, regardless of the sensitivity to bortezomib, via the inhibition of the AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	Similar to normal developmental programs, oncogenic functions of STAT5 rely on molecular crosstalk with PI3K/AKT signaling for the initiation, and in some instances the progression, of breast cancer. (Review)
Literature	GENERIF_SUMMARY	our findings suggested that MCM7 promoted tumor cell proliferation, colony formation and migration of ESCC cells via activating AKT1/mTOR signaling pathway
Literature	GENERIF_SUMMARY	The authors conclude that the PTEN-Akt1-mTOR pathway does not appear to play a central role in mitochondrial efficiency in ocular hypertension.
Literature	GENERIF_SUMMARY	In summary, we demonstrate that ADAMTS18 silencing in breast cancer is significantly correlated with promoter CpG methylation. ADAMTS18 acts as an antagonist of AKT and NF-kappaB signaling, further suppressing EMT and metastasis of breast cancer cells.
Literature	GENERIF_SUMMARY	This study for the first time demonstrated that HDAC8 activity determines susceptibility to cell cycle arrest induced by Anthrax Lethal Toxin, through regulating the PI3K-PTEN-AKT signaling axis.
Literature	GENERIF_SUMMARY	we studied S100A11 and noted knockdown of S100A11 using short hairpin RNA, which inhibited proliferation, invasion, and migration of renal carcinoma cells as well as increased expression of E-cadherin and decreased expression of epidermal growth factor receptor/Akt in renal carcinoma cells. Therefore, S100A11 may be a key molecular target for treating renal carcinoma.
Literature	GENERIF_SUMMARY	grape seed proanthocyanidin dose-dependently suppressed the activation of MAP kinases (JNK, ERK and p38) and Akt in LPS-stimulated hepatic stellate cells.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with

Main Category	Category	Terms
		non-small cell lung cancer metastasis.
Literature	GENERIF_SUMMARY	AKT1 overexpression is associated with colorectal cancer metastasis.
Literature	GENERIF_SUMMARY	These findings suggest that future trials of everolimus in biliary tract cancer would benefit from preselecting patients based on their K-Ras and PI3K/mTOR pathway activation status.
Literature	GENERIF_SUMMARY	BTK-mediated signaling was found to be highly attenuated accompanied by a shift in PI3K/AKT and apoptosis regulation-associated genes/proteins. Cytotoxicity studies using the AKT inhibitor, MK2206+/-ibrutinib, and the Bcl-2-specific inhibitor, venetoclax+/-ibrutinib, demonstrated synergistic loss of cell viability
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	This review summarises and discusses the consequences of genetic deletions of Akt isoforms in adult mice and their implications for cancer therapy. Whereas combined Akt1 and Akt2 rapidly induced mortality, hepatic Akt inhibition induced liver injury that promotes hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	findings uncover a new function of p53 in the regulation of Akt signaling and reveal how p53, ASS1, and Akt are interrelated to each other.
Literature	GENERIF_SUMMARY	findings suggest that the PI3K/Akt/mTOR signaling pathway participates in the process of autophagy induced by CVB3 infection. This finding may provide a new perspective of CVB3-induced autophagy.
Literature	GENERIF_SUMMARY	melatonin stimulates PINK1 expression via an MT2 /Akt/NF-kappaB pathway, and such stimulation is important for the prevention of neuronal cell apoptosis under high glucose conditions.
Literature	GENERIF_SUMMARY	the results indicated that butein has antiproliferative and proapoptotic properties through the suppression of NF-kappaB, AP-1 and Akt signaling in HTLV-1-infected T cells, both in vitro and in vivo, suggesting its therapeutic potential against HTLV-1-associated diseases including adult T-cell leukemia/lymphoma
Literature	GENERIF_SUMMARY	High expression of AKT1 is associated with nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	eIF3d promotes gallbladder cancer (GBC) progression mainly via eIF3d-GRK2-AKT axis

Main Category	Category	Terms
		and it may be used as a prognostic factor. The therapeutic targeting of eIF3d-GRK2 axis may be a potential treatment approach for GBC.
Literature	GENERIF_SUMMARY	the signaling pathway study confirmed that Sox9 protected MRC-5 from lipopolysaccharide injury through the AKT/GSK3beta pathway.
Literature	GENERIF_SUMMARY	Lysophosphatidylcholine activates the Akt pathway to upregulate extracellular matrix protein production in human aortic valve cells and may contribute to the mechanism of valvular sclerosis associated with the development and progression of aortic stenosis.
Literature	GENERIF_SUMMARY	Suppression of Akt1-beta-catenin pathway in advanced prostate cancer promotes TGFbeta1-mediated epithelial to mesenchymal transition and metastasis.
Literature	GENERIF_SUMMARY	Findings demonstrate that COX-2 and p-Akt1 play an important combined role during melanoma progression and are associated with highly metastatic tumors and survival rates in patients with MM.
Literature	GENERIF_SUMMARY	This study presents further evidence to show that TRPM4 regulates beta-catenin signaling and enhances the proliferation of prostate cancer cell lines, through a calcium-dependent regulation of Akt1 and GSK-3beta activity.
Literature	GENERIF_SUMMARY	We conclude that the predisposition to thrombosis is likely to be multifaceted with risk factors including vascular malformations, immobility, surgery, additional prothrombotic factors, and possible pathophysiologic effects of the somatic AKT1 mutation on platelet function or the vascular endothelium.
Literature	GENERIF_SUMMARY	We further investigated whether curcumin exerts its antitumor activities via suppressing NEDD4 expression. We found that curcumin reduced the expression of NEDD4 and Notch1 and pAKT, leading to glioma cell growth inhibition, apoptosis, and suppression of migration and invasion. Moreover, deletion of NEDD4 expression enhanced the sensitivity of glioma cells to curcumin treatment
Literature	GENERIF_SUMMARY	these results demonstrate that the oncogenic signal transducer and activator of transcription 5/Akt pathway is a cellular target for indole-3-carbinol in chronic myeloid leukemia cells. Thus, this clinically

Main Category	Category	Terms
		tested natural compound can be a potential candidate in the treatment of chronic myeloid leukemia following confirmation with clinical studies
Literature	GENERIF_SUMMARY	Data suggest that CAMKK2 is highly expressed in high-grade ovarian cancer and ovarian cancer cell lines; CAMKK2 directly activates Akt1 by phosphorylation at Thr-308 in a Ca2+/calmodulin-dependent manner; CAMKK2 knockdown or inhibition decreases Akt1 phosphorylation at Thr-308 and Ser-473. (CAMKK2 = calcium/calmodulin dependent protein kinase kinase 2; AKT1 = AKT serine/threonine kinase 1)
Literature	GENERIF_SUMMARY	these results suggest that MEG3 might suppress the tumor growth and angiogenesis via AKT signaling pathway and MEG3 may serve as a potential novel diagnostic and therapeutic target of breast cancer.
Literature	GENERIF_SUMMARY	Mechanistic investigations found that quercetin suppressed Snail-dependent Akt activation by upregulating maspin and Snail-independent a disintegrin and metalloproteinase (ADAM) 9 expression pathways to modulate the invasive ability of NSCLC cells
Literature	GENERIF_SUMMARY	These results suggested that AKT pathway was the dominant pathway involved in Cx43-mediated chronic cadmium toxicity.
Literature	GENERIF_SUMMARY	tumor cells incubated with LGRFYAASG-pen showed disruption of filamentous actin, focal adhesions and caveolae-mediated membrane trafficking, resulting in impaired cell adhesion and migration in vitro. These effects were paralleled by a decrease in the phosphorylation of both focal adhesion kinase (Fak) and protein kinase B (Akt).
Literature	GENERIF_SUMMARY	T-type channel signaling is redirected towards the activation of the kinase Akt1, leading to increased expression of the anti-apoptotic protein survivin, and a decrease in the pro-apoptotic mediator FoxO3A. Finally, in iPAH cells, Akt1 is no longer able to regulate caspase 9 activation, whereas T-type channel overexpression reverses PP2A defect in iPAH cells but reinforces the deleterious effects of Akt1 activation
Literature	GENERIF_SUMMARY	inhibition of Akt and 14-3-3 function efficiently reduces cell proliferation in 786-0 cells characterized by hyperactivation of beta-catenin signaling due to pVHL loss.
Literature	GENERIF_SUMMARY	mutp53 augments insulin-induced AKT1

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		activation by binding and inhibiting the tumor suppressor DAB2IP (DAB2-interacting protein) in the cytoplasm
Literature	GENERIF_SUMMARY	In the present study, galangin was found to suppress laryngeal cancer cell proliferation. Also, flow cytometry, immunohistochemical and western blot analysis indicated that cell apoptosis was induced for galangin administration, promoting caspase-3 expression through regulating PI3K/AKT/NF-kappaB.
Literature	GENERIF_SUMMARY	The results suggest that p53 simultaneously controls multiple pathways to induce cellular senescence through p21 and Akt.
Literature	GENERIF_SUMMARY	IGF-1R and AKT inhibitors further increased apoptosis by Nutlin-3a in parental MHM cells and the cisplatin-resistant clones, confirming IGF-1R/AKT signaling promotes apoptosis resistance.
Literature	GENERIF_SUMMARY	Taken together, MREG regulates thyroid cancer cell invasion and proliferation through PI3K/Akt-mTOR signaling pathway. MREG may serve as a promising therapeutic strategy for thyroid cancer.
Literature	GENERIF_SUMMARY	Rictor regulates the vasculogenic mimicry of melanoma and determines the patients' survival via the AKT-MMP2-MMP9 pathway.
Literature	GENERIF_SUMMARY	These findings provide a new understanding of the effects of oxLDL on endothelial proliferation, which is essential for developing new treatments against neovascularization and progression of atherosclerosis.
Literature	GENERIF_SUMMARY	reduction of Alex3 correlates with the development of non-small cell lung cancer and predicts adverse clinical outcome of non-small cell lung cancer patients. The effect of Alex3 on inhibiting invasion and migration may attribute to upregulation of E-cadherin expression through AKT-Slug pathway inactivation.
Literature	GENERIF_SUMMARY	The ectopic overexpression of miR-379 inhibited cell migration, invasion and EMT progress, while downregulated miR-379 reversed the effect. In addition, miR-379 regulated the focal adhesion kinase (FAK) by directly binding to its 3'-UTR, resulting in suppression of AKT signaling. In clinical samples of gastric cancer (GC), miR-379 inversely correlated with FAK, which was upregulated in GC.
Literature	GENERIF_SUMMARY	The findings indicated that miR-135a

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		promotes cell apoptosis and inhibits cell proliferation, migration, invasion and tumor angiogenesis by targeting IGF-1 gene through the IGF-1/PI3K/Akt signaling pathway in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with Lung cancer.
Literature	GENERIF_SUMMARY	For the first time, the present study showed that anacardic acid induces cell apoptosis of prostatic cancer through autophagy by ER stress/DAPK3/Akt signaling pathway.
Literature	GENERIF_SUMMARY	MicroRNA-302b-3p suppresses cell proliferation through AKT pathway by targeting IGF-1R in human gastric cancer.
Literature	GENERIF_SUMMARY	These results reveal a novel RTK-AKT-p300-ADA3 signaling pathway involved in growth factor-induced cell cycle progression.
Literature	GENERIF_SUMMARY	Our findings suggest that RACK1 promotes proliferation and radioresistance in ESCC cells by activating the AKT pathway, upregulating Bcl-2 expression and downregulating protein levels of Bim. Our study fills in gaps in the field of RACK1 and radiation resistance and may provide new possibilities for improving strategies of radiotherapy in esophageal cancer.
Literature	GENERIF_SUMMARY	Data suggest that NOX5 expression in melanoma cells could contribute to cell proliferation due, in part, to the generation of high local concentrations of extracellular ROS that modulate multiple pathways that regulate HIF-1alpha and networks that signal through Akt/GSK3beta/p27(Kip1) .
Literature	GENERIF_SUMMARY	Visfatin increases the nuclear translocation of Akt and its binding with ABCC1. These data showed that visfatin can decrease doxorubicin sensitivity of Non-small-cell lung cancer cells via activation of Akt/MRP1.
Literature	GENERIF_SUMMARY	These first-in-human data potentially qualify AKT1(low) quiescent cancer cells as a non-genetic cell state that persists after neoadjuvant chemotherapy in triple negative breast cancer patients and warrants further study.
Literature	GENERIF_SUMMARY	This study unravels an anti-metastatic role of AKT1 in the NSCLC cells with KRAS or EGFR mutations, and establishes an AKT1-MARCKS-LAMC2 feedback loop in this regulation.
Literature	GENERIF_SUMMARY	study established that the Akt pathway regulates JCVirus DNA replication and that

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		JCVirus DNA replication can be inhibited by MK2206, a compound that is specific for Akt
Literature	GENERIF_SUMMARY	data suggest that both FUT5 and FUT6 can promote the development of colorectal cancer (CRC) via the PI3K/Akt signalling pathway, which is regulated by miR-125a-3p. miR-125a-3p may serve as a predictive biomarker and a potential therapeutic target in CRC treatment.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with melanoma.
Literature	GENERIF_SUMMARY	These data demonstrated that the knockdown of BZW2 suppresses protein phosphorylation in the Akt/mTOR signaling pathway. These observations suggest that BZW2 is upregulated and has a pro-tumor effect in osteosarcoma via activation of the Akt/mTOR signaling pathway and thus is a potential target for gene therapy.
Literature	GENERIF_SUMMARY	Preoperative plasma hyperfibrinogenemia might serve as an independent predictor of unfavorable survival in ESCC. Furthermore, fibrinogen may promote cell motility by inducing EMT via the p-AKT/p-mTOR pathway.
Literature	GENERIF_SUMMARY	ablation of Glut1 attenuated apoptosis and increased drug resistance via upregulation of p-Akt/p-GSK-3beta (Ser9)/beta-catenin/survivin.
Literature	GENERIF_SUMMARY	NHE1 and Akt are downregulated in human umbilical vein endothelial cells by tumor microenvironment conditions, more potently than by hypoxia alone. This inhibits endothelial cell migration and growth in a manner likely modulated by the cancer cell secretome.
Literature	GENERIF_SUMMARY	Building upon previous work suggesting that FAK-Akt1 binding is mediated by the FAK F1 lobe, we demonstrated that independently expressing the F1 domain in human Caco-2 or murine CT-26 colon cancer cells by transient or stable inducible plasmid expression respectively prevents the stimulation of cancer cell adhesion by increased extracellular pressure.
Literature	GENERIF_SUMMARY	AKT inhibition led to decreased FMRP levels, as expected due to the known FMR1/FMRP negative feedback loop. But rFSH and the mTOR inhibition increased them, indicating a decoupling of this FMR1/FMRP negative feedback loop in our model system
Literature	GENERIF_SUMMARY	The results of the present study revealed a

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		correlation between BAFF and the PI3K/Akt/mTOR signaling pathway, and it is hypothesized that they are involved in the pathogenesis of lupus nephritis
Literature	GENERIF_SUMMARY	PPAR-gamma and Akt regulate GLUT1 and GLUT3 surface localization during Mycobacterium tuberculosis infection.
Literature	GENERIF_SUMMARY	These data indicate that increased irisin levels may have protective roles in liver cancer cells through partial activation of the PI3K/AKT pathway, which may facilitate liver cancer progression and decrease the sensitivity to chemotherapy.
Literature	GENERIF_SUMMARY	Our results showed that TSN inhibits pancreatic cancer progression via downregulating Akt/mTOR signaling.
Literature	GENERIF_SUMMARY	resistant tumors display high Akt activation and a combined treatment with AKT inhibitors causes synergistic tumour growth inhibition in vitro and in vivo
Literature	GENERIF_SUMMARY	Data show that LINC00963 (MetaLnc9) interacted with the glycolytic kinase PGK1 and prevented its ubiquitination in non-small cell lung cancer (NSCLC) cells, leading to activation of the oncogenic AKT/mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Low Frequency Magnetic Fields Induce Autophagy-associated Cell Death in Lung Cancer through miR-486-mediated Inhibition of Akt/mTOR Signaling Pathway.
Literature	GENERIF_SUMMARY	ILT3 may functionally contribute to a regulatory network controlling tumor progression by suppressing the Akt pathway.
Literature	GENERIF_SUMMARY	Report frequency of genetic variation in Akt1 and discuss link to cancer.
Literature	GENERIF_SUMMARY	AKT role in HPV16 RNA polyadenylation and splicing
Literature	GENERIF_SUMMARY	Authors show here that cyclin F, a substrate receptor F-box protein for the SCF (Skp1/Cul1/F-box) family of E3 ubiquitin ligases, is a bona fide AKT substrate.
Literature	GENERIF_SUMMARY	CXCR2 promotes breast cancer metastasis and chemoresistance via suppressing AKT1 and activating COX2.
Literature	GENERIF_SUMMARY	TMCO1 recruited the PH domain and leucine-rich repeat protein phosphatase 2 (PHLPP2) to dephosphorylate pAKT1(serine 473) (S473). Mutagenesis at S60 of the TMCO1 protein released TMCO1-induced

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		cell-cycle arrest and restored the AKT pathway in BFTC905 cells. Stable TMCO1 (wild-type) overexpression suppressed, whereas T33A and S60A mutants recovered, tumor size in xenograft mice.
Literature	GENERIF_SUMMARY	The role of AKT signaling pathway in the fibronectin and HIF-1alpha synthesis
Literature	GENERIF_SUMMARY	Mycobacterial trehalose 6,6'-dimycolate modulates transendothelial migration of neutrophils upon mycobacterial infection through prolonged AKT phosphorylation.
Literature	GENERIF_SUMMARY	Overexpression of KAT6A or TRIM24 promoted PIK3CA expression, AKT phosphorylation, and cell proliferation.
Literature	GENERIF_SUMMARY	AKT1 and AKT2 isoforms have opposing roles in smooth muscle cell proliferation, migration, differentiation, and rapamycin response in vitro and in vascular injury in vivo.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	Authors find that selective depletion of AKT1(low) slow proliferators actually reduces the growth of a molecularly diverse panel of human cancer cell xenograft models without globally altering cell proliferation or survival in vivo.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with glioblastoma.
Literature	GENERIF_SUMMARY	AKT pathway is activated by CBX8 in in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	frequent upregulation of MIF is implicated in the development and progression ofesophageal squamous cell carcinoma (ESCC).
Literature	GENERIF_SUMMARY	The essential role of AKT in the endocrine therapy resistance in estrogen receptor-positive, HER2-negative breast cancer. [review]
Literature	GENERIF_SUMMARY	SMOX plays a central role in the formation of bile canalicular lumen in liver cells by activating Akt pathway through acrolein production.
Literature	GENERIF_SUMMARY	Studies indicate the role of the insulin-like growth factor 1 (IGF1) - phosphoinositide 3-kinase (PI3K)-Akt signaling pathway in regulating exercise-induced physiological cardiac hypertrophy and cardiac protection [Review].

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Literature	GENERIF_SUMMARY	High phosphorylated AKT expression is associated with Esophageal Squamous Cell Cancer.
Literature	GENERIF_SUMMARY	Receptor tyrosine kinase activation of RhoA is mediated by AKT phosphorylation of DLC1.
Literature	GENERIF_SUMMARY	based on this novel role of Akt1 in homologous recombination(HR) and the previously described role of Akt1 in NHEJ, we propose that targeting Akt1 could be an effective approach to selectively improve the killing of tumor cells by DSB-inducing cytotoxic agents, such as ionizing radiation
Literature	GENERIF_SUMMARY	AKT serine/threonine kinase 1 (AKT1) gene silencing can enhance the sensitivity of gastric cancer xenografts to doxorubicin through promoting cell apoptosis.
Literature	GENERIF_SUMMARY	A reduction in invasion was also observed in the cancer stem cell compartment. Targeting AKT activity in the clinical setting may slow the growth and spread of aggressive thyroid neoplasms, and target the tumor stem cell compartment.
Literature	GENERIF_SUMMARY	data show that ErbB2 and Akt are important for CNC migration and implicate other ErbB receptors and Akt-independent signaling pathways. Our findings provide the first example of a functional interaction between the extracellular domain of a type II classical cadherin and growth factor receptors
Literature	GENERIF_SUMMARY	The optimal melatonin concentration (3 mM) significantly decreased the intracellular reactive oxygen species levels, caspase-3 activity and the percentage of both dead and apoptotic-like sperm cells and increased the vitality, progressive motility and total motility and AKT phosphorylation compared with the control group.
Literature	GENERIF_SUMMARY	intact keratin filaments are regulators for PKB/Akt and p44/42 activity, basal and in response to stretch.
Literature	GENERIF_SUMMARY	Taken together, our findings suggested that miR-126 alleviates ox-LDL-induced HUVECs injury through restoring autophagy flux via repressing PI3K/Akt/mTOR pathway, and further implicate the potential therapeutic targets to reverse atherosclerosis.
Literature	GENERIF_SUMMARY	Knockdown of MTDH expression also upregulated PTEN and suppressed pAKT protein expression in Caki2 cells. In conclusion, the present study is the first, to the best of our knowledge, to provide

Main Category	Category	Terms
		evidence suggesting that miRNA30a5p suppresses tumor human renal cancer cell proliferation via the MTDH/PTEN/AKT pathway
Literature	GENERIF_SUMMARY	RIO kinase 3 (RIOK3) positively regulates the activity of the AKT/mTOR pathway in glioma cells.
Literature	GENERIF_SUMMARY	Akt1-Akt3 activity (according to Thr308 phosphorylation) is not associated with proliferative processes in the tumor tissue of the thyroid.
Literature	GENERIF_SUMMARY	By monitoring single-cell dynamics in each of these contexts, the authors identified PI3K/Akt regulation of glycolysis as a multifaceted modulator of single-cell metabolic dynamics that is required to maintain metabolic stability in proliferating cells.
Literature	GENERIF_SUMMARY	Results show that AKT is a direct target of miR-564 in osteosarcoma (OS) cells inhibiting the glycolysis.
Literature	GENERIF_SUMMARY	We conclude that KGF-KGFR activates Akt-mTORC1 downstream Nrf2 signaling to protect RPE cells from UV radiation.
Literature	GENERIF_SUMMARY	EGFR is confirmed a poor prognostic factor in resected gastric cancers. We firstly describe a mutually exclusive overexpression of EGFR and AKT1 with potential prognostic implications, suggesting the relevance of this pathway for the growth of gastric cancers.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with invasion and metastasis in tongue squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Data suggest that PLSCR1 is involved in viral entry; here, herpes simplex viruses activate PLSCR1 and up-regulate flipping of phosphatidylserines and AKT to outside of plasma membrane shortly following HSV-1 or HSV-2 exposure. (PLSCR1 = phospholipid scramblase 1; AKT = proto-oncogene c-Akt)
Literature	GENERIF_SUMMARY	Germline variants in AKT1 gene are associated with prostate cancer.
Literature	GENERIF_SUMMARY	a deletion in the amino terminus of glycoprotein K (gK) inhibits gB binding to Akt-1(S473), the release of intracellular calcium, and virus entry via fusion of the viral envelope with cellular plasma membranes.
Literature	GENERIF_SUMMARY	These results indicate that BSNQ and OSNQ induce apoptosis in human hepatoma

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		Hep3B cells via ROS-mediated p38/MAPK, Akt and STAT3 signaling pathways, suggesting that these 1,4-naphthoquinone derivatives may provide promising new anticancer agents to treat HCC.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with cervical cancer.
Literature	GENERIF_SUMMARY	EGFR inhibition can only sensitize EGFR-high cells for chemotherapy, while AKT inhibition increases chemosensitivity in EGFR-low cells. By understanding these mechanisms, we can take advantage of the cellular context to individualize antineoplastic therapy. Finally, our data also suggest targeting of EFFRI1 in EGFR-low cancer as a promising therapeutic approach
Literature	GENERIF_SUMMARY	A double-negative feedback loop between Akt and AMPK controls the switch between matrix-attached and matrix-detached states needed to coordinate cell growth and survival during breast cancer metastasis.
Literature	GENERIF_SUMMARY	results reveal how alterations in FBXO31 phosphorylation, mediated by AKT and ATM, underlie physiological regulation of FBXO31 levels in unstressed and genotoxically stressed cells
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with drug resistance in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	results suggest that HBP1 phosphorylation by AKT blocks its functions as transcriptional regulator and tumor suppressor.
Literature	GENERIF_SUMMARY	Dengue virus capsid expression activates Akt by sequestering phosphatases that downregulate phospho-Akt.
Literature	GENERIF_SUMMARY	that Akt1 was a novel target for miR-637, and its knockdown also induced cell growth inhibition and apoptosis in pancreatic ductal adenocarcinoma cells
Literature	GENERIF_SUMMARY	Knockdown studies revealed that AKT sustained BCL-2 expression in uterine leiomyoma.
Literature	GENERIF_SUMMARY	the results indicated that JB inhibits glycolysis by down-regulating HK2 expression through inactivating the Akt/mTOR pathway in non-small cell lung cancer (NSCLC) cells, suggesting that JB might be a potential therapeutic agent for the treatment of NSCLC.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with

Main Category	Category	Terms
		Pancreatic Ductal Adenocarcinoma Metastasis.
Literature	GENERIF_SUMMARY	this is the first report showing long-duration exposure to nicotine causes increased proliferation of human kidney epithelial cells through activation of AKT pathway.
Literature	GENERIF_SUMMARY	The findings indicate that SPRY4 and SPRY4-IT1 may act as oncogenes in testicular germ cell tumors via activation of the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	PI3K-Akt pathway inhibitors, Akti-1/2 and LY294002, reduced PFKFB3 gene induction by PHA, as well as Fru-2,6-P2 and lactate production. Moreover, both inhibitors blocked activation and proliferation in response to PHA, showing the importance of PI3K/Akt signaling pathway in the antigen response of T-lymphocytes.
Literature	GENERIF_SUMMARY	The long non-coding RNA FOXD2-AS1 promotes bladder cancer progression and recurrence through a positive feedback loop with Akt and E2F1.
Literature	GENERIF_SUMMARY	STK17B promotes carcinogenesis and metastasis via AKT/GSK-3beta/Snail signaling in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Inflammation-induced TNF promotes hepatocellular CCR10 expression and downstream PI3K/Akt-mediated hepatocarcinogenesis.
Literature	GENERIF_SUMMARY	The present study demonstrated that the oncostatic effects of melatonin on SGC-7901 GC cells are mediated via the blockade of the AKT/MDM2 intracellular pathway.
Literature	GENERIF_SUMMARY	these results indicate that miR124 transection inhibits the growth and aggressive of osteosarcoma, potentially via suppression of TGFbetamediated AKT/GSK3beta/snail family transcriptional repressor 1 (SNAIL1) signaling, suggesting miR124 may be a potential anticancer agent/target for osteosarcoma therapy.
Literature	GENERIF_SUMMARY	piperine reduced the expression of pAkt, MMP9 and pmTOR. Together, these data indicated that piperine may serve as a promising novel therapeutic agent to better overcome prostate cancer metastasis.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with head and neck carcinoma.
Literature	GENERIF_SUMMARY	we review the current experience with anticancer therapies that target the PI3K-

Main Category	Category	Terms
		AKT-mTOR pathway.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with proliferation, migration, invasion and epithelial-mesenchymal transition of esophageal cancer.
Literature	GENERIF_SUMMARY	AKT1 mutations were detected in 3.6% of a cohort of Chinese patients with breast neoplasm, and 4.9% in the luminal subtype. These AKT1 mutations were found to be significantly associated with clinicopathological and molecular characteristics of breast tumors.
Literature	GENERIF_SUMMARY	AKT role in the melanoma drug resistance.T
Literature	GENERIF_SUMMARY	LncARSR promotes hepatic lipogenesis via Akt/SREBP-1c pathway and contributes to the pathogenesis of nonalcoholic steatohepatitis.
Literature	GENERIF_SUMMARY	WT PDCD5 competitively inhibited interaction between histone deacetylase 3 (HDAC3) and AKT, but PDCD5(L6R), an HDAC3-binding-deficient mutant, did not. Knockdown of PDCD5 accelerated HDAC3-AKT interaction, AKT and eNOS phosphorylation, and nitric oxide (NO) production
Literature	GENERIF_SUMMARY	S100A8 gene knockdown reduced cell proliferation in the HEC-1A cells compared with control cells, induced cell apoptosis, inhibited the phosphorylation of protein kinase B (Akt), and induced the expression of pro-apoptotic genes.
Literature	GENERIF_SUMMARY	Radiation resistance tumors have upregulated Onzin and POU5F1 expression.
Literature	GENERIF_SUMMARY	RBL2/p130 and AKT1 physically interact and AKT phosphorylates RBL2/p130 Ser941, located in the pocket domain, but not when this residue is mutated into Alanine.
Literature	GENERIF_SUMMARY	a model for the activation and inactivation of Akt by an ordered series of membrane binding, phosphorylation, dissociation, and dephosphorylation events.
Literature	GENERIF_SUMMARY	Vav1, ectopically expressed in invasive breast tumors derived cells, downmodulates Akt acting at expression and/or activation levels depending on tumor subtype.
Literature	GENERIF_SUMMARY	This review highlights the current status of PI3K/Akt/mTOR inhibitors in B-ALL, with an emphasis on emerging evidence of the superior efficacy of synergistic combinations involving the use of traditional chemotherapeutics or other novel, targeted

Main Category	Category	Terms
		agents.
Literature	GENERIF_SUMMARY	Results suggest that the enhanced akt proto-oncogene rotein AKT phosphorylation (pAKT) in circulating B cells may be useful for the discrimination of immunologic deficiency syndromes APDS1, APDS2, and APDS-L from other antibody deficiencies.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with hypercholesterolemia.
Literature	GENERIF_SUMMARY	The overexpression of CHIP significantly increased the migration and invasion of the DU145 cells, which is possible due to activation of the AKT signaling pathway and upregulation of vimentin. The expression level of CHIP was observed to be increased in human prostate cancer tissues compared with the adjacent normal tissue.
Literature	GENERIF_SUMMARY	Genistein (GE) inhibited the growth of human Cholangiocarcinoma (CCA) cell lines by reducing the activation of EGFR and AKT, and by attenuating the production of IL6. E2 and ER were also involved in the growth-inhibitory effect of GE in CCA cells.
Literature	GENERIF_SUMMARY	The effects of oridonin were verified in patient-derived xenograft tumors expressing high levels of AKT. In summary, our results indicate that oridonin acts as an AKT inhibitor to suppress the growth of esophageal squamous cell carcinoma (ESCC) by attenuating AKT signaling.
Literature	GENERIF_SUMMARY	Study demonstrated that expression of RhoB was decreased in osteosarcoma primary samples and cell lines. miR-19a, which exhibits abnormal expression in highly metastatic osteosarcoma cell lines, could inhibit the expression of RhoB and promote the lung metastasis of osteosarcoma cells in vivo. Results indicate that miR-19a-mediated RhoB is a critical regulator for the dephosphorylation of AKT1 in osteosarcoma cells.
Literature	GENERIF_SUMMARY	CIP2A- and SETBP1-mediated PP2A inhibition reveals AKT S473 phosphorylation to be a new biomarker in AML.
Literature	GENERIF_SUMMARY	that AKT and mTOR proteins are involved in oral squamous cell carcinoma biology and that GSK3beta itself may drive cervical lymph node metastatic spread of oral squamous cell carcinoma cells
Literature	GENERIF_SUMMARY	50 diffuse-type gastric tumors were analyzed for AKT1 exon 2 and 14 mutations. The novel mutations, 15553T >A and

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		25376C >G might affect the exonic splicing enhancers and silencers. Significant decline was observed in the S-phase population in the tumor cells with 15553T >A and 15579G >C mutations suggesting the arrest of G1 phase.
Literature	GENERIF_SUMMARY	CREBRF promotes the proliferation of human gastric cancer cells via the AKT signaling pathway.
Literature	GENERIF_SUMMARY	Report AKT1 p.Glu17Lys mutations occuring in intraductal papillary mucinous neoplasms of minor salivary glands.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with metastasis in ovarian cancer.
Literature	GENERIF_SUMMARY	The role of USP18 in breast cancer provides a novel insight into the clinical application of the USP18/AKT/Skp2 pathway.
Literature	GENERIF_SUMMARY	Study shows that the lack of Akt1 in human lung endothelial cells promotes prostate cancer cell transendothelial migration in vitro and metastasis to the mouse lungs in vivo involving increased nuclear localisation of beta-catenin and suppression of endothelial tight-junction proteins.
Literature	GENERIF_SUMMARY	Linc-ROR is highly expressed in the ectopic endometrium of adenomyosis, and it can promote the proliferative activity of endometrial cells by activating the PI3K-Akt pathway.
Literature	GENERIF_SUMMARY	In MCF-7 cells, AIB1 overexpression increases p-AKT (Ser 473) activity.In both T47D and MCF-7 cells overexpressing A1B1, p-AKT (Ser 473) expression was significantly increased in the presence or absence of IGF-1,but increased more in presence of IGF-1.
Literature	GENERIF_SUMMARY	These results demonstrated that shikonin inhibits CML proliferation and migration and induces apoptosis by the PTEN/PI3K/AKT pathway, revealing the effects of shikonin therapy on CML.
Literature	GENERIF_SUMMARY	miR-195 suppresses cell proliferation of ovarian cancer cells through regulation of VEGFR2 and AKT signaling pathways.
Literature	GENERIF_SUMMARY	enigma siRNA-transfected cells, cell viability, and the protein levels of AKT and survivin decreased.
Literature	GENERIF_SUMMARY	LINC00470 bound to FUS and AKT to form a ternary complex, anchoring FUS in the cytoplasm to increase AKT activity, promoting glioblastoma progression.

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Literature	GENERIF_SUMMARY	We performed quantitative mass spectrometry of IAV1918-infected cells to measure host protein dysregulation. Selected proteins were validated by immunoblotting and phosphorylation levels of members of the PI3K/AKT/mTOR pathway were assessed.
Literature	GENERIF_SUMMARY	CircIBTK was downregulated in SLE and might regulate DNA demethylation and the AKT serine/threonine kinase 1 (AKT) signaling pathway via binding to miR-29b in systemic lupus erythematosus (SLE).
Literature	GENERIF_SUMMARY	These data revealed that CXCR4-mediated Akt activation and signaling require endocytosis and the Rab5-effector EEA1 but not APPL1, suggesting a role for EEA1-positive endosomes in Akt signaling.
Literature	GENERIF_SUMMARY	TERT could induce thyroid carcinoma cell proliferation mainly through the PTEN/AKT signaling pathway.
Literature	GENERIF_SUMMARY	RBAP48 overexpression contributes to the radiosensitivity of AGS gastric cancer cells via phosphoinositide3kinase/protein kinase B pathway suppression.
Literature	GENERIF_SUMMARY	in our study, we found that PI3K/AKT/mTOR pathway was involved in osteosarcoma progression. Knockdown of LINC00968 inactivated PI3K/AKT/mTOR signaling pathway in vitro. Subsequently, in vivo tumor xenografts were established using 143-B cells to investigate whether LINC00968 can induce osteosarcoma development in vivo
Literature	GENERIF_SUMMARY	2-ME reduced the production of CTGF and collagen I in SSc fibroblasts induced by hypoxia through PI3K/Akt/mTOR/ HIF-1alpha signalling and inhibited the proliferation of fibroblasts. These findings suggested that 2-ME could be employed as a promising antifibrotic therapy for SSc
Literature	GENERIF_SUMMARY	Results suggest that transient receptor potential vanilloid 4 (TRPV4) accelerates glioma migration and invasion through the AKT/Rac1 signaling, and TRPV4 might be considered as a potential target for glioma therapy.
Literature	GENERIF_SUMMARY	LHPP suppresses cell proliferation and metastasis in cervical cancer, and promotes apoptosis by suppressing AKT activation.
Literature	GENERIF_SUMMARY	this study identifies ORP2 as a new regulatory nexus of Akt signaling, cellular energy metabolism, actin cytoskeletal function, cell migration, and proliferation.

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Literature	GENERIF_SUMMARY	These results suggest that the classical metabolic PI3K pathway and not the canonical proliferation ERK pathway is involved in the insulin/insulin-like growth factor-1-induced increase in crypt proliferation in obese humans, which may contribute to abnormal tissue renewal and function.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with cisplatinresistant oral cancer.
Literature	GENERIF_SUMMARY	SIRT6 inhibited proliferation, migration, and invasion of colon cancer cells by up-regulating PTEN expression and down-regulating AKT1 expression.
Literature	GENERIF_SUMMARY	results demonstrated that PLAUR induces geftinib-resistance through EGFR/p-AKT/survivin signaling pathway in gefitinib-resistant human lung adenocarcinoma cells. PLAUR could be a novel therapeutic target for gefitinib-resistant NSCLC patients.
Literature	GENERIF_SUMMARY	we demonstrated that overexpression of CHP2 enhanced G1-S phase transition mediated by AKT activation, which subsequently induced phosphorylation and inactivation of FOXO3a in breast cancer cells.
Literature	GENERIF_SUMMARY	High AKT1 phosphorylation is associated with colorectal carcinoma.
Literature	GENERIF_SUMMARY	Combined targeting of AKT and SRC resulted in a synergistic efficacy against human pancreatic cancer growth and metastasis
Literature	GENERIF_SUMMARY	LASP1 may function as an oncogene in glioblastoma and regulate cell proliferation and chemosensitivity in a PI3K/AKT-dependent mechanism.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with angiogenesis of esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with cell growth, aggressiveness, metastasis in gastric cancer.
Literature	GENERIF_SUMMARY	Data indicate that the telomerase reverse transcriptase (TERT)(TERT)-DNA methyltransferase 3B (DNMT3B)-PTEN-AKT axis provides an explanation for multi-oncogenic activities of TERT and may be exploited in hepatocellular carcinoma (HCC) treatment.
Literature	GENERIF_SUMMARY	FAL1 may work as a ceRNA to modulate AKT1 expression via competitively binding

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		to miR-637 in HSCR.
Literature	GENERIF_SUMMARY	he intronic variant, AKT1 rs3803304, may act as a predictive biomarker in the risk of developing breast cancer in the high altitude Ecuadorian mestizo population.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with metastasis via epithelialmesenchymal transition carcinoma in colorectal cancer.
Literature	GENERIF_SUMMARY	The findings suggest that HIV-1 may alter AKT phosphorylation to inhibit the translocation of IRF-7 into plasmacytoid dendritic cell nucleus, leading to IFN-alpha suppression, and this may be the reason for IFN-alpha abrogation observed in recently infected HIV patients.
Literature	GENERIF_SUMMARY	The current study showed that miR-152 could be a novel therapeutic small molecule to suppress CRC cell proliferation, survival, and migration by suppressing AKT-ERK signaling pathways.
Literature	GENERIF_SUMMARY	JPX and XIST play a suppressive role in HCC. JPX increases expression levels of XIST in HCC cells, which suppresses HCC development by sponging the cancer promoting miR-155-5p.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with tumor-node-metastasis in nonsmall cell lung cancer.
Literature	GENERIF_SUMMARY	Circ-CFH promotes glioma progression by sponging miR-149 and regulating the AKT1 signaling pathway.
Literature	GENERIF_SUMMARY	Overexpression of ATF3 increases radioresistance of breast cancer cells by up-regulating the level of phosphorylation of Akt in the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	DDX5 promotes hepatocellular carcinoma tumorigenesis via Akt signaling pathway.
Literature	GENERIF_SUMMARY	Akt signaling pathway role in multiple myeloma.miR-30d suppresses the PI3K/Akt signaling pathway by targeting MTDH.
Literature	GENERIF_SUMMARY	AKT signaling pathway is regulated by ASF1B in prostate cancer.
Literature	GENERIF_SUMMARY	study identifies USP43 to be an H2BK120 deubiquitinase and a potential tumor suppressor and reveals a reciprocally inhibitory loop between USP43 and EGFR/PI3K/AKT, whose imbalance drives breast carcinogenesis
Literature	GENERIF_SUMMARY	findings reveal that Akt activity determines

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		the phosphorylation status of TBC1D7 at the phospho-switch Ser-124, which governs binding to either 14-3-3 or beta-TrCP2, resulting in increased or decreased stability of TBC1D7, respectively.
Literature	GENERIF_SUMMARY	These results highlight an Arf GAP-independent function of ARAP2 in regulating Akt activity.
Literature	GENERIF_SUMMARY	Results show that AKT1 was associated with hypertension in Mexican Mestizos but not Mexican Amerindians.
Literature	GENERIF_SUMMARY	miR-1179 inhibits the growth and invasion of non-small cell lung cancer cells by targeting SPAG5 and inhibiting Akt
Literature	GENERIF_SUMMARY	AKT kinase pathway is regulated by SPC24 in breast cancer.
Literature	GENERIF_SUMMARY	miRNA-520a reduced hepatitis b virus transcription and replication by reducing AKT levels.
Literature	GENERIF_SUMMARY	miR-935 is down-regulated in non-small cell lung cancer tissue, is linked to poor outcome, and acts on signal transduction mediator E2F7 and the AKT pathway.
Literature	GENERIF_SUMMARY	findings provided a comprehensive mutation profiling of AKT1, PIK3CA, PTEN and TP53 genes in Chinese breast cancer patients, which have potential implications in clinical management.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with periodontitis.
Literature	GENERIF_SUMMARY	In this study, we used the Ion Personal Genome Machine (PGM) and Ion Torrent Ampliseq Cancer panel to sequence hotspot regions from PIK3CA, AKT and PTEN genes to identify genetic mutations in 39 samples of TNBC subtype from Moroccan patients and to correlate the results with clinical-pathologic data
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with Epithelial-mesenchymal transition in gastric carcinoma.
Literature	GENERIF_SUMMARY	over-expression of DIRAS3 promotes EGFR-AKT signaling activation at the downstream of EGFR and increases AKT phosphorylation
Literature	GENERIF_SUMMARY	NT5E expression was upregulated in sunitinib-resistant RCC tissues and cells. NT5E downregulation repressed RCC cell

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		proliferation, migration as well as invasion. EMT course and AKT/GSK-3beta signal pathway both in vitro and in vivo in sunitinib environment was suppressed to varying degrees via NT5E inhibition.
Literature	GENERIF_SUMMARY	The results indicated that FOXO6 knockdown inhibited colorectal cancer cell proliferation, migration, invasion, and glycolysis via the PI3K/Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	AKT1 expression was significantly increased in prostate cancer (PCa) tissues and this expression is associated with poor patients survival. Additionally, microRNA-373 targeted the 3'UTR of AKT1 and inhibited its expression.
Literature	GENERIF_SUMMARY	miR-129-5p inhibited autophagy and apoptosis in H2O2-induced H9c2 cells partly by down-regulation of ATG14 through the activation of PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with cholangiocarcinoma metastasis.
Literature	GENERIF_SUMMARY	TP53I11 functions as a mediator to balance activation of AKT and AMPK to adapt cells to different cellular contexts such as extracellular matrix-attachment and -detachment.
Literature	GENERIF_SUMMARY	APE1 stimulates EGFR-tyrosine kinase inhibitors resistance by activating Akt signaling through a redox-dependent mechanism in lung adenocarcinoma.
Literature	GENERIF_SUMMARY	MEG3 overexpression inhibited the tumorigenesis of breast cancer by downregulating miR-21 through the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	High AKT expression is associated with metastasis of hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	AMPK is an essential regulator for Akt activation by various stresses.
Literature	GENERIF_SUMMARY	Data show that Y-box binding protein-1 (YBX1) promoted tumorigenesis and progression in spinal chordoma via the epidermal growth factor receptor (EGFR)/proto-oncogene proteins c-akt (AKT) pathway.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with progression of endometrial cancer.
Literature	GENERIF_SUMMARY	EGFR-mediated beta-catenin nuclear accumulation is critical for Akt1 inhibition-induced breast cancer metastasis

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Literature	GENERIF_SUMMARY	High AKT1 expression is associated with lung cancer.
Literature	GENERIF_SUMMARY	we found that RBM10 activated key proliferative signaling pathways [such as the epidermal growth factor receptor (EGFR), mitogenactivated protein kinase (MAPK) and phosphoinositide 3kinase (PI3K)AKT pathways] and inhibited apoptotic pathways. In addition, we demonstrated that a high expression of RBM10 protein in patient tissue samples was associated with a shorter overall survival time and a poor prognosis
Literature	GENERIF_SUMMARY	AKT phosphorylation sites of Ser473 and Thr308 regulate AKT degradation
Literature	GENERIF_SUMMARY	The mechanism underlying the antiapoptotic and antioxidative role of SAC is the induction of the heme oxygenase-1 (HO-1) gene in an NF-E2-related factor-2 (Nrf-2) and Akt dependent manner.
Literature	GENERIF_SUMMARY	this study shows that the IL-15-AKT-XBP1s signaling pathway contributes to effector functions and survival in human NK cells
Literature	GENERIF_SUMMARY	LncRNA TP73-AS1 promoted the progression of lung adenocarcinoma via PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Data suggest that overexpression of phosphatidylinositol 3-kinase (PI3K), protein kinase B (PKB), and mammalian target of rapamycin (mTOR) proteins is an important indicator of poor survival in nasopharyngeal carcinoma (NPC).
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with Hodgkin's lymphoma.
Literature	GENERIF_SUMMARY	The major role in stimulation of the synthesis of granulocytic CSF during stressful stimulation is played by PI3K/Akt signaling cascade.
Literature	GENERIF_SUMMARY	The role of AKT signaling in the cervical cancer cell proliferation.
Literature	GENERIF_SUMMARY	verall, the present study uncovered a yet unrecognized nuclear coupling of AKT and provides insights into the involvement of AKT in the interaction network of nuclear actin.
Literature	GENERIF_SUMMARY	IL-17 can promote the proliferation and migration of glioma cells via PI3K/Akt1/NF-kappaB-p65 activation.

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Literature	GENERIF_SUMMARY	Cx32 expression may induce cisplatin resistance by modulating drug efflux transporter expression and activating the EGFRprotein kinase B signalling pathway in ovarian cancer cells.
Literature	GENERIF_SUMMARY	PIK3CA silencing inhibited Nalm-6 cell proliferation and invasion, and promoted their apoptosis and sensitivity to chemotherapeutic drugs, potentially through regulation of the PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	MAPK4 can promote cancer by activating the AKT/mTOR signaling pathway and that targeting MAPK4 may provide a novel therapeutic approach for cancer.
Literature	GENERIF_SUMMARY	WTIP impairs AKT phosphorylation and activation, leading to enhanced expression and transcriptional activity of FOXO1, which further increases p21Cip1 and p27Kip1, and decreases cyclin D1, which consequently results in cell cycle arrest.
Literature	GENERIF_SUMMARY	Data suggest that targeting SET domain, bifurcated 1 protein (SETDB1) signalling could be a potential therapeutic strategy for combatting hyperactive AKT serine/threonine kinase 1 (AKT)-driven cancers.
Literature	GENERIF_SUMMARY	Findings indicate complicated layers of serine/threonine kinase Akt (Akt) activation regulation coordinated by SET domain, bifurcated 1 protein (SETDB1)-mediated Akt K64 methylation.
Literature	GENERIF_SUMMARY	results uncover that miR-21 targets PTEN/Akt signaling pathway and regulates cell proliferation, migration and apoptosis in human gastric cancer cells.
Literature	GENERIF_SUMMARY	LINC00520-targeted EGFR inhibition might result in the inactivation of the PI3K/Akt pathway, thus inhibiting cutaneous squamous cell carcinoma (cSCC) development.
Literature	GENERIF_SUMMARY	Circular RNA circNRIP1 acts as a microRNA-149-5p sponge to promote gastric cancer progression via the AKT1/mTOR pathway.
Literature	GENERIF_SUMMARY	MiR-637 inhibits proliferation and invasion of hepatoma cells by targeted degradation of AKT1.
Literature	GENERIF_SUMMARY	Collectively, these data uncover a functional key role of the PI3K/mTORC2/AKT signaling cascade for viral oncogene repression in hypoxic human papilloma virus-positive cancer cells and provide new insights into

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		the poorly understood cross talk between oncogenic human papilloma viruses and their host cells under hypoxia.
Literature	GENERIF_SUMMARY	this study suggests that miR-633 deletion can activate the AKT/mTOR pathway by targeting AKT1 to participate in the pathogenesis of systemic lupus erythematosus.
Literature	GENERIF_SUMMARY	LncRNA PCAT1 activates AKT and NF-kappaB signaling in castration-resistant prostate cancer by regulating the PHLPP/FKBP51/IKKalpha complex.
Literature	GENERIF_SUMMARY	Urinary exosome as a potential biomarker for urinary tract infection.
Literature	GENERIF_SUMMARY	SGK1 activation occurs at a distinct subcellular compartment from that of Akt and suggests a mechanism for the selective activation of these functionally distinct mTORC2 targets through subcellular partitioning of mTORC2 activity.
Literature	GENERIF_SUMMARY	Upregulation of AKT1 is associated with Angiogenesis in Degenerated Discs.
Literature	GENERIF_SUMMARY	Authors found that Ser389 phosphorylation of p53 was a downstream event of ROS-inactivated AKT pathway, which was critical to p53 mitochondrial trafficking during HMS stimuli. Transfecting p53-shRNA C2C12s with the mutant p53 (S389A) that was unable to target p53 to mitochondria underwent significantly lower apoptosis than transfection with wild-type p53
Literature	GENERIF_SUMMARY	miR493 acted as a onco-miR in prostate cancer cells and promoted cell proliferation via inhibiting tumor suppresser PHLPP2 expression and activating Akt signaling pathway.
Literature	GENERIF_SUMMARY	A novel gene fusion between the LAMTOR1 and AKT1 was identified in a histopathologically indeterminate epithelioid neoplasm of a patient. The patient initially responded to treatment with an AKT inhibitor ipatasertib with eventual drug resistance and disease progression.
Literature	GENERIF_SUMMARY	New insight into the mechanism of SOX2 overexpression in cancer and evidence for targeting AKT as a potential therapeutic strategy for SOX2-positive cancers, as AKT drives SOX2 overexpression and cancer cell stemness.
Literature	GENERIF_SUMMARY	Results suggest that tenascin-C (TNC) may enhance the cancer stem-like properties and promote epithelial-to-mesenchymal

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		transition (EMT)-like changes via the proto-oncogene proteins c-akt (Akt)/hypoxia-inducible factor 1, alpha subunit (HIF1alpha) axis.
Literature	GENERIF_SUMMARY	study revealed that berberine not only up-regulates mRNA and protein levels of p21/cip1 and p27/kip1 but also increases their nuclear localization and post-translational protein stability. Further, Akt inhibition was found to mediate berberine-mediated up-regulation of p21/cip1 but not the p27/kip1.
Literature	GENERIF_SUMMARY	MiR-155-5p inhibits the proliferation and migration of VSMCs and HUVECs in atherosclerosis by targeting AKT1.
Literature	GENERIF_SUMMARY	Stem cells features and aberrant metabolic reprogramming of esophageal cancer stem cells (CSC) depend on the heat shock 27 kDa protein (Hsp27)-proto-oncogene protein Akt (AKT)-HXK2 hexokinase 2 (HK2) pathway.
Literature	GENERIF_SUMMARY	Suppression of host DNA-repair pathways that remove TOP2bcc trigger HSV-1 reactivation. Moreover, perturbation of AKT phosphorylation dynamics by downregulating the PHLPP1 phosphatase led to AKT mis-localization and disruption of DSB-induced HSV-1 reactivation.
Literature	GENERIF_SUMMARY	High expressions of AKT1 is associated with non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Tumor necrosis factor receptor-associated factor 6 (TRAF6)-mediated proto-oncogene protein Akt (AKT) ubiquitination and subsequent phosphorylation played an essential role in the control of tumor cell malignant behavior.
Literature	GENERIF_SUMMARY	Increased AKT1 expression is associated with Colorectal Cancer.
Literature	GENERIF_SUMMARY	The resistant effect of SIRT1 in oxidative stress-induced senescence of rat nucleus pulposus cell is regulated by Akt-FoxO1 pathway.
Literature	GENERIF_SUMMARY	Silencing of Akt1 in human umbilical vein endothelial cells significantly inhibited vascular endothelial growth factor (VEGF)-induced capillary-like tube formation. Mice lacking Akt1 exhibited impaired retinal angiogenesis with delayed endothelial cell (EC) proliferation.
Literature	GENERIF_SUMMARY	The continuous interaction of AKT/PKB with PI3,4,5P3 or PI3,4P2 in a lipid membrane is required for its activation.[review]

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Literature	GENERIF_SUMMARY	Mechanism for IL-15-Driven B Cell Chronic Lymphocytic Leukemia Cycling
Literature	GENERIF_SUMMARY	Advances in the role of HCV nonstructural protein 5a (NS5A) of 3a genotype in inducing insulin resistance by possible phosphorylation of AKT/PKB.
Literature	GENERIF_SUMMARY	USP49 inhibits non-small cell lung cancer cell growth by suppressing PIK3R2/AKT signaling.
Literature	GENERIF_SUMMARY	The data suggest that tumourigenic alterations in the PI3K/AKT-pathway are indispensable in hidradenoma papilliferum (HP) and establish a homogenous morphomolecular entity with a functionally converging and selecting tumourigenic mechanism.
Literature	GENERIF_SUMMARY	K1 gene transformation activities in AIDS-related and classic type Kaposi's sarcoma
Literature	GENERIF_SUMMARY	The data demonstrate a novel phosphorylation-dependent mechanism to regulate Gld2 activity, revealing tumour suppressor miRNAs(miR-122) as a previously unknown target of Akt1-dependent signalling
Literature	GENERIF_SUMMARY	MAPK, Akt, and STAT3 pathways might play diverse roles in oral carcinogenesis
Literature	GENERIF_SUMMARY	Hepatic expression of Akt (Thr308), GSK3beta (Ser9), and GSK3alpha (Ser21) did not correspond to progression of chronic hepatitis C to liver steatosis.
Literature	GENERIF_SUMMARY	miR-615-5p as an inhibitor of VEGF-AKT/eNOS-mediated endothelial cell angiogenic responses.
Literature	GENERIF_SUMMARY	Breast adenomyoepitheliomas are genetically heterogeneous exhibiting recurrent mutually exclusive mutations of PIK3CA and AKT1 in a majority of cases.
Literature	GENERIF_SUMMARY	EMP1 regulates cell proliferation, migration, and stemness in gliomas through PI3K-AKT signaling and CD44.
Literature	GENERIF_SUMMARY	Inhibition of CTRP6 prevented survival and migration in hepatocellular carcinoma through inactivating the AKT signaling pathway.
Literature	GENERIF_SUMMARY	the data of the present study support that miR3383p inhibits lung cancer cell invasion and proliferation by downregulating AKT/bcatenin signaling pathway. The present findings supported the potential use of miR3383p in the treatment of lung cancer

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Literature	GENERIF_SUMMARY	The findings of this study suggested L-THP increased the sensitivity of ovarian cancer cells to cisplatin via modulating miR-93/ PTEN/AKT pathway in A2780/DDP ovarian cancer cell line.
Literature	GENERIF_SUMMARY	Overexpressed lncRNA H19 promotes Hodgkin's lymphoma (HL) development by stimulating proliferation of HL cells via AKT pathway.
Literature	GENERIF_SUMMARY	Lysosome positioning influences mTORC2 and AKT signaling.
Literature	GENERIF_SUMMARY	In an established melanoma model, mice harboring tumors expressing AKT1(E17K) had the highest incidence of brain metastasis and lowest mean survival. Tumors expressing AKT1(E17K) displayed elevated levels of focal adhesion factors, enhanced phosphorylation, and increased invasion. This study suggests that AKT1(E17K) promotes melanoma brain metastasis through activation of FAK.
Literature	GENERIF_SUMMARY	Multiplex PCR analysis of apocrine lesions shows frequent PI3K-AKT pathway mutations in both benign and malignant apocrine breast tumors.
Literature	GENERIF_SUMMARY	circZFR promotes hepatocellular carcinoma progression by directly down-regulating miR-511 to activate AKT1 signaling, suggesting that circZFR is a potential target in HCC treatment.
Literature	GENERIF_SUMMARY	The effects of CAPON-L overexpression on glioma cell proliferation are dependent on the AKT/mTOR/P53 activity. The overexpression of CAPON inhibits U251 cell proliferation through the AKT/mTOR signaling pathway, while overexpressing CAPON-L promoted U87 cell proliferation, possibly through down-regulating the P53 level.
Literature	GENERIF_SUMMARY	NFIA enhances cell radiosensitivity by downregulating p-AKT and p-ERK in non-small cell lung cancer
Literature	GENERIF_SUMMARY	Absent in melanoma 2 suppresses epithelial-mesenchymal transition via Akt and inflammasome pathways in human colorectal cancer cells.
Literature	GENERIF_SUMMARY	ROS act in concert with BLM to facilitate Prostate cancer (PC) oncogenesis, potentially via further enhancing AKT signaling and downregulating PTEN expression. Importantly, inhibiting the BLM-AKT-PRAS40 axis induced PC cell apoptosis.

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Literature	GENERIF_SUMMARY	The PI3K/Akt pathway is not a main driver in HDL-mediated cell protection.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with glioblastoma progression.
Literature	GENERIF_SUMMARY	role of Akt1 and 2 isoforms have been elucidated in oral cancer with in-depth mechanistic analysis
Literature	GENERIF_SUMMARY	The results suggested that FoxM1 is critical downstream gene of AKT and functions to overcome BRAF V600E-induced senescence.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with colorectal cancer angiogenesis.
Literature	GENERIF_SUMMARY	The inhibition of AKT attenuated EYA2-induced Bcl-2 upregulation. In conclusion, our data demonstrated that Eya2 was upregulated in prostate cancers. EYA2 promotes cell proliferation and invasion as well as cancer progression by regulating docetaxel sensitivity and mitochondrial membrane potential, possibly via the AKT/Bcl-2 axis.
Literature	GENERIF_SUMMARY	Results showed that the mRNA and protein expression levels of AKT1 were downregulated in the melanoma cell lines when miR429 was overexpressed according to qRT-PCR and western bolt, indicating MicroRNA-429 may directly target AKT1 in melanoma.
Literature	GENERIF_SUMMARY	targeting of the HER2-AKT1 cascade augments damage-induced cellular senescence and apoptosis, and enhances STING-mediated antiviral and antitumour immunity
Literature	GENERIF_SUMMARY	Knocking down Akt1 and Jak1 genes using siRNAs could increase levels of apoptosis.
Literature	GENERIF_SUMMARY	mechanistic investigations suggested that CKIP-1 sharply suppressed the activity of AKT by inhibiting the phosphorylation, markedly downregulated the phosphorylated GSK3beta at Ser9, and promoted beta-catenin degradation.
Literature	GENERIF_SUMMARY	Liraglutide inhibits the apoptosis of human nucleus pulposus cells induced by high glucose through PI3K/Akt/caspase-3 signaling pathway.
Literature	GENERIF_SUMMARY	we revealed that CPA4 downregulation may induce apoptosis and G1-S arrest by suppressing the protein kinase B/c-MYC pathway. These results suggest that CPA4 has an oncogenic effect on lung cancer

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		growth. Taken together, we identified a novel gene in lung cancer that might provide a basis for new therapeutic targets.
Literature	GENERIF_SUMMARY	Intracellular accumulation of phosphorylated AKT in the cumulus cells of corresponding cumulus-oocyte complexes may be considered as a marker of oocyte competence, leading to development of embryos of presumed good quality.
Literature	GENERIF_SUMMARY	MAT1 serves as a promoter in the lung metastasis of osteosarcoma through increasing AKT1 expression.
Literature	GENERIF_SUMMARY	Microgravity inhibits decidualization via decreasing Akt activity and FOXO3a expression in human endometrial stromal cells.
Literature	GENERIF_SUMMARY	Depletion of PIK3CD attenuates vitreous-induced activation of AKT and MDM2 and decrease of p53.
Literature	GENERIF_SUMMARY	study provides evidence that Sec3 is involved in TGF-beta induced cell migration and epithelial-mesenchymal transition processes, presumably through the regulation of PI3K/Akt signaling activation in A549 cancer cells
Literature	GENERIF_SUMMARY	an AKT inhibitor (LY294002) only effected on MMP9 and p-AKT, not on c-Met. These data collectively indicated that calycosin possibility targeting on c-Met and exert an anti-tumor role via MMP9 and AKT.
Literature	GENERIF_SUMMARY	PTEN loss regulates alveolar epithelial cell senescence in pulmonary fibrosis depending on Akt activation.
Literature	GENERIF_SUMMARY	AKT1 internal tandem duplications and point mutations are the genetic hallmarks of sclerosing pneumocytoma.
Literature	GENERIF_SUMMARY	Data indicate a mechanism of USP46 regulation in renal cell carcinoma (RCC), and that USP46 is a tumor suppressor in RCC via proto-oncogene protein Akt (AKT) signaling pathway inactivation.
Literature	GENERIF_SUMMARY	Downregulation of PTEN and phosphorylated AKT is associated with refractory acute myeloid leukemia (AML) and resistance to idarubicin and cytarabine treatment.
Literature	GENERIF_SUMMARY	GL-V9 exerts anti-T cell malignancies effects via promoting lysosome-dependent AKT1 degradation and activating AKT1/FOXO3A/BIM axis.

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Literature	GENERIF_SUMMARY	DJ-1 may play a role in protecting RRPs from high glucose induced-oxidative injury. DJ-1 might improve mitochondrial function, inhibit ROS production and enhance antioxidant capacity to reduce apoptosis of retinal pericytes through the PI3K/AKT/mTOR signaling pathway which may be related to early pathogenesis of diabetic retinopathy.
Literature	GENERIF_SUMMARY	NUSAP1 knockdown inhibits cell growth and metastasis of non-small-cell lung cancer through regulating BTG2/PI3K/Akt signaling.
Literature	GENERIF_SUMMARY	CPA6 promotes the proliferation and migration of hepatocellular carcinoma by up-regulating AKT signaling pathway.
Literature	GENERIF_SUMMARY	Differential localization of PD-L1 and Akt-1 involvement in radioresistant and radiosensitive cell lines of head and neck squamous cell carcinoma.
Literature	GENERIF_SUMMARY	No Association Between AKT1 Polymorphisms and Methamphetamine Addiction in Iranian Population.
Literature	GENERIF_SUMMARY	FKBP4 connects mTORC2 and PI3K to activate the PDK1/Akt-dependent cell proliferation signaling in breast cancer.
Literature	GENERIF_SUMMARY	PRMT5 Promotes Human Lung Cancer Cell Apoptosis via Akt/Gsk3beta Signaling Induced by Resveratrol.
Literature	GENERIF_SUMMARY	Akt1 and Akt2 Isoforms Play Distinct Roles in Regulating the Development of Inflammation and Fibrosis Associated with Alcoholic Liver Disease.
Literature	GENERIF_SUMMARY	Integrative transcriptomic and proteomic analysis reveals CD9/ITGA4/PI3K-Akt axis mediates trabecular meshwork cell apoptosis in human glaucoma.
Literature	GENERIF_SUMMARY	A computational approach for investigating the mutational landscape of RAC-alpha serine/threonine-protein kinase (AKT1) and screening inhibitors against the oncogenic E17K mutation causing breast cancer.
Literature	GENERIF_SUMMARY	these results indicated a prooncogenic role of DLEU1 in the progression and development of renal cell carcinoma via modulation of the Akt pathway and EMT phenotype.
Literature	GENERIF_SUMMARY	AKT, which mediates Ser432-USP14 phosphorylation, was required for IRIF formation by USP14. Similar to USP14 block, AKT inhibition rescued the activity of NHEJ-DDR proteins in autophagy- and PTEN-

Main Category	Category	Terms
		deficient cells. These findings reveal a novel negative PTEN/Akt-dependent regulation of NHEJ by
Literature	GENERIF_SUMMARY	PTEN and AKT1 Variations in Childhood T-Cell Acute Lymphoblastic Leukemia
Literature	GENERIF_SUMMARY	Knockdown of Ski decreases osteosarcoma cell proliferation and migration by suppressing the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	High AKT1 expression is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	Tannic acid and vitamin E loaded PLGA nanoparticles ameliorate hepatic injury in a chronic alcoholic liver damage model via EGFR-AKT-STAT3 pathway.
Literature	GENERIF_SUMMARY	Mechanistic analysis of the redox regulation of Akt identified two cysteine residues in the pleckstrin homology domain (C60 and C77) to be reversibly oxidized.
Literature	GENERIF_SUMMARY	Leishmania aethiopica cell-to-cell spreading involves caspase-3, AkT, and NF-kappaB but not PKC-delta activation and involves uptake of LAMP-1-positive bodies containing parasites.
Literature	GENERIF_SUMMARY	LINC01518 knockdown inhibited tumorigenicity in ESCC cells by suppression of PIK3CA/Akt pathway through upregulating miR-1-3p
Literature	GENERIF_SUMMARY	AKT1 and genetic vulnerability to bipolar disorder.
Literature	GENERIF_SUMMARY	Bladder cancer cellsecreted exosomal miR21 activates the PI3K/AKT pathway in macrophages to promote cancer progression.
Literature	GENERIF_SUMMARY	this study demonstrated that SSRP1 silencing influenced the proliferation and apoptosis of colorectal cancer cells via the AKT signaling pathway.
Literature	GENERIF_SUMMARY	Long noncoding RNA FOXD2-AS1 promoted hepatocellular carcinoma development by regulation miR-185/AKT axis.
Literature	GENERIF_SUMMARY	Inhibition of NHEJ and HR using DNA-PK and Rad51 antagonists resulted in stronger radiosensitivity of AKT1 and AKT2 knockout cells versus wild type cells.
Literature	GENERIF_SUMMARY	Relationship between polycomb-group protein BMI-1 and phosphatases regulating AKT phosphorylation level in endometrial cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Melanoma migration is promoted by prion protein via Akt-hsp27 signaling axis.
Literature	GENERIF_SUMMARY	A novel miRNA located in the GATA4 gene regulates the expression of IGF-1R and AKT1/2 genes and controls cell proliferation.
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	The GATOR2-mTORC2 axis mediates Sestrin2-induced AKT Ser/Thr kinase activation.
Literature	GENERIF_SUMMARY	SRYrelated HMG box2 role in anaplastic thyroid cancer aggressiveness is related to the fibronectin 1 and PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	The p-4E-BP1 expression, especially in Combined hepatocellular carcinoma (HCC) and cholangiocarcinoma (CC) (cHCC-CC) cases with a cholangiocarcinoma (CC) component, was a notable Akt/mTOR pathway-related factor associated with poor prognosis.
Literature	GENERIF_SUMMARY	MAGI1-IT1 stimulates proliferation in non-small cell lung cancer by upregulating AKT1 as a ceRNA.
Literature	GENERIF_SUMMARY	overexpression of (phosphorylated)p-Akt, p-mTOR and p-eIF4E proteins is associated with metastasis and poor prognosis of non-small cell lung cancer patients.
Literature	GENERIF_SUMMARY	Sulfhydration of total or phosphorylated AKT inhibits its interaction with GSK3beta and subsequently down-regulates the phosphorylation and inactivation of GSK3beta.
Literature	GENERIF_SUMMARY	Overexpression of tripartite motif containing 26 inhibits non-small cell lung cancer cell growth by suppressing PI3K/AKT signaling.
Literature	GENERIF_SUMMARY	Putative functional variants of PI3K/AKT/mTOR pathway are associated with knee osteoarthritis susceptibility.
Literature	GENERIF_SUMMARY	FCX-146, a potent allosteric inhibitor of Akt kinase in cancer cells
Literature	GENERIF_SUMMARY	A Rapid Robust Method for Subgrouping Non-NF2 Meningiomas According to Genotype and Detection of Lower Levels of M2 Macrophages in AKT1 E17K Mutated Tumours.
Literature	GENERIF_SUMMARY	Amino Acids Regulate Glycolipid Metabolism and Alter Intestinal Microbial Composition.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Phosphorylation-Dependent Differences in CXCR4-LASP1-AKT1 Interaction between Breast Cancer and Chronic Myeloid Leukemia.
Literature	GENERIF_SUMMARY	Analysis of variants of AKT1 in schizophrenia multiplex families.
Literature	GENERIF_SUMMARY	Juvenile papillomatosis of the breast (Swiss cheese disease) has frequent associations with PIK3CA and/or AKT1 mutations.
Literature	GENERIF_SUMMARY	Sin1 promotes proliferation and invasion of prostate cancer cells by modulating mTORC2-AKT and androgen receptor signaling cascades.
Literature	GENERIF_SUMMARY	Targeting of AKT improves treatment outcomes for solid tumors. (Review)
Literature	GENERIF_SUMMARY	AKT Regulates Mitotic Progression of Mammalian Cells by Phosphorylating MASTL, Leading to Protein Phosphatase 2A Inactivation.
Literature	GENERIF_SUMMARY	Overexpression of IFIT2 inhibits the proliferation of chronic myeloid leukemia cells by regulating the BCRABL/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	PTEN Loss and Cyclin A2 Upregulation Define a PI3K/AKT Pathway Activation in Helicobacter pylori-induced MALT and DLBCL Gastric Lymphoma With Features of MALT.
Literature	GENERIF_SUMMARY	Lactic acidosis induces resistance to the pan-Akt inhibitor uprosertib in colon cancer cells.
Literature	GENERIF_SUMMARY	MiR-433 Regulates Myocardial Ischemia Reperfusion Injury by Targeting NDRG4 Via the PI3K/Akt Pathway.
Literature	GENERIF_SUMMARY	MIIP inhibits malignant progression of hepatocellular carcinoma through regulating AKT.
Literature	GENERIF_SUMMARY	Diosmetin Induces Apoptosis by Downregulating AKT Phosphorylation via P53 Activation in Human Renal Carcinoma ACHN Cells.
Literature	GENERIF_SUMMARY	LHPP inhibits cell growth and migration and triggers autophagy in papillary thyroid cancer by regulating the AKT/AMPK/mTOR signaling pathway.
Literature	GENERIF_SUMMARY	BMAL1 Suppresses Proliferation, Migration, and Invasion of U87MG Cells by Downregulating Cyclin B1, Phospho-AKT, and Metalloproteinase-9.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Hexosamine pathway inhibition overcomes pancreatic cancer resistance to gemcitabine through unfolded protein response and EGFR-Akt pathway modulation.
Literature	GENERIF_SUMMARY	Altered expression of AKT1 and P38A in the colons of patients with Hirschsprung's disease.
Literature	GENERIF_SUMMARY	RUNX1-activated upregulation of lncRNA RNCR3 promotes cell proliferation, invasion, and suppresses apoptosis in colorectal cancer via miR-1301-3p/AKT1 axis in vitro and in vivo.
Literature	GENERIF_SUMMARY	ROS accumulation contributes to abamectin-induced apoptosis and autophagy via the inactivation of PI3K/AKT/mTOR pathway in TM3 Leydig cells.
Literature	GENERIF_SUMMARY	Gain of function mutant p53 protein activates AKT through the Rac1 signaling to promote tumorigenesis.
Literature	GENERIF_SUMMARY	Effects of the mTOR and AKT genes polymorphisms on systemic lupus erythematosus risk.
Literature	GENERIF_SUMMARY	Allelic heterogeneity of Proteus syndrome.
Literature	GENERIF_SUMMARY	Insulin receptor substrate 1 overexpression promotes survival of glioblastoma cells through AKT1 activation.
Literature	GENERIF_SUMMARY	The Key Role of Akt Protein Kinase in Metabolic-Inflammatory Pathways Cross-Talk
Literature	GENERIF_SUMMARY	Phosphorylation-dependent substrate selectivity of protein kinase B (AKT1).
Literature	GENERIF_SUMMARY	Do AKT1, COMT and FAAH influence reports of acute cannabis intoxication experiences in patients with first episode psychosis, controls and young adult cannabis users?
Literature	GENERIF_SUMMARY	Overexpression of long non-coding RNA ANRIL promotes post-ischaemic angiogenesis and improves cardiac functions by targeting Akt.
Literature	GENERIF_SUMMARY	AKT Ser/Thr kinase increases V-ATPase-dependent lysosomal acidification in response to amino acid starvation in mammalian cells.
Literature	GENERIF_SUMMARY	TRPC1 inhibits the proliferation and migration of estrogen receptor-positive Breast cancer and gives a better prognosis by inhibiting the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Activating AKT1 and PIK3CA Mutations in

Main Category	Category	Terms
		Metastatic Castration-Resistant Prostate Cancer.
Literature	GENERIF_SUMMARY	AKT drives sustained motility following MEK inhibition via promoting SNAIL and AXL in MDA-MB-231 LM2.
Literature	GENERIF_SUMMARY	FOXD3 inhibits cell proliferation, migration, and invasion in nasopharyngeal carcinoma through regulation of the PI3K-Akt pathway.
Literature	GENERIF_SUMMARY	CFTR promotes malignant glioma development via up-regulation of Akt/Bcl2-mediated anti-apoptosis pathway.
Literature	GENERIF_SUMMARY	USP53 promotes apoptosis and inhibits glycolysis in lung adenocarcinoma through FKBP51-AKT1 signaling.
Literature	GENERIF_SUMMARY	The effect of the AKT1 gene and cannabis use on cognitive performance in healthy subjects.
Literature	GENERIF_SUMMARY	Association of a genetic variant in the AKT gene locus and cardiovascular risk factors.
Literature	GENERIF_SUMMARY	Lysosome AKT Targeting in Metastatic Cancer.
Literature	GENERIF_SUMMARY	Human Cytomegalovirus Mediates Unique Monocyte-to-Macrophage Differentiation through the PI3K/SHIP1/Akt Signaling Network.
Literature	GENERIF_SUMMARY	MiRNA-620 promotes TGF-beta1-induced proliferation of airway smooth muscle cell through controlling PTEN/AKT signaling pathway.
Literature	GENERIF_SUMMARY	Heat shock protein 47 promotes tumor survival and therapy resistance by modulating AKT signaling via PHLPP1 in colorectal cancer.
Literature	GENERIF_SUMMARY	In this study, the association of AKT1 polymorphism with congenital heart disease susceptibility were researched and 3 SNPs (rs1130214, rs2494732, rs3803300) were selected. At the same time, the interaction of gene polymorphism and environmental factor was also analyzed to clarify the mechanism of CHD and provide basis for genetic intervention.
Literature	GENERIF_SUMMARY	Speedy/RINGO protein interacts with ERK/MAPK and PI3K/AKT pathways in SH-SY5Y neuroblastoma cells.
Literature	GENERIF_SUMMARY	MicroRNA-92b acts as an oncogene by targeting PTEN/AKT in NSCLC.
Literature	GENERIF_SUMMARY	Isoliquiritigenin inhibits TGF-beta1-induced

Main Category	Category	Terms
		fibrogenesis through activating autophagy via PI3K/AKT/mTOR pathway in MRC-5 cells.
Literature	GENERIF_SUMMARY	GLUT5 regulation by AKT1/3-miR-125b-5p downregulation induces migratory activity and drug resistance in TLR-modified colorectal cancer cells.
Literature	GENERIF_SUMMARY	Tetrahedral framework nucleic acids promote scarless healing of cutaneous wounds via the AKT-signaling pathway.
Literature	GENERIF_SUMMARY	Circular RNA HIPK3 regulates human lens epithelial cell dysfunction by targeting the miR-221-3p/PI3K/AKT pathway in age-related cataract.
Literature	GENERIF_SUMMARY	Shenmai Injection Supresses Glycolysis and Enhances Cisplatin Cytotoxicity in Cisplatin-Resistant A549/DDP Cells via the AKT-mTOR-c-Myc Signaling Pathway.
Literature	GENERIF_SUMMARY	Celecoxib combined with salirasib strongly inhibits pancreatic cancer cells in 2D and 3D cultures.
Literature	GENERIF_SUMMARY	Metabolic stress regulates genome-wide transcription in a PTEN-dependent manner.
Literature	GENERIF_SUMMARY	Light-emitting diode irradiation induces AKT/mTOR-mediated apoptosis in human pancreatic cancer cells and xenograft mouse model.
Literature	GENERIF_SUMMARY	PD-L1, PDK-1 and p-Akt are correlated in patients with papillary thyroid carcinoma.
Literature	GENERIF_SUMMARY	LncRNA EIF3J-AS1 enhanced esophageal cancer invasion via regulating AKT1 expression through sponging miR-373-3p.
Literature	GENERIF_SUMMARY	Accumulation of Asn450Tyr mutant myocilin in ER promotes apoptosis of human trabecular meshwork cells.
Literature	GENERIF_SUMMARY	AKT1 Mutations in Peripheral Bronchiolar Papilloma
Literature	GENERIF_SUMMARY	Is REDD1 a metabolic double agent? Lessons from physiology and pathology.
Literature	GENERIF_SUMMARY	Aberrant JmjC domain-containing protein 8 (JMJD8) expression promotes activation of AKT and tumor epithelial-mesenchymal transition.
Literature	GENERIF_SUMMARY	AKT signaling restrains tumor suppressive functions of FOXO transcription factors and GSK3 kinase in multiple myeloma.
Literature	GENERIF_SUMMARY	CircAKT1 acts as a sponge of miR-338-3p to

Main Category	Category	Terms
		facilitate clear cell renal cell carcinoma progression by up-regulating CAV1.
Literature	GENERIF_SUMMARY	Zingerone suppresses cell proliferation via inducing cellular apoptosis and inhibition of the PI3K/AKT/mTOR signaling pathway in human prostate cancer PC-3 cells.
Literature	GENERIF_SUMMARY	THRIL mediates endothelial progenitor cells autophagy via AKT pathway and FUS.
Literature	GENERIF_SUMMARY	AKT/GSK-3beta/beta-catenin signaling pathway participates in erythropoietin-promoted glioma proliferation.
Literature	GENERIF_SUMMARY	PTPN13 acts as a tumor suppressor in clear cell renal cell carcinoma by inactivating Akt signaling.
Literature	GENERIF_SUMMARY	Endogenous Cyclin D1 Promotes the Rate of Onset and Magnitude of Mitogenic Signaling via Akt1 Ser473 Phosphorylation.
Literature	GENERIF_SUMMARY	BDNF/TrkB/Akt Signaling Pathway Epithelial Odontogenic Tumors and Keratocyst
Literature	GENERIF_SUMMARY	Inactivation of AKT/NFkappaB signaling by eurycomalactone decreases human NSCLC cell viability and improves the chemosensitivity to cisplatin.
Literature	GENERIF_SUMMARY	RP11-462C24.1 suppresses proliferation and invasion of colorectal carcinoma cells by regulating HSP70 through PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	Expression of HGF, pMet, and pAkt is related to benefit of radiotherapy after breast-conserving surgery
Literature	GENERIF_SUMMARY	Evaluation of B-cell intracellular signaling by monitoring the PI3K-Akt axis in patients with common variable immunodeficiency and activated phosphoinositide 3-kinase delta syndrome.
Literature	GENERIF_SUMMARY	Long noncoding RNA PTCSC1 drives esophageal squamous cell carcinoma progression through activating Akt signaling.
Literature	GENERIF_SUMMARY	De novo induction of lineage plasticity from human prostate luminal epithelial cells by activated AKT1 and c-Myc.
Literature	GENERIF_SUMMARY	Inhibitory Effect of a Human MicroRNA, miR-6133-5p, on the Fibrotic Activity of Hepatic Stellate Cells in Culture.
Literature	GENERIF_SUMMARY	Mutant SF3B1 promotes AKT- and NF-kappaB-driven mammary tumorigenesis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Expression of Akt1 and p-Akt1 in peripheral T cell subsets of multiple sclerosis patients.
Literature	GENERIF_SUMMARY	Phenotypic and genetic spectrum of isolated macrodactyly
Literature	GENERIF_SUMMARY	Synthesis, In Silico and In Vitro Assessment of New Quinazolinones as Anticancer Agents via Potential AKT Inhibition.
Literature	GENERIF_SUMMARY	LncRNA INHBA-AS1 promotes colorectal cancer cell proliferation by sponging miR-422a to increase AKT1 axis.
Literature	GENERIF_SUMMARY	Effects of the MTOR and AKT1 genes polymorphisms on papillary thyroid cancer development.
Literature	GENERIF_SUMMARY	SP3 is associated with migration, invasion, and Akt/PKB signalling in MDA-MB-231 breast cancer cells.
Literature	GENERIF_SUMMARY	Guanine nucleotide binding protein like-1 (GNL1) promotes cancer cell proliferation and survival through AKT/p21 (CIP1) signaling cascade.
Literature	GENERIF_SUMMARY	KIF3C Promotes Proliferation, Migration, and Invasion of Glioma Cells by Activating the PI3K/AKT Pathway and Inducing EMT.
Literature	GENERIF_SUMMARY	Loss of STAT5A promotes glucose metabolism and tumor growth through miRNA-23a-AKT signaling in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	A New Twist in Protein Kinase B/Akt Signaling
Literature	GENERIF_SUMMARY	Intracellular and extracellular S100A9 trigger epithelial-mesenchymal transition and promote the invasive phenotype of pituitary adenoma through activation of AKT1.
Literature	GENERIF_SUMMARY	Eldecalcitol induces apoptosis and autophagy in human osteosarcoma MG-63 cells by accumulating ROS to suppress the PI3K/Akt/mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Metformin induces apoptosis and inhibits migration by activating the AMPK/p53 axis and suppressing PI3K/AKT signaling in human cervical cancer cells.
Literature	GENERIF_SUMMARY	Scutellarin regulates osteoarthritis in vitro by inhibiting the PI3K/AKT/mTOR signaling pathway.
Literature	GENERIF_SUMMARY	lncRNA EZRAS1 knockdown represses proliferation, migration and invasion of cSCC via the PI3K/AKT signaling pathway.

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Literature	GENERIF_SUMMARY	AKT1 Polymorphism (rs10138227) and Risk of Colorectal Cancer in Moroccan Population
Literature	GENERIF_SUMMARY	Alternative AKT2 splicing produces protein lacking the hydrophobic motif regulatory region.
Literature	GENERIF_SUMMARY	Implication of integrins alpha3beta1 and alpha5beta1 in invasion and anoikis of SK-Mel-147 human melanoma cells
Literature	GENERIF_SUMMARY	Inhibition of AKT-Signaling Sensitizes Soft Tissue Sarcomas (STS) and Gastrointestinal Stromal Tumors (GIST) to Doxorubicin via Targeting of Homology-Mediated DNA Repair.
Literature	GENERIF_SUMMARY	The Role of the PI3K/AKT/mTOR Signalling Pathway in Male Reproduction.
Literature	GENERIF_SUMMARY	MicroRNA-139 inhibits pancreatic-cancer carcinogenesis by suppressing RalB via the Ral/RAC/PI3K pathway.
Literature	GENERIF_SUMMARY	AKT1 E17K Inhibits Cancer Cell Migration by Abrogating beta-Catenin Signaling.
Literature	GENERIF_SUMMARY	EGFR-rich extracellular vesicles derived from highly metastatic nasopharyngeal carcinoma cells accelerate tumour metastasis through PI3K/AKT pathway-suppressed ROS.
Literature	GENERIF_SUMMARY	CAMTA1, a novel antitumor gene, regulates proliferation and the cell cycle in glioma by inhibiting AKT phosphorylation.
Literature	GENERIF_SUMMARY	Upregulation of PI3K/AKT/PTEN pathway is correlated with glucose and glutamine metabolic dysfunction during tamoxifen resistance development in MCF-7 cells.
Literature	GENERIF_SUMMARY	Vicenin-2 inhibits the Helicobacterium pylori infection associated gastric carcinogenic events through modulation of PI3K/AKT and Nrf2 signaling in GES-1 cells.
Literature	GENERIF_SUMMARY	Purpurin binding interacts with LHPP protein that inhibits PI3K/AKT phosphorylation and induces apoptosis in colon cancer cells HCT-116.
Literature	GENERIF_SUMMARY	Knockout of Akt1/2 suppresses the metastasis of human prostate cancer cells CWR22rv1 in vitro and in vivo.
Literature	GENERIF_SUMMARY	Inhibition of USP1 induces apoptosis via ID1/AKT pathway in B-cell acute lymphoblastic leukemia cells.
Literature	GENERIF_SUMMARY	Proteus syndrome caused by novel somatic

Main Category	Category	Terms
		AKT1 duplication.
Literature	GENERIF_SUMMARY	DNA damage promotes microtubule dynamics through a DNA-PK-AKT axis for enhanced repair.
Literature	GENERIF_SUMMARY	Akt-Fas to Quell Aberrant T Cell Differentiation and Apoptosis in Covid-19.
Literature	GENERIF_SUMMARY	Follicular Lymphoma-associated BTK Mutations are Inactivating Resulting in Augmented AKT Activation.
Literature	GENERIF_SUMMARY	Ionizing Radiation-Induced Extracellular Vesicle Release Promotes AKT-Associated Survival Response in SH-SY5Y Neuroblastoma Cells.
Literature	GENERIF_SUMMARY	Downregulation of microRNA253p inhibits the proliferation and promotes the apoptosis of multiple myeloma cells via targeting the PTEN/PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	KNSTRN promotes tumorigenesis and gemcitabine resistance by activating AKT in bladder cancer.
Literature	GENERIF_SUMMARY	Cartilage endplate stem cells inhibit intervertebral disc degeneration by releasing exosomes to nucleus pulposus cells to activate Akt/autophagy.
Literature	GENERIF_SUMMARY	A disorder-related variant (E420K) of a PP2A-regulatory subunit (PPP2R5D) causes constitutively active AKT-mTOR signaling and uncoordinated cell growth.
Literature	GENERIF_SUMMARY	ROR1-AS1 knockdown inhibits growth and invasion and promotes apoptosis in NSCLC cells by suppression of the PI3K/Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	SCARA5 suppresses the proliferation and migration, and promotes the apoptosis of human retinoblastoma cells by inhibiting the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Tomentosin inhibits cell proliferation and induces apoptosis in MOLT-4 leukemia cancer cells through the inhibition of mTOR/PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	A growth-factor-activated lysosomal K(+) channel regulates Parkinson's pathology.
Literature	GENERIF_SUMMARY	Low Dose of Trichostatin A Improves Radiation Resistance by Activating Akt/Nrf2-Dependent Antioxidation Pathway in Cancer Cells.
Literature	GENERIF_SUMMARY	Regulation of valproic acid induced EMT by

Main Category	Category	Terms
		AKT/GSK3beta/beta-catenin signaling pathway in triple negative breast cancer.
Literature	GENERIF_SUMMARY	Active Akt signaling triggers CLL toward Richter transformation via overactivation of Notch1.
Literature	GENERIF_SUMMARY	AKT1-CREB stimulation of PDGFRalpha expression is pivotal for PTEN deficient tumor development.
Literature	GENERIF_SUMMARY	Leptin activates Akt in oesophageal cancer cells via multiple atorvastatin-sensitive small GTPases.
Literature	GENERIF_SUMMARY	MAPK4 promotes prostate cancer by concerted activation of androgen receptor and AKT.
Literature	GENERIF_SUMMARY	2,6-DMBQ suppresses cell proliferation and migration via inhibiting mTOR/AKT and p38 MAPK signaling pathways in NSCLC cells.
Literature	GENERIF_SUMMARY	Akt-independent effects of triciribine on ACE2 expression in human lung epithelial cells
Literature	GENERIF_SUMMARY	Silencing of lncRNA LBX2-AS1 suppresses glioma cell proliferation and metastasis through the Akt/GSK3beta pathway in vitro.
Literature	GENERIF_SUMMARY	MiR-335-5p inhibits proliferation of Huh-7 liver cancer cells via targeting the Oct4/Akt pathway.
Literature	GENERIF_SUMMARY	AKT-mediated regulation of chromatin ubiquitylation and tumorigenesis through Mel18 phosphorylation.
Literature	GENERIF_SUMMARY	PRKAR1B-AS2 Long Noncoding RNA Promotes Tumorigenesis, Survival, and Chemoresistance via the PI3K/AKT/mTOR Pathway.
Literature	GENERIF_SUMMARY	Targeting Akt in Hepatocellular Carcinoma and Its Tumor Microenvironment.
Literature	GENERIF_SUMMARY	Molecular Biomarkers for Contemporary Therapies in Hormone Receptor-Positive Breast Cancer.
Literature	GENERIF_SUMMARY	Cell specific tumor suppressor effect of Hsa-miR-1226-3p through downregulation of HER2, PIK3R2, and AKT1 genes.
Literature	GENERIF_SUMMARY	PTEN/AKT upregulation of TMSB10 contributes to lung cancer cell growth and predicts poor survival of the patients.
Literature	GENERIF_SUMMARY	Hypoxia-induced CNPY2 upregulation promotes glycolysis in cervical cancer through activation of AKT pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Phyllanthin inhibits MOLT-4 leukemic cancer cell growth and induces apoptosis through the inhibition of AKT and JNK signaling pathway.
Literature	GENERIF_SUMMARY	Salivary Mucinous Adenocarcinoma Is a Histologically Diverse Single Entity With Recurrent AKT1 E17K Mutations
Literature	GENERIF_SUMMARY	Focal Adhesion Kinase Inhibitor Inhibits the Oxidative Damage Induced by Central Venous Catheter via Abolishing Focal Adhesion Kinase-Protein Kinase B Pathway Activation.
Literature	GENERIF_SUMMARY	Primary cilia and the reciprocal activation of AKT and SMAD2/3 regulate stretch-induced autophagy in trabecular meshwork cells.
Literature	GENERIF_SUMMARY	FAM84B acts as a tumor promoter in human glioma via affecting the Akt/GSK-3beta/beta-catenin pathway.
Literature	GENERIF_SUMMARY	Syringic acid regulates suppression of the STAT3/JNK/AKT pathway via inhibition of human ovarian teratoma cancer cell (PA-1) growth-in vitro study.
Literature	GENERIF_SUMMARY	SH3BP4 promotes neuropilin-1 and alpha5-integrin endocytosis and is inhibited by Akt.
Literature	GENERIF_SUMMARY	Therapeutic Potential of RTA 404 in Human Brain Malignant Glioma Cell Lines via Cell Cycle Arrest via p21/AKT Signaling.
Literature	GENERIF_SUMMARY	CCL26 is upregulated by nab-paclitaxel in pancreatic cancer-associated fibroblasts and promotes PDAC invasiveness through activation of the PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	Hypoxia-induced cofilin 1 promotes hepatocellular carcinoma progression by regulating the PLD1/AKT pathway.
Literature	GENERIF_SUMMARY	PI3Kalpha inhibitor impairs AKT phosphorylation and synergizes with novel angiogenesis inhibitor AL3810 in human hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	FoxO1-GAB1 axis regulates homing capacity and tonic AKT activity in chronic lymphocytic leukemia.
Literature	GENERIF_SUMMARY	Cell competition between anaplastic thyroid cancer and normal thyroid follicular cells exerts reciprocal stress response defining tumor suppressive effects of normal epithelial tissue.
Literature	GENERIF_SUMMARY	Extracellular vesicles-derived microRNA-222 promotes immune escape via interacting with ATF3 to regulate AKT1 transcription in

Main Category	Category	Terms
		colorectal cancer.
Literature	GENERIF_SUMMARY	Anti-DLBCL efficacy of DCZ0825 in vitro and in vivo
Literature	GENERIF_SUMMARY	Role of AMPK and Akt in triple negative breast cancer lung colonization.
Literature	GENERIF_SUMMARY	IRS4 promotes the progression of non-small cell lung cancer and confers resistance to EGFR-TKI through the activation of PI3K/Akt and Ras-MAPK pathways.
Literature	GENERIF_SUMMARY	Oncogenic SNORD12B activates the AKT-mTOR-4EBP1 signaling in esophageal squamous cell carcinoma via nucleus partitioning of PP-1alpha.
Literature	GENERIF_SUMMARY	Spectrum of PIK3CA/AKT mutations across molecular subtypes of triple-negative breast cancer.
Literature	GENERIF_SUMMARY	Dasatinib inhibits proliferation of liver cancer cells, but activation of Akt/mTOR compromises dasatinib as a cancer drug.
Literature	GENERIF_SUMMARY	A novel lncRNA LOLA1 may predict malignant progression and promote migration, invasion, and EMT of oral leukoplakia via the AKT/GSK-3beta pathway.
Literature	GENERIF_SUMMARY	SGK1 mutations in DLBCL generate hyperstable protein neoisoforms that promote AKT independence.
Literature	GENERIF_SUMMARY	ARHGAP25 Inhibits Pancreatic Adenocarcinoma Growth by Suppressing Glycolysis via AKT/mTOR Pathway.
Literature	GENERIF_SUMMARY	GTSE1 Facilitates the Malignant Phenotype of Lung Cancer Cells via Activating AKT/mTOR Signaling.
Literature	GENERIF_SUMMARY	An AKT/PRMT5/SREBP1 axis in lung adenocarcinoma regulates de novo lipogenesis and tumor growth.
Literature	GENERIF_SUMMARY	LncRNA RUSC1-AS1 promotes osteosarcoma progression through regulating the miR-340-5p and PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	delta-Catenin Participates in EGF/AKT/p21(Waf) Signaling and Induces Prostate Cancer Cell Proliferation and Invasion.
Literature	GENERIF_SUMMARY	Patient with Lobular Carcinoma of the Breast and Activating AKT1 E17K Variant.
Literature	GENERIF_SUMMARY	Collective ERK/Akt activity waves orchestrate epithelial homeostasis by driving apoptosis-induced survival.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	PRMT5-mediated arginine methylation activates AKT kinase to govern tumorigenesis.
Literature	GENERIF_SUMMARY	Combined inhibition of DDR1 and CDK4/6 induces synergistic effects in ER-positive, HER2-negative breast cancer with PIK3CA/AKT1 mutations.
Literature	GENERIF_SUMMARY	Targeted PI3K/AKT-hyperactivation induces cell death in chronic lymphocytic leukemia.
Literature	GENERIF_SUMMARY	M2 macrophages, but not M1 macrophages, support megakaryopoiesis by upregulating PI3K-AKT pathway activity.
Literature	GENERIF_SUMMARY	miR-424 induces apoptosis in glioblastoma cells and targets AKT1 and RAF1 oncogenes from the ERBB signaling pathway.
Literature	GENERIF_SUMMARY	Akt scaffold proteins
Literature	GENERIF_SUMMARY	LINC00665 promotes the viability, migration and invasion of T cell acute lymphoblastic leukemia cells by targeting miR-101 via modulating PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	AKT1 and PTEN show the highest affinities among phosphoinositide binding proteins for the second messengers PtdIns(3,4,5)P3 and PtdIns(3,4)P2.
Literature	GENERIF_SUMMARY	Glioma stem cell-derived exosomal miR-944 reduces glioma growth and angiogenesis by inhibiting AKT/ERK signaling.
Literature	GENERIF_SUMMARY	MicroRNA-373-3p inhibits the growth of cervical cancer by targeting AKT1 both in vitro and in vivo.
Literature	GENERIF_SUMMARY	CLIC1 promotes the progression of cervical cancer through the PTEN/PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	PTEN suppresses tumorigenesis by directly dephosphorylating Akt.
Literature	GENERIF_SUMMARY	Plasma Cell-Free DNA Profiling of PTEN-PI3K-AKT Pathway Aberrations in Metastatic Castration-Resistant Prostate Cancer.
Literature	GENERIF_SUMMARY	Implication of integrin alpha2beta1 in senescence of SK-Mel-147 human melanoma cells.
Literature	GENERIF_SUMMARY	Copper Promotes Tumorigenesis by Activating the PDK1-AKT Oncogenic Pathway in a Copper Transporter 1 Dependent Manner.
Literature	GENERIF_SUMMARY	RRM2 Regulates Sensitivity to Sunitinib and PD-1 Blockade in Renal Cancer by

Main Category	Category	Terms
		Stabilizing ANXA1 and Activating the AKT Pathway.
Literature	GENERIF_SUMMARY	Cx43 deficiency confers EMT-mediated tamoxifen resistance to breast cancer via c-Src/PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	FBXL16 modulates the proliferation and autophagy in breast cancer cells via activating SRC-3-AKT signaling pathway.
Literature	GENERIF_SUMMARY	CCDC65 as a new potential tumor suppressor induced by metformin inhibits activation of AKT1 via ubiquitination of ENO1 in gastric cancer.
Literature	GENERIF_SUMMARY	Exosomal microRNA-15a from ACHN cells aggravates clear cell renal cell carcinoma via the BTG2/PI3K/AKT axis.
Literature	GENERIF_SUMMARY	Long non-coding RNA ABHD11-AS1 promotes colorectal cancer progression and invasion through targeting the integrin subunit alpha 5/focal adhesion kinase/phosphoinositide 3 kinase/Akt signaling pathway.
Literature	GENERIF_SUMMARY	Regulation of the voltage-dependent sodium channel NaV1.1 by AKT1.
Literature	GENERIF_SUMMARY	MiR-149-3p can improve the osteogenic differentiation of human adipose-derived stem cells via targeting AKT1.
Literature	GENERIF_SUMMARY	The PI3K/Akt signaling axis in Alzheimer's disease
Literature	GENERIF_SUMMARY	Identification of a novel inhibitor of liver cancer cell invasion and proliferation through regulation of Akt and Twist1.
Literature	GENERIF_SUMMARY	The enhancement of Tetrandrine to gemcitabine-resistant PANC-1 cytochemical sensitivity involves the promotion of PI3K/Akt/mTOR-mediated apoptosis and AMPK-regulated autophagy.
Literature	GENERIF_SUMMARY	Tumor suppressive effect of scavenger receptor class A member 5 overexpression in colorectal cancer by regulating PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	PI3K/PTEN/AKT Signaling Pathways in Germ Cell Development and Their Involvement in Germ Cell Tumors and Ovarian Dysfunctions.
Literature	GENERIF_SUMMARY	A PLCB1-PI3K-AKT Signaling Axis Activates EMT to Promote Cholangiocarcinoma Progression.
Literature	GENERIF_SUMMARY	Progesterone affects clinic oocyte yields by

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		coordinating with follicle stimulating hormone via PI3K/AKT and MAPK pathways.
Literature	GENERIF_SUMMARY	Exosomal lncRNA FOXD3-AS1 upregulates ELAVL1 expression and activates PI3K/Akt pathway to enhance lung cancer cell proliferation, invasion, and 5-fluorouracil resistance.
Literature	GENERIF_SUMMARY	A role for AKT1 in nonsense-mediated mRNA decay.
Literature	GENERIF_SUMMARY	Neuroprotective Properties of Kempferol Derivatives from Maesa membranacea against Oxidative Stress-Induced Cell Damage
Literature	GENERIF_SUMMARY	Adenylate Kinase 4-A Key Regulator of Proliferation and Metabolic Shift in Human Pulmonary Arterial Smooth Muscle Cells via Akt and HIF-1alpha Signaling Pathways.
Literature	GENERIF_SUMMARY	Hotspot mutation profiles of AKT1 in Asian women with breast and endometrial cancers.
Literature	GENERIF_SUMMARY	PD-L1 Status in Gastric Cancers, Association with the Transcriptional, Growth Factors, AKT/mTOR Components Change, and Autophagy Initiation.
Literature	GENERIF_SUMMARY	The sialidase NEU1 directly interacts with the juxtamembranous segment of the cytoplasmic domain of mucin-1 to inhibit downstream PI3K-Akt signaling.
Literature	GENERIF_SUMMARY	Circ_0009910 sponges miR-491-5p to promote acute myeloid leukemia progression through modulating B4GALT5 expression and PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	CBX2 Induces Glioma Cell Proliferation and Invasion Through the Akt/PI3K Pathway.
Literature	GENERIF_SUMMARY	Akt-mediated Ephexin1-Ras interaction promotes oncogenic Ras signaling and colorectal and lung cancer cell proliferation.
Literature	GENERIF_SUMMARY	Heparanase-Induced Activation of AKT Stabilizes beta-Catenin and Modulates Wnt/beta-Catenin Signaling during Herpes Simplex Virus 1 Infection.
Literature	GENERIF_SUMMARY	Proteotoxic Stress as an Exploitable Vulnerability in Cells with Hyperactive AKT.
Literature	GENERIF_SUMMARY	Correlation of the AKT/mTOR signaling pathway with the clinicopathological features and prognosis of nasopharyngeal carcinoma.

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Literature	GENERIF_SUMMARY	Downregulation of Williams syndrome transcription factor (WSTF) suppresses glioblastoma cell growth and invasion by inhibiting PI3K/AKT signal pathway.
Literature	GENERIF_SUMMARY	Crosstalk between circRNAs and the PI3K/AKT signaling pathway in cancer progression.
Literature	GENERIF_SUMMARY	Substrate stiffening promotes VEGF-A functions via the PI3K/Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	Akt acts as a switch for GPCR transactivation of the TGF-beta receptor type 1.
Literature	GENERIF_SUMMARY	TRAM2 promotes the malignant progression of glioma through PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	Effects of Cynaroside on Cell Proliferation, Apoptosis, Migration and Invasion though the MET/AKT/mTOR Axis in Gastric Cancer.
Literature	GENERIF_SUMMARY	Fluctuations in AKT and PTEN Activity Are Linked by the E3 Ubiquitin Ligase cCBL.
Literature	GENERIF_SUMMARY	5-epi-Sinuleptolide from Soft Corals of the Genus Sinularia Exerts Cytotoxic Effects on Pancreatic Cancer Cell Lines via the Inhibition of JAK2/STAT3, AKT, and ERK Activity.
Literature	GENERIF_SUMMARY	MicroRNA-4317 suppresses the progression of hepatocellular carcinoma by targeting ZNF436-mediated PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	MiR-1246 regulates the PI3K/AKT signaling pathway by targeting PIK3AP1 and inhibits thyroid cancer cell proliferation and tumor growth.
Literature	GENERIF_SUMMARY	TMEM176B Regulates AKT/mTOR Signaling and Tumor Growth in Triple-Negative Breast Cancer.
Literature	GENERIF_SUMMARY	Canagliflozin Modulates Hypoxia-Induced Metastasis, Angiogenesis and Glycolysis by Decreasing HIF-1alpha Protein Synthesis via AKT/mTOR Pathway.
Literature	GENERIF_SUMMARY	Akt Signaling Pathway Is Activated in the Minor Salivary Glands of Patients with Primary Sjogren's Syndrome.
Literature	GENERIF_SUMMARY	Malignant growth of arsenic-transformed cells depends on activated Akt induced by reactive oxygen species.
Literature	GENERIF_SUMMARY	Mevalonate pathway orchestrates insulin signaling via RAB14 geranylgeranylation-

Main Category	Category	Terms
		mediated phosphorylation of AKT to regulate hepatic glucose metabolism.
Literature	GENERIF_SUMMARY	[Mechanism of ring finger protein 11 regulating Akt signaling pathway to promote osteogenic differentiation of bone marrow mesenchymal stem cells].
Literature	GENERIF_SUMMARY	HERV-W Envelope Triggers Abnormal Dopaminergic Neuron Process through DRD2/PP2A/AKT1/GSK3 for Schizophrenia Risk.
Literature	GENERIF_SUMMARY	2-Methoxyestradiol combined with ascorbic acid facilitates the apoptosis of chronic myeloid leukemia cells via the microRNA-223/Fms-like tyrosine kinase 3/phosphatidylinositol-3 kinase/protein kinase B axis.
Literature	GENERIF_SUMMARY	Supplementing cultured human myotubes with hibernating bear serum results in increased protein content by modulating Akt/FOXO3a signaling.
Literature	GENERIF_SUMMARY	Bak interacts with AKT and is involved in TNFalpha/CHX-induced apoptosis.
Literature	GENERIF_SUMMARY	Down-regulation of PBK inhibits proliferation of human endometrial stromal cells in thin endometrium.
Literature	GENERIF_SUMMARY	Protein Kinase B/Akt1 Phosphorylates Dysbindin-1A at Serine 10 to Regulate Neuronal Development.
Literature	GENERIF_SUMMARY	Loss of ASAP1 in the MMTV-PyMT model of luminal breast cancer activates AKT, accelerates tumorigenesis, and promotes metastasis.
Literature	GENERIF_SUMMARY	Calpain7 negatively regulates human endometrial stromal cell decidualization in EMs by promoting FoxO1 nuclear exclusion via hydrolyzing AKT1.
Literature	GENERIF_SUMMARY	Component Analysis and Anti-Colorectal Cancer Mechanism via AKT/mTOR Signalling Pathway of Sanghuangporus vaninii Extracts.
Literature	GENERIF_SUMMARY	AKT1 and PIK3CA activating mutations in Moroccan bladder cancer patients biopsies and matched urine.
Literature	GENERIF_SUMMARY	Rab22a promotes the proliferation, migration, and invasion of lung adenocarcinoma via up-regulating PI3K/Akt/mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Discoidin domain receptor 1 promotes lung adenocarcinoma migration via the AKT/

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		snail signaling axis.
Literature	GENERIF_SUMMARY	Endocytic trafficking of GAS6-AXL complexes is associated with sustained AKT activation.
Literature	GENERIF_SUMMARY	ARHGAP9 inhibits colorectal cancer cell proliferation, invasion and EMT via targeting PI3K/AKT/mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Porphyromonas gingivalis lipopolysaccharide affects the angiogenic function of endothelial progenitor cells via Akt/FoxO1 signaling.
Literature	GENERIF_SUMMARY	Combination of AKT1 and CDH1 mutations predicts primary resistance to immunotherapy in dMMR/MSI-H gastrointestinal cancer.
Literature	GENERIF_SUMMARY	Clinicopathological Significance of AKT1 and PLK1 Expression in Oral Squamous Cell Carcinoma.
Literature	GENERIF_SUMMARY	The Role of AKT in Soft Tissue Sarcoma
Literature	GENERIF_SUMMARY	AKT1 Transcriptomic Landscape in Breast Cancer Cells.
Literature	GENERIF_SUMMARY	Increased Phospho-AKT in Blood Cells from LRRK2 G2019S Mutation Carriers.
Literature	GENERIF_SUMMARY	Impact of AKT1 polymorphism on DNA damage, BTG2 expression, and risk of colorectal cancer development.
Literature	GENERIF_SUMMARY	Effect of AKT silence on malignant biological behavior of renal cell carcinoma cells.
Literature	GENERIF_SUMMARY	CDC42 governs normal oviduct multiciliogenesis through activating AKT to ensure timely embryo transport.
Literature	GENERIF_SUMMARY	Phospholipid Scramblase 4 (PLSCR4) Regulates Adipocyte Differentiation via PIP3-Mediated AKT Activation.
Literature	GENERIF_SUMMARY	Mechanically activated ion channel Piezo1 contributes to melanoma malignant progression through AKT/mTOR signaling.
Literature	GENERIF_SUMMARY	Hyperactive Akt1 Signaling Increases Tumor Progression and DNA Repair in Embryonal Rhabdomyosarcoma RD Line and Confers Susceptibility to Glycolysis and Mevalonate Pathway Inhibitors.
Literature	GENERIF_SUMMARY	Identification of NHLRC1 as a Novel AKT Activator from a Lung Cancer Epigenome-Wide Association Study (EWAS).

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Literature	GENERIF_SUMMARY	Non-genomic activation of the AKT-mTOR pathway by the mitochondrial stress response in thyroid cancer.
Literature	GENERIF_SUMMARY	Phosphatase and tensin homologue determine inflammatory status by differentially regulating the expression of Akt1 and Akt2 in macrophage alternative polarization of periodontitis.
Literature	GENERIF_SUMMARY	Reducing Akt2 in retinal pigment epithelial cells causes a compensatory increase in Akt1 and attenuates diabetic retinopathy.
Literature	GENERIF_SUMMARY	Hsa_circ_0000851 promotes PDK1/p-AKT-mediated cell proliferation and migration by regulating miR-1183 in triple-negative breast cancer.
Literature	GENERIF_SUMMARY	miR-451a suppresses papillary thyroid cancer cell proliferation and invasion and facilitates apoptosis through targeting DCBLD2 and AKT1.
Literature	GENERIF_SUMMARY	AKT supports the metabolic fitness of multiple myeloma cells by restricting FOXO activity.
Literature	GENERIF_SUMMARY	Silencing of lncRNA SNHG17 inhibits the tumorigenesis of epithelial ovarian cancer through regulation of miR-485-5p/AKT1 axis.
Literature	GENERIF_SUMMARY	Mechanism of Akt regulation of the expression of collagens and MMPs in conjunctivochalasis.
Literature	GENERIF_SUMMARY	Long non-coding RNA ncRuPAR regulates gastric cancer cell proliferation and apoptosis via phosphoinositide 3-kinase/protein kinase B signaling.
Literature	GENERIF_SUMMARY	Monocytes educated by cancer-associated fibroblasts secrete exosomal miR-181a to activate AKT signaling in breast cancer cells.
Literature	GENERIF_SUMMARY	JCAD promotes arterial thrombosis through PI3K/Akt modulation
Literature	GENERIF_SUMMARY	AKT/GSK-3beta/VEGF signaling is involved in P2RY2 activation-induced the proliferation and metastasis of gastric cancer.
Literature	GENERIF_SUMMARY	AKT1 phosphorylates RBM17 to promote Sox2 transcription by modulating alternative splicing of FOXM1 to enhance cancer stem cell properties in colorectal cancer cells.
Literature	GENERIF_SUMMARY	HMGB1 promotes Ox-LDL-induced endothelial cell damage by inhibiting PI3K/Akt signaling pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	TAGLN2 Promotes the Proliferation, Migration, Invasion, and EMT of Clear Cell Renal Cell Carcinoma Through the PI3K/Akt Signaling Pathway.
Literature	GENERIF_SUMMARY	PI3K-Akt signaling regulates BMP2-induced osteogenic differentiation of mesenchymal stem cells (MSCs)
Literature	GENERIF_SUMMARY	PLAC8 contributes to the malignant behaviors of cervical cancer cells by activating the SOX4-mediated AKT pathway.
Literature	GENERIF_SUMMARY	The deubiquitinase OTUD1 noncanonically suppresses Akt activation through its N-terminal intrinsically disordered region.
Literature	GENERIF_SUMMARY	TRIM15 forms a regulatory loop with the AKT/FOXO1 axis and LASP1 to modulate the sensitivity of HCC cells to TKIs.
Literature	GENERIF_SUMMARY	First Report of FARSA in the Regulation of Cell Cycle and Survival in Mantle Cell Lymphoma Cells via PI3K-AKT and FOXO1-RAG1 Axes.
Literature	GENERIF_SUMMARY	SKP2 Contributes to AKT Activation by Ubiquitination Degradation of PHLPP1, Impedes Autophagy, and Facilitates the Survival of Thyroid Carcinoma.
Literature	GENERIF_SUMMARY	rhMYDGF Alleviates I/R-induced Kidney Injury by Inhibiting Inflammation and Apoptosis via the Akt Pathway.
Literature	GENERIF_SUMMARY	MLL4 Regulates the Progression of Non-Small-Cell Lung Cancer by Regulating the PI3K/AKT/SOX2 Axis.
Literature	GENERIF_SUMMARY	Hypo-osmolarity induces apoptosis resistance via TRPV2-mediated AKT-Bcl-2 pathway.
Literature	GENERIF_SUMMARY	Low concentrations of tumor necrosis factor-alpha promote human periodontal ligament stem cells osteogenic differentiation by activation of autophagy via inhibition of AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	AR and PI3K/AKT in Prostate Cancer
Literature	GENERIF_SUMMARY	Mitochondrial Hydrogen Peroxide Activates PTEN and Inactivates Akt Leading to Autophagy Inhibition-Dependent Cell Death in Neuronal Models of Parkinson's Disease.
Literature	GENERIF_SUMMARY	NAT10 promotes osteogenic differentiation of periodontal ligament stem cells by regulating VEGFA-mediated PI3K/AKT signaling pathway through ac4C modification.

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Literature	GENERIF_SUMMARY	mTOR pathway candidate genes and physical activity interaction on breast cancer risk in black women from the women's circle of health study.
Literature	GENERIF_SUMMARY	STIM1-regulated exosomal EBV-LMP1 empowers endothelial cells with an aggressive phenotype by activating the Akt/ERK pathway in nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	PRMT5/FGFR3/AKT Signaling Axis Facilitates Lung Cancer Cell Metastasis.
Literature	GENERIF_SUMMARY	CDK12 Promotes the Proliferation, Migration, and Angiogenesis of Gastric Carcinoma via Activating the PI3K/AKT/mTOR Signaling Pathway.
Literature	GENERIF_SUMMARY	Correlated with better prognosis, CSTA inhibits metastasis of nasopharyngeal carcinoma cells via suppressing AKT signaling through promoting METTL3 degradation.
Literature	GENERIF_SUMMARY	LZTFL1 inhibits kidney tumor cell growth by destabilizing AKT through ZNRF1-mediated ubiquitin proteosome pathway.
Literature	GENERIF_SUMMARY	VEGF Stimulates Activation of ERK5 in the Absence of C-Terminal Phosphorylation Preventing Nuclear Localization and Facilitating AKT Activation in Endothelial Cells.
Literature	GENERIF_SUMMARY	Circ_0067997 boosted the growth while repressed the apoptosis of SGC-7901/DDP cells via repressing miR-615-5p/AKT1 pathway.
Literature	GENERIF_SUMMARY	AKT1 participates in ferroptosis vulnerability by driving autophagic degradation of FTH1 in cisplatin-resistant ovarian cancer.
Literature	GENERIF_SUMMARY	Association between the AKT1 single nucleotide polymorphism (rs2498786, rs2494752 and rs5811155) and microscopic polyangiitis risk in a Chinese population.
Literature	GENERIF_SUMMARY	Inflammation and DKK1-induced AKT activation contribute to endothelial dysfunction following NR2F2 loss.
Literature	GENERIF_SUMMARY	Downregulation of BUD31 Promotes Prostate Cancer Cell Proliferation and Migration via Activation of p-AKT and Vimentin In Vitro.
Literature	GENERIF_SUMMARY	[DTX2 overexpression promotes migration and invasion of colorectal cancer cells through the Notch2/Akt axis].
Literature	GENERIF_SUMMARY	lncNALT knockdown ameliorates

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		hypertensive retinopathy via PTEN/PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Knockdown of CENPN Inhibits Glucose Metabolism and Induces G1 Arrest in Esophageal Cancer Cells by Regulating PI3K/AKT Signaling Pathway.
Literature	GENERIF_SUMMARY	Ring finger protein 12 activates AKT signalling to promote the progression of liver cancer by interacting with EGFR.
Literature	GENERIF_SUMMARY	Long noncoding RNA FTX promotes epithelial-mesenchymal transition of epithelial ovarian cancer through modulating miR-7515/TPD52 and activating Met/Akt/mTOR.
Literature	GENERIF_SUMMARY	YTHDC2 Retards Cell Proliferation and Triggers Apoptosis in Papillary Thyroid Cancer by Regulating CYLD-Mediated Inactivation of Akt Signaling.
Literature	GENERIF_SUMMARY	AQP3-Dependent PI3K/Akt Modulation in Breast Cancer Cells.
Literature	GENERIF_SUMMARY	The activity and surface presence of organic cation/carnitine transporter OCTN2 (SLC22A5) in breast cancer cells depends on AKT kinase.
Literature	GENERIF_SUMMARY	YKL-40 promotes proliferation and invasion of HTR-8/SVneo cells by activating akt/MMP9 signalling in placenta accreta spectrum disorders.
Literature	GENERIF_SUMMARY	miR-433 Inhibits Glioblastoma Progression by Suppressing the PI3K/Akt Signaling Pathway Through Direct Targeting of ERBB4.
Literature	GENERIF_SUMMARY	Targeted hyperactivation of AKT through inhibition of ectopic expressed SHIP1 induces cell death in colon carcinoma cells and derived metastases.
Literature	GENERIF_SUMMARY	TLR4 inhibited autophagy by modulating PI3K/AKT/mTOR signaling pathway in Gastric cancer cell lines.
Literature	GENERIF_SUMMARY	RBFOX2 confers tumor growth by PI3K/AKT and MAPK signaling in gastric cancer.
Literature	GENERIF_SUMMARY	Upregulation of SEMP1 Contributes to Improving the Biological Functions of Trophoblast via the PI3K/AKT Pathway in Preeclampsia.
Literature	GENERIF_SUMMARY	Immunosensitivity mediated by downregulated AKT1-SKP2 induces anti-PD-1-associated thyroid immune injury.
Literature	GENERIF_SUMMARY	HSPA4 regulated glioma progression via

Main Category	Category	Terms
		activation of AKT signaling pathway.
Literature	GENERIF_SUMMARY	Interferon-gamma inducible protein 30 promotes the epithelial-mesenchymal transition-like phenotype and chemoresistance by activating EGFR/AKT/GSK3beta/beta-catenin pathway in glioma.
Literature	GENERIF_SUMMARY	IGF2BP3-EGFR-AKT axis promotes breast cancer MDA-MB-231 cell growth.
Literature	GENERIF_SUMMARY	Tumor-suppressive action of miR-30a-5p in lung adenocarcinoma correlates with ABL2 inhibition and PI3K/AKT pathway inactivation.
Literature	GENERIF_SUMMARY	Association of PI3K/AKT/mTOR pathway autophagy-related gene polymorphisms with pulmonary tuberculosis susceptibility in a Chinese population.
Literature	GENERIF_SUMMARY	Lipid-correlated alterations in the transcriptome are enriched in several specific pathways in the postmortem prefrontal cortex of Japanese patients with schizophrenia.
Literature	GENERIF_SUMMARY	The roles of stress-induced premature senescence and Akt/FoxO1 signaling in periapical lesions.
Literature	GENERIF_SUMMARY	Cooperative activation of PDK1 and AKT by MAPK4 enhances cancer growth and resistance to therapy.
Literature	GENERIF_SUMMARY	CircCDK1 blocking IGF2BP2-mediated m6A modification of CPPED1 promotes laryngeal squamous cell carcinoma metastasis via the PI3K/AKT signal pathway.
Literature	GENERIF_SUMMARY	Loss of RACK1 promotes glutamine addiction via activating AKT/mTOR/ASCT2 axis to facilitate tumor growth in gastric cancer.
Literature	GENERIF_SUMMARY	Knockdown of BUB1 inhibits tumor necrosis factor-alpha-induced proliferation and migration of rheumatoid arthritis synovial fibroblasts by regulating PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	In vitro siRNA-mediated GPX4 and AKT1 silencing in oxaliplatin resistance cancer cells induces ferroptosis and apoptosis.
Literature	GENERIF_SUMMARY	Cytochrome b561 regulates iron metabolism by activating the Akt/mTOR pathway to promote Breast Cancer Cells proliferation.
Literature	GENERIF_SUMMARY	STMN1 promotes cell malignancy and bortezomib resistance of multiple myeloma cell lines via PI3K/AKT signaling.

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Literature	GENERIF_SUMMARY	SEM1 promotes tumor progression of glioblastoma via activating the akt signaling pathway.
Literature	GENERIF_SUMMARY	Somatic mutation spectrum of a Chinese cohort of pediatrics with vascular malformations.
Literature	GENERIF_SUMMARY	FUNDC2, a mitochondrial outer membrane protein, mediates triple-negative breast cancer progression via the AKT/GSK3beta/GLI1 pathway.
Literature	GENERIF_SUMMARY	Spinster homolog 2 reduces malignancies of glioblastoma via PTEN/PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Role of PI3K/Akt/mTOR pathway-mediated macrophage autophagy in affecting the phenotype transformation of lung fibroblasts induced by silica dust exposure.
Literature	GENERIF_SUMMARY	Differential activation of AKT isoforms by growth factors in human myotubes.
Literature	GENERIF_SUMMARY	KIRREL promotes the proliferation of gastric cancer cells and angiogenesis through the PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	KLF4 suppresses the proliferation of perihilar cholangiocarcinoma by negatively regulating GDF15 and phosphorylating AKT.
Literature	GENERIF_SUMMARY	DUSP22 suppresses tumor progression by directly dephosphorylating AKT in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Extracellular vesicle-encapsulated microRNA-296-3p from cancer-associated fibroblasts promotes ovarian cancer development through regulation of the PTEN/AKT and SOCS6/STAT3 pathways.
Literature	GENERIF_SUMMARY	A multi-layered network model identifies Akt1 as a common modulator of neurodegeneration.
Literature	GENERIF_SUMMARY	Hyperactive Rac stimulates cannibalism of living target cells and enhances CAR-M-mediated cancer cell killing.
Literature	GENERIF_SUMMARY	KLK10 promotes the progression of KRAS mutant colorectal cancer via PAR1-PDK1-AKT signaling pathway.
Literature	GENERIF_SUMMARY	Polymorphisms in PI3K/AKT genes and gene-smoking interaction are associated with susceptibility to tuberculosis.
Literature	GENERIF_SUMMARY	Akt Is Controlled by Bag5 through a Monoubiquitination to Polyubiquitination Switch.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	SOD3 regulates FLT1 to affect bone metabolism by promoting osteogenesis and inhibiting adipogenesis through PI3K/AKT and MAPK pathways.
Literature	GENERIF_SUMMARY	IRS1 promotes thyroid cancer metastasis through EMT and PI3K/AKT pathways.
Literature	GENERIF_SUMMARY	FOXM1 transcriptional regulation of RacGAP1 activates the PI3K/AKT signaling pathway to promote the proliferation, migration, and invasion of cervical cancer cells.
Literature	GENERIF_SUMMARY	PAX2 is regulated by estrogen/progesterone through promoter methylation in endometrioid adenocarcinoma and has an important role in carcinogenesis via the AKT/mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Akt1 is involved in HCV release by promoting endoplasmic reticulum-to-endosome transition of infectious virions.
Literature	GENERIF_SUMMARY	ABHD6 suppresses colorectal cancer progression via AKT signaling pathway.
Literature	GENERIF_SUMMARY	EID3 inhibits the osteogenic differentiation of periodontal ligament stem cells and mediates the signal transduction of TAZ-EID3-AKT/MTOR/ERK.
Literature	GENERIF_SUMMARY	AKT1 interacts with DHX9 to Mitigate R Loop-Induced Replication Stress in Ovarian Cancer.
Literature	GENERIF_SUMMARY	Breast cancer malignancy is governed by regulation of the macroH2A2/TM4SF1 axis, the AKT/NF-kappaB pathway, and elevated MMP13 expression.
Literature	GENERIF_SUMMARY	Insulin-like growth factor 2 mRNA-binding protein 3 enhanced melanoma migration through regulation of AKT1 and RELA expression.
Literature	GENERIF_SUMMARY	Polo-like kinase 4 promotes tumorigenesis and glucose metabolism in glioma by activating AKT1 signaling.
Literature	GENERIF_SUMMARY	A gain of function mutation in AKT1 increases hexokinase 2 and diminishes oxidative stress in meningioma.
Literature	GENERIF_SUMMARY	EMX2 inhibits clear cell renal cell carcinoma progress via modulating Akt/FOXO3a pathway.
Literature	GENERIF_SUMMARY	The novel miR-873-5p-YWHAE-PI3K/AKT axis is involved in non-small cell lung cancer progression and chemoresistance by mediating autophagy.

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Literature	GENERIF_SUMMARY	All Three AKT Isoforms Can Upregulate Oxygen Metabolism and Lactate Production in Human Hepatocellular Carcinoma Cell Lines.
Literature	GENERIF_SUMMARY	GSG2 promotes thyroid cancer via stabilizing AURKB and activating AKT pathway.
Literature	GENERIF_SUMMARY	Enhanced lipid biosynthesis in oral squamous cell carcinoma cancer-associated fibroblasts contributes to tumor progression
Literature	GENERIF_SUMMARY	Differential prognostic values of the three AKT isoforms in acute myeloid leukemia.
Literature	GENERIF_SUMMARY	FNDC5 inhibits malignant growth of human cervical cancer cells via restraining PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	SUMOylation of annexin A6 retards cell migration and tumor growth by suppressing RHOU/AKT1-involved EMT in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Promoter hypomethylated PDZK1 acts as a tumorigenic gene in glioma by interacting with AKT1.
Literature	GENERIF_SUMMARY	DNAJC24 acts directly with PCNA and promotes malignant progression of LUAD by activating phosphorylation of AKT.
Literature	GENERIF_SUMMARY	Delivery of AKT1 phospho-forms to human cells reveals differential substrate selectivity.
Literature	GENERIF_SUMMARY	A miRNA-21-Mediated PTEN/Akt/NF-kappaB Axis Promotes Chronic Obstructive Pulmonary Disease Pathogenesis.
Literature	GENERIF_SUMMARY	Aberrant methylation and expression of TNXB promote chondrocyte apoptosis and extracullar matrix degradation in hemophilic arthropathy via AKT signaling.
Literature	GENERIF_SUMMARY	Pleckstrin-2 Mediates the Activation of AKT in Prostate Cancer and Is Repressed by Androgen Receptor.
Literature	GENERIF_SUMMARY	CRLF1 bridges AKT and mTORC2 through SIN1 to inhibit pyroptosis and enhance chemo-resistance in ovarian cancer.
Literature	GENERIF_SUMMARY	NaV1.1 contributes to the cell cycle of human mesenchymal stem cells by regulating AKT and CDK2.
Literature	GENERIF_SUMMARY	The mitochondrial redistribution of ENOS is regulated by AKT1 and dimer status.

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Pathways	KEGG_PATHWAY	EGFR tyrosine kinase inhibitor resistance
Pathways	KEGG_PATHWAY	Endocrine resistance
Pathways	KEGG_PATHWAY	Platinum drug resistance
Pathways	KEGG_PATHWAY	MAPK signaling pathway
Pathways	KEGG_PATHWAY	ErbB signaling pathway
Pathways	KEGG_PATHWAY	Ras signaling pathway
Pathways	KEGG_PATHWAY	Rap1 signaling pathway
Pathways	KEGG_PATHWAY	cGMP-PKG signaling pathway
Pathways	KEGG_PATHWAY	cAMP signaling pathway
Pathways	KEGG_PATHWAY	Chemokine signaling pathway
Pathways	KEGG_PATHWAY	HIF-1 signaling pathway
Pathways	KEGG_PATHWAY	FoxO signaling pathway
Pathways	KEGG_PATHWAY	Sphingolipid signaling pathway
Pathways	KEGG_PATHWAY	Phospholipase D signaling pathway
Pathways	KEGG_PATHWAY	Autophagy - animal
Pathways	KEGG_PATHWAY	mTOR signaling pathway
Pathways	KEGG_PATHWAY	PI3K-Akt signaling pathway
Pathways	KEGG_PATHWAY	AMPK signaling pathway
Pathways	KEGG_PATHWAY	Apoptosis
Pathways	KEGG_PATHWAY	Longevity regulating pathway
Pathways	KEGG_PATHWAY	Longevity regulating pathway - multiple species
Pathways	KEGG_PATHWAY	Cellular senescence
Pathways	KEGG_PATHWAY	Adrenergic signaling in cardiomyocytes
Pathways	KEGG_PATHWAY	VEGF signaling pathway
Pathways	KEGG_PATHWAY	Apelin signaling pathway
Pathways	KEGG_PATHWAY	Osteoclast differentiation
Pathways	KEGG_PATHWAY	Focal adhesion
Pathways	KEGG_PATHWAY	IgSF CAM signaling
Pathways	KEGG_PATHWAY	Signaling pathways regulating pluripotency of stem cells
Pathways	KEGG_PATHWAY	Platelet activation
Pathways	KEGG_PATHWAY	Neutrophil extracellular trap formation

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Pathways	KEGG_PATHWAY	Toll-like receptor signaling pathway
Pathways	KEGG_PATHWAY	C-type lectin receptor signaling pathway
Pathways	KEGG_PATHWAY	JAK-STAT signaling pathway
Pathways	KEGG_PATHWAY	T cell receptor signaling pathway
Pathways	KEGG_PATHWAY	B cell receptor signaling pathway
Pathways	KEGG_PATHWAY	Fc epsilon RI signaling pathway
Pathways	KEGG_PATHWAY	Fc gamma R-mediated phagocytosis
Pathways	KEGG_PATHWAY	TNF signaling pathway
Pathways	KEGG_PATHWAY	Neurotrophin signaling pathway
Pathways	KEGG_PATHWAY	Cholinergic synapse
Pathways	KEGG_PATHWAY	Dopaminergic synapse
Pathways	KEGG_PATHWAY	Regulation of actin cytoskeleton
Pathways	KEGG_PATHWAY	Insulin signaling pathway
Pathways	KEGG_PATHWAY	Progesterone-mediated oocyte maturation
Pathways	KEGG_PATHWAY	Estrogen signaling pathway
Pathways	KEGG_PATHWAY	Prolactin signaling pathway
Pathways	KEGG_PATHWAY	Thyroid hormone signaling pathway
Pathways	KEGG_PATHWAY	Adipocytokine signaling pathway
Pathways	KEGG_PATHWAY	Glucagon signaling pathway
Pathways	KEGG_PATHWAY	Regulation of lipolysis in adipocytes
Pathways	KEGG_PATHWAY	Relaxin signaling pathway
Pathways	KEGG_PATHWAY	GnRH secretion
Pathways	KEGG_PATHWAY	Insulin resistance
Pathways	KEGG_PATHWAY	Non-alcoholic fatty liver disease
Pathways	KEGG_PATHWAY	AGE-RAGE signaling pathway in diabetic complications
Pathways	KEGG_PATHWAY	Growth hormone synthesis, secretion and action
Pathways	KEGG_PATHWAY	Alcoholic liver disease
Pathways	KEGG_PATHWAY	Carbohydrate digestion and absorption
Pathways	KEGG_PATHWAY	Alzheimer disease
Pathways	KEGG_PATHWAY	Spinocerebellar ataxia
Pathways	KEGG_PATHWAY	Shigellosis

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Pathways	KEGG_PATHWAY	Salmonella infection
Pathways	KEGG_PATHWAY	Yersinia infection
Pathways	KEGG_PATHWAY	Chagas disease
Pathways	KEGG_PATHWAY	Toxoplasmosis
Pathways	KEGG_PATHWAY	Tuberculosis
Pathways	KEGG_PATHWAY	Hepatitis C
Pathways	KEGG_PATHWAY	Hepatitis B
Pathways	KEGG_PATHWAY	Measles
Pathways	KEGG_PATHWAY	Human cytomegalovirus infection
Pathways	KEGG_PATHWAY	Influenza A
Pathways	KEGG_PATHWAY	Human papillomavirus infection
Pathways	KEGG_PATHWAY	Human T-cell leukemia virus 1 infection
Pathways	KEGG_PATHWAY	Kaposi sarcoma-associated herpesvirus infection
Pathways	KEGG_PATHWAY	Herpes simplex virus 1 infection
Pathways	KEGG_PATHWAY	Epstein-Barr virus infection
Pathways	KEGG_PATHWAY	Human immunodeficiency virus 1 infection
Pathways	KEGG_PATHWAY	Pathways in cancer
Pathways	KEGG_PATHWAY	Proteoglycans in cancer
Pathways	KEGG_PATHWAY	Chemical carcinogenesis - receptor activation
Pathways	KEGG_PATHWAY	Chemical carcinogenesis - reactive oxygen species
Pathways	KEGG_PATHWAY	Colorectal cancer
Pathways	KEGG_PATHWAY	Renal cell carcinoma
Pathways	KEGG_PATHWAY	Pancreatic cancer
Pathways	KEGG_PATHWAY	Endometrial cancer
Pathways	KEGG_PATHWAY	Glioma
Pathways	KEGG_PATHWAY	Prostate cancer
Pathways	KEGG_PATHWAY	Melanoma
Pathways	KEGG_PATHWAY	Chronic myeloid leukemia
Pathways	KEGG_PATHWAY	Acute myeloid leukemia
Pathways	KEGG_PATHWAY	Small cell lung cancer
Pathways	KEGG_PATHWAY	Non-small cell lung cancer

Main Category	Category	Terms
Pathways	KEGG_PATHWAY	Breast cancer
Pathways	KEGG_PATHWAY	Hepatocellular carcinoma
Pathways	KEGG_PATHWAY	Gastric cancer
Pathways	KEGG_PATHWAY	Central carbon metabolism in cancer
Pathways	KEGG_PATHWAY	Choline metabolism in cancer
Pathways	KEGG_PATHWAY	PD-L1 expression and PD-1 checkpoint pathway in cancer
Pathways	KEGG_PATHWAY	Diabetic cardiomyopathy
Pathways	KEGG_PATHWAY	Lipid and atherosclerosis
Pathways	KEGG_PATHWAY	Fluid shear stress and atherosclerosis
Pathways	REACTOME_PATHWAY	Apoptosis
Pathways	REACTOME_PATHWAY	Hemostasis
Pathways	REACTOME_PATHWAY	Intrinsic Pathway for Apoptosis
Pathways	REACTOME_PATHWAY	Activation of BAD and translocation to mitochondria
Pathways	REACTOME_PATHWAY	Activation of BH3-only proteins
Pathways	REACTOME_PATHWAY	Signaling by ERBB2
Pathways	REACTOME_PATHWAY	PIP3 activates AKT signaling
Pathways	REACTOME_PATHWAY	Developmental Biology
Pathways	REACTOME_PATHWAY	Cytokine Signaling in Immune system
Pathways	REACTOME_PATHWAY	Adaptive Immune System
Pathways	REACTOME_PATHWAY	Downregulation of ERBB2:ERBB3 signaling
Pathways	REACTOME_PATHWAY	Metabolism
Pathways	REACTOME_PATHWAY	Translocation of SLC2A4 (GLUT4) to the plasma membrane
Pathways	REACTOME_PATHWAY	Tetrahydrobiopterin (BH4) synthesis, recycling, salvage and regulation
Pathways	REACTOME_PATHWAY	Signaling by NOTCH
Pathways	REACTOME_PATHWAY	Signal Transduction
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	Disease
Pathways	REACTOME_PATHWAY	MTOR signalling
Pathways	REACTOME_PATHWAY	Immune System
Pathways	REACTOME_PATHWAY	Regulation of beta-cell development

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Signaling by VEGF
Pathways	REACTOME_PATHWAY	Signaling by WNT
Pathways	REACTOME_PATHWAY	Metabolism of vitamins and cofactors
Pathways	REACTOME_PATHWAY	AKT phosphorylates targets in the cytosol
Pathways	REACTOME_PATHWAY	AKT phosphorylates targets in the nucleus
Pathways	REACTOME_PATHWAY	Negative regulation of the PI3K/AKT network
Pathways	REACTOME_PATHWAY	Membrane Trafficking
Pathways	REACTOME_PATHWAY	TCF dependent signaling in response to WNT
Pathways	REACTOME_PATHWAY	Metabolism of nitric oxide: NOS3 activation and regulation
Pathways	REACTOME_PATHWAY	eNOS activation
Pathways	REACTOME_PATHWAY	Regulation of gene expression in beta cells
Pathways	REACTOME_PATHWAY	AKT-mediated inactivation of FOXO1A
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway
Pathways	REACTOME_PATHWAY	PI3K/AKT Signaling in Cancer
Pathways	REACTOME_PATHWAY	Cellular responses to stress
Pathways	REACTOME_PATHWAY	Integrin signaling
Pathways	REACTOME_PATHWAY	Transcriptional Regulation by TP53
Pathways	REACTOME_PATHWAY	Signaling by GPCR
Pathways	REACTOME_PATHWAY	Deactivation of the beta-catenin transactivating complex
Pathways	REACTOME_PATHWAY	GPCR downstream signalling
Pathways	REACTOME_PATHWAY	Regulation of T cell activation by CD28 family
Pathways	REACTOME_PATHWAY	Co-stimulation by CD28
Pathways	REACTOME_PATHWAY	CD28 dependent PI3K/Akt signaling
Pathways	REACTOME_PATHWAY	Co-inhibition by CTLA4
Pathways	REACTOME_PATHWAY	G beta:gamma signalling through PI3Kgamma
Pathways	REACTOME_PATHWAY	G-protein beta:gamma signalling
Pathways	REACTOME_PATHWAY	VEGFA-VEGFR2 Pathway
Pathways	REACTOME_PATHWAY	Signaling by Interleukins
Pathways	REACTOME_PATHWAY	Butyrate Response Factor 1 (BRF1) binds

Main Category	Category	Terms
		and destabilizes mRNA
Pathways	REACTOME_PATHWAY	Regulation of mRNA stability by proteins that bind AU-rich elements
Pathways	REACTOME_PATHWAY	KSRP (KHSRP) binds and destabilizes mRNA
Pathways	REACTOME_PATHWAY	Mitotic G1 phase and G1/S transition
Pathways	REACTOME_PATHWAY	VEGFR2 mediated vascular permeability
Pathways	REACTOME_PATHWAY	Programmed Cell Death
Pathways	REACTOME_PATHWAY	TP53 Regulates Metabolic Genes
Pathways	REACTOME_PATHWAY	Regulation of TP53 Activity
Pathways	REACTOME_PATHWAY	Vesicle-mediated transport
Pathways	REACTOME_PATHWAY	Diseases of signal transduction by growth factor receptors and second messengers
Pathways	REACTOME_PATHWAY	Infectious disease
Pathways	REACTOME_PATHWAY	Constitutive Signaling by AKT1 E17K in Cancer
Pathways	REACTOME_PATHWAY	Interleukin-4 and Interleukin-13 signaling
Pathways	REACTOME_PATHWAY	Regulation of TP53 Degradation
Pathways	REACTOME_PATHWAY	Regulation of TP53 Activity through Acetylation
Pathways	REACTOME_PATHWAY	Regulation of TP53 Activity through Association with Co-factors
Pathways	REACTOME_PATHWAY	Regulation of TP53 Expression and Degradation
Pathways	REACTOME_PATHWAY	PTEN Regulation
Pathways	REACTOME_PATHWAY	PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling
Pathways	REACTOME_PATHWAY	Cyclin E associated events during G1/S transition
Pathways	REACTOME_PATHWAY	G1/S Transition
Pathways	REACTOME_PATHWAY	S Phase
Pathways	REACTOME_PATHWAY	Cell Cycle, Mitotic
Pathways	REACTOME_PATHWAY	Cyclin A:Cdk2-associated events at S phase entry
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)
Pathways	REACTOME_PATHWAY	Platelet activation, signaling and

Main Category	Category	Terms
		aggregation
Pathways	REACTOME_PATHWAY	Platelet Aggregation (Plug Formation)
Pathways	REACTOME_PATHWAY	Signaling by PTK6
Pathways	REACTOME_PATHWAY	PTK6 Regulates RTKs and Their Effectors AKT1 and DOK1
Pathways	REACTOME_PATHWAY	Downregulation of ERBB2 signaling
Pathways	REACTOME_PATHWAY	RAB GEFs exchange GTP for GDP on RABs
Pathways	REACTOME_PATHWAY	Transcriptional regulation by RUNX2
Pathways	REACTOME_PATHWAY	ESR-mediated signaling
Pathways	REACTOME_PATHWAY	RUNX2 regulates genes involved in cell migration
Pathways	REACTOME_PATHWAY	Regulation of PTEN stability and activity
Pathways	REACTOME_PATHWAY	Metabolism of RNA
Pathways	REACTOME_PATHWAY	Cellular responses to stimuli
Pathways	REACTOME_PATHWAY	Metabolism of cofactors
Pathways	REACTOME_PATHWAY	Intracellular signaling by second messengers
Pathways	REACTOME_PATHWAY	Signaling by Non-Receptor Tyrosine Kinases
Pathways	REACTOME_PATHWAY	Signaling by Nuclear Receptors
Pathways	REACTOME_PATHWAY	Signaling by Receptor Tyrosine Kinases
Pathways	REACTOME_PATHWAY	Rab regulation of trafficking
Pathways	REACTOME_PATHWAY	Extra-nuclear estrogen signaling
Pathways	REACTOME_PATHWAY	Signaling by NOTCH4
Pathways	REACTOME_PATHWAY	Negative regulation of NOTCH4 signaling
Pathways	REACTOME_PATHWAY	FLT3 Signaling
Pathways	REACTOME_PATHWAY	FOXO-mediated transcription
Pathways	REACTOME_PATHWAY	Regulation of localization of FOXO transcription factors
Pathways	REACTOME_PATHWAY	Estrogen-dependent nuclear events downstream of ESR-membrane signaling
Pathways	REACTOME_PATHWAY	SARS-CoV Infections
Pathways	REACTOME_PATHWAY	SARS-CoV-2 Infection
Pathways	REACTOME_PATHWAY	SARS-CoV-2-host interactions
Pathways	REACTOME_PATHWAY	Cellular response to chemical stress

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	KEAP1-NFE2L2 pathway
Pathways	REACTOME_PATHWAY	SARS-CoV-2 targets host intracellular signalling and regulatory pathways
Pathways	REACTOME_PATHWAY	Viral Infection Pathways
Pathways	REACTOME_PATHWAY	Mitochondrial unfolded protein response (UPRmt)
Pathways	REACTOME_PATHWAY	Cellular responses to mechanical stimuli
Pathways	REACTOME_PATHWAY	High laminar flow shear stress activates signaling by PIEZO1 and PECAM1:CDH5:KDR in endothelial cells
Pathways	REACTOME_PATHWAY	Mechanical load activates signaling by PIEZO1 and integrins in osteocytes
Pathways	REACTOME_PATHWAY	Response of endothelial cells to shear stress
Pathways	WIKIPATHWAYS	IL5 signaling
Pathways	WIKIPATHWAYS	Androgen receptor signaling
Pathways	WIKIPATHWAYS	AMP activated protein kinase signaling
Pathways	WIKIPATHWAYS	Regulation of toll like receptor signaling
Pathways	WIKIPATHWAYS	Target of rapamycin signaling
Pathways	WIKIPATHWAYS	Angiogenesis
Pathways	WIKIPATHWAYS	MicroRNAs in cardiomyocyte hypertrophy
Pathways	WIKIPATHWAYS	Transcription factors regulate miRNAs related to cardiac hypertrophy
Pathways	WIKIPATHWAYS	Integrin mediated cell adhesion
Pathways	WIKIPATHWAYS	IL1 signaling
Pathways	WIKIPATHWAYS	Integrated cancer pathway
Pathways	WIKIPATHWAYS	Sterol regulatory element binding proteins SREBP signaling
Pathways	WIKIPATHWAYS	Integrated breast cancer pathway
Pathways	WIKIPATHWAYS	Angiogenesis overview
Pathways	WIKIPATHWAYS	RANKL RANK signaling
Pathways	WIKIPATHWAYS	Thyroid stimulating hormone TSH signaling
Pathways	WIKIPATHWAYS	Leptin signaling
Pathways	WIKIPATHWAYS	Follicle stimulating hormone FSH signaling
Pathways	WIKIPATHWAYS	TNF related weak inducer of apoptosis TWEAK signaling

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Prolactin signaling
Pathways	WIKIPATHWAYS	Microtubule cytoskeleton regulation
Pathways	WIKIPATHWAYS	IL7 signaling
Pathways	WIKIPATHWAYS	Alzheimer 39 s disease and miRNA effects
Pathways	WIKIPATHWAYS	IL17 signaling
Pathways	WIKIPATHWAYS	Thymic stromal lymphopoietin TSLP signaling
Pathways	WIKIPATHWAYS	Glioblastoma signaling
Pathways	WIKIPATHWAYS	Androgen receptor network in prostate cancer
Pathways	WIKIPATHWAYS	B cell receptor signaling
Pathways	WIKIPATHWAYS	TNF alpha signaling
Pathways	WIKIPATHWAYS	AGE RAGE pathway
Pathways	WIKIPATHWAYS	IL11 signaling
Pathways	WIKIPATHWAYS	Corticotropin releasing hormone signaling
Pathways	WIKIPATHWAYS	Butyrate induced histone acetylation
Pathways	WIKIPATHWAYS	Oncostatin M signaling
Pathways	WIKIPATHWAYS	Brain derived neurotrophic factor BDNF signaling
Pathways	WIKIPATHWAYS	Alpha 6 beta 4 signaling
Pathways	WIKIPATHWAYS	Apoptosis
Pathways	WIKIPATHWAYS	Primary focal segmental glomerulosclerosis FSGS
Pathways	WIKIPATHWAYS	T cell receptor and co stimulatory signaling
Pathways	WIKIPATHWAYS	S1P receptor signal transduction
Pathways	WIKIPATHWAYS	Cardiac hypertrophic response
Pathways	WIKIPATHWAYS	Mammary gland development embryonic development stage 1 of 4
Pathways	WIKIPATHWAYS	IL3 signaling
Pathways	WIKIPATHWAYS	Apoptosis related network due to altered Notch3 in ovarian cancer
Pathways	WIKIPATHWAYS	Extracellular vesicle mediated signaling in recipient cells
Pathways	WIKIPATHWAYS	Kit receptor signaling
Pathways	WIKIPATHWAYS	Focal adhesion

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Copper homeostasis
Pathways	WIKIPATHWAYS	RAC1 PAK1 p38 MMP2 pathway
Pathways	WIKIPATHWAYS	Ovarian infertility
Pathways	WIKIPATHWAYS	MECP2 and associated Rett syndrome
Pathways	WIKIPATHWAYS	Wnt signaling NetPath
Pathways	WIKIPATHWAYS	IL6 signaling
Pathways	WIKIPATHWAYS	Hepatitis C and hepatocellular carcinoma
Pathways	WIKIPATHWAYS	Pathways affected in adenoid cystic carcinoma
Pathways	WIKIPATHWAYS	Wnt beta catenin signaling in leukemia
Pathways	WIKIPATHWAYS	BDNF TrkB signaling
Pathways	WIKIPATHWAYS	Physico chemical features and toxicity associated pathways
Pathways	WIKIPATHWAYS	MAPK signaling
Pathways	WIKIPATHWAYS	PI3K AKT mTOR signaling and therapeutic opportunities in prostate cancer
Pathways	WIKIPATHWAYS	Factors and pathways affecting insulin like growth factor IGF1 Akt signaling
Pathways	WIKIPATHWAYS	TGF beta signaling in thyroid cells for epithelial mesenchymal transition
Pathways	WIKIPATHWAYS	T cell antigen receptor TCR pathway during Staphylococcus aureus infection
Pathways	WIKIPATHWAYS	Apoptosis regulation by parathyroid hormone related protein
Pathways	WIKIPATHWAYS	RIOK1 and RIOK2 in EGFR and PI3K mediated tumorigenesis
Pathways	WIKIPATHWAYS	4 hydroxytamoxifen dexamethasone and retinoic acids regulation of p27 expression
Pathways	WIKIPATHWAYS	VEGFA VEGFR2 signaling
Pathways	WIKIPATHWAYS	Angiopoietin like protein 8 regulatory pathway
Pathways	WIKIPATHWAYS	Chemokine signaling
Pathways	WIKIPATHWAYS	Embryonic stem cell pluripotency pathways
Pathways	WIKIPATHWAYS	Focal adhesion PI3K Akt mTOR signaling
Pathways	WIKIPATHWAYS	Leptin insulin signaling overlap
Pathways	WIKIPATHWAYS	IL4 signaling
Pathways	WIKIPATHWAYS	Lipid metabolism pathway

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Clear cell renal cell carcinoma pathways
Pathways	WIKIPATHWAYS	Fibrin complement receptor 3 signaling
Pathways	WIKIPATHWAYS	PI3K AKT mTOR vitamin D3 signaling
Pathways	WIKIPATHWAYS	Endometrial cancer
Pathways	WIKIPATHWAYS	PI3K Akt signaling
Pathways	WIKIPATHWAYS	Somatroph axis GH and its relationship to dietary restriction and aging
Pathways	WIKIPATHWAYS	Caloric restriction and aging
Pathways	WIKIPATHWAYS	MET in type 1 papillary renal cell carcinoma
Pathways	WIKIPATHWAYS	Chromosomal and microsatellite instability in colorectal cancer
Pathways	WIKIPATHWAYS	Ebola virus infection in host
Pathways	WIKIPATHWAYS	Ras signaling
Pathways	WIKIPATHWAYS	Epithelial to mesenchymal transition in colorectal cancer
Pathways	WIKIPATHWAYS	Non small cell lung cancer
Pathways	WIKIPATHWAYS	Breast cancer pathway
Pathways	WIKIPATHWAYS	Pancreatic adenocarcinoma pathway
Pathways	WIKIPATHWAYS	Acute viral myocarditis
Pathways	WIKIPATHWAYS	Neovascularization processes
Pathways	WIKIPATHWAYS	EGF EGFR signaling
Pathways	WIKIPATHWAYS	Nonalcoholic fatty liver disease
Pathways	WIKIPATHWAYS	Resistin as a regulator of inflammation
Pathways	WIKIPATHWAYS	Relationship between inflammation COX 2 and EGFR
Pathways	WIKIPATHWAYS	Synaptic signaling associated with autism spectrum disorder
Pathways	WIKIPATHWAYS	Fragile X syndrome
Pathways	WIKIPATHWAYS	Neural crest cell migration during development
Pathways	WIKIPATHWAYS	Neural crest cell migration in cancer
Pathways	WIKIPATHWAYS	Translation inhibitors in chronically activated PDGFRA cells
Pathways	WIKIPATHWAYS	Small cell lung cancer
Pathways	WIKIPATHWAYS	Gastrin signaling

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Hepatitis B infection
Pathways	WIKIPATHWAYS	Head and neck squamous cell carcinoma
Pathways	WIKIPATHWAYS	Melanoma
Pathways	WIKIPATHWAYS	Endochondral ossification
Pathways	WIKIPATHWAYS	Netrin UNC5B signaling
Pathways	WIKIPATHWAYS	EGFR tyrosine kinase inhibitor resistance
Pathways	WIKIPATHWAYS	Endochondral ossification with skeletal dysplasias
Pathways	WIKIPATHWAYS	Insulin signaling
Pathways	WIKIPATHWAYS	Regucalcin in proximal tubule epithelial kidney cells
Pathways	WIKIPATHWAYS	Host pathogen interaction of human coronaviruses apoptosis
Pathways	WIKIPATHWAYS	Kisspeptin kisspeptin receptor system in the ovary
Pathways	WIKIPATHWAYS	IL2 signaling
Pathways	WIKIPATHWAYS	Perturbations to host cell autophagy induced by SARS CoV 2 proteins
Pathways	WIKIPATHWAYS	Burn wound healing
Pathways	WIKIPATHWAYS	Modulators of TCR signaling and T cell activation
Pathways	WIKIPATHWAYS	Kallmann syndrome
Pathways	WIKIPATHWAYS	Neuroinflammation and glutamatergic signaling
Pathways	WIKIPATHWAYS	Pleural mesothelioma
Pathways	WIKIPATHWAYS	T cell activation SARS CoV 2
Pathways	WIKIPATHWAYS	Network map of SARS CoV 2 signaling
Pathways	WIKIPATHWAYS	Alzheimer 39 s disease
Pathways	WIKIPATHWAYS	NRP1 triggered signaling in pancreatic cancer
Pathways	WIKIPATHWAYS	Urotensin II mediated signaling
Pathways	WIKIPATHWAYS	Roles of ceramides in development of insulin resistance
Pathways	WIKIPATHWAYS	Resolvin E1 and resolvin D1 signaling decrease inflammation
Pathways	WIKIPATHWAYS	Modulation of PI3K Akt mTOR signaling by bioactive sphingolipids

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Dravet syndrome
Pathways	WIKIPATHWAYS	Clock controlled autophagy in bone metabolism
Pathways	WIKIPATHWAYS	Targeted agents in triple negative breast cancer
Pathways	WIKIPATHWAYS	Warburg effect modulated by deubiquitinating enzymes and their substrates
Pathways	WIKIPATHWAYS	Acute myeloid leukemia
Pathways	WIKIPATHWAYS	Sildenafil treatment
Pathways	WIKIPATHWAYS	IL26 signaling
Pathways	WIKIPATHWAYS	Macrophage stimulating protein MSP signaling
Pathways	WIKIPATHWAYS	JAK STAT signaling in the regulation of beta cells
Pathways	WIKIPATHWAYS	Involvement of secretase in neurodegenerative diseases
Pathways	WIKIPATHWAYS	Osteoarthritic chondrocyte hypertrophy
Pathways	WIKIPATHWAYS	Growth factors and hormones in beta cell proliferation
Pathways	WIKIPATHWAYS	ADHD and autism ASD pathways
Pathways	WIKIPATHWAYS	IL19 signaling
Pathways	WIKIPATHWAYS	HDAC6 interactions in the central nervous system
Pathways	WIKIPATHWAYS	Cancer pathways
Pathways	WIKIPATHWAYS	GNAQ pathways in port wine stain
Pathways	WIKIPATHWAYS	Artemisinin inhibition of uveal melanoma
Pathways	WIKIPATHWAYS	Effect of omega 3 PUFA on Huntington 39 s disease pathways
Pathways	WIKIPATHWAYS	Catabolism of skeletal muscle in cachexia
Pathways	WIKIPATHWAYS	Hallmark of cancer sustaining proliferative signaling
Pathways	WIKIPATHWAYS	Molecular pathway for oxidative stress
Pathways	WIKIPATHWAYS	Inhibition of GNAQ regulated signaling in uveal melanoma
Pathways	WIKIPATHWAYS	Effect of dasatinib on the BCR ABL signaling pathway
Pathways	WIKIPATHWAYS	Tyrosine kinase inhibitors in glioblastoma

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Targeted therapy in breast cancer
Pathways	WIKIPATHWAYS	Cyclin dependent kinase 4 6 inhibitors in breast cancer
Pathways	WIKIPATHWAYS	AKT signaling and ARTD family members
Pathways	WIKIPATHWAYS	BDNF signaling in neurodevelopment
Pathways	WIKIPATHWAYS	Deregulating cellular metabolism through non genotoxic carcinogens
Pathways	WIKIPATHWAYS	Sodium channel in epilepsy
Pathways	WIKIPATHWAYS	EPO receptor signaling
Pathways	WIKIPATHWAYS	Notch signaling
Pathways	WIKIPATHWAYS	ErbB signaling
Pathways	WIKIPATHWAYS	T cell receptor signaling
Pathways	WIKIPATHWAYS	DNA damage response
Pathways	WIKIPATHWAYS	DNA damage response only ATM dependent
Pathways	WIKIPATHWAYS	Estrogen signaling
Pathways	WIKIPATHWAYS	Toll like receptor signaling
Protein Domains	INTERPRO	Prot_kinase_dom
Protein Domains	INTERPRO	AGC-kinase_C
Protein Domains	INTERPRO	PH_domain
Protein Domains	INTERPRO	Ser/Thr_kinase_AS
Protein Domains	INTERPRO	Kinase-like_dom_sf
Protein Domains	INTERPRO	PH-like_dom_sf
Protein Domains	INTERPRO	Protein_kinase_ATP_BS
Protein Domains	INTERPRO	Pkinase_C
Protein Domains	INTERPRO	Akt1
Protein Domains	INTERPRO	PH_PKB
Protein Domains	INTERPRO	PI-Interact_SigTrans_Reg

Main Category	Category	Terms
Protein Domains	SMART	S_TK_X
Protein Domains	SMART	S_TKc
Protein Domains	SMART	PH